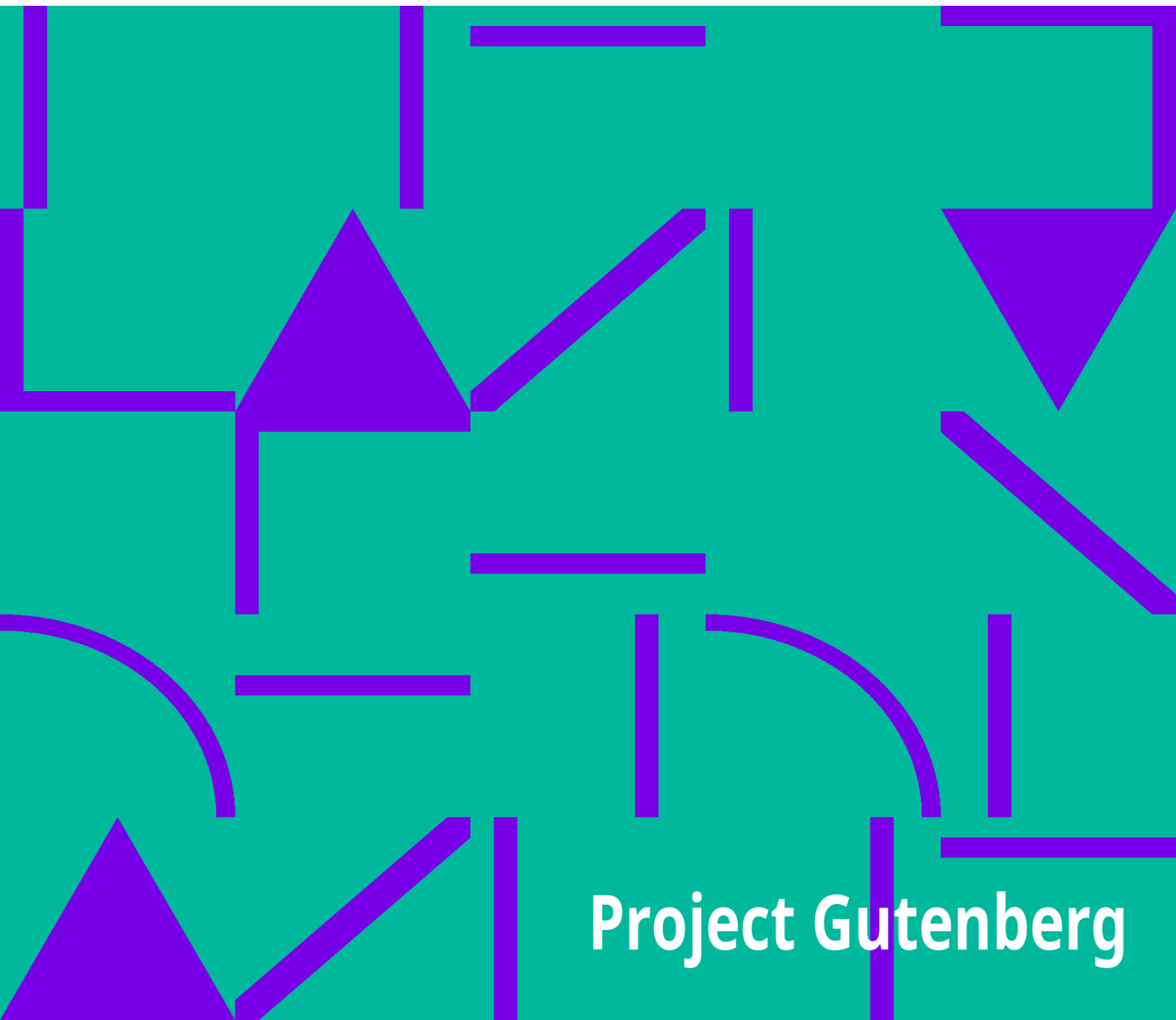


The Plants of Michigan

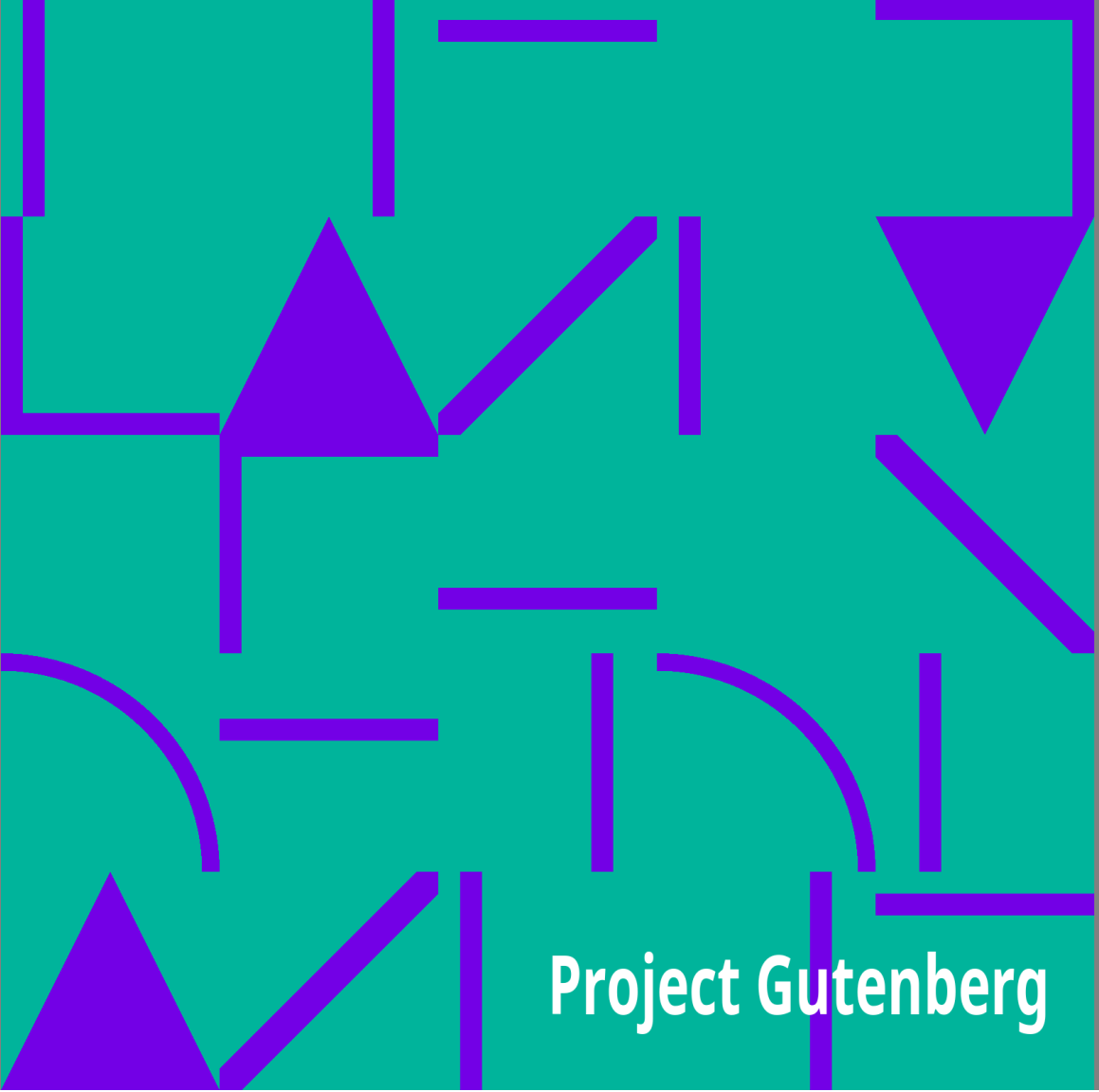
Henry A. Gleason

The background of the lower half of the image is a solid teal color. Overlaid on this teal background is a complex, abstract pattern of yellow geometric shapes. These shapes include various sizes of triangles, rectangles, and curved segments, some of which are nested or overlapping, creating a dynamic and modern visual effect.

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The Plants of Michigan

Henry A. Gleason

The lower half of the cover features a complex, abstract geometric pattern. It consists of various shapes including triangles, rectangles, and curved lines in shades of teal, purple, and grey, creating a modern, geometric aesthetic.

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*** START OF THE PROJECT GUTENBERG EBOOK THE PLANTS
OF MICHIGAN ***

THE PLANTS OF MICHIGAN

**SIMPLE KEYS FOR THE
IDENTIFICATION OF THE NATIVE SEED
PLANTS
OF THE STATE**

**By HENRY ALLAN GLEASON, Ph. D.
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1918

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PREFACE

This book is not intended for the expert botanist. He should consult one of the regular Manuals which give full descriptions of each species of plant.

Neither is it intended for the merely curious. Only those who have sufficient interest in a plant to observe it can find its name by this book.

Furthermore, it is not a textbook. It does not attempt to convey botanical information, but offers merely an opportunity to learn the names of plants.

Its mission is fully accomplished if, through its use, students, vacationists, and plant-lovers in general are able to recognize by name the plants about them.

HOW TO USE THE BOOK

One recognizes a plant by the presence of structural features peculiar to itself, and not found on any other kind of plant. In such a book as this, these characters are given one or a few at a time, and contrasted with the characters which other sorts of plants possess. Such a presentation is called a Key, and by its proper use the name may be learned of any plant considered in it. This process is called Identification.

Keys are constructed in several different ways, although the principle of all is the same. In this book, the user will begin with lines 1a and 1b on the page headed Key to the Groups. Each of these lines includes some descriptive matter, but only one of them can apply to the plant being identified. For example, if the plant to be identified is an Oak, line 1a will apply perfectly, and the same line will also apply to any other kind of tree or to any shrub. But if the plant is a Violet, a Buttercup, or any other herb, line 1b agrees and line 1a will not apply. At the end of each line is a reference to be consulted next. If the plant is a tree or shrub, one turns accordingly to Group 1, on page ix, and begins again at the first number

given. If the plant is an herb, he follows the reference to line 2, just below, and again compares the plants with lines 2a and 2b.

Under every number at least two lines of description are given, designated a and b, and under a few numbers additional lines appear, designated c, d, etc. In every case, the user of the key will select from the different lines under the same number that particular line which agrees with the structure of the plant, and follow up the reference given at the end of that line. Eventually one finds at the end of a line, instead of a number, the name of a family of plants, to which this particular plant belongs, and then turns over to the page where this family is treated. Under each family is a similar key, to be followed in exactly the same way, until finally one finds instead of a number the common name and the scientific name of the plant in hand. The process of identification is now completed, and the student has found the name of the plant.

In some cases, a reference is made in the key to a particular portion of the family key. One then turns directly to this particular number in the family key, and continues his identification in the usual way.

As a definite example of the use of the key, suppose that one has in hand a branch of the White Oak, and that he does not know its name. To determine its name, he will trace it through the following steps in the key. Under the Key to the Groups, it agrees with line 1a, which refers to Group 1, Woody Plants. Under this group it agrees in structure with line 1a, which refers to 2; with line 2b, referring to 21; with line 21b, referring to 22; with 22b, referring to 29; with 29b, referring to 32; with 32c, referring to 47; with 47b, referring to 48; with 48b, referring to 51; with 51b, referring to number 1b in the Beech Family. Turning to the proper number in this family, the plant is referred to line 3; it agrees with line 3c, referring to 10; with 10a, referring to 11; and with 11b, which gives the name of the plant. White Oak, *Quercus alba*.

As a second example, suppose one has a common yellow-flowered plant blooming on lawns and roadsides in spring. Under the Key to the Groups, it agrees with 1b, referring to 2; with 2b, referring to 3; and its net-veined leaves place it in 3b, referring to Group 4, Dicotyledones. Under this group, its basal leaves place it in 1b, referring to 2; its simple leaves in 2b,

referring to 18; the absence of stem-leaves places it in 18b, referring to 23; its solitary flowers on each flower-stalk place it in 23b, referring to 24; its yellow flowers agree with 24a, referring to 25; and its milky juice refers the plant to number 16, in the Composite Family. In the key to this family, its lobed leaves agree with 16b, referring on to 17; its large flowers with 17b, giving one the common name Dandelion, and referring on to 18 to determine which kind of Dandelion the plant may be.

At some point in the key there will be found for each plant a statement in parentheses. This is general information concerning the height of the plant, the color of the flowers, or the season of bloom. It must be remembered that the height of plants is subject to great variation; that most plants have white-flowered varieties; and that the month of bloom depends largely on the latitude and the climate. Therefore this general information should not be used as means of distinguishing species.

The names of plants. Each plant bears a scientific name. This is composed of two parts and is usually of Latin or Greek derivation. In some cases these names are taken directly from the Latin language, as *Quercus*, the Oak, or *Acer*, the Maple. In other cases the name may indicate some characteristic feature of the plant, as *Polygonum*, many joints, for the Knotweed, or *Ammophila*, sand-loving, for the Beach Grass.

An English name is also given for almost every kind of plant. In a few cases there is no accepted English name, and none has been given. In many cases the same English name applies to several kinds of plants and has been repeated for each. When this is so, the common name is given in the key in parentheses before the scientific name is reached. Thus, if one is satisfied to know merely that his plant is a dandelion, he learns it in line 17b of the key to the Composite Family, but to discover which kind of a dandelion he has, he must follow through the key and use the scientific name.

There is in this book, therefore, no necessity of learning or using scientific names. The less critical may be satisfied with an English name, and others may use the scientific names as they see fit.

Possible Errors. In using this book, care must be taken to compare all the lines under each number with the plant, and to use judgment in selecting the right one. While faulty observation or poor judgment may lead to error, a

mistake is usually due to carelessness in not following correctly the reference at the end of the line chosen. If one reaches a number in which none of the lines of description agrees with the plant, it is very probable that he has made a mistake at an earlier stage of the identification, and he should then begin anew.

It has been the intention of the author to make the key as nearly as possible proof against errors of judgment. For example, the Indian Turnip may be sought under either Group 3 or Group 4; the Matrimony Vine may be identified either as a shrub or as an herb, and numerous other similar examples may be discovered.

Botanical Information Needed. It is presumed that those using the book will be familiar with the parts of the flower and with the commoner descriptive terms applied to leaves. Unusual terms have been avoided as far as possible, but those which do occur, as well as the simpler ones, are explained in the glossary.

In general, only those characters have been used in the keys which can be observed without a magnifying glass and without dissection of the flower.

In several groups of plants, reference is made to the fuller descriptions to be found in the Manuals. The standard manuals for Michigan are Gray's New Manual of Botany, 7th edition (American Book Company, \$2.50), and Britton and Brown, Illustrated Flora of the Northern States and Canada (Chas. Scribner's Sons, \$13.50). These books may be consulted in most school or public libraries.

All dimensions are expressed in the metric system. For convenience, it may be stated that 25 millimeters (mm.) are about equal to one inch; 1 centimeter (cm.) to two-fifths of an inch; 1 decimeter (dm.) to 4 inches; and 1 meter (m.) is a little more than 3 feet.

KEY TO GROUPS

1a. Trees, shrubs, or woody climbers,
with stems which last from
year to year

Group 1, **WOODY**
PLANTS, p. ix

1b. Herbaceous plants, with stems
which live above ground only
a single season — 2.

2a. Plants with unusual
habits or structures,
including leafless,
colorless,
submerged,
floating, parasitic,
or hollow-leaved
plants

Group 2, **UNUSUAL**
PLANTS, p. xxiii.

2b. Ordinary terrestrial or
swamp plants,
without unusual
structural
peculiarities — 3.

3a. Leaves parallel-veined (or net-
veined in a few species); parts
of the flower usually in threes
or sixes, never in fives; wood-
fibers scattered through the
stem; seed with one
cotyledon. All plants with
definitely parallel-veined
leaves may be identified
through this division, unless
the parts of the flower are
distinctly in fives.

Group 3,
MONOCOTYLEDONES,
p. xxvii.

3b. Leaves net-veined (or parallel-
veined in a few species); parts
of the flower usually in fours
or fives; wood-fibers arranged
in a circle in the stem; seeds
with two cotyledons. All

Group 4,
DICOTYLEDONES,
p. xxx.

plants with definitely net-veined leaves may be identified through this division.

Note:—In order to avoid possible chances of error, many plants have been treated under both of the above groups. The following hints may also be useful in distinguishing Groups 3 and 4:

All herbaceous plants with deeply lobed, dissected or compound leaves may be sought under the Dicotyledones.

All herbaceous plants with five stamens in each flower, or with seven or more stamens and one ovary in each flower, may be sought under Dicotyledones.

GROUP 1, WOODY PLANTS

- 1a. Trees, with erect stem and central trunk, attaining a height of 6 m. (20 ft.) or more [—2.](#)
- 1b. Shrubs or woody vines, without true tree habit, or attaining heights of less than 6 m. (20 ft.) [—52.](#)
 - 2a. Key for use in earliest spring, for trees which have flowers but no leaves [—3.](#)
 - 2b. Key for use with trees bearing leaves [—21.](#)
- 3a. Flowers in catkins, without brightly colored or petal-like parts [—4.](#)
- 3b. Flowers not in catkins, either with or without petals [—14.](#)

- 4a. Leaf-scars and lateral buds 2-ranked, i. e., in two longitudinal rows with the third leaf above the first —5.
- 4b. Leaf-scars and buds in three or more longitudinal rows —9.
- 5a. From 1 to 3 bud-scales visible on each leaf-bud 2b, in FAGACEAE, p. 22.
- 5b. From 4 to 7 bud-scales visible on each leaf-bud —6.
- 5c. From 8 to 20 bud-scales visible on each leaf-bud; buds long and slender; bark of the trunk smooth —8.
- 6a. Bundle-scars 5 or more 2b, in URTICACEAE, p. 23.
- 6b. Bundle-scars 3 —7.
- 7a. Twigs bearing numerous dwarf branches thickly covered with crowded leaf-scars (Birch) BETULACEAE, p. 21.
- 7b. Twigs without dwarf branches (Ironwood) 9a, in BETULACEAE, p. 22.
- 8a. Trunk cylindrical or nearly so 2a, in FAGACEAE, p. 22.
- 8b. Trunk prominently fluted with longitudinal ridges 1b, in BETULACEAE, p. 21.
- 9a. Bundle-scars 3 in each leaf-scar —10.
- 9b. Bundle scars more than 3 in each leaf-scar —12.
- 10a. Pith divided into separate cavities by 1a, in JUGLANDACEAE, p.

- transverse partitions 21.
- 10b. Pith not partitioned
— 11.
- 11a. Buds small, with only one external bud-scale [1b, in SALICACEAE](#), p. 19.
- 11b. Buds with more than one outer bud-scale [1a, in SALICACEAE](#), p. 19.
- 12a. Buds clustered near the tips of the twigs [1b, in FAGACEAE](#), p. 22.
- 12b. Buds not clustered at the tips of the twigs
— 13.
- 13a. Buds with about 3 visible bud-scales [2b, in FAGACEAE](#), p. 22.
- 13b. Terminal bud large, with 4 or more visible bud-scales [1b, in JUGLANDACEAE](#), p. 21.
- 14a. Flowers conspicuous, brightly colored, at least 8 mm. wide. with both calyx and corolla — 15.
- 14b. Flowers inconspicuous, seldom brightly colored, and then less than 8 mm. wide
— 17.
- 15a. Flowers irregular, pink or red [3a, in LEGUMINOSAE](#), p. 58.
- 15b. Flowers regular, white — 16.
- 16a. Ovary one, superior, in the center of the flower [32b, in ROSACEAE](#), p. 54.
- 16b. Ovary inferior, appearing as a [42b, in ROSACEAE](#), p. 55.

swelling below the
calyx at the summit of
the pedicel

17a. Leaf-scars and buds opposite

— 18.

17b. Leaf-scars and buds alternate

— 19.

18a. Bundle-scar one in each
leaf-scar

1b, in OLEACEAE, p. 88.

18b. Bundle-scars 3 or more
in each leaf-scar

ACERACEAE, p. 70.

19a. Bundle-scar 1 in each leaf-scar

1b, in LAURACEAE, p.
41.

19b. Bundle-scars 3 in each leaf-scar

— 20.

19c. Bundle-scars 5 in each leaf-scar

7a, in
LEGUMINOSAE, p.
58.

20a. Branches thorny

5a, in
LEGUMINOSAE, p.
58.

20b. Branches not thorny

3a, in URTICACEAE,
p. 24.

— 21—

21a. Leaves narrow, needle-like or
scale-like; trees mostly
evergreen

PINACEAE, p. 1.

21b. Leaves broader, flat, never needle-
like or scale-like, falling in
winter — 22.

22a. Leaves compound — 23.

22b. Leaves simple — 29.

23a. Leaves opposite — 24.

23b. Leaves alternate — 26.

24a. Leaves palmately
compound with 5-7

SAPINDACEAE, p. 70.

- leaflets
- 24b. Leaves pinnately compound — [25](#).
- 25a. Leaflets 3-5 [1a, in ACERACEAE](#), p. 70.
- 25b. Leaflets 7-11 [1b, in OLEACEAE](#), p. 88.
- [4a, in](#)
- 26a. Stem or branches thorny [LEGUMINOSAE](#), p. 58.
- 26b. Stem or branches not thorny — [27](#).
- [7a, in](#)
- 27a. Leaflets entire [LEGUMINOSAE](#), p. 58.
- 27b. Leaflets entire except for a few large glandular teeth near their base [SIMARUBACEAE](#), p. 65.
- 27c. Leaflets serrate their entire length — [28](#).
- 28a. Upper leaflets less than 25 mm. wide [4b, in ROSACEAE](#), p. 51.
- 28b. Upper leaflets more than 25 mm. wide [JUGLANDACEAE](#), p. 21.
- 29a. Leaves opposite — [30](#).
- 29b. Leaves alternate — [32](#).
- 30a. Leaves entire [CORNACEAE](#), p. 83.
- 30b. Leaves toothed or lobed, not entire — [31](#).
- 31a. Leaves lobed [1b, in ACERACEAE](#), p. 70.
- [27b, in](#)
- 31b. Leaves merely toothed [CAPRIFOLIACEAE](#), p. 111.
- 32a. Leaves entire — [33](#).

- 32b. Leaves toothed — [36.](#)
- 32c. Leaves lobed — [47.](#)
- 33a. Leaves heart-shape [3a, in LEGUMINOSAE](#), p. 58.
- 33b. Leaves not heart-shape — [34.](#)
 - 34a. Twigs and foliage spicy-aromatic [1b, in LAURACEAE](#), p. 41.
 - 34b. Twigs and foliage not aromatic — [35.](#)
- 35a. Pith 5-angled; fruit an acorn [3a, in FAGACEAE](#), p. 22.
- 35b. Pith not 5-angled; fruit a berry [1a, in CORNACEAE](#), p. 83.
 - 36a. Leaves oblique at base, i. e., one side of the leaf larger than the other — [37.](#)
 - 36b. Leaves symmetrical, not oblique at base — [38.](#)
- 37a. Leaves heart-shape, about as broad as long [TILIACEAE](#), p. 72.
- 37b. Leaves oval or ovate, much longer than wide [1a, in URTICACEAE](#), p. 23.
- 38a. Stems thorny [41b, in ROSACEAE](#), p. 54.
- 38b. Stems not thorny — [39.](#)
- 39a. Leaves finely toothed, with 3-many teeth per centimeter of margin — [40.](#)
- 39b. Leaves coarsely toothed, with 1-2 teeth per centimeter of margin — [46.](#)
 - 40a. Petioles laterally compressed [2a, in SALICACEAE](#), p. 19.

- 40b. Petioles not compressed
— [41.](#)
- 41a. Leaves, or many of them, crowded
on short spur-like branches
— [42.](#)
- 41b. Leaves scattered, not on short
spur-like branches — [43.](#)
 - 42a. Bark of the trunks
separating in thin
papery or leathery
sheets [BETULACEAE](#), p. 21.
 - 42b. Bark of the trunk not
papery or leathery [24b, in ROSACEAE](#), p.
53.
- 43a. Lateral leaf-veins straight and
parallel, and terminating in the
teeth [BETULACEAE](#), p. 21.
- 43b. Lateral veins more or less curved,
and not ending in the teeth
— [44.](#)
 - 44a. Leaves palmately
veined, about as broad
as long; juice
somewhat milky [2b, in URTICACEAE](#),
p. 23.
 - 44b. Leaves pinnately veined;
juice not milky — [45.](#)
- 45a. Willows, with slender leaves and
brittle twigs [8a, in SALICACEAE](#), p.
19.
- 45b. Trees with lanceolate, ovate, or
oblong leaves and tough twigs [30, in ROSACEAE](#), p.
54.
- 45c. Cottonwoods, with broad, heart-
shape or rounded leaves [1a, in SALICACEAE](#), p.
19.
- 46a. Petioles laterally
compressed [2a, in SALICACEAE](#), p.
19.
- 46b. Petioles not compressed;
lateral veins straight [FAGACEAE](#), p. 22.

and parallel, running
directly to the teeth

- 47a. Stem thorny [41b, in ROSACEAE](#), p. 54.
- 47b. Stem not thorny [— 48.](#)
- 48a. Leaves palmately veined
 [— 49.](#)
- 48b. Leaves pinnately veined
 [— 51.](#)
- 49a. Lobes of the leaf entire [1b, in LAURACEAE](#), p. 41.
- 49b. Lobes of the leaf serrate [— 50.](#)
- 50a. Juice somewhat milky;
 lateral buds visible [2b, in URTICACEAE](#), p. 23.
- 50b. Juice not milky; lateral
 buds covered by the [PLATANACEAE](#), p. 51.
 base of the petiole
- 51a. Leaves with 4 large entire lobes;
 stem marked with a ring at each
 node [MAGNOLIACEAE](#), p. 40.
- 51b. Leaves with many lobes; stem not
 ringed [1b, in FAGACEAE](#), p. 22.
- 52—
- 52a. For specimens bearing
 leaves only [— 53.](#)
- 52b. For specimens bearing
 flowers only [— 140.](#)
- 52c. For specimens with both
 leaves and flowers
 [— 155.](#)
- 53a. Leaves narrow, needle-like or
 scale-like, mostly evergreen
 [— 54.](#)
- 53b. Leaves broader, flat or rolled, but
 not needle-like or scale-like
 [— 56.](#)

- 54a. Foliage densely gray-pubescent; low bushy shrubs with yellow flowers [2a, in CISTACEAE](#), p. 74.
- 54b. Foliage green [— 55.](#)
- 55a. Leaves opposite or whorled [PINACEAE](#), p. 1.
- 55b. Leaves scattered on the stem [TAXACEAE](#), p. 2.
- 56a. Twining or climbing vines [— 57.](#)
- 56b. Not climbing or twining [— 66.](#)
- 57a. Thorny vines [— 58.](#)
- 57b. Not thorny [— 60.](#)
- 58a. Climbing by tendrils at the base of the leaves [4b, in LILIACEAE](#), p. 11.
- 58b. Tendrils none [— 59.](#)
- 59a. Leaves simple [5a, in SOLANACEAE](#), p. 100.
- 59b. Leaves compound [8a, in ROSACEAE](#), p. 52.
- 60a. Leaves compound [— 61.](#)
- 60b. Leaves simple [— 63.](#)
- 61a. Leaves with 5 leaflets [1a, in VITACEAE](#), p. 71.
- 61b. Leaves with 3 leaflets [— 62.](#)
- 62a. Plant climbing by tendril-like leaf-stalks [1a, in RANUNCULACEAE](#), p. 35.
- 62b. Plant climbing by hold-fast roots [5b, in ANACARDIACEAE](#), p. 69.
- 63a. Leaves opposite [5a, in CAPRIFOLIACEAE](#), p. 110.
- 63b. Leaves alternate [— 64.](#)

- 64a. Plants climbing by tendrils [1b, in VITACEAE](#), p. 71.
- 64b. Twining plants [— 65.](#)
- 65a. Leaves ovate-oblong, attached by the base [1a, in CELASTRACEAE](#), p. 69.
- 65b. Leaves almost round, peltate near the edge [MENISPERMACEAE](#) p. 40.
- 66a. Leaves opposite [— 67.](#)
- 66b. Leaves alternate [— 84.](#)
- 67a. Leaves compound [— 68.](#)
- 67b. Leaves simple [— 69.](#)
- 68a. Leaflets 3 [STAPHYLEACEAE](#), p. 69.
- 68b. Leaflets 7 or more [1a, in CAPRIFOLIACEAE](#), p. 109.
- 69a. Leaves palmately veined, or at least with a pair of prominent lateral veins from the base [— 70.](#)
- 69b. Leaves pinnately veined [— 72.](#)
- 70a. Leaves not lobed [1b, in RHAMNACEAE](#), p. 71.
- 70b. Leaves more or less lobed [— 71.](#)
- 71a. Leaves with stipules [22a, in CAPRIFOLIACEAE](#), p. 111.
- 71b. Leaves without stipules [2a, in ACERACEAE](#), p. 70.
- 72a. Leaves serrate [— 73.](#)
- 72b. Leaves entire [— 77.](#)
- 73a. Stem thorny [1b, in RHAMNACEAE](#), p. 71.

73b. Stem not thorny — [74.](#)

74a. Bark of the ripe twigs
green

[1b, in](#)
[CELASTRACEAE](#), p.
69.

74b. Bark of the ripe twigs
brown, reddish, or
gray — [75.](#)

75a. Twigs with a prominent hairy
ridge extending downward from
the middle of the line
connecting the petiole bases

[13a, in](#)
[CAPRIFOLIACEAE](#),
p. 110.

75b. Twigs without any pubescent ridge
— [76.](#)

76a. Erect shrubs; leaves
obviously toothed

[11b, in](#)
[CAPRIFOLIACEAE](#),
p. 110.

76b. Spreading shrubs; most
of the leaves entire,
and only here and
there some with
serrate margins

[18b, in](#)
[CAPRIFOLIACEAE](#),
p. 110.

77a. Leaves silvery beneath with a
dense coating of scales

[ELAEAGNACEAE](#), p.
77.

77b. Leaves green beneath, or
somewhat hairy and light-
colored — [78.](#)

78a. Aquatic plant with
lanceolate leaves, and
stems bending over
and into the water

[1a, in](#) [LYTHRACEAE](#),
p. 77.

78b. Not truly aquatic,
although frequently in
wet places — [79.](#)

79a. Leaves evergreen, as shown by
their presence on the older
stems

[18a, in](#) [ERICACEAE](#), p.
85.

- 79b. Leaves deciduous each year — 80.
- 80a. Leaves dotted with translucent dots, easily seen when the leaf is held to the light 1a, in [HYPERICACEAE](#), p. 73.
- 80b. Leaves not dotted with translucent dots — 81.
- 81a. Lateral veins curved forward and running almost parallel to the leaf-margin CORNACEAE, p. 83.
- 81b. Lateral veins spreading, and not paralleling the leaf-margin — 82.
- 82a. Leaves with stipules 1a, in [RUBIACEAE](#), p. 108.
- 82b. Leaves with a prominent ridge connecting them at the base — 83.
- 82c. Leaves with neither stipules nor connecting ridges 1a, in [OLEACEAE](#), p. 88.
- 83a. Bundle-scar one in each leaf-scar 18b, in [CAPRIFOLIACEAE](#), p. 110.
- 83b. Bundle-scars three in each leaf-scar (Honeysuckle) 12, in [CAPRIFOLIACEAE](#), p. 110.
- 84a. Leaves compound — 85.
- 84b. Leaves simple — 95.
- 85a. Stems prickly or thorny — 86.
- 85b. Stems without prickles or thorns — 88.
- 86a. Leaves twice-pinnate 1a, in [ARALIACEAE](#), p. 80.

- 86b. Leaves evenly pinnate [5b, in LEGUMINOSAE](#), p. 58.
- 86c. Leaves odd-pinnate or trifoliate [— 87.](#)
- 87a. Leaflets entire 1a, [1a, in RUTACEAE](#), p. 65.
- 87b. Leaflets serrate 3b, [3b, in ROSACEAE](#), p. 51.
- 88a. Leaflets 3 [— 89.](#)
- 88b. Leaflets 5 to many [— 91.](#)
- 89a. Tall shrubs; leaflets entire or minutely toothed [1b, in RUTACEAE](#), p. 65.
- 89b. Shrubs 2 m. high or less; leaves conspicuously toothed [— 90.](#)
- 90a. Stipules present [17b, in ROSACEAE](#), p. 53.
- 90b. Stipules none [1b, in ANACARDIACEAE](#), p. 68.
- 91a. Leaflets 6-25 mm. long [— 92.](#)
- 91b. Leaflets 30 mm. long, or more [— 93.](#)
- 92a. Leaflets mostly 5, rarely 3 or 7 [7a, in ROSACEAE](#), p. 52.
- 92b. Leaflets mostly 9-19, 15 mm. long or more [26a, in LEGUMINOSAE](#), p. 60.
- 92c. Leaflets mostly 25-45, less than 15 mm. long [8a, in LEGUMINOSAE](#), p. 58.
- 93a. Leaflets entire [3a, in ANACARDIACEAE](#), p. 68.

- 93b. Leaflets entire, except for 1-4
large glandular teeth near their
base [SIMARUBACEAE](#), p.
65.
- 93c. Leaflets toothed throughout [— 94.](#)
- 94a. Juice milky [1a, in](#)
[ANACARDIACEAE](#), p.
68.
- 94b. Juice not milky [3a, in ROSACEAE](#), p.
51.
- 95a. Leaves minute, scale-like,
appressed [2a, in CISTACEAE](#), p.
74.
- 95b. Leaves 3-10 mm. long, spreading,
completely rolled into a tube [EMPETRACEAE](#), p.
68.
- 95c. Leaves normal in shape, green in
color [— 96.](#)
- 96a. Leaves evergreen, as
shown by their [4b, in ERICACEAE](#), p.
presence on the older 84.
parts of the stem
- 96b. Leaves deciduous [— 97.](#)
- 97a. Stems or branches thorny [— 98.](#)
- 97b. Stems or branches without thorns
or thorny leaves [— 103.](#)
- 98a. Leaves conspicuously
palmately veined
[— 99.](#)
- 98b. Leaves pinnately veined,
or sometimes with
smaller lateral veins
arising from the end
of the petiole [— 100.](#)
- 99a. Leaves 5 cm. wide or less [1a, in](#)
[SAXIFRAGACEAE](#), p.
49.
- 99b. Leaves 15 cm. wide or more [1a, in ARALIACEAE](#),
p. 80.

- 100a. Leaves entire [5a, in SOLANACEAE](#),
p. 100.
- 100b. Leaves toothed or
somewhat lobed
[— 101.](#)
- 101a. Leaves with bristly margins;
some of the thorns three-pointed [2a, in
BERBERIDACEAE](#), p.
40.
- 101b. Leaves not bristly on the margin;
thorns not branched [— 102.](#)
 - 102a. Only lateral thorns
present [41b, in ROSACEAE](#), p.
54.
 - 102b. Only terminal thorns
present [2a, in RHAMNACEAE](#),
p. 71.
- 103a. Leaves palmately veined, or with
one or more pairs of lateral
veins from the base of the leaf
[— 104.](#)
- 103b. Leaves pinnately veined [— 111.](#)
 - 104a. Leaves entire [— 105.](#)
 - 104b. Leaves toothed [— 106.](#)
 - 104c. Leaves palmately lobed
[— 108.](#)
- 105a. Foliage fragrant when crushed;
leaves ovate [1b, in LAURACEAE](#), p.
41.
- 105b. Foliage not aromatic; leaves
heart-shape [3b, in
LEGUMINOSAE](#), p.
58.
- 106a. Lateral veins straight
and parallel, running
to the teeth of the leaf [3a, in BETULACEAE](#),
p. 22.
- 106b. Lateral veins curved or
branched, and not
running straight to the
teeth [— 107.](#)

- 107a. Tall shrubs with somewhat milky juice and broadly ovate leaves [2b, in URTICACEAE](#), p. 23.
- 107b. Low shrubs less than 1 m. high with watery juice [1b, in RHAMNACEAE](#), p. 71.
 - 108a. Leaves with 2 or 3 entire lobes, spicy-fragrant when crushed [1b, in LAURACEAE](#), p. 41.
 - 108b. Leaves with milky juice [2b, in URTICACEAE](#), p. 23.
 - 108c. Leaves neither spicy-fragrant nor with milky juice [— 109.](#)
- 109a. Stem covered with brown bristles [29, in ROSACEAE](#), p. 54.
- 109b. Stem not bristly [— 110.](#)
 - 110a. Sides of the petiole strongly decurrent on the stem; bundle-scars crowded or nearly in contact in the leaf-scars [28b, in ROSACEAE](#), p. 53.
 - 110b. Sides of petiole little decurrent or not at all; bundle-scars distinctly separate [2b, in SAXIFRAGACEAE](#), p. 49.
- 111a. Leaves aromatically fragrant when crushed [— 112.](#)
- 111b. Leaves not aromatically fragrant when crushed [— 113.](#)
 - 112a. Leaves broadly obovate, entire; a common woodland shrub [1a, in LAURACEAE](#), p. 41.
 - 112b. Leaves linear-lanceolate or oblanceolate, [MYRICACEAE](#), p. 20.

- conspicuously toothed
or entire
- 113a. Leaves entire — [114.](#)
- 113b. Leaves toothed or lobed — [121.](#)
- 114a. Base of the petiole
covering the axillary
buds; twigs marked
with rings [THYMELAEACEAE](#),
p. 77.
- 114b. Base of petiole not
covering the bud, and
twigs not marked with
rings — [115.](#)
- 115a. Pith with prominent partitions;
tall shrubs with fetid bark;
leaves obovate, 10 cm. long or
larger [ANONACEAE](#), p. 40.
- 115b. Pith not partitioned; leaves
smaller — [116.](#)
- 116a. Leaves waxy or
resinous underneath [24b, in ERICACEAE](#), p.
86.
- 116b. Leaves not waxy or
resinous — [117.](#)
- 117a. Lateral veins curved forward and
almost parallel to the margin of
the leaf [2a, in CORNACEAE](#), p.
83.
- 117b. Lateral veins spreading — [118.](#)
- 118a. Leaves lanceolate or
linear, much longer
than wide — [119.](#)
- 118b. Leaves ovate or
elliptical — [120.](#)
- 119a. Stem weak, spreading or trailing [5a, in SOLANACEAE](#),
p. 100.
- 119b. Stems, or some of them, erect [3b, in
LEGUMINOSAE](#), p.
58.

- 120a. Leaves with purple petioles, which are at least one-fourth as long as the leaf-blade [1a, in AQUIFOLIACEAE](#), p. 69.
- 120b. Leaves with short petioles or sessile [24c, in ERICACEAE](#), p. 86.
- 121a. Lateral veins straight and parallel, mostly ending in the teeth of the leaf [— 122.](#)
- 121b. Lateral veins not straight and parallel [— 129.](#)
 - 122a. Leaves 2-ranked, i. e., in two longitudinal rows, with the third leaf directly above the first [— 123.](#)
 - 122b. Leaves not 2-ranked [— 127.](#)
- 123a. Leaves unsymmetrical, oblique at the base, i. e., with one side of the leaf larger than the other [HAMAMELIDACEAE](#), p. 51.
- 123b. Leaves symmetrical or nearly so at the base [— 124.](#)
 - 124a. Lateral leaf-veins ending in the leaf-teeth [— 125.](#)
 - 124b. Lateral veins not ending in the teeth [42b, in ROSACEAE](#), p. 55.
- 125a. Bark smooth and fluted on the large stems [1b, in BETULACEAE](#), p. 21.
- 125b. Bark rough or warty or glandular [— 126.](#)
 - 126a. Leaves 4 cm. long or less [3a, in BETULACEAE](#), p. 22.
 - 126b. Leaves 5 cm. long or more [9b, in BETULACEAE](#), p. 22.

- 127a. Pith 3-angled [6a, in BETULACEAE](#), p. 22.
- 127b. Pith 5-angled [13b, in FAGACEAE](#), p. 23.
- 127c. Pith cylindrical [— 128.](#)
- 128a. Leaves finely serrate [42b, in ROSACEAE](#), p. 55.
- 128b. Leaves coarsely or doubly serrate [41a, in ROSACEAE](#), p. 54.
- 129a. Leaves coarsely or doubly serrate [— 130.](#)
- 129b. Leaves simply serrate [— 131.](#)
- 130a. Leaves 2-ranked, i. e., in two longitudinal rows, with the third leaf directly over the first [9b, in BETULACEAE](#), p. 22.
- 130b. Leaves not 2-ranked [41a, in ROSACEAE](#), p. 54.
- 131a. Leaves with glands on the petiole or at the base of the leaf-blade [— 132.](#)
- 131b. Leaves with small dark glands on the upper side of the mid-vein [42a, in ROSACEAE](#), p. 54.
- 131c. Leaves without glands [— 133.](#)
- 132a. Willows; with usually slender leaves, frequently conspicuous broad stipules, and lateral buds protected by a single external bud-scale [13, in SALICACEAE](#), p. 20.
- 132b. Plums and cherries; with leaves lanceolate or broader, and [30, in ROSACEAE](#), p. 54.

stipules minute and
falling early in the
season; lateral buds
with more than one
bud-scale

133a. With stipules or with stipular
scars indicating where stipules
have been detached [— 134.](#)

133b. Without stipules or stipular scars
[— 138.](#)

134a. With three bundle-scars
in each leaf-scar
[— 135.](#)

134b. With one bundle-scar
in each leaf-scar
[— 136.](#)

135a. Willows; usually with slender
leaves and twigs and frequently
with large conspicuous stipules; [13, in SALICACEAE,](#)
lateral buds covered by a single p. 20.
external bud-scale

135b. Juneberries; with oblong or ovate
leaves and small stipules which [42b, in ROSACEAE,](#) p.
fall early; lateral buds with 55.
more than one external scale

136a. Leaves mostly entire, [1a, in](#)
only a few here and [AQUIFOLIACEAE,](#) p.
there with low teeth 69.

136b. Leaves sharply toothed
[— 137.](#)

137a. Axillary buds superposed, i. e., [1b, in](#)
with a second one just above the [AQUIFOLIACEAE,](#) p.
first 69.

137b. Axillary buds not superposed [2b, in RHAMNACEAE,](#)
p. 71.

- 138a. Leaves with purple petioles, which are at least one-fourth as long as the blade [1a, in AQUIFOLIACEAE](#), p. 69.
- 138b. Leaves short-petioled or sessile [— 139.](#)
- 139a. Stems erect and straight, unbranched or with very few branches [27, in ROSACEAE](#), p. 53.
- 139b. Stems more or less crooked and freely branched, making a spreading shrub [25, in ERICACEAE](#), p. 86.
- 140 —
- 140a. Flowers appearing in autumn, after the leaves have fallen [HAMAMELIDACEAE](#), p. 51.
- 140b. Flowers appearing in spring, before the leaves have opened [— 141.](#)
- 141a. Flowers in catkins, without brightly colored or petal-like parts [— 142.](#)
- 141b. Flowers not in catkins, either with or without brightly colored or petal-like parts [— 149.](#)
- 142a. Leaves 2-ranked, as shown by the arrangement of buds and leaf-scars in two longitudinal rows, so that the third bud is directly over the first [— 143.](#)
- 142b. Leaves and leaf-scars not 2-ranked [— 146.](#)

- 143a. Bundle-scars three in each leaf-scar — [144.](#)
- 143b. Bundle-scars several in each leaf-scar [2b, in URTICACEAE](#), p. 23.
- 144a. Leaf-buds with only 1-3 visible bud-scales [10, in BETULACEAE](#), p. 22.
- 144b. Leaf-buds with more than 3 visible scales — [145.](#)
- 145a. Bark of the branches smooth and dark gray, the larger stems fluted with projecting longitudinal ridges [1b, in BETULACEAE](#), p. 21.
- 145b. Branches without projecting ridges [4, in BETULACEAE](#), p. 22.
- 146a. Pith 3-angled [8, in BETULACEAE](#), p. 22.
- 146b. Pith 5-angled [13b, in FAGACEAE](#), p. 23.
- 146c. Pith cylindrical — [147.](#)
- 147a. Visible outer bud-scales 2 or more — [148.](#)
- 147b. Buds with a single visible outer scale [13, in SALICACEAE](#), p. 20.
- 148a. Bundle-scars 3 in each leaf-scar [MYRICACEAE](#), p. 20.
- 148b. Bundle-scars more than 3 in each leaf-scar [2b, in URTICACEAE](#), p. 23.
- 149a. Flowers dark red-purple, about 3 cm. wide [ANONACEAE](#), p. 40.
- 149b. Flowers bright pink, irregular, about 1 cm. wide [3a, in LEGUMINOSAE](#), p. 58.
- 149c. Flowers white, with 5 conspicuous petals — [150.](#)

- 149d. Flowers greenish or yellowish,
small, inconspicuous — [152.](#)
- 150a. Ovary 1, superior, i. e.,
in the center of the flower and not
attached to surrounding parts [36, in ROSACEAE](#), p. 54.
- 150b. Ovary inferior,
appearing as a swelling below the
calyx at the apex of the pedicel — [151.](#)
- 151a. Flowers in racemes or solitary [42b, in ROSACEAE](#), p. 55.
- 151b. Flowers in flattened or rounded
branching clusters [42a, in ROSACEAE](#), p. 55.
- 152a. Stems thorny [1a, in RUTACEAE](#), p. 65.
- 152b. Stems not thorny
— [153.](#)
- 153a. Bark pleasantly aromatic when
crushed; perianth with 6 parts
— [154.](#)
- 153b. Bark not pleasantly aromatic;
perianth with 4 short lobes [THYMELAEACEAE](#), p. 77.
- 154a. Flowers in sessile
lateral clusters [1a, in LAURACEAE](#), p. 41.
- 154b. Flowers in peduncled
terminal clusters [1b, in LAURACEAE](#), p. 41.
- 155—
- 155a. Leaves narrow, needle-like or
scale-like, mostly evergreen
— [54.](#)
- 155b. Leaves broader, flat or rolled, but
not needle-like or scale-like
— 156.

- 156a. Twining or climbing vines — [56.](#)
- 156b. Not twining or climbing — [157.](#)
- 157a. Leaves opposite — [158.](#)
- 157b. Leaves alternate — [165.](#)
- 158a. Leaves compound — [68.](#)
- 158b. Leaves simple — [159.](#)
- 159a. Leaves broad, palmately lobed — [160.](#)
- 159b. Leaves not palmately lobed — [161.](#)
- 160a. Flowers greenish-yellow, in racemes or panicles [2a, in ACERACEAE](#), p. 70.
- 160b. Flowers white, in rounded or flattened clusters [22a, in CAPRIFOLIACEAE](#), p. 111.
- 161a. Flowers in close clusters, subtended by four large white petal-like bracts [4b, in CORNACEAE](#), p. 83.
- 161b. Flowers small, white, in dense spherical heads [1a, in RUBIACEAE](#), p. 108.
- 161c. Flowers not in dense heads, and not subtended by petal-like bracts — [162.](#)
- 162a. Petals separate from each other — [163.](#)
- 162b. Petals united with each other — [164.](#)
- 163a. Flowers dark purple-red [1b, in CELASTRACEAE](#), p. 69.
- 163b. Flowers bright yellow [1a, in HYPERICACEAE](#), p.

- 73.
- 163c. Flowers pink-purple; aquatic shrubs [1a, in LYTHRACEAE](#), p. 77.
- 163d. Flowers white [5, in CORNACEAE](#), p. 83.
- 163e. Flowers minute, greenish; twigs usually thorny [1b, in RHAMNACEAE](#), p. 71.
- 163f. Flowers small, yellowish; leaves silvery beneath [ELAEAGNACEAE](#), p. 77.
- 164a. Stamens 2 [1a, in OLEACEAE](#), p. 88.
- 164b. Stamens 4 or 5 [11, in CAPRIFOLIACEAE](#), p. 110.
- 164c. Stamens 10 [18a, in ERICACEAE](#), p. 85.
- 165a. Leaves once-compound — [166](#).
- 165b. Leaves simple — [172](#).
- 165c. Leaves twice- or thrice-compound [4a, in ARALIACEAE](#), p. 80.
- 166a. Flowers small, greenish or greenish-yellow — [167](#).
- 166b. Flowers conspicuous, white, or brightly colored — [170](#).
- 167a. Stems thorny [1a, in RUTACEAE](#), p. 65.
- 167b. Stems not thorny — [168](#).
- 168a. Leaflets 3, entire or minutely toothed [1b, in RUTACEAE](#), p. 65.
- 168b. Leaflets 3, conspicuously toothed [1b, in ANACARDIACEAE](#), p. 68.
- 168c. Leaflets more than 3 — [169](#).

- 169a. Leaflets entire except for 1-4 large glandular teeth near their base [SIMARUBACEAE](#), p. 65.
- 169b. Leaflets entire, or toothed for their entire length [ANACARDIACEAE](#), p. 68.
- 170a. Flowers with a single bright-blue petal [8a, in LEGUMINOSAE](#), p. 58.
- 170b. Flowers with several petals [— 171.](#)
- 171a. Flowers regular [3, in ROSACEAE](#), p. 51.
- 171b. Flowers irregular, the upper petal the largest [4, in LEGUMINOSAE](#), p. 58.
- 172a. Leaves minute, scale-like, appressed [2a, in CISTACEAE](#), p. 74.
- 172b. Leaves 3-10 mm. long, spreading, completely rolled into a tube [EMPETRACEAE](#), p. 68.
- 172c. Leaves normal in shape, green in color [— 173.](#)
- 173a. Leaves evergreen, as shown by their presence on the older parts of the stem [4b, in ERICACEAE](#), p. 84.
- 173b. Leaves deciduous [— 174.](#)
- 174a. Flowers in catkins, without petal-like parts [— 142.](#)
- 174b. Flowers not in catkins, either with or without petal-like parts [— 175.](#)
- 175a. Flowers small, inconspicuous, yellowish or greenish in color [— 176.](#)

- 175b. Flowers white or colored, with
conspicuous petals — [180.](#)
- 176a. Leaves broad and
palmately lobed; stem
thorny [1a, in ARALIACEAE,](#)
p. 80.
- 176b. Leaves entire, toothed,
or with 2-3 entire
lobes; stem not thorny
— [177.](#)
- 177a. Flowers in small axillary clusters;
foliage not spicy-aromatic
— [178.](#)
- 177b. Flowers in clusters terminating
last year's twigs; foliage spicy-
aromatic [1b, in LAURACEAE,](#) p.
41.
- 178a. Petals none; sepals
present [2b, in RHAMNACEAE,](#)
p. 71.
- 178b. Petals present but small
— [179.](#)
- 179a. A stamen in front of each petal [2a, in RHAMNACEAE,](#)
p. 71.
- 179b. A stamen between each two
petals [AQUIFOLIACEAE,](#) p.
69.
- 180a. Petals united with each
other; stamens 8-10 [24, in ERICACEAE,](#) p.
86.
- 180b. Petals separate from
each other — [181.](#)
- 181a. Petals 4 — [182.](#)
- 181b. Petals 5 — [183.](#)
- 181c. Petal-like sepals 6; real petals
none; foliage spicy-aromatic [1b, in LAURACEAE,](#) p.
41.
- 181d. Petals 6; sepals also present
— [187.](#)
- 182a. Flowers in late spring
or early summer [2a, in CORNACEAE,](#) p.
83.

- 182b. Flowers in autumn [HAMAMELIDACEAE](#), p. 51.
- 183a. Corolla irregular, the upper petal largest [3b, in LEGUMINOSAE](#), p. 58.
- 183b. Corolla regular, all petals alike in size and shape [— 184.](#)
- 184a. Stamens 5 [— 185.](#)
- 184b. Stamens 10 or more [24, in ROSACEAE](#), p. 53.
- 185a. Flowers in loose racemes or axillary clusters [2, in SAXIFRAGACEAE](#), p. 49.
- 185b. Flowers in branching clusters [— 186.](#)
- 186a. Leaves palmately lobed; stem thorny [1a, in ARALIACEAE](#), p. 80.
- 186b. Leaves not lobed; stem not thorny [1b, in RHAMNACEAE](#), p. 71.
- 187a. Flowers dark red-purple [ANONACEAE](#), p. 40.
- 187b. Flowers yellow [2a, in BERBERIDACEAE](#), p. 40.

GROUP 2, UNUSUAL PLANTS

- 1a. Small brown leafless plants, growing as parasites on the tamarack or black spruce [LORANTHACEAE](#), p. 25.
- 1b. Aquatic plants, with all or most of the leaves submerged, or leafless [— 2.](#)
- 1c. Aquatic plants, with the leaves or the whole plant floating on or

- near the surface — [20](#).
- 1d. Terrestrial or marsh plants,
without floating or
submerged leaves — [29](#).
- 2a. Submerged aquatics,
without leaves
— [3](#).
- 2b. Submerged aquatics,
with the leaves
linear or dissected
— [4](#).
- 3a. Flowers showy, yellow or purple [LENTIBULARIACEAE](#),
p. 105.
- 3b. Flowers small and [3a, in](#)
inconspicuous, sessile, [HALORAGIDACEAE](#), p.
purplish or greenish 79.
- 4a. Leaves linear or
lanceolate, not
lobed or dissected
— [5](#).
- 4b. Leaves more or less
lobed or dissected
— [13](#).
- 5a. Leaves all basal — [6](#).
- 5b. Stem-leaves present — [7](#).
- 6a. Flowers blue, 1 cm.
long or more;
leaves cylindrical, [1a, in LOBELIACEAE](#), p.
blunt, hollow, 113.
partitioned
lengthwise
- 6b. Flowers yellow; leaves [LENTIBULARIACEAE](#),
minute p. 105.
- 6c. Flowers white, in [1b, in ALISMACEAE](#), p.
clusters; leaves 3.

- linear-lanceolate,
acute, not hollow
- 6d. Flowers greenish,
solitary at the end
of elongated
peduncles; leaves [1a, in
HYDROCHARITACEAE](#),
very long and
ribbon-like, flat or
trough-shape p. 3.
- 6e. Flowers minute,
whitish or lead-
color, in heads [ERIOCAULACEAE](#), p. 9.
- 7a. Leaves alternate [— 8.](#)
- 7b. Leaves opposite [— 10.](#)
- 7c. Leaves whorled [— 12.](#)
 - 8a. Leaves with thin
sheathing stipules [1b, in NAJADACEAE](#), p.
2.
 - 8b. Leaves without
stipules [— 9.](#)
- 9a. Flowers greenish, in a head [SPARGANIACEAE](#), p. 2.
[1b, in](#)
- 9b. Flowers pale yellow [PONTEDERIACEAE](#), p.
10.
- 10a. Leaves serrate [2b, in NAJADACEAE](#), p.
2.
- 10b. Leaves entire [— 11.](#)
- 11a. Leaves 2 cm. long or less [CALLITRICHACEAE](#), p.
68.
- 11b. Leaves thread-like, 2-8 cm.
long [2a, in NAJADACEAE](#), p.
2.
- 11c. Leaves linear to elliptical, more
than 2 cm. long [1b, in NAJADACEAE](#), p.
2.
- 12a. Leaves in whorls of 3,
abruptly widened
at the base [2b, in NAJADACEAE](#), p.
2.

- 12b. Leaves in whorls of 3,
widest near the middle [1b, in HYDROCHARITACEAE](#), p. 3.
- 12c. Stems straight and erect, at leaves the flowers emerged; leaves in whorls of 4 or more [2a, in HALORAGIDACEAE](#), p. 79.
- 13a. Leaves with numerous small bladders attached, each bladder 1-3 mm. long [LENTIBULARIACEAE](#), p. 105.
- 13b. Leaves without bladders [— 14.](#)
- 14a. Leaves alternate [— 15.](#)
- 14b. Leaves opposite or whorled [— 18.](#)
- 15a. Delicate and rare plants growing attached to stones in running water [PODOSTEMACEAE](#), p. 48.
- 15b. Plants 2 dm. high or more, with roots in mud or sand [— 16.](#)
- 16a. Leaves once-pinnate [— 17.](#)
- 16b. Leaves 2-3 times pinnate [15b, in UMBELLIFERAE](#), p. 81.
- 16c. Leaves palmately dissected [1b, in RANUNCULACEAE](#), p. 35.
- 17a. Taste peppery or mustardy [34, in CRUCIFERAE](#), p. 45.
- 17b. Taste not peppery or mustardy [1a, in HALORAGIDACEAE](#), p. 79.
- 18a. Leaves pinnately compound [3b, in HALORAGIDACEAE](#), p. 79.

- 18b. Leaves palmately compound — [19](#).
- 19a. Leaves opposite or in whorls of four [119a, in COMPOSITAE](#), p. 123.
- 19b. Leaves in whorls of 5-12 [CERATOPHYLLACEAE](#), p. 34.
- 20a. Plants small, flattened, rounded or ovate, without distinction of stem and leaf; the whole plant floating on or near the surface [LEMNACEAE](#), p. 9.
- 20b. Plant attached to the soil, with differentiated stem and leaves — [21](#).
- 21a. Leaves on long slender stalks, which bear also a cluster of slender tubers near the leaf-base; leaves heart-shape [1b, in GENTIANACEAE](#), p. 88.
- 21b. Leaf-stalks without a cluster of tubers — [22](#).
- 22a. Leaves all basal — [23](#).
- 22b. Stem-leaves present — [24](#).
- 23a. Leaves parallel-veined [1b, in ALISMACEAE](#), p. 3.
- 23b. Leaves net-veined [1b, in NYMPHAEACEAE](#), p. 35.
- 24a. Leaves opposite or whorled — [25](#).
- 24b. Leaves alternate — [26](#).

- 25a. Leaves less than 2 cm. long [**CALLITRICHACEAE**](#), p. 68.
- 25b. Leaves more than 2 cm. long [**1b, in NAJADACEAE**](#), p. 2.
- 26a. Leaves attached by the center to the stalk [**2b, in NYMPHAEACEAE**](#), p. 35.
- 26b. Leaves attached by the margin [**— 27.**](#)
- 27a. Leaves parallel-veined [**— 28.**](#)
- 27b. Leaves net-veined, with a single mid-vein; lanceolate or elliptical in outline [**29b, in POLYGONACEAE**](#), p. 28.
- 28a. Leaves not over 2 dm. long [**1b, in NAJADACEAE**](#), p. 2.
- 28b. Leaves very long and grass-like [**14a, in GRAMINEAE**](#), p. 5.
- 29a. Brown, yellow, or white plants, without green color [**— 30.**](#)
- 29b. Plants with normal green color, at least in some parts [**— 33.**](#)
- 30a. Stemless and leafless plants, consisting of flowers only and partly underground [**4a, in ARACEAE**](#), p. 8.
- 30b. Stems climbing on other plants [**1b, in CONVULVULACEAE**](#), p. 91.
- 30c. Stem and flower-stalks erect, not climbing [**— 31.**](#)
- 31a. Corolla regular; stamens 6-12 [**1a, in ERICACEAE**](#), p. 84.
- 31b. Corolla irregular [**— 32.**](#)
- 32a. Sepals and petals each 3; flowers in simple racemes [**5a, in ORCHIDACEAE**](#), p. 16.

- 32b. Sepals 5; corolla of united petals [OROBANCHACEAE](#), p. 106.
- 33a. Stem thick and fleshy, leafless, thorny [CACTACEAE](#), p. 77.
- 33b. Stem not thorny [— 34.](#)
 - 34a. Leaves none [— 35.](#)
 - 34b. Leaves reduced to small scales [— 40.](#)
 - 34c. Leaves thick and fleshy [— 44.](#)
 - 34d. Leaves hollow [— 45.](#)
 - 34e. Leaves small, all basal, bearing large glandular hairs on the upper surface [DROSERACEAE](#), p. 48.
- 35a. Stem none, the flowers appearing at or partially beneath the surface of the soil [4a, in ARACEAE](#), p. 8.
- 35b. Stem present [— 36.](#)
 - 36a. Stem freely branched [2a, in LILIACEAE](#), p. 10.
 - 36b. Stem unbranched, except possibly in the flower-clusters [— 37.](#)
- 37a. Flowers greenish or brownish, without obvious colored petals [— 38.](#)
- 37b. Flowers with conspicuous white or colored petals [— 39.](#)
 - 38a. Each flower with 6 small chaffy petals [4, in JUNCACEAE](#), p. 10.
 - 38b. Petals none; each flower in the axil of a single chaffy bract [2a, in CYPERACEAE](#), p. 7.

- 39a. Flowers regular [34a, in LILIACEAE](#), p. 11.
- 39b. Flowers irregular [10a, in ORCHIDACEAE](#),
p. 16.
- 40a. Stem erect,
unbranched or with
one or two
branches only
[— 41.](#)
- 40b. Stem freely branched
[— 42.](#)
- 41a. Plants of moist soil, with
opposite scales; corolla
regular, with 4 petal-like
lobes [1a, in GENTIANACEAE](#),
p. 88.
- 41b. Swamp plants, with a few
alternate scales; corolla
irregular [LENTIBULARIACEAE](#),
p. 105.
- 42a. Leaves numerous and
close, concealing
the stem [2a, in CISTACEAE](#), p. 74
- 42b. Leaves spreading, not
concealing the
stem [— 43.](#)
- 43a. Leaf-scales in small clusters;
flowers greenish, with 6
petals [2a, in LILIACEAE](#), p. 10.
- 43b. Leaf-scales distinctly opposite;
petals 5, yellow [7a, in HYPERICACEAE](#),
p. 74.
- 44a. Sepals 2 [3, in PORTULACACEAE](#),
p. 34.
- 44b. Sepals 4 or 5 [CRASSULACEAE](#), p. 48.
- 45a. Leaves pitcher-shape, open at
the top [SARRACENIACEAE](#), p.
48.
- 45b. Leaves tubular, closed at the
end [37b, in LILIACEAE](#), p. 13.

GROUP 3, MONOCOTYLEDONES

- 1a. Twining plants, with flowers in panicles or racemes [DIOSCOREACEAE](#), p. 14.
- 1b. Plants with milky juice [13a, in COMPOSITAE](#), p. 115.
- 1c. Plants not twining (some climb by tendrils) and not with milky juice. [—2.](#)
 - 2a. Flowers in close spikes or heads, surrounded or subtended by a green or colored bract, the whole resembling a single flower; petals minute or wanting; leaves broad, not grass-like, linear, or sword-shape [ARACEAE](#), p. 8.
 - 2b. Plants with narrow, linear, grass-like, or sword-shape leaves (a few species of *Carex* have broader, lanceolate to ovate leaves); flowers greenish, yellowish, or brownish, never brightly colored, and frequently dry or chaffy in texture; perianth small or wanting; individual

flowers
inconspicuous in
size, but sometimes
grouped into
conspicuous clusters
— 3.

2c. Plants with leaves of
various widths, but
the flowers petaloid,
i. e., with a white or
colored, more or less
conspicuous
perianth, and never
chaffy in texture. In
a few cases the
flowers are greenish,
but the size and
conspicuousness of
the perianth
identifies them in
this class — 9.

3a. Flowers in the axils of dry,
membranous or chaffy scales,
which are regularly arranged
into spikes or spikelets of
uniform size and structure,
which are variously grouped
or clustered; fruit an achene;
grasses and sedges, with
joined stems and sheathing
leaves, or leafless and the
stems not jointed — 4.

3b. Flowers not subtended
individually by dry,
membranous, or chaffy scales,
and otherwise not agreeing
with 3a — 5.

- 4a. Leaf-sheaths split on the side opposite the leaf; leaves usually 2-ranked, i. e., in 2 longitudinal rows with the third leaf above the first; stems rounded or flat, never triangular, usually hollow [GRAMINEAE](#), p. 4.
- 4b. Leaf-sheaths closed into a continuous tube; leaves usually 3-ranked; stems frequently triangular, usually solid [CYPERACEAE](#), p. 7.
- 5a. Flowers in dense spikes [— 6.](#)
- 5b. Flowers in heads, racemes, or panicles [— 7.](#)
 - 6a. Spike terminal, with pistillate flowers at the base and staminate ones at the apex [TYPHACEAE](#), p. 2.
 - 6b. Spike short, apparently lateral, near the apex of the stem [3a, in ARACEAE](#), p. 8.
- 7a. Flowers in globose heads which are arranged in spikes, the lowest heads pistillate, the upper staminate; ovary 1-celled [SPARGANIACEAE](#), p. 2.
- 7b. Flowers in globose woolly heads terminating leafless unbranched stalks [ERIOCAULACEAE](#), p. 9.

- 7c. Flowers in a spike-like raceme;
ovaries 3-6, separate or nearly
so [JUNCAGINACEAE](#), p. 3.
- 7d. Flowers in heads or panicles, all
perfect, not woolly, with one
ovary [—8.](#)
- 8a. Leaves less than 1 cm.
wide, or none;
divisions of the
perianth 6 [JUNCACEAE](#), p. 10.
- 8b. Leaves 2 cm. wide or
more; petals 5 [2a, in UMBELLIFERAE](#),
p. 80.
- 9a. Flowers regular, with all the petals
of approximately the same
size and shape [—10.](#)
- 9b. Flowers irregular, with the petals
of each flower not of the same
size or shape [—22.](#)
- 10a. Ovaries 3 or more,
separate or barely
united with each
other at the base
[—11.](#)
- 10b. Ovary one in each
flower [—12.](#)
- 11a. Ovaries 3-6 in number; flowers
in spikes or racemes; leaves
linear [JUNCAGINACEAE](#), p. 3.
- 11b. Ovaries more than 6 [ALISMACEAE](#), p. 3.
- 12a. Flowers or flower-
clusters lateral,
axillary or
apparently so [—13.](#)
- 12b. Flowers or flower-
clusters terminal or

- on leafless stalks
— [14.](#)
- 13a. Leaves minute and scale-like;
flowers greenish-yellow [2a, in LILIACEAE](#), p. 10.
- 13b. Leaves linear, grass-like [3b, in IRIDACEAE](#), p. 13.
- 13c. Leaves lanceolate or broader, not
grass-like or scale-like [2b, in LILIACEAE](#), p. 10.
- 14a. Divisions of the
perianth 5-12 cm.
long — [15.](#)
- 14b. Divisions of the
perianth less than 5
cm. long — [16.](#)
- 15a. Flowers blue, or blue marked
with yellow [1a, in IRIDACEAE](#), p. 15.
- 15b. Flowers not blue [12a, in LILIACEAE](#), p. 11.
- 16a. Flowers solitary — [17.](#)
- 16b. Flowers 2 or more, in
some kind of a
cluster — [18.](#)
- 17a. Leaves 2, broadly heart-shape,
basal, on long stalks [1b, in ARISTOLOCHIACEAE](#), p. 25.
- 17b. Leaves not heart-shape 20, [20, in LILIACEAE](#), p. 12.
- 18a. Divisions of the
perianth (4 to 6) all
essentially alike
— [19.](#)
- 18b. Perianth differentiated
into sepals and 2 or
3 colored petals
— [21.](#)
- 19a. Flowers in umbels — [20.](#)

- 19b. Flowers in dense round heads; petals 5 [2a, in UMBELLIFERAE](#), p. 80.
- 19c. Flowers in spikes, racemes, or panicles [38, in LILIACEAE](#), p. 13.
- 20a. Ovary inferior, appearing below the perianth as a swelling at the apex of the stalk; flowers bright yellow; leaves linear [AMARYLLIDACEAE](#), p. 14.
- 20b. Ovary inferior; flowers blue or white, terminating a flattened winged leafless flower-stalk [3a, in IRIDACEAE](#), p. 15.
- 20c. Ovary superior, i. e., in the center of the flower and separate from the perianth [31, in LILIACEAE](#), p. 13.
- 21a. Flowers in dense heads, yellow, 1 cm. wide or smaller [XYRIDACEAE](#), p. 9.
- 21b. Flowers in umbels, blue or white, 2 cm. wide or larger [COMMELINACEAE](#), p. 9.
- 22a. Flowers blue; ovary superior (defined under [20c](#)); stamens distinct from the other parts of the flower [— 23.](#)
- 22b. Ovary inferior (defined under [20a](#)); floral structure complex; stamens attached to other parts of the flower and not

resembling ordinary
stamens in form or
structure

- 23a. Leaves triangular-heart-shape at
base; marsh plants [1a, in](#)
[PONTEDERIACEAE](#), p.
10.
- 23b. Leaves not heart-shape at base [COMMELINACEAE](#), p.
9.

GROUP 4, DICOTYLEDONES

- 1a. Foliage leaves all or principally basal; flower-stalk either completely leafless, or bearing a single pair of opposite leaves only. Bracts and scale-leaves are not considered foliage leaves — 2.
- 1b. Stem-leaves present on the stem, either one or more in number, and not limited to a single opposite pair — 3.
 - 2a. Leaves compound — 7.
 - 2b. Leaves simple — 18.
- 3a. Stem-leaves all or chiefly opposite or whorled (the bracts of the flower clusters may be alternate) — 4.
- 3b. Stem-leaves all or chiefly alternate — 5.
 - 4a. Flowers small and inconspicuous, the perianth none or greenish or chaffy, and never petal-like in appearance — 50.
 - 4b. Flowers with a white or colored petal-like perianth — 66.

5a. Flowers small and inconspicuous,
without any white or colored
petal-like perianth — [118](#).

5b. Flowers large or small, but with a
white or colored petal-like
perianth — [6](#).

6a. Flowers small, not
exceeding 3 mm. in
length or breadth
— [144](#).

6b. Flowers larger, more
than 3 mm. in
length or breadth
— [166](#).

— 7 —

7a. Leaves twice to three times
compound or dissected — [8](#).

7b. Leaves once-compound — [11](#).

8a. Flowers in racemes

[2b, in FUMARIACEAE](#),
p. 41.

8b. Flowers in umbels — [9](#).

9a. Flowers about 3 mm. wide, white
or greenish-white — [10](#).

9b. Flowers 10-20 mm. wide,
conspicuous

[45b, in
RANUNCULACEAE](#), p.
39.

10a. Leaflets 5-15 cm. long

[5b, in ARALIACEAE](#), p.
80.

10b. Leaflets not over 2
cm. long

[16a, in
UMBELLIFERAE](#), p. 81.

11a. Leaflets 2

[3a, in
BERBERIDACEAE](#), p.
40.

11b. Leaflets three or more — [12](#).

12a. Leaflets entire or very
finely toothed

— 13.

12b. Leaflets coarsely
toothed or lobed

— 15.

13a. Flowers irregular, in dense head-
like umbels [37a, in LEGUMINOSAE](#),
p. 61.

13b. Flowers consisting of a greenish
or purplish bract wholly or
partly enclosing a fleshy spike [2, in ARACEAE](#), p. 8.

13c. Flowers regular, solitary or in
loose clusters — 14.

14a. Leaflets reverse heart-
shape, not over 2
cm. long [OXALIDACEAE](#), p. 64.

14b. Leaflets 4-10 cm. long [1c, in GENTIANACEAE](#),
p. 88.

15a. Flowers with colored or white
petals and green or colored
sepals — 16.

15b. Flowers with one kind of
perianth only (calyx), with
broad and petal-like parts [32, in
RANUNCULACEAE](#), p.
38.

15c. Flowers with white petal-like
sepals and small
inconspicuous petals [38b, in
RANUNCULACEAE](#), p.
39.

16a. Petals and sepals each
4 [31, in CRUCIFERAE](#), p.
44.

16b. Petals and sepals each
5 or more — 17.

17a. Leaves with stipules [ROSACEAE](#), p. 51.

17b. Leaves without stipules [23b, in
RANUNCULACEAE](#), p.
37.

— 18—

18a. Stem-leaves a single
opposite pair (basal

- leaves may also be present) — 19.
- 18b. Stem-leaves none — 23.
- 19a. Leaves entire — 20.
- 19b. Leaves toothed or lobed — 21.
 - 20a. Flowers solitary; leaves broadly kidney-shape ARISTOLOCHIACEAE, p. 25.
 - 20b. Flowers in racemes; leaves linear or lanceolate PORTULACACEAE, p. 34.
- 21a. Flowers in racemes; petals deeply toothed 23b, in SAXIFRAGACEAE, p. 51.
- 21b. Flowers solitary or few in a cluster; petals entire or nearly so — 22.
 - 22a. Petals 5 GERANIACEAE, p. 64.
 - 22b. Petals 6 or more 2b, in BERBERIDACEAE, p. 40.
- 23a. Flowers or flower-clusters sessile, at or partly beneath the surface of the ground 4a, in ARACEAE, p. 8.
- 23b. Flowers or flower-heads solitary at the ends of the flower-stalks — 24.
- 23b. Flowers or flower-heads numerous or several on each flower-stalk — 38.
 - 24a. Flowers yellow — 25.
 - 24b. Flowers not yellow — 29.
- 25a. Juice milky 16, in COMPOSITAE, p. 115.

- 25b. Juice not milky — [26.](#)
- 26a. Aquatic or mud plants,
with large entire
leaves 1-4 dm. wide [NYMPHAEACEAE](#), p.
35.
- 26b. Land plants, with
smaller or lobed
leaves — [27.](#)
- 27a. Flower-stalk scaly [110, in COMPOSITAE](#), p.
122.
- 27b. Flower-stalk bare or with 1 to 2
minute bracts — [28.](#)
- 28a. Flowers regular [10, in](#)
[RANUNCULACEAE](#), p.
36.
- 28b. Flowers irregular, with
a spur [3a, in VIOLACEAE](#), p.
75.
- 29a. Flowers obviously irregular,
with a spur — [30.](#)
- 29b. Flowers regular or nearly so,
without a spur — [31.](#)
- 30a. Leaves sessile or
nearly so; stamens 2 [LENTIBULARIACEAE](#),
p. 105.
- 30b. Leaves petioled;
stamens 5 [VIOLACEAE](#), p. 75.
- 31a. Flowers 3 mm. wide or smaller;
leaves thread-like [16a, in](#)
[SCROPHULARIACEAE](#),
p. 103.
- 31b. Flowers 6 mm. wide or larger
— [32.](#)
- 32a. Ovary 1 — [33.](#)
- 32b. Ovaries numerous
— [37.](#)
- 33a. Leaves lobed or cleft — [34.](#)
- 33b. Leaves entire or toothed — [35.](#)
- 34a. Leaves deeply 2-cleft;
stamens 8 [3a, in](#)
[BERBERIDACEAE](#), p.

- 40.
- 34b. Leaf 1, palmately lobed; stamens numerous [1a, in PAPAVERACEAE](#), p. 41.
- 35a. Flowers dull red [ARISTOLOCHIACEAE](#), p. 25.
- 35b. Flowers white or pinkish [— 36.](#)
- 36a. Leaves entire; stamens with good anthers 5 [13, in SAXIFRAGACEAE](#), p. 50.
- 36b. Leaves minutely toothed; stamens 8-10 [14a, in ERICACEAE](#), p. 85.
- 37a. Leaves lobed or divided [RANUNCULACEAE](#), p. 35.
- 37b. Leaves crenate or toothed [64a, in ROSACEAE](#), p. 56.
- 38—
- 38a. Flowers of the composite type, with several or many small flowerets closely aggregated into a dense head surrounded by a calyx-like involucre of small bracts [10, in COMPOSITAE](#), p. 114.
- 38b. Flowers separate; variously clustered, but never crowded into involucre heads [— 39.](#)
- 39a. Flowers in dense close spikes [— 40.](#)

- 46b. Sepals and petals each
5 — 47.
- 46c. Sepals 6; petals none;
flowers minute,
green; leaves
frequently lobed at
the base 4, in **POLYGONACEAE**,
p. 25.
- 47a. Petals united with each other in
an irregular corolla **LENTIBULARIACEAE**,
p. 105.
- 47b. Petals separate from each other
— 48.
- 48a. Stamens 10; style 1 **ERICACEAE**, p. 84.
- 48b. Stamens 5 or 10;
styles 2 **SAXIFRAGACEAE**, p.
49.
- 48c. Stamens and pistils
each very numerous
— 49.
- 49a. Flowers white 64a, in **ROSACEAE**, p.
56.
- 49b. Flowers yellow 13a, in
RANUNCULACEAE, p.
36.
- 50—
- 50a. Leaves compound or
deeply lobed — 51.
- 50b. Leaves entire or
toothed — 54.
- 51a. Flowers in axillary racemes,
spikes, or panicles **URTICACEAE**, p. 23.
- 51b. Flowers terminal, or in terminal
clusters — 52.
- 52a. Flowers in umbels;
leaves palmately
compound **ARALIACEAE**, p. 80.
- 52b. Flowers in racemes or
spikes, or solitary

— 53.

- 53a. Leaves deeply pinnatifid; swamp plants with flowers in spikes or solitary [HALORAGIDACEAE](#), p. 79.
- 53b. Leaves lobed, or irregularly pinnately cut or dissected; weedy plants with flowers in racemes [2, in COMPOSITAE](#), p. 114.
- 54a. Juice milky [3, in EUPHORBIACEAE](#), p. 67.
- 54b. Juice not milky or colored — 55.
- 55a. Leaves whorled — 56.
- 55b. Leaves all opposite — 58.
- 56a. Aquatic or mud plants with erect stems [2a, in HALORAGIDACEAE](#), p. 79.
- 56b. Prostrate weedy terrestrial plants [AIZOACEAE](#), p. 31.
- 56c. Erect or ascending terrestrial plants — 57.
- 57a. Leaves more than 2.5 cm. long [4a, in CORNACEAE](#), p. 83.
- 57b. Leaves 2 cm. long or less [3a, in PRIMULACEAE](#), p. 87.
- 58a. Flowers in terminal or axillary spikes, racemes, or other clusters — 59.
- 58b. Flowers solitary or few in the axils of the leaves — 61.
- 59a. Leaves less than 5 mm. long [1a, in GENTIANACEAE](#), p. 88.

- 59b. Leaves more than 2 cm. long
 — 60.
- 60a. Inflorescence chiefly terminal, paniced 2, in COMPOSITAE, p. 114.
- 60b. Inflorescence chiefly axillary URTICACEAE, p. 23.
- 61a. Flowers on long pedicels 2, in CARYOPHYLLACEAE, p. 31.
- 61b. Flowers on short pedicels or sessile — 62.
- 62a. Stem erect, repeatedly branched; leaves linear to oblong ILLECEBRACEAE, p. 31.
- 62b. Stem decumbent or prostrate — 63.
- 63a. Leaves round, ovate, or kidney-shape, rounded at the base, crenate or lobed 12a, in SAXIFRAGACEAE, p. 50.
- 63b. Leaves of a narrower shape, entire, tapering toward the base — 64.
- 64a. Principal leaves 2-3 cm. long 1a, in ONAGRACEAE, p. 78.
- 64b. Principal leaves 1 cm. long or less — 65.
- 65a. Petals present ELATINACEAE, p. 74.
- 65b. Petals none CALLITRICHACEAE, p. 68.
- 66—
- 66a. Plants of the composite type, with several or many small flowerets closely aggregated into a

dense head
surrounded or
subtended by a
calyx-like involucre
of small bracts
[— 67.](#)

66b. Flowers solitary or
variously clustered,
but not in
involucrated heads
[— 70.](#)

67a. Involucre of 4 conspicuous
white bracts, much larger than
the small flower-cluster
[4a, in CORNACEAE](#), p. 83.

67b. Involucral bracts green or
somewhat colored [— 68.](#)

68a. Stem more or less
prickly [DIPSACACEAE](#), p. 112.

68b. Stem and leaves not
prickly [— 69.](#)

69a. Stem square; foliage aromatic
when crushed [LABIATAE](#), p. 95.

69b. Stem not square; foliage not
with the odor of mint [COMPOSITAE](#), p. 113.

70a. The conspicuous
portion of the
perianth^[1]
composed of
separate parts
[— 71.](#)

70b. The conspicuous
portion of the
perianth^[1]
composed of united
parts [— 97.](#)

- 71a. Stem-leaves compound, or cleft to the very base [— 72.](#)
- 71b. Stem-leaves simple [— 78.](#)
 - 72a. Stamens more than 10 in each flower [— 73.](#)
 - 72b. Stamens 5-10 in each flower [— 75.](#)
- 73a. Ovary 1 in each flower [2a, in PAPAVERACEAE,](#) p. 41.
- 73b. Ovaries several in each flower [— 74.](#)
 - 74a. Leaves pinnately compound [68a, in ROSACEAE,](#) p. 57.
 - 74b. Leaves palmately compound [RANUNCULACEAE,](#) p. 35.
- 75a. Stamens 6 [31, in CRUCIFERAE,](#) p. 44.
- 75b. Stamens 5 or 10 [— 76.](#)
 - 76a. Leaflets reverse heart-shape; flowers yellow [OXALIDACEAE,](#) p. 64.
 - 76b. Leaflets not reverse heart-shape; flowers white, greenish, or pink [— 77.](#)
- 77a. Flowers greenish or white, about 2 mm. broad [ARALIACEAE,](#) p. 80.
- 77b. Flowers 5 mm. broad or larger [GERANIACEAE,](#) p. 64.
 - 78a. Juice milky [— 79.](#)
 - 78b. Juice not milky [— 80.](#)
- 79a. A 3-lobed ovary with 3 short styles visible in some of the flowers [EUPHORBIACEAE,](#) p. 66.
- 79b. Ovaries 2 in the center of each flower (sometimes concealed

- by other organs); sepals and petals each 5 — [104.](#)
- 80a. Perianth with one circle of parts only — [81.](#)
- 80b. Each flower with both calyx and corolla — [82.](#)
- 81a. Stem prostrate; flowers only 2 mm. broad [AIZOACEAE](#), p. 31.
- 81b. Stem erect; flowers at least 20 mm. wide [4a, in CORNACEAE](#), p. 83.
- 82a. Petals 2 or 4 — [83.](#)
- 82b. Petals 3 [22, in LILIACEAE](#), p. 12.
- 82c. Petals 5 or 6 — [84.](#)
- 83a. Leaves with 3-5 principal veins [MELASTOMACEAE](#), p. 77.
- 83b. Leaves with 1 principal mid-vein [ONAGRACEAE](#), p. 78.
- 84a. Leaves palmately lobed — [85.](#)
- 84b. Leaves entire or toothed, or with 1-2 small lobes near the base only — [86.](#)
- 85a. Petals entire or notched at the end [GERANIACEAE](#), p. 64.
- 85b. Petals conspicuously fringed [23b, in SAXIFRAGACEAE](#), p. 51.
- 86a. Ovary 1 — [87.](#)
- 86b. Ovaries 2, surrounded and concealed by other organs [ASCLEPIADACEAE](#), p. 90.
- 86c. Ovaries 4-5 [2b, in CRASSULACEAE](#), p. 48.

- 87a. Leaves dotted with translucent dots (easily seen when the leaf is held to the light) [HYPERICACEAE](#), p. 73.
- 87b. Leaves not dotted with translucent dots [— 88.](#)
 - 88a. Leaves only 1-3 mm. long, closely appressed and concealing the stem [2a, in CISTACEAE](#), p. 74.
 - 88b. Leaves larger, not concealing the stem [— 89.](#)
- 89a. Style 1 or none [— 90.](#)
- 89b. Styles 2 to 7 [— 94.](#)
 - 90a. Stamens 5 [— 91.](#)
 - 90b. Stamens 10 [— 92.](#)
 - 90c. Stamens neither 5 nor 10 [— 93.](#)
- 91a. Flowers blue, sessile in terminal spikes with leaf-like bracts [LYTHRACEAE](#), p. 77.
- 91b. Flowers not in terminal bracted spikes [PRIMULACEAE](#), p. 87.
 - 92a. Leaves entire [LYTHRACEAE](#), p. 77.
 - 92b. Leaves toothed or crenate [ERICACEAE](#), p. 84.
- 93a. Flowers irregular; petals 3 [POLYGALACEAE](#), p. 65.
- 93b. Flowers regular; petals 5 or more [LYTHRACEAE](#), p. 77.
 - 94a. Sepals 2, partly attached to the ovary [PORTULACACEAE](#), p. 34.
 - 94b. Sepals 5, free from the ovary [— 95.](#)
- 95a. Stamens 5 [— 96.](#)

- 95b. Stamens not 5 [CARYOPHYLLACEAE](#), p. 31.
- 96a. Flowers blue or yellow [LINACEAE](#), p. 63.
- 96b. Flowers white or pinkish [CARYOPHYLLACEAE](#), p. 31.
- 97 —
- 97a. Perianth with but one circle of floral leaves [— 98.](#)
- 97b. Perianth consisting of both calyx and corolla [— 101.](#)
- 98a. Flowers small, in dense heads subtended by conspicuous bracts [— 99.](#)
- 98b. Flowers 2-5, in a colored spreading 5-lobed involucre [NYCTAGINACEAE](#), p. 31.
- 98c. Flowers in various sorts of clusters or solitary, but never in heads with a conspicuous involucre [— 100.](#)
- 99a. Bracts 4, white and conspicuous; stem not thorny [4a, in CORNACEAE](#), p. 83.
- 99b. Bracts green; stem thorny [DIPSACACEAE](#), p. 112.
- 100a. Stamens 3 [VALERIANACEAE](#), p. 111.
- 100b. Stamens 4 or 5 [RUBIACEAE](#), p. 108.
- 101a. Anthers more numerous than the lobes of the corolla [— 102.](#)
- 101b. Anthers just as many as the lobes of the corolla [— 103.](#)

- 101c. Anthers fewer than the lobes of the corolla — [110](#).
- 102a. Leaves simple [POLYGALACEAE](#), p. 65.
- 102b. Leaves compound with 3 leaflets [OXALIDACEAE](#), p. 64.
- 102c. Leaves finely dissected; stems climbing [2a, in FUMARIACEAE](#), p. 41.
- 103a. Ovaries 2 — [104](#).
- 103b. Ovary 1, but very deeply 4-lobed, with a single style [LABIATAE](#), p. 95.
- 103c. Ovary 1, not deeply lobed — [105](#).
- 104a. Stamens united, surrounding and more or less concealing the ovaries, and not resembling ordinary stamens [ASCLEPIADACEAE](#), p. 90.
- 104b. Stamens separate, of ordinary structure [APOCYNACEAE](#), p. 90.
- 105a. Ovary inferior, appearing as a swelling below the calyx at the apex of the pedicel — [106](#).
- 105b. Ovary superior, located in the center of the flower — [107](#).
- 106a. Corolla 3-4-lobed [RUBIACEAE](#), p. 108.
- 106b. Corolla 5-lobed [3b, in CAPRIFOLIACEAE](#), p. 109.
- 107a. Leaves toothed or deeply cut [VERBENACEAE](#), p. 99.
- 107b. Leaves entire — [108](#).
- 108a. A stamen in front of the middle of each [PRIMULACEAE](#), p. 87.

petal

108b. Stamens located
between the petals
or lobes of the
corolla, or else so
far down in the
tubular corolla that
their position is not
easily ascertained
— 109.

109a. Corolla salver-form, with a
very slender tube, and
abruptly spreading lobes
[POLEMONIACEAE](#), p.
92.

109b. Corolla salver-form, with a
wide tube and fringed blue
petals
[GENTIANACEAE](#), p. 88.

109c. Corolla rotate, funnel-form, or
bell-shape
[GENTIANACEAE](#), p. 88.

110a. Ovary deeply 4-
lobed, appearing
like 4 separate
ovaries; style 1
[LABIATAE](#), p. 95.

110b. Ovary not 4-lobed
— 111.

111a. Stamens 2 — 112.

111b. Stamens 3
[VALERIANACEAE](#), p.
111.

111c. Stamens 4 — 113.

112a. Flowers in dense
heads
[1a, in ACANTHACEAE](#),
p. 107.

112b. Flowers solitary or in
loose clusters
[SCROPHULARIACEAE](#),
p. 101.

113a. Corolla distinctly 2-lipped and
irregular — 114.

113b. Corolla not distinctly 2-lipped,
its 5 lobes all alike or nearly

so — 115.

114a. Calyx 2-lipped; the
upper lip with 3
awl-shape teeth, the
lower with 2 short PHRYMACEAE, p. 107.
teeth; flowers in
slender terminal
spikes

114b. Calyx not obviously
2-lipped, its teeth
equal or nearly so SCROPHULARIACEAE,
p. 101.

115a. Flowers sessile or nearly so, in
spikes — 116.

115b. Flowers in nodding pairs at the
top of a slender stalk 3a, in
CAPRIFOLIACEAE, p.
109.

115c. Flowers solitary or in clusters;
not in spikes or nodding pairs
— 117.

116a. Corolla not over 1
cm. long VERBENACEAE, p. 99.

116b. Corolla 1.5 cm. long
or more SCROPHULARIACEAE,
p. 101.

117a. Calyx-lobes 15-25 mm. long;
flowers blue, 3-5 cm. long;
calyx without an obvious tube 2, in ACANTHACEAE, p.
107.

117b. Calyx-lobes united below into
an obvious calyx-tube SCROPHULARIACEAE,
p. 101.

— 118—

118a. Leaves deeply lobed,
compound, or
dissected — 119.

118b. Leaves simple and
not deeply lobed
— 129.

- 119a. Leaves once-pinnately compound or lobed [— 120.](#)
- 119b. Leaves once-palmately compound or lobed [— 124.](#)
- 119c. Leaves dissected or 2-3 times compound [— 125.](#)
 - 120a. Leaves merely lobed [— 121.](#)
 - 120b. Leaves actually compound [— 123.](#)
- 121a. Flowers axillary; marsh or swamp plants [HALORAGIDACEAE](#), p. 79.
- 121b. Flowers in terminal clusters [— 122.](#)
 - 122a. Stamens 2-6; taste mustard-like [52b, in CRUCIFERAE](#), p. 46.
 - 122b. Stamens 10 or more [RESEDACEAE](#), p. 48.
- 123a. Leaflets entire [LIMNANTHACEAE](#), p. 68.
- 123b. Leaflets serrate [70a, in ROSACEAE](#), p. 57.
- 124a. Flowers solitary [31a, in RANUNCULACEAE](#), p. 38.
 - 124b. Flowers in dense terminal umbels or heads [UMBELLIFERAE](#), p. 80.
 - 124c. Flowers in terminal racemes [2a, in COMPOSITAE](#), p. 114.
 - 124d. Flowers in axillary spikes or panicles [9b, in URTICACEAE](#), p. 24.
- 125a. Leaves merely dissected, not truly compound with distinct leaflets [— 126.](#)
- 125b. Leaves truly compound, with distinct petiolate leaflets

- [127.](#)
- 126a. Flowers in axillary clusters [CHENOPODIACEAE](#), p. 28.
- 126b. Flowers in terminal clusters [COMPOSITAE](#), p. 113.
- 127a. Stamens 5; flowers in umbels — [128.](#)
- 127b. Stamens 6; flowers in small clusters [3b, in BERBERIDACEAE](#), p. 40.
- 127c. Stamens many; flowers in racemes or panicles [RANUNCULACEAE](#), p. 35.
- 128a. Styles 5 [ARALIACEAE](#), p. 80.
- 128b. Styles 2 [UMBELLIFERAE](#), p. 80.
- 129a. Flowers in dense cottony heads; plants more or less white-woolly [79, in COMPOSITAE](#), p. 120.
- 129b. Pistillate flowers in an ovoid spiny involucre, ripening into a bur [3a, in COMPOSITAE](#), p. 114.
- 129c. Flowers minute, subtended by palmately cleft axillary bracts [1a, in EUPHORBIACEAE](#), p. 66.
- 129d. Plants without any of the preceding characters — [130.](#)
- 130a. Plants with milky or colored juice [10, in EUPHORBIACEAE](#), p. 67.
- 130b. Plants with tendrils, at least on the upper leaves [4a, in LILIACEAE](#), p. 11.
- 130c. Plants of nettle-like character, with stinging hairs [8b, in URTICACEAE](#), p. 24.
- 130d. Plants with sheathing stipules, [POLYGONACEAE](#), p. 25.

surrounding the
stem above the base
of each leaf

130e. Plants with smooth,
pale, juicy, almost
translucent stems [12a, in URTICACEAE](#), p.
24.

130f. Plants without any of
the preceding
structures or habits
[— 131.](#)

131a. Flowers axillary, solitary or in
few-flowered clusters [— 132.](#)

131b. Flowers in terminal, or
terminal and axillary clusters
[— 138.](#)

132a. Leaves linear [CHENOPODIACEAE](#), p.
28.

132b. Leaves of a broader
shape than linear
[— 133.](#)

133a. Flower-clusters with bracts as
long as or longer than the
flowers [— 134.](#)

133b. Flowers without conspicuous
bracts [— 135.](#)

134a. Leaves broadest
below the middle [8a, in URTICACEAE](#), p.
24.

134b. Leaves broadest
above the middle [2, in
AMARANTHACEAE](#), p.
30.

135a. Principal leaves 3 cm. long or
more [— 136.](#)

135b. Principal leaves 2.5 cm. long
or less [— 137.](#)

136a. Flowers nodding in
the axils of the [1a, in VIOLACEAE](#), p.
75.

- leaves
 - 136b. Flowers erect in the axils [2a, in ONAGRACEAE](#), p. 78.
 - 137a. Leaves narrowly oblong [1a, in HALORAGIDACEAE](#), p. 79.
 - 137b. Leaves roundish and somewhat heart-shape [12a, in SAXIFRAGACEAE](#), p. 50.
 - 138a. Flowers in racemes or simple spikes [— 139.](#)
 - 138b. Flowers in panicles or other branched clusters [— 141.](#)
 - 139a. Leaves toothed or lobed [52b, in CRUCIFERAE](#), p. 46.
 - 139b. Leaves entire [— 140.](#)
 - 140a. Flowers sessile; leaves heart-shape [PIPERACEAE](#), p. 19.
 - 140b. Flowers pedicelled; leaves obovate [8b, in PRIMULACEAE](#), p. 87.
 - 141a. Individual flowers distinct from each other, on pedicels [— 142.](#)
 - 141b. Individual flowers crowded in close clusters, or separate and sessile [— 143.](#)
 - 142a. Leaves finely serrate [3b, in CRASSULACEAE](#), p. 49.
 - 142b. Leaves entire [1b, in CISTACEAE](#), p. 74.
 - 143a. Flower-clusters mingled with sharp-pointed bracts [AMARANTHACEAE](#), p. 30.
 - 143b. Flower-clusters without bracts, or (rarely) with bracts which are not sharp-pointed [CHENOPODIACEAE](#), p. 28.

- 144a. Juice not milky
— 145.
- 144b. Juice milky; apparent
flowers consisting
of a few white or
colored petal-like
bracts, inclosing a
few inconspicuous
flowers without
petals. In some of
them a 3-lobed
ovary with 3 styles
may be plainly seen
- 145a. Plants of the composite type,
with several or many small
flowerets closely aggregated
into a dense head subtended
by a calyx-like involucre of
small bracts — 146.
- 145b. Flowers solitary or in clusters,
but not in involucred heads
— 147.
- 146a. Leaves compound
with 3 leaflets;
stipules present
- 146b. Stipules none
- 147a. Leaves compound or deeply
lobed — 148.
- 147b. Leaves simple and not deeply
lobed — 154.
- 148a. Flowers irregular
- 148b. Flowers regular
— 149.
- 149a. Petals 3
- EUPHORBIACEAE**, p. 66.
- LEGUMINOSAE**, p. 58.
- COMPOSITAE**, p. 113.
- LEGUMINOSAE**, p. 58.
- LIMNANTHACEAE**, p. 68.

- 149b. Petals 4 — [150.](#)
- 149c. Petals 5 or more — [151.](#)
- 150a. Stamens 4 [70a, in ROSACEAE](#), p. 57.
- 150b. Stamens 6 [CRUCIFERAE](#), p. 42.
[43, in](#)
- 150c. Stamens numerous [RANUNCULACEAE](#), p. 39.
- 151a. Flowers in heads or umbels — [152.](#)
- 151b. Flowers in slender spikes or racemes — [153.](#)
- 152a. Styles 2 [UMBELLIFERAE](#), p. 80.
- 152b. Styles 5 [ARALIACEAE](#), p. 80.
[43, in](#)
- 153a. Leaves ternately compound [RANUNCULACEAE](#), p. 39.
- 153b. Leaves once-pinnately compound [50b, in ROSACEAE](#), p. 55.
- 154a. Flowers irregular — [155.](#)
- 154b. Flowers regular — [156.](#)
[40b, in](#)
- 155a. Stamens 2 [SCROPHULARIACEAE](#), p. 104.
- 155b. Stamens 5 or 10 [SAXIFRAGACEAE](#), p. 49.
- 155c. Stamens 6 or 8 [POLYGALACEAE](#), p. 65.
- 156a. With sheathing stipules surrounding the stem at the base of each leaf [POLYGONACEAE](#), p. 24.
- 156b. Stipules not encircling the stem, or none — [157.](#)

- 157a. Petals 3 [CISTACEAE](#), p. 74.
- 157b. Petals 4 [— 158.](#)
- 157c. Petals 5 [— 161.](#)
- 158a. Stamens 2 [— 159.](#)
- 158b. Stamens 6 [CRUCIFERAE](#), p. 42.
- 158c. Stamens 8 [— 160.](#)
- 159a. Corolla perfectly regular;
flowers in terminal clusters without bracts, becoming racemes; taste peppery [52a, in CRUCIFERAE](#), p. 46.
- 159b. Corolla slightly irregular;
flowers in bracted clusters, or axillary; taste not peppery [40b, in SCROPHULARIACEAE](#), p. 104.
- 160a. Delicate trailing evergreen, with flowers solitary in the axils and nearly or quite hidden beneath the leaves [22b, in ERICACEAE](#), p. 85.
- 160b. Erect or nearly so; flowers in terminal clusters [12b, in ONAGRACEAE](#), p. 78.
- 161a. Leaves sword-shape, finely parallel-veined, with bristly margins [2a, in UMBELLIFERAE](#), p. 80.
- 161b. Leaves not sword-shape [— 162.](#)
- 162a. Leaves toothed or lobed [— 163.](#)
- 162b. Leaves entire [— 164.](#)
- 163a. Stems creeping [2b, in UMBELLIFERAE](#), p. 80.
- 163b. Stems erect or nearly so [SAXIFRAGACEAE](#), p. 49.
- 164a. Flowers in open panicles; leaves [SAXIFRAGACEAE](#), p. 49.

- principally basal
 - 164b. Flowers in rounded or flattened clusters; leaves principally on the stem [SANTALACEAE](#), p. 24.
 - 164c. Flowers in racemes, which become elongated at maturity [— 165.](#)
- 165a. Foliage glabrous [8b, in PRIMULACEAE](#), p. 87.
- 165b. Foliage pubescent [BORAGINACEAE](#), p. 93.
- 166—
- 166a. Juice milky or colored [— 167.](#)
- 166b. Juice watery, not colored [— 172.](#)
- 167a. Apparent flowers consisting of a few petal-like bracts, inclosing a few inconspicuous flowers without petals. In some of them a 3-lobed ovary with 3 styles may be seen [EUPHORBIACEAE](#), p. 66.
- 167b. Plants of the composite type, with several or many small flowerets closely aggregated in dense heads subtended by a calyx-like involucre of small bracts [11, in COMPOSITAE](#), p. 115.
- 167c. Flowers never aggregated in involucred clusters resembling a single flower [— 168.](#)
- 168a. Corolla very irregular; stamens protruding [LOBELIACEAE](#), p. 113

- 168b. Corolla regular
— [169.](#)
- 169a. Stamens with ordinary visible
anthers; ovary 1 — [170.](#)
- 169b. Stamens so grown together and
to the stigma as to be almost
unrecognizable; ovaries 2 [ASCLEPIADACEAE](#), p.
90.
- 170a. Petals separate [PAPAVERACEAE](#), p. 41.
- 170b. Petals united — [171.](#)
- 171a. Stamens attached to the tube of [CONVOLVULACEAE](#), p.
the corolla 91.
- 171b. Stamens attached at the very [CAMPANULACEAE](#), p.
base of the corolla 112.
- 172a. Plants of the
composite type,
with several or
many small
flowerets closely
aggregated into
dense heads
subtended by a
calyx-like involucre
of small bracts
— [173.](#)
- 172b. Flowers solitary or
clustered, but not in
involucred heads
— [174.](#)
- 173a. True composites, without a
normal calyx [COMPOSITAE](#), p. 113.
- 173b. A normal calyx with each
floweret; leaves compound
with 3 leaflets [LEGUMINOSAE](#), p. 58.
- 174a. Flowers irregular, i.
e., the conspicuous
lobes of the

- perianth unlike in size or shape
— [175.](#)
 - 174b. Flowers regular
— [185.](#)
- 175a. Stamens 2 or 4 [SCROPHULARIACEAE](#), p. 101.
- 175b. Stamens 5 — [176.](#)
- 175c. Stamens 6 — [181.](#)
- 175d. Stamens 8; leaves simple [POLYGALACEAE](#), p. 65.
- 175e. Stamens 10; leaves usually compound [LEGUMINOSAE](#), p. 58.
- 175f. Stamens more than 10 — [183.](#)
 - 176a. Petals separate
— [177.](#)
 - 176b. Petals united — [179.](#)
- 177a. Flowers greenish or purplish, in a panicle [SAXIFRAGACEAE](#), p. 49.
- 177b. Flowers white or purplish, in compound umbels [UMBELLIFERAE](#), p. 80.
- 177c. Flowers solitary, or in few-flowered clusters — [178.](#)
 - 178a. Flowers blue, yellow, or white, in spring [VIOLACEAE](#), p. 75.
 - 178b. Flowers red-orange or yellow, in summer [BALSAMINACEAE](#), p. 71.
 - 178c. Flowers small and greenish, in spring [1a, in VIOLACEAE](#), p. 75.
- 179a. Anthers united; stamens protruding from the very irregular corolla [LOBELIACEAE](#), p. 113.
- 179b. Anthers separate; corolla almost regular — [180.](#)
 - 180a. Corolla rotate; some or all filaments [2, in SCROPHULARIACEAE](#),

- hairy p. 101.
- 180b. Corolla funnel-form,
dull yellow and [12a, in SOLANACEAE](#),
purple; filaments p. 101.
not hairy
- 180c. Corolla funnel-form,
blue or violet; [7b, in BORAGINACEAE](#),
filaments not hairy p. 94.
- 181a. Leaves compound or dissected [FUMARIACEAE](#), p. 41.
- 181b. Leaves simple [— 182.](#)
- 182a. Flowers solitary [ARISTOLOCHIACEAE](#),
p. 25.
- 182b. Flowers in spikes,
heads, or racemes [POLYGALACEAE](#), p. 65.
- 183a. Leaves truly compound [— 184.](#)
- 183b. Leaves palmately cleft; flowers [6a, in](#)
15 mm. wide or larger [RANUNCULACEAE](#), p.
36.
- 183c. Leaves irregularly cleft;
flowers about 5 mm. wide [RESEDACEAE](#), p. 48.
- 184a. Petals prolonged [6b, in](#)
backward into [RANUNCULACEAE](#), p.
hollow spurs 36.
- 184b. Petals not prolonged [CAPPARIDACEAE](#), p.
into spurs 47.
- [— 185—](#)
- 185a. Perianth consisting of one
circle of parts only (usually
considered to be the calyx)
[— 186.](#)
- 185b. Perianth consisting of both
calyx and corolla [— 192.](#)
- 186a. Leaves with [POLYGONACEAE](#), p.
sheathing stipules 25.
encircling the stem

- above the base of every leaf
 - 186b. Leaves without sheathing stipules
— [187.](#)
- 187a. Leaves entire — [188.](#)
- 187b. Leaves toothed, lobed, or compound — [190.](#)
 - 188a. Stamens 5 [SANTALACEAE](#), p. 24.
 - 188b. Stamens 6 — [189.](#)
 - 188c. Stamens 10 [PHYTOLACCACEAE](#), p. 30.
- 189a. Perianth-lobes 3; perianth-tube curved [ARISTOLOCHIACEAE](#), p. 25.
- 189b. Perianth-lobes 6, spreading [4a, in LILIACEAE](#), p. 11.
- 190a. Stamens 5 [UMBELLIFERAE](#), p. 80.
 - 190b. Stamens 6 — [191.](#)
 - 190c. Stamens more than 6 [RANUNCULACEAE](#), p. 35.
- 191a. Climbing plant with heart-shape perianth [2a, in FUMARIACEAE](#), p. 41.
- 191b. Erect plant, with spreading perianth [3b, in BERBERIDACEAE](#), p. 40.
- 192a. Corolla composed of united petals
— [193.](#)
 - 192b. Corolla composed of separate petals
— [208.](#)
- 193a. Stamens 2 or 4 [SCROPHULARIACEAE](#), p. 101.
- 193b. Stamens 3; climbing vines [CUCURBITACEAE](#), p. 112.
- 193c. Stamens 5 — [194.](#)

- 193d. Stamens 6 [2a, in FUMARIACEAE](#), p. 41.
- 193e. Stamens 8 [22b, in ERICACEAE](#), p. 85.
- 193f. Stamens 10 [— 207.](#)
- 193g. Stamens very numerous [MALVACEAE](#), p. 72.
- 194a. Ovaries 2; flowers orange-red, in umbels [7a, in ASCLEPIADACEAE](#), p. 91.
- 194b. Ovary 1, deeply 4-lobed; flowers in racemes [BORAGINACEAE](#), p. 93.
- 194c. Ovary 1, not deeply lobed [— 195.](#)
- 195a. Climbing or scrambling vines [— 196.](#)
- 195b. Not climbing or scrambling [— 198.](#)
- 196a. Flowers about 1 cm. wide [— 197.](#)
- 196b. Flowers 2-8 cm. wide [CONVOLVULACEAE](#), p. 91.
- 197a. Leaves 1 cm. wide or less [2b, in CAMPANULACEAE](#), p. 112.
- 197b. Leaves 2 cm. wide or more [4a, in SOLANACEAE](#), p. 100.
- 198a. Flowers solitary, either terminal or axillary [— 199.](#)
- 198b. Flowers in terminal clusters [— 201.](#)
- 199a. Calyx concealed by 2 bracts [CONVOLVULACEAE](#), p. 91.
- 199b. Calyx not completely concealed by bracts [— 200.](#)

- 200a. Ovary inferior,
appearing as a
swelling below the
calyx at the base of
the flower, 3-celled;
stigma 3-lobed [CAMPANULACEAE](#), p. 112.
- 200b. Ovary superior,
located in the center
of the flower [SOLANACEAE](#), p. 100.
- 201a. Some or all filaments hairy
— [202.](#)
- 201b. Filaments not hairy — [203.](#)
 - 202a. Leaves lobed or
divided [HYDROPHYLLACEAE](#),
p. 93.
[2, in](#)
 - 202b. Leaves not lobed [SCROPHULARIACEAE](#),
p. 101.
- 203a. Anthers close together, longer
than the filaments [2, in SOLANACEAE](#), p. 100.
- 203b. Anthers separate from each
other — [204.](#)
 - 204a. Leaves compound or
very deeply lobed
— [205.](#)
 - 204b. Leaves simple or
with shallow lobes
only — [206.](#)
- 205a. Leaf-segments linear or
oblong, irregular [HYDROPHYLLACEAE](#),
p. 93.
[1a, in](#)
- 205b. Leaves truly compound with
separate leaflets [POLEMONIACEAE](#), p. 92.
- 206a. Flowers greenish-
yellow [SOLANACEAE](#), p. 100.
- 206b. Flowers blue, violet,
white, or [CAMPANULACEAE](#), p. 112.

- intermediate tints
- 207a. Leaves simple [ERICACEAE](#), p. 84.
- 207b. Leaves compound [OXALIDACEAE](#), p. 64.
- 208 —
- 208a. Petals 4 [— 209.](#)
- 208b. Petals 5 [— 212.](#)
- 208c. Petals 6 or more
[— 227.](#)
- 209a. Stamens 4 or 8 [— 210.](#)
- 209b. Stamens 6, 4 long and 2 short [CRUCIFERAE](#), p. 42.
- 209c. Stamens 9 or more [CAPPARIDACEAE](#), p. 47.
- 210a. Leaves compound [ROSACEAE](#), p. 51.
- 210b. Leaves simple
[— 211.](#)
- 211a. Ovary 1, inferior, appearing as a swelling below the calyx [ONAGRACEAE](#), p. 78.
- 211b. Ovaries 4 or 5, in the center of the flower [CRASSULACEAE](#), p. 48.
- 212a. Stamens with good anthers 5 [— 213.](#)
- 212b. Stamens 6 to 10
[— 217.](#)
- 212c. Stamens more than 10 [— 222.](#)
- 213a. Flowers solitary, terminating the stem [12b, in SAXIFRAGACEAE](#), p. 50.
- 213b. Flowers axillary, solitary or in small clusters [— 214.](#)
- 213c. Flowers several, in loose irregular terminal clusters; leaves simple [LINACEAE](#), p. 63.
- 213d. Flowers in slender spike-like racemes; leaves compound [50b, in ROSACEAE](#), p. 55.

- 213e. Flowers in panicles [20a, in SAXIFRAGACEAE](#), p. 50.
- 213f. Flowers in umbels [— 216.](#)
- 214a. Leaves compound [25b, in LEGUMINOSAE](#), p. 60.
- 214b. Leaves simple [— 215.](#)
- 215a. Flowers blue or yellow, erect or spreading [LINACEAE](#), p. 63.
- 215b. Flowers greenish, nodding [1a, in VIOLACEAE](#), p. 75.
- 216a. Flowers pink or purple [GERANIACEAE](#), p. 64.
- 216b. Flowers yellow or white [UMBELLIFERAE](#), p. 80.
- 217a. Leaves compound with 3 leaflets [— 218.](#)
- 217b. Leaves pinnately compound [— 219.](#)
- 217c. Leaves deeply palmately lobed [GERANIACEAE](#), p. 64.
- 217d. Leaves simple and not deeply lobed [— 220.](#)
- 218a. Leaflets reverse heart-shape; flowers yellow [OXALIDACEAE](#), p. 64.
- 218b. Leaflets taper-pointed; flowers white or pink [73, in ROSACEAE](#), p. 57.
- 219a. Leaflets entire [24, in LEGUMINOSAE](#), p. 60.
- 219b. Leaflets toothed [50b, in ROSACEAE](#), p. 55.
- 220a. Prostrate plants, with thick, fleshy, entire leaves [3, in PORTULACACEAE](#), p. 34.

- 220b. Bushy branched plants, with small gray leaves concealing the stem [2a, in CISTACEAE](#), p. 74.
- 220c. Erect or spreading plants [— 221.](#)
- 221a. Ovary 1, style 1 [4a, in ERICACEAE](#), p. 84.
- 221b. Ovary with 2 distinct styles [SAXIFRAGACEAE](#), p. 49.
- 221c. Ovaries 5; styles 5 [CRASSULACEAE](#), p. 48.
- 222a. Stamens united by their filaments into a tube [MALVACEAE](#), p. 72.
- 222b. Stamens separate from each other [— 223.](#)
- 223a. Leaves with stipules; ovaries more than 1 [ROSACEAE](#), p. 51.
- 223b. Leaves without stipules [— 224.](#)
- 224a. Leaves toothed, deeply lobed, or compound [RANUNCULACEAE](#), p. 35.
- 224b. Leaves entire [— 225.](#)
- 225a. Ovaries numerous [RANUNCULACEAE](#), p. 35.
- 225b. Ovary 1 [— 226.](#)
- 226a. Sepals 2 [PORTULACACEAE](#), p. 34.
- 226b. Sepals 3 or 5 [CISTACEAE](#), p. 74.
- 227—
- 227a. Leaves entire [LYTHRACEAE](#), p. 77.
- 227b. Leaves lobed, divided, dissected, or compound [— 228.](#)

228a. Flowers in slender racemes [RESEDACEAE](#), p. 48.

228b. Flowers solitary or clustered, but not in slender racemes
[— 229.](#)

229a. Stamens 6 [3b, in](#)
[BERBERIDACEAE](#), p. 40.

229b. Stamens numerous [RANUNCULACEAE](#), p. 35.

THE PLANTS OF MICHIGAN

PINACEAE, the Pine Family

Trees or shrubs, usually evergreen, with needle-like or scale-like leaves; fruit a cone or berry.

- 1a. Leaves in clusters of 2-5 —2.
- 1b. Leaves mostly in clusters of 10 or more,
on short lateral wart-like branches,
deciduous each autumn **Tamarack, *Larix laricina*.**
- 1c. Leaves not in clusters —4.
 - 2a. Leaves in clusters of 5 **White Pine, *Pinus strobus*.**
 - 2b. Leaves in clusters of 2 or 3
—3.
 - 3a. Leaves 8-15 cm. long **Norway Pine, *Pinus resinosa*.**
 - 3b. Leaves 2-4 cm. long **Jack Pine, *Pinus banksiana*.**
 - 4a. Leaves alternate or scattered
—5.
 - 4b. Leaves opposite or whorled
—8.
 - 5a. Leaves four-sided —6.
 - 5b. Leaves flattened —7.
 - 6a. Leaves 6-12 mm. long **Black Spruce, *Picea mariana*.**
 - 6b. Leaves 15-25 mm. long **White Spruce, *Picea canadensis*.**

- | | |
|--|---|
| 7a. Leaves short-stalked, 15 mm. long or less | Hemlock, <i>Tsuga canadensis</i>. |
| 7b. Leaves sessile, 15-30 mm. long | Balsam, <i>Abies balsamea</i>. |
| 8a. Leafy twigs soft and flattened | White Cedar, <i>Thuja occidentalis</i>. |
| 8b. Leafy twigs not distinctly flattened <u>— 9.</u> | |
| 9a. Leaves opposite <u>— 10.</u> | |
| 9b. Leaves in whorls of three <u>— 11.</u> | |
| 10a. Erect shrub or tree | Red Cedar, <i>Juniperus virginiana</i>. |
| 10b. Prostrate or spreading shrub | Creeping Cedar, <i>Juniperus horizontalis</i>. |
| 11a. Erect shrub or small tree | Juniper, <i>Juniperus communis</i>. |
| 11b. Spreading or ascending shrub, growing in dense mats | Low Juniper, <i>Juniperus communis</i> var. <i>depressa</i>. |

TAXACEAE, the Yew Family

Shrubs, with needle-like evergreen leaves; fruit red and berry-like.

- | | |
|---|---|
| One species in Michigan; straggling shrub 1-3 m. high | Ground Hemlock, <i>Taxus canadensis</i>. |
|---|---|

TYPHACEAE, the Cat-tail Family

Erect plants 1-2 m. high, with linear leaves and terminal spikes of brown flowers, appearing in summer.

1a. Staminate and pistillate portions of the flower-spike contiguous, the latter 2.5 cm. in diameter

**Common Cat-tail,
Typha latifolia.**

1b. Staminate and pistillate portions of the spike separated, the latter 2 cm. or less in diameter

**Narrow-leaved
Cat-tail, *Typha angustifolia*.**

SPARGANIACEAE, the Bur-reed Family

Marsh plants with linear leaves and spherical heads of inconspicuous greenish flowers, appearing in summer.

About 5 species occur in Michigan, of which the commonest is

**Bur-reed,
Sparganium eurycarpum.**

NAJADACEAE, the Pondweed Family

Aquatic plants with submerged or floating leaves and inconspicuous flowers in summer.

1a. Leaves opposite or whorled —2.

1b. Leaves alternate —2c.

2a. Leaves thread-like, 3-8 cm.
long

**Horned
Pondweed,
Zannichellia palustris.**

2b. Leaves linear, toothed, abruptly
dilated at the base, 3 cm.
long or less (Naiad) —3.

- | | |
|--|--|
| 2c. Leaves entire, not abruptly
dilated at base | ^[2] Pondweed,
Potamogeton spp. |
| 3a. Leaves about 2 mm. wide, sharply and
coarsely toothed | Naiad, Naias
marina. |
| 3b. Leaves very narrowly linear, with
numerous minute teeth | Naiad, Naias
flexilis. |

JUNCAGINACEAE, the Arrow Grass Family

Marsh plants, with linear cylindrical leaves and inconspicuous flowers in spikes or racemes, appearing in early summer.

- | | |
|--|--|
| 1a. Leaves all basal; flowers numerous in a
spike-like raceme (Arrow Grass)
<u>—2.</u> | |
| 1b. Stem-leaves present; flowers in a loose
bracted raceme (1-3 dm. high). | Scheuchzeria,
Scheuchzeria
palustris. |
| 2a. Fruit (usually to be seen at the
base of the raceme) ovoid
or oblong, rounded at the
base | Arrow Grass,
Triglochin
maritima. |
| 2b. Fruit linear, narrowed at the
base (1-5 dm. high) | Arrow Grass,
Triglochin
palustris. |

ALISMACEAE, the Water Plantain Family

Marsh plants, with scape-like stems; flowers with 3 green sepals, 3 white petals, 6 or more stamens, and several separate pistils.

- | | |
|---|--|
| 1a. Ovaries in a ring; flowers in panicles (2-8 dm. high, summer) | Water Plantain,
Alisma Plantago-aquatica. |
| 1b. Ovaries in a head; flowers in racemes or umbels <u>—2.</u> | |
| 2a. Flowers all perfect, in a single umbel of 2-8 flowers; stamens 9 (leaves lanceolate; 15 cm. high or less; summer) | Dwarf Water Plantain,
Echinodorus tenellus. |
| 2b. Flowers in a raceme of 3-flowered whorls, the lower pistillate, the upper staminate; stamens usually more than nine (1-10 dm. high, summer) (Arrow-head) <u>—3.</u> | |
| 3a. Leaves ovate to linear, not sagittate at base <u>—4.</u> | |
| 3b. Leaves broad or narrow, sagittate at base <u>—5.</u> | |
| 4a. Pistillate (basal) flowers sessile or nearly so (2-8 dm. high, summer) | Arrow-head,
Sagittaria heterophylla. |
| 4b. Pistillate flowers with obvious pedicels | Arrow-head,
Sagittaria graminea. |
| 5a. Basal lobes of the leaf conspicuous, triangular, almost or quite as long as the terminal portion <u>—6.</u> | |
| 5b. Basal lobes small, short, linear <u>—4b.</u> | |
| 6a. Beak of the achene very short and erect; rare species | Arrow-head,
Sagittaria arifolia. |
| 6b. Beak of the achene sharp, incurved at right angles to the body; common species | Arrow-head,
Sagittaria latifolia. |

HYDROCHARITACEAE, the Frog's Bit Family

Submerged aquatics, with inconspicuous flowers in summer.

- 1a. Leaves all from the base, 2 dm. long or more

**Eel Grass,
Vallisneria
spiralis.**

- 1b. Leaves on the stem, 2 cm. long or less

**Water-weed,
Elodea canadensis.**

GRAMINEAE, the Grass Family

Grasses, with linear or narrow sheathing leaves, and very small flowers without perianth in the axils of chaffy bracts, appearing in late spring and summer.

Of the large number (over 150) of grasses in Michigan, only the commonest are included here, and the student is referred to the Manuals for a full treatment of them.

Their classification depends chiefly upon the structure and arrangement of the spikelets. These consist typically of a short axis, the rachilla, almost or quite concealed by several chaffy bracts. The two lower bracts are termed glumes, and have no flowers in their axils. Above the glumes are two or more other bracts, the lemmas. In the axil of each lemma, and usually concealed by it, is a smaller bract, the palea, and between the lemma and the palea is a single flower. The number of flowers in a spikelet is therefore normally equal to the number of lemmas. The spikelets are grouped in racemes, spikes, or panicles of various size.

- 1a. Spikelets one-flowered —2.

- 1b. Spikelets with 2 or more flowers —24.

- 2a. Spikelets grouped into dense solitary cylindrical spikes —3.
- 2b. Spikelets arranged in panicles or in panicle spikes —8.
- 3a. Spikelets without awns or bristles, or with short awns not more than 3 mm. long —4.
- 3b. Spikelets with awns 2-5 cm. long, terminating the bracts
- 3c. Bracts of the spikelet without terminal awns, but the spikelets with one or more long bristles arising from their base —6.
 - 4a. Spike-like panicle thickened in the middle, more than 1 cm. thick
 - 4b. Spike little or not at all thickened in the middle, less than 1 cm. thick —5.
- 5a. Lower bracts awned; stem erect, unbranched
- 5b. Lower scales unawned; stem branched at the base
 - 6a. Bristles 5 or more at the base of each spikelet
 - 6b. Bristles 1-3 at the base of each spikelet —7.
- 7a. Spikelets about 2 mm. long; bristles not much longer, green
- 7b. Spikelets about 3 mm. long; bristles much longer, usually purple
 - 8a. Spikelets numerous, in long slender symmetrical spikes

**Squirrel-tail,
Hordeum
jubatum.**

**Beach Grass,
Ammophila
arenaria.**

**Timothy, Phleum
pratense.**

**Floating Foxtail,
Alopecurus
geniculatus.**

**Yellow Foxtail,
Setaria glauca.**

**Green Foxtail,
Setaria viridis.**

**Millet, Setaria
italica.**

— 9.

8b. Spikelets in panicles, racemes,
or loose spikes — 13.

9a. Spikelets without awns; plants 8 dm.
high or less (Crab Grass) — 10.

9b. Spikelets with awns; plants 12 dm. high
or more — 12.

10a. Leaf-sheaths all glabrous.

10b. Lower leaf-sheaths hairy
— 11.

11a. Axis of the spike flat, with wing-like
margins

11b. Axis of the spike slender, without
winged margins

12a. Spikes numerous, appressed
to the axis of the panicle;
tall marsh grass

12b. Spikes 2-6, widely divergent;
plant of dry ground

13a. Spikelets subtended by an ovoid thorny
involucre 3-8 mm. wide

13b. Spikelets without a thorny involucre
— 14.

14a. Lower branches of the panicle
spreading, bearing
staminate flowers, the
upper branches erect, with
pistillate flowers; aquatic
or marsh grass 2-4 m. high

**Crab Grass,
Digitaria
humifusa.**

**Crab Grass,
Digitaria
sanguinalis.**

**Crab Grass,
Digitaria
filiformis.**

**Slough Grass,
Spartina
michauxiana.**

**Blue-joint,
Andropogon
furcatus.**

**Sand Bur,
Cenchrus
carolinianus.**

**Wild Rice, Zizania
aquatica.**

- 14b. Panicle uniform throughout
— [15.](#)
- 15a. Spikelets with awns 2 mm. or more long
— [16.](#)
- 15b. Spikelets not awned, or with short
inconspicuous awns — [18.](#)
 - 16a. Spikelets in solitary raceme-
like spikes; awn about 1
cm. long or more **Beard Grass,**
Andropogon
scoparius.
 - 16b. Spikelets in branching
clusters; awn less than 1
cm. long — [17.](#)
- 17a. Leaf-blade 4 mm. wide or narrower;
panicle slender **Drop-seed,**
Muhlenbergia
schreberi.
- 17b. Leaf-blade 6 mm. wide or more; panicle
stout and coarse **Barnyard Grass,**
Echinochloa crus-
galli.
- 18a. Spikelet plump and compact,
its bracts closely folded
about each other — [19.](#)
- 18b. Spikelet very flat, its two
bracts closely folded
together **Cut-grass, Leersia**
oryzoides.
- 18c. Spikelet loose and open,
somewhat flattened, its 3
bracts ascending or
spreading and not closely
folded about each other
— [20.](#)
- 19a. Panicle about half as long as the entire
plant; leaves copiously hairy **Witch Grass,**
Panicum capillare.
- 19b. Panicle of smaller size **Panic-grasses,**
various species of
Panicum.

- 20a. Panicle strongly contracted or spike-like; plants of sand-dunes — 4a.
- 20b. Panicle spreading or slightly contracted; axis of the spikelet beset with bristles; leaves 2 dm. long or more; marsh grass
**Reed Grass,
Calamagrostis canadensis.**
- 20c. Panicle spreading or somewhat contracted, but not spike-like; axis of the spikelet without bristles
— [21.](#)
- 21a. Panicle-branches erect or ascending
— [22.](#)
- 21b. Panicle-branches strongly spreading
— [23.](#)
- 22a. The two outer scales of the spikelet one-fourth as long as the third scale, or sometimes one of them absent — 17a.
- 22b. The glumes at least half as long as the lemma
**Wood-grass,
Muhlenbergia mexicana.**
- 23a. The chief lateral branches of the panicle dividing and bearing flowers below their middle
Red-top, Agrostis alba.
- 23b. The chief branches of the panicle dividing only beyond the middle
**Hair Grass,
Agrostis hyemalis.**
- 24a. Spikelets arranged in two rows to form a definite spike — [25.](#)
- 24b. Spikelets in panicles, never in definite rows — [29.](#)

- 25a. Spikelets in a single row on one side of the axis, forming a one-sided spike **Yard Grass, *Eleusine indica*.**
- 25b. Spikelets alternating on opposite sides of the axis, forming a two-rowed spike — [26](#).
- 26a. Spikelets in pairs at each joint, forming a dense spike (Wild Rye) — [27](#).
- 26b. Spikelets single at each joint, forming a loose, open or interrupted spike — [28](#).
- 27a. Glumes lanceolate **Wild Rye, *Elymus canadensis*.**
- 27b. Glumes narrowly subulate **Wild Rye, *Elymus virginicus*.**
- 28a. Spikelets with their edges toward the axis of the spike **Rye Grass, *Lolium perenne*.**
- 28b. Spikelets with their sides toward the axis of the spike **Quack Grass, *Agropyron repens*.**
- 29a. Glumes longer than the lemmas **Oats, *Avena sativa*.**
- 29b. Glumes shorter than the lemmas — [30](#).
- 30a. Axis of the spikelet beset with conspicuous long hairs about equaling the lemmas; tall marsh grass 1-4 m. high **Reed, *Phragmites communis*.**
- 30b. Spikelets without conspicuous long hairs — [31](#).
- 31a. Spikelets sessile or nearly so, forming crowded or spike-like panicles — [32](#).
- 31b. Spikelets distinctly panicked — [33](#).
- 32a. Spikelets in dense one-sided clusters at the ends of the panicle branches **Orchard Grass, *Dactylis glomerata*.**
- 32b. Spikelets in an erect spike-like cluster **Prairie June-grass, *Koeleria cristata*.**

- 33a. Lemmas, exclusive of the awn when present, 8 mm. long or more — [34.](#)
- 33b. Lemmas, exclusive of the awn when present, 6 mm. long or less — [35.](#)
- 34a. Awns on the lemmas 12 mm. long or more **Brome-grass, *Bromus tectorum.***
- 34b. Awns on the lemmas 8 mm. long or less, or none **Cheat, *Bromus secalinus.***
- 35a. Lemmas with 7 sharp conspicuous veins from base to apex **Manna Grass, *Glyceria nervata.***
- 35b. Lemmas with 3-5 inconspicuous veins — [36.](#)
- 36a. Spikelets with 5 flowers or more — [37.](#)
- 36b. Spikelets with 2-4 (rarely 5) flowers — [40.](#)
- 37a. Stems tufted and decumbent at base (Love Grass) — [38.](#)
- 37b. Stems erect (Fescue Grass) — [39.](#)
- 38a. Spikelets 1.5 mm. wide **Love Grass, *Eragrostis pilosa.***
- 38b. Spikelets 3 mm. wide **Love Grass, *Eragrostis megastachya.***
- 39a. Lemmas with conspicuous awns **Fescue Grass, *Festuca octoflora.***
- 39b. Lemmas without awns **Fescue Grass, *Festuca elatior.***
- 40a. Tufted annual grass **Spear Grass, *Poa annua.***
- 40b. Perennials, with erect flowering stems — [41.](#)
- 41a. Stems round **Blue Grass, *Poa pratensis.***
- 41b. Stems strongly flattened **Canadian Blue Grass, *Poa***

compressa.

CYPERACEAE, the Sedge Family

Grass-like or rush-like plants, with linear leaves or leafless, and inconspicuous flowers in small chaffy spikes.

Over 200 species occur in Michigan, of which only the commonest are included here. For the remaining species the Manuals should be consulted.

1a. Spikes all alike [— 2.](#)

1b. The uppermost spike or spikes wholly staminate, the lower one or more pistillate; ovary and achene surrounded by a sac, the perigynium. Mature fruit is necessary for satisfactory identification (Sedge) [— 12.](#)

2a. Stems leafless, bearing one or more spikes at or near the top [— 3.](#)

2b. Stems leafy [— 6.](#)

3a. Spike one, terminal and erect (Spike Rush) [— 4.](#)

3b. Spikes usually more than one, lateral and spreading [— 5.](#)

4a. Annual, with fibrous roots

**Spike Rush,
Eleocharis obtusa.**

4b. Perennial, with a running rootstock

**Spike Rush,
Eleocharis
palustris.**

5a. Stem round

**Bulrush, Scirpus
validus.**

5b. Stem 3-cornered

**Three-square,
Scirpus
americanus.**

- | | |
|---|---|
| 6a. Spikes axillary along the side of the stem | Dulichium,
Dulichium,
arundinaceum. |
| 6b. Spikes terminal <u>—7.</u> | |
| 7a. Spikes subtended by long conspicuous leaves which greatly exceed the flower clusters <u>—8.</u> | |
| 7b. Spikes not conspicuously exceeded by the bract-like leaves <u>—10.</u> | |
| 8a. Spikes in a dense head-like cluster, white-woolly at maturity | Cotton Grass,
Eriophorum
virginicum. |
| 8b. Spikes not in dense heads, nor white-woolly at maturity <u>—9.</u> | |
| 9a. Perennial by a creeping rootstock | Nut Grass,
Cyperus
esculentus. |
| 9b. Perennial by hard basal corms | Cyperus, Cyperus
strigosus. |
| 10a. Bracts of the spike chestnut-brown | Twig Rush,
Cladium
mariscoides. |
| 10b. Bracts of the spike green or straw-color (Sedge) <u>—11.</u> | |
| 11a. Spikes 3-8, separate | Sedge, Carex
straminea. |
| 11b. Spikes very numerous and densely crowded | Sedge, Carex
vulpinoidea. |
| 12a. Achenes flattened | Sedge, Carex
crinita. |
| 12b. Achenes 3-angled <u>—13.</u> | |
| 13a. Perigynium tipped with a sharp straight 2-toothed beak <u>—14.</u> | |
| 13b. Perigynium with a short soft beak <u>—17.</u> | |

- | | |
|--|--|
| 14a. Perigynium thin and papery,
loosely enclosing the
achene — 15 . | |
| 14b. Perigynium firm, closely
enclosing the achene — 16 . | |
| 15a. Perigynium less than 1 cm. long | Sedge, Carex
hystericina. |
| 15b. Perigynium more than 1 cm. long | Sedge, Carex
lupulina. |
| 16a. Perigynium smooth | Sedge, Carex
riparia. |
| 16b. Perigynium hairy | Sedge, Carex
filiformis. |
| 17a. Beak of the perigynium bent abruptly to
one side | Sedge, Carex
laxiflora. |
| 17b. Beak of the perigynium straight | Sedge, Carex
pennsylvanica. |

ARACEAE, the Arum Family

Individual flowers small, but crowded on a fleshy spadix to form a conspicuous spike, usually surrounded by a green or colored spathe.

- | | |
|---|--|
| 1a. Leaves compound — 2 . | |
| 1b. Leaves simple — 3 . | |
| 2a. Leaflets 3, spathe pale green or
purple (3-6 dm. high;
spring) | Indian Turnip,
Arisaema
triphyllum. |
| 2b. Leaflets 7-11; spathe green;
spadix long and slender (3-
8 dm. high; late spring) | Dragon Root,
Arisaema
dracontium. |
| 3a. Leaves linear, sword-shape; spathe none
(5-15 dm. high; early summer) | Sweet Flag, Acorus
calamus. |

- 3b. Leaves broader than linear; spathe present —4.
- 4a. Flower clusters partly underground, appearing in earliest spring **Skunk Cabbage, *Symplocarpus foetidus*.**
- 4b. Flower clusters peduncled, in early summer —5.
- 5a. Leaves broadly ovate-cordate; spathe white **Water Arum, *Calla palustris*.**
- 5b. Leaves more or less sagittate; spathe green **Arrow Arum, *Peltandra virginica*.**

LEMNACEAE, the Duckweed Family

Minute leafless plants floating on quiet water; flowers exceedingly small and seldom seen.

- 1a. Plant thick, ovoid, less than 2 mm. long; roots none. Two species are reported from Michigan ***Wolffia* spp.**
- 1b. Plant flattened, with short roots —2.
- 2a. Roots several from each rounded plant **Duckweed, *Spirodela polyrhiza*.**
- 2b. Root single from each rounded plant **Duckweed, *Lemna* spp.**
- Three species are reported from Michigan, of which the commonest is *Lemna minor*.

ERIOCAULACEAE, the Pipewort Family

Bog or marsh herbs, with small flowers in heads terminating long slender scapes.

One species in Michigan; leaves linear and basal; flower-stalk 5-15 cm. high; flower-heads whitish or lead-color

**Pipewort,
Eriocaulon
articulatum.**

XYRIDACEAE, the Yellow-eyed Grass Famil

Small herbs with basal leaves and erect flower-stalks bearing a head of perfect yellow flowers, in summer.

- 1a. Base of plant bulbous-thickened (3-6 dm. high) **Yellow-eyed Grass,
Xyris flexuosa.**
- 1b. Base of plant not bulbous-thickened (1-3 dm. high) **Yellow-eyed Grass,
Xyris montana.**

COMMELINACEAE, the Spiderwort Family

Leafy-stemmed herbs; flowers with 3 sepals, 3 petals, and 6 stamens, lasting but a single day; petals blue.

- 1a. Stamens 6; petals all equal (Spiderwort)
—2.
- 1b. Perfect stamens 3, sterile stamens 3; two of the petals larger than the third (3-6 dm. high; summer)
 - 2a. Sepals villous (3-10 dm. high; late spring) **Day-flower,
Commelina
virginica.
Spiderwort,
Tradescantia
virginiana.**

2b. Sepals glabrous, or with a tuft
of hairs at the apex (4-10
dm. high; late spring)

**Spiderwort,
Tradescantia
reflexa.**

PONTEDERIACEAE, the Pickerel-weed Family

Aquatic herbs, with 6 rather conspicuous petals; flowers in summer.

1a. Flowers blue; leaves cordate-sagittate (3-
10 dm. high)

**Pickerel-weed,
Pontederia
cordata.**

1b. Flowers yellow; leaves linear
(submerged)

**Mud Plantain,
Heteranthera
dubia.**

JUNCACEAE, the Rush Family

Grass-like or rush-like plants, with inconspicuous greenish or brownish
flowers, of 3 chaffy or scale-like sepals and as many similar petals.

1a. Leaf-sheaths closed; capsule 1-celled and
3-seeded; stem or leaves usually
hairy at or near the base (1-4 dm.
high). (Wood Rush) 2.

1b. Leaf-sheaths open; capsule many-seeded;
plants never hairy 4.

2a. Flowers solitary at the ends of
the branches of the umbel-
like cluster (spring)

**Wood Rush,
Luzula saltuensis.**

2b. Flowers in spikes or dense
clusters 3.

3a. Flower-cluster spike-like, nodding at the tip (summer)

**Wood Rush,
Luzula spicata.**

3b. Flower-cluster umbel-like (spring)

**Wood Rush,
Luzula campestris
var. *multiflora*.**

4. The genus *Juncus*, or Rush, contains about 25 species in Michigan, blooming in summer or autumn. For their identification the Manuals should be consulted. One of the commonest species is *Juncus effusus*, growing in marshes, with erect leafless cylindrical stems, bearing a lateral cluster of flowers near the summit. Another common species is *Juncus tenuis*, with slender stems and linear leaves, growing in hard ground, especially in woodland paths.

LILIACEAE, the Lily Family

Herbs or twining shrubs, with generally conspicuous flowers; sepals and petals each 3, and usually colored alike, stamens 6, ovary 3-celled, superior. In one species the perianth is 4-parted and the stamens are 4.

1a. Flowers or flower-clusters lateral,
axillary or apparently so [— 2.](#)

1b. Flowers or flower-clusters scapose or
terminal [— 12.](#)

- | | |
|--|---|
| 2a. Leaves minute and scale-like
(7-15 dm. high; flowers
greenish-yellow, June) | Asparagus,
Asparagus
officinalis. |
| 2b. Leaves broad and flat, not
scale-like <u>—3.</u> | |
| 3a. Flowers numerous in rounded umbels;
perianth-segments nearly separate;
leaves long-petioled <u>—4.</u> | |
| 3b. Flowers in clusters of 1-8; leaves short-
petioled, sessile, or clasping <u>—8.</u> | |
| 4a. Stems herbaceous (flowers
greenish-yellow, ill-
scented) (Carrion-flower)
<u>—5.</u> | |
| 4b. Stems woody, thorny, climbing
(flowers greenish-yellow,
early summer) (Green
Brier) <u>—7.</u> | |
| 5a. Stems climbing by tendrils <u>—6.</u> | |
| 5b. Stem not climbing; only the upper
leaves, or none, with tendrils | Carrion-flower,
Smilax ecirrhata. |
| 6a. Leaves smooth beneath. | Carrion-flower,
Smilax herbacea. |
| 6b. Leaves minutely pubescent
beneath when mature | Carrion-flower,
Smilax herbacea
var. pulverulenta. |
| 7a. Leaves with 5 principal veins. | Green Brier,
Smilax
rotundifolia. |
| 7b. Leaves with 7 principal veins. | Green Brier,
Smilax hispida. |
| 8a. Perianth-segments united into a
tube (flowers axillary, late
spring) (Solomon's Seal)
<u>—9.</u> | |

- 8b. Perianth-segments separate (2-8 dm. high; spring)
(Twisted-stalk) — [10.](#)
- 9a. Leaves minutely pubescent beneath; filaments rough, inserted at three-fourths the length of the perianth (3-8 dm. high) **Small Solomon's Seal, *Polygonatum biflorum*.**
- 9b. Leaves smooth beneath; filaments smooth, inserted at the middle of the perianth tube (5-15 dm. high) **Great Solomon's Seal, *Polygonatum commutatum*.**
 - 10a. Leaves distinctly clasping the stem; flowers greenish-white **Twisted-stalk, *Streptopus amplexifolius*.**
 - 10b. Leaves closely sessile; flowers reddish to purple — [11.](#)
- 11a. Rootstock short and thick; berries spherical **Twisted-stalk, *Streptopus roseus*.**
- 11b. Rootstock long and slender; berries 3-angled **Twisted-stalk, *Streptopus longipes*.**
 - 12a. Perianth-segments 5-12 cm. long — [13.](#)
 - 12b. Perianth-segments shorter than 5 cm. — [19.](#)
- 13a. Leaves all or chiefly basal, stem-leaves bract-like or none — [14.](#)
- 13b. Leaves chiefly or entirely on the stem — [15.](#)
 - 14a. Leaves numerous, linear or sword-shape (flowers orange, summer) **Day Lily, *Hemerocallis fulva*.**
 - 14b. Leaves a single pair, oblong or lanceolate — [21.](#)
- 15a. Leaves a single whorl of 3 (Wake Robin, Trillium) — [22.](#)

- 15b. Leaves numerous (6-12 dm. high; flowers yellow, orange, or red, in summer) (Lily) [— 16.](#)
 - 16a. Flowers erect [— 17.](#)
 - 16b. Flowers nodding [— 18.](#)
- 17a. Leaves narrowly lanceolate, 6-15 mm. wide, mostly whorled **Wood Lily, *Lilium philadelphicum*.**
- 17b. Leaves linear, 5 mm. wide or less, almost all alternate **Wood Lily, *Lilium philadelphicum* var. *andinum*.**
 - 18a. Perianth-segments strongly revolute **Turk's-cap Lily, *Lilium superbum*.**
 - 18b. Perianth-segments half-recurved **Yellow Lily, *Lilium canadense*.**
- 19a. Flower solitary [— 20.](#)
- 19b. Flowers in clusters, not solitary [— 31.](#)
 - 20a. Stem with a single pair of basal leaves (Dog's-tooth Violet) [— 21.](#)
 - 20b. Stem with a single whorl of 3 leaves (1-3 dm. high; spring) (Wake Robin, Trillium) [— 22.](#)
 - 20c. Stem leafy (Bellwort) [— 29.](#)
- 21a. Perianth yellow **Dog's-tooth Violet, *Erythronium americanum*.**
- 21b. Perianth white, bluish, or pinkish **Dog's-tooth Violet, *Erythronium albidum*.**
 - 22a. Flower sessile, red or brown [— 23.](#)
 - 22b. Flower peduncled, white or pink, rarely red [— 24.](#)
- 23a. Leaves sessile, sepals spreading **Wake Robin, *Trillium sessile*.**

23b. Leaves short-petioled, sepals reflexed	Wake Robin, Trillium recurvatum.
24a. Ovary with 6 distinct wing-like angles —25.	
24b. Ovary obtusely 3-angled or lobed —28.	
25a. Stamens distinctly longer than the pistil —26.	
25b. Stamens equaling or shorter than the pistil —27.	
26a. Stigmas erect or nearly so, slender	Wake Robin, Trillium grandiflorum.
26b. Stigmas strongly recurved or spreading	Wake Robin, Trillium erectum.
27a. Filaments about as long as the anthers	Wake Robin, Trillium cernuum.
27b. Filaments half as long as the anthers or shorter	Wake Robin, Trillium declinatum.
28a. Leaves obtuse; petals obtuse, white	Dwarf White Trillium, Trillium nivale.
28b. Leaves acuminate; petals acute, purple-striped at base	Painted Trillium, Trillium undulatum.
29a. Leaves sessile (3-6 dm. high; flowers yellow, spring)	Bellwort, Oakesia sessilifolia.
29b. Leaves perfoliate (4-6 dm. high; flowers yellow, in spring) —30.	
30a. Leaves minutely pubescent beneath	Bellwort, Uvularia grandiflora.
30b. Leaves glabrous and all glaucous	Bellwort, Uvularia perfoliata.

- 31a. Stem bearing 2 whorls of 3-9 leaves (3-6 dm. high; flowers pale yellow, early summer) **Indian Cucumber-root, *Medeola virginiana*.**
- 31b. Stem-leaves not whorled, or all leaves basal [— 32.](#)
- 32a. Flowers in umbels [— 33.](#)
- 32b. Flowers in racemes or panicles [— 38.](#)
- 33a. Plant with the odor of onions or garlic (leaves all or chiefly basal; flower-stalks 2-8 dm. high, late spring or summer) [— 34.](#)
- 33b. Plant not with the odor of onions (leaves basal; flower-stalks 2-3 dm. high, with an umbel of 3-6 greenish-yellow flowers in late spring) **Clintonia, *Clintonia borealis*.**
- 34a. Leaves oblong, 2-5 cm. wide, not present when the plants are in bloom (greenish-white flowers) **Wild Leek, *Allium tricoccum*.**
- 34b. Leaves linear, present with the flowers [— 35.](#)
- 35a. Umbel nodding or horizontal (petals rose-color) **Wild Onion, *Allium cernuum*.**
- 35b. Umbel erect [— 36.](#)
- 36a. Pedicels longer than the flowers [— 37.](#)
- 36b. Pedicels equaling or shorter than the flowers (petals rose-purple) **Wild Chives, *Allium schoenoprasum* var. *sibiricum*.**
- 37a. Leaves flattened; flowers pink to white **Wild Onion, *Allium canadense*.**
- 37b. Leaves cylindrical; flowers greenish to purple **Field Garlic, *Allium vineale*.**

- 38a. Leaves lanceolate to ovate,
not more than 8 times as
long as broad — [39](#).
- 38b. Leaves linear or grass-like, at
least 12 times as long as
broad — [45](#).
- 39a. Principal leaves all basal, stem-leaves
none or bract-like — [40](#).
- 39b. Principal leaves on the stem — [41](#).
 - 40a. Flowers in a spike-like
raceme (4-10 dm. high;
small white flowers in
summer) **Colic-root, Aletris
farinosa.**
 - 40b. Flowers in an umbel-like
cluster — 34b.
- 41a. Perianth-segments 4 (1-2 dm. high;
flowers white, early summer) **Wild Lily of the
Valley,
Maianthemum
canadense.**
- 41b. Perianth-segments 6 — [42](#).
 - 42a. Styles 3; flowers dioecious (3-
10 dm. high; flowers white,
early summer) **Blazing Star,
Chamaelirium
luteum.**
 - 42b. Style 1; flowers perfect,
white, in spring (False
Solomon's Seal) — [43](#).
 - 43a. Flowers panicled (3-6 dm. high) **False Solomon's
Seal, Smilacina
racemosa.**
 - 43b. Flowers racemed — [44](#).
 - 44a. Leaves 2-4, usually 3 (1-2
dm. high) **False Solomon's
Seal, Smilacina
trifolia.**
 - 44b. Leaves 5-12 (2-5 dm. high) **False Solomon's
Seal, Smilacina
stellata.**

- | | |
|---|--|
| 45a. Flowers bright blue; perianth-segments united (2-3 dm. high, spring) | Grape Hyacinth,
Muscari
botryoides. |
| 45b. Flowers blue, greenish, yellowish, or white; perianth-segments separate
<u>— 46.</u> | |
| 46a. Flowers 1 cm. wide, or smaller (white or greenish, in racemes, late spring or summer) (False Asphodel)
<u>— 47.</u> | |
| 46b. Flowers 1.5 cm. wide, or larger <u>— 48.</u> | |
| 47a. Stem glabrous (2 dm. high or less) | False Asphodel,
Tofieldia palustris. |
| 47b. Stem viscid-pubescent (1-5 dm. high) | False Asphodel,
Tofieldia glutinosa. |
| 48a. Perianth-segments with 2 glands near the base (3-8 dm. high; greenish-white paniced flowers in summer) | Zygadenus,
Zygadenus
chloranthus. |
| 48b. Perianth-segments without glands <u>— 49.</u> | |
| 49a. Plant 3-5 dm. tall; flowers blue or nearly white, in long racemes (early summer) | Wild Hyacinth,
Camassia
esculenta. |
| 49b. Plant 1-3 dm. tall; flowers greenish-white, in short corymb-like racemes (spring) | Star of Bethlehem,
Ornithogalum
umbellatum. |

DIOSCOREACEAE, the Yam Family

Twining herbs with net-veined leaves and greenish or white flowers in panicles or racemes.

One species in Michigan; leaves ovate-cordate; flowers in summer

**Wild Yam,
Dioscorea villosa.**

AMARYLLIDACEAE, the Amaryllis Family

Plants with linear basal leaves, and perfect flowers, with 6-parted perianth, inferior ovary, and 6 stamens.

One species in Michigan; 1-2 dm. high;
flowers yellow, 1 cm. wide, in spring

**Star Grass,
Hypoxis hirsuta.**

IRIDACEAE, the Iris Family

Herbs, with 6-parted perianth, inferior ovary, and 3 stamens.

1a. Flowers blue, 5 cm. wide or larger —2.

1b. Flowers about 1 cm. wide (blue or white,
from a spathe terminating a 2-edged
stem 2-5 dm. high, spring and early
summer) (Blue-eyed Grass) —3.

2a. Flowering stems 4-8 dm. high
(early summer)

**Blue Flag, Iris
versicolor.**

2b. Flowering stems 2 dm. or less
high (spring)

**Dwarf Iris, Iris
lacustris.**

3a. Spathes terminal, sessile —4.

3b. Spathes long-peduncled, axillary —8.

4a. Spathe single —5.

4b. Spathes 2 on each flowering
stem —7.

- | | |
|--|---|
| 5a. Pedicels much longer than the inner (shorter) bract | Blue-eyed Grass,
Sisyrinchium
mucronatum. |
| 5b. Pedicels equaling or barely exceeding the inner bract 6. | |
| 6a. Capsule brown; common species | Blue-eyed Grass,
Sisyrinchium
angustifolium. |
| 6b. Capsule green or yellowish; rare species | Blue-eyed Grass,
Sisyrinchium
montanum. |
| 7a. Leaves folded lengthwise; stems narrowly winged | Blue-eyed Grass,
Sisyrinchium
hastile. |
| 7b. Leaves flat; stem broadly winged, 2-3 mm. wide | Blue-eyed Grass,
Sisyrinchium
albidum. |
| 8a. Capsules pale straw-color or whitish 9. | |
| 8b. Capsules brown, or tinged with purple 10. | |
| 9a. Plant with straight fibrous bristles at base; pedicels long-exserted | Blue-eyed Grass,
Sisyrinchium
farwellii. |
| 9b. Plants not bristly at base; pedicels barely exserted | Blue-eyed Grass,
Sisyrinchium
strictum. |
| 10a. Pedicels scarcely exceeding the inner bract 6a. | |
| 10b. Pedicels much exceeding the inner bract 11. | |
| 11a. Stem 2-6 mm. wide; bracts 1.5-2 cm. long | Blue-eyed Grass,
Sisyrinchium
gramineum. |
| 11b. Stem 1-2 mm. wide; bracts 1-1.5 cm. long | Blue-eyed Grass,
Sisyrinchium |

apiculatum.

ORCHIDACEAE, the Orchis Family

Herbs, with irregular flowers, one petal, the lip, differing from the others in size and shape, inferior ovary, and one or two stamens adherent to the style.

1a. Flowers in a spike-like obviously twisted raceme; small, yellowish or greenish-white, in late summer and autumn (except 4a) (Ladies' Tresses) [2.](#)

1b. Flowers solitary or in clusters, but never in a twisted raceme [5.](#)

2a. Flowers in 1 row

**Ladies' Tresses,
Spiranthes
gracilis.**

2b. Flowers in several rows [3.](#)

3a. Lip constricted near the apex

**Ladies' Tresses,
Spiranthes
romanzoffiana.**

3b. Lip not constricted [4.](#)

4a. Lip yellow; flowers in spring and early summer

**Ladies' Tresses,
Spiranthes lucida.**

4b. Lip white

**Ladies' Tresses,
Spiranthes cernua.**

5a. Brown, purple, or yellow plants, without green color, with scale-like leaves (1-4 dm. high; summer) (Coral Root) [6.](#)

5b. Plants with normal green color [9.](#)

6a. Lip white, not spotted

**Coral Root,
Corallorrhiza
trifida.**

- 6b. Lip white, spotted with red
— [7.](#)
- 7a. Lip distinctly 3-lobed **Coral Root,
Corallorrhiza
maculata.**
- 7b. Lip entire, or barely toothed — [8.](#)
 - 8a. Flower, exclusive of ovary, 4
mm. long **Coral Root,
Corallorrhiza
odontorrhiza.**
 - 8b. Flower about 10 mm. long. **Coral Root,
Corallorrhiza
striata.**
- 9a. Leaf 1 or none at flowering time — [10.](#)
- 9b. Leaves a single pair, basal, or opposite
on the stem; never alternate on the
stem — [19.](#)
- 9c. Leaves several, all basal, prominently
net-veined, and frequently blotched
with white (scape 1-4 dm. high;
flowers whitish, pubescent, in
summer) (Rattlesnake Plantain)
— [26.](#)
- 9d. Leaves 2 or more, on the stem — [28.](#)
 - 10a. Foliage leaf absent or
undeveloped at flowering
time, or merely persisting
through the winter from the
previous year — [11.](#)
 - 10b. Foliage leaf present at
flowering time — [12.](#)
- 11a. Flower rose-purple, 3-5 cm. long,
solitary or two (1-3 dm. high, early
summer) **Arethusa,
Arethusa bulbosa.**
- 11b. Flowers purplish-green, in racemes,
with a spur 2 cm. long **Crane-fly Orchis,
Tipularia discolor.**

- 11c. Flowers yellowish, purple tinged, in racemes; spur none (3-4 dm. high, early summer).
- 12a. Leaf linear or linear-lanceolate — [13](#).
- 12b. Leaf of a broader shape — [14](#).
- 13a. Flower solitary or two; leaf just below the flower — [11a](#).
- 13b. Flowers in a loose raceme, sometimes only 2; leaf basal
- 14a. Flowers greenish, yellowish, or white — [15](#).
- 14b. Flowers pink to purple, often variegated — [17](#).
- 15a. Flowers 6-10 mm. wide, with a spur about the same length (1-4 dm. high, summer) (Rein Orchis) — [35](#).
- 15b. Flowers 5 mm. wide or less; spur none (1-2 dm. high; summer) (Adder's Mouth) — [16](#).
- 16a. Pedicels less than 5 mm. long; lip broadest below the middle
- 16b. Pedicels more than 5 mm. long; lip broadest near the apex
- 17a. Flowers spicate; lip distinctly 3-lobed (1-2 dm. high; early summer) — [24](#).
- 17b. Flowers solitary or two — [18](#).
- 18a. Leaf on the stem, lanceolate to ovate — [34a](#).
- 18b. Leaf basal, round-ovate (2 dm. high or less; early summer)
- Putty Root,
Aplectrum
hyemale.**
- Calopogon,
Calopogon
pulchellus.**
- Adder's Mouth,
Microstylis
monophyllos.**
- Adder's Mouth,
Microstylis
unifolia.**
- Calypso, Calypso
bulbosa.**

- 19a. Leaves opposite and sessile near the middle of the stem (1-3 dm. high; flowers in summer) (Tway-blade) — [20.](#)
- 19b. Leaves basal — [21.](#)
- 20a. Lip deeply 2-cleft (flowers purplish) **Tway-blade, *Listera cordata.***
- 20b. Lip wedge-shape, with 2 round shallow lobes (flowers greenish-yellow) **Tway-blade, *Listera convallarioides.***
- 21a. Lip an inflated sac about 4 cm. long — [29d.](#)
- 21b. Lip not sac-like — [22.](#)
- 22a. Flower with a spur 15-50 mm. long — [23.](#)
- 22b. Flower not spurred (1-2 dm. high; early summer) (Tway-blade) — [25.](#)
- 23a. Flowers purple or magenta, or with white markings — [24.](#)
- 23b. Flowers greenish, yellowish, or white (1-4 dm. high; summer) (Rein Orchis) — [36.](#)
- 24a. Leaf 1 (1-2 dm. high; early summer) **Round-leaved Orchis, *Orchis rotundifolia.***
- 24b. Leaves 2 (-20 cm. high; late spring) **Showy Orchis, *Orchis spectabilis.***
- 25a. Lip about 10 mm. long, purple **Tway-blade, *Liparis liliifolia.***
- 25b. Lip about 5 mm. long, yellowish-green **Tway-blade, *Liparis loeselii.***
- 26a. Perianth 8-10 mm. long; lip with elongated point **Rattlesnake Plantain, *Epipactis decipiens.***

- 26b. Perianth 4-6 mm. long; lip
sack-like — 27.
- 27a. Raceme loosely flowered, one-sided
- 27b. Raceme closely flowered, not one-sided
- 28a. Lip conspicuously sack-like,
inflated (late spring and
early summer) (Lady's
Slipper) — 29.
- 28b. Lip not sack-like nor inflated
— 32.
- 29a. Lip white (1-3 dm. high)
- 29b. Lip yellow (2-7 dm. high) — 30.
- 29c. Lip white, with crimson or purple
markings — 31.
- 29d. Lip pink (1-4 dm. high, late spring)
- 30a. Lip 2-3 cm. long.
- 30b. Lip 3.5-5 cm. long
- Rattlesnake
Plantain, *Epipactis
repens* var.
ophioides.
Rattlesnake
Plantain, *Epipactis
pubescens*.**
- White Lady's
Slipper,
*Cypripedium
candidum*.**
- Stemless Lady's
Slipper,
*Cypripedium
acaule*.
Small Yellow
Lady's Slipper,
*Cypripedium
parviflorum*.
Large Yellow
Lady's Slipper,
*Cypripedium
parviflorum* var.
pubescens.**

- | | |
|--|--|
| 31a. Lip 2 cm. long or less; sepals separate
(1.5-3 dm. high) | Ram's Head
Lady's Slipper,
Cypripedium
arietinum. |
| 31b. Lip 3 cm. long or more; the 2 lower
sepals united | Showy Lady's
Slipper,
Cypripedium
hirsutum. |
| 32a. Flowers solitary in the axils,
or solitary and terminal; not
spurred — 33. | |
| 32b. Flowers in terminal racemes,
spurred — 38. | |
| 33a. Leaves a whorl of 5 (2-3 dm. high;
petals greenish; late summer) | Whorled Pogonia,
Pogonia
verticillata. |
| 33b. Leaves alternate — 34. | |
| 34a. Leaves lanceolate or narrowly
ovate, 2-8 cm. long,
narrowed at base; flower
terminal (1-4 dm. high;
flowers pink purple, early
summer) | Snake Mouth,
Pogonia
ophioglossoides. |
| 34b. Leaves ovate, 1-2 cm. long,
clasping; flowers axillary
(5-20 cm. high; flowers
purple, summer) | Nodding Pogonia,
Pogonia
trianthophora. |
| 35a. Leaf basal; spur nearly straight | Rein Orchis,
Habenaria
obtusata. |
| 35b. Leaf on the stem; spur strongly curved | Rein Orchis,
Habenaria
clavellata. |
| 36a. Flower-stalk without bracts
below the raceme; flowers
yellowish-green | Rein Orchis,
Habenaria
hookeri. |

- 36b. Flower-stalk bearing bracts
below the raceme; flowers
greenish-white [—37.](#)
- 37a. Spur 15-25 mm. long
- 37b. Spur 30-50 mm. long
- 38a. Lip fringed (Fringed Orchis)
[—43.](#)
- 38b. Lip not fringed (Rein Orchis)
[—39.](#)
- 39a. Lip with 2-3 evident teeth at apex
[—40.](#)
- 39b. Lip without apical teeth [—41.](#)
- 40a. Stem-leaves 3 or more
- 40b. Stem-leaves 2 — 35b.
- 41a. Flowers white
- 41b. Flowers greenish-yellow [—42.](#)
- 42a. Lip lanceolate, tapering
toward the apex
- 42b. Lip oblong, truncate at the
apex
- 43a. Lip deeply 3-lobed, toothed or fringed
[—44.](#)
- 43b. Lip not 3-lobed, but deeply fringed
[—46.](#)
- 44a. Flowers purple
- Rein Orchis,
Habenaria
orbiculata.
Rein Orchis,
Habenaria
macrophylla.**
- Rein Orchis,
Habenaria
bracteata.**
- Rein Orchis,
Habenaria
dilatata.**
- Rein Orchis,
Habenaria
hyperborea.
Rein Orchis,
Habenaria flava.**
- Purple Fringed
Orchis, Habenaria
psycodes.**

44b. Flowers white or nearly so
— 45.

45a. Spur 3 cm. long or more

**Prairie Fringed
Orchis, Habenaria
leucophaea.**

45b. Spur 1-1.5 cm. long

**Ragged Fringed
Orchis, Habenaria
lacera.**

46a. Flowers yellow

**Yellow Fringed
Orchis, Habenaria
ciliaris.**

46b. Flowers white

**White Fringed
Orchis, Habenaria
blephariglottis.**

PIPERACEAE, the Pepper Family

Herbaceous plants with alternate leaves, and flowers without either calyx or corolla.

One species in Michigan, a marsh plant with
heart-shape leaves and slender
racemes of white flowers

**Lizard's Tail,
Saururus cernuus.**

SALICACEAE, the Willow Family

Trees or shrubs, with dioecious flowers in catkins.

1a. Leaves less than twice as long as broad,
on petioles 3 cm. long or more — 2.

1b. Leaves more than twice as long as broad,
on petioles 2.5 cm. long or less. (The

genus *Salix*, or Willow, contains about 30 species in Michigan, of which only the commoner are mentioned here. For the others the Manuals should be consulted.) —8.

2a. Petioles strongly flattened laterally —3.

2b. Petioles not flattened laterally —6.

3a. Leaves broadly ovate or nearly circular —4.

3b. Leaves broadly triangular or deltoid in shape —5.

4a. Leaves coarsely toothed

4b. Leaves finely crenulate or serrate

5a. Tree with narrow spire-shape crown

5b. Tree with spreading crown

6a. Lower side of leaf densely tomentose

6b. Lower side of leaf glabrous or nearly so —7.

7a. Petioles glabrous

7b. Petioles ciliate

8a. Trees —9.

8b. Shrubs —13.

**Large-toothed
Aspen, *Populus
grandidentata.***

**Quaking Aspen,
*Populus
tremuloides.***

**Lombardy Poplar,
Populus nigra var.
*italica.***

**Cottonwood,
*Populus deltoides.***

**White Poplar,
*Populus alba.***

**Balsam Poplar,
*Populus
balsamifera.***

**Balm of Gilead,
*Populus
candicans.***

- 9a. Petioles without glands — [10](#).
- 9b. Petioles with glands — [11](#).
 - 10a. Petiole short (about 5 mm.), broad and flat **Black Willow, *Salix nigra*.**
 - 10b. Petiole slender, about 10-20 mm. long **Peach-leaved Willow, *Salix amygdaloides*.**
- 11a. Leaves green beneath **Crack Willow, *Salix fragilis*.**
- 11b. Leaves pale beneath — [12](#).
 - 12a. Branches and twigs conspicuously drooping **Weeping Willow, *Salix babylonica*.**
 - 12b. Branches and twigs not conspicuously drooping, yellow **Yellow Willow, *Salix alba* var. *vitellina*.**
- 13a. Shrubs of bogs — [14](#).
- 13b. Plants of sand-dunes along the Great Lakes — [15](#).
- 13c. Plants of dry upland hills — [16](#).
- 13d. Plants of wet ground, river-banks, and swamps — [17](#).
 - 14a. Leaves densely white-tomentose beneath **Willow, *Salix candida*.**
 - 14b. Leaves pale beneath but not tomentose **Willow, *Salix serissima*.**
 - 14c. Leaves glabrous and green beneath **Willow, *Salix pedicellaris*.**
- 15a. Leaves linear **Willow, *Salix longifolia*.**
- 15b. Leaves ovate-lanceolate, tomentose beneath **Willow, *Salix syrticola*.**
- 15c. Leaves ovate-lanceolate, glabrous **Willow, *Salix glaucophylla*.**
 - 16a. Leaves about 3 times as long as broad **Willow, *Salix rostrata*.**

16b. Leaves narrower, nearly sessile	Willow, <i>Salix tristis</i>.
16c. Leaves narrower, distinctly petioled	Willow, <i>Salix humilis</i>.
17a. Leaves linear or nearly so	Willow, <i>Salix longifolia</i>.
17b. Leaves shining	Willow, <i>Salix lucida</i>.
17c. Leaves silky	Willow, <i>Salix sericea</i>.
17d. Leaves not as in the preceding 3 species <u>— 18.</u>	
18a. Leaves rounded at base	Willow, <i>Salix cordata</i>.
18b. Leaves acute at base <u>— 19.</u>	
19a. Leaves finely serrulate	Willow, <i>Salix petiolaris</i>.
19b. Leaves remotely serrate or nearly entire	Willow, <i>Salix discolor</i>.

MYRICACEAE, the Sweet Gale Family

Shrubs, with monoecious or dioecious flowers in catkins, and aromatic foliage.

1a. Leaves pinnately lobed	Sweet Fern, <i>Myrica asplenifolia</i>.
1b. Leaves merely serrate — 2.	
2a. Shrub of sandy soil, shore of Lake Erie	Bayberry, <i>Myrica carolinensis</i>.
2b. Shrub of bogs and shores, northern half of state	Sweet Gale, <i>Myrica gale</i>.

JUGLANDACEAE, the Walnut Family

Trees with alternate pinnately compound leaves and flowers in catkins.

- 1a. Leaflets 11-23; pith divided by partitions into chambers —2.
- 1b. Leaflets 5-11; pith not partitioned (Hickory) —3.
 - 2a. Pith brown; bark with flat longitudinal ridges **Butternut, *Juglans cinerea*.**
 - 2b. Pith cream-color; bark of trunk without flat ridges **Black Walnut, *Juglans nigra*.**
- 3a. Bark of the trunk essentially smooth, not deeply furrowed or shaggy —4.
- 3b. Bark of the trunk deeply furrowed or shaggy —6.
 - 4a. Leaflets glabrous beneath; buds greenish —5.
 - 4b. Leaflets somewhat pubescent beneath; buds bright yellow **Bitter Nut, *Carya cordiformis*.**
Small-fruited Hickory, *Carya microcarpa*.
- 5a. Twigs hairy **Pignut Hickory, *Carya glabra*.**
- 5b. Twigs smooth
 - 6a. Twigs and leaves both pubescent —7.
 - 6b. Twigs nearly smooth; leaves smooth beneath **Shag-bark Hickory, *Carya ovata*.**
Mocker-nut Hickory, *Carya alba*.
- 7a. Twigs brownish; buds densely hairy

7b. Twigs orange; buds very slightly hairy

**King-nut Hickory,
Carya laciniosa.**

BETULACEAE, the Birch Family

Trees or shrubs with alternate simple leaves and inconspicuous monoecious flowers, the staminate flowers in catkins, and the pistillate in catkins or small clusters.

1a. Trees, with white or yellowish bark
exfoliating in thin papery plates or
scales —2.

1b. Tree or shrub, with smooth, dark gray
bark; trunk fluted with prominent
longitudinal ridges

**Hornbeam,
Carpinus
caroliniana.**

1c. Trees or shrubs; the bark more or less
roughened, but not exfoliating; trunk
not fluted —3.

2a. Bark white or chalky

**Paper Birch,
Betula alba var.
papyrifera.**

2b. Bark yellowish

**Yellow Birch,
Betula lutea.**

3a. Shrubs, with leaves 4 cm. long or less
—4.

3b. Shrubs or trees, with leaves 5 cm. long or
more —5.

4a. Twigs glandular-warty

**Dwarf Birch,
Betula glandulosa.**

4b. Twigs not glandular

**Swamp Birch,
Betula pumila.**

5a. Twigs and bark with the odor of
wintergreen

**Sweet Birch,
Betula lenta.**

- 5b. Twigs and bark without odor of wintergreen —6.
 - 6a. Fruit clusters woody, persistent on the plant for a long time —7.
 - 6b. Fruit clusters herbaceous, dropping in late autumn —9.
- 7a. Leaves rusty or whitish beneath, and pubescent at least on the veins **Speckled Alder, *Alnus incana*.**
- 7b. Leaves green beneath, and either pubescent or smooth —8.
 - 8a. Leaves broadest at or below the middle **Mountain Alder, *Alnus crispa*.**
 - 8b. Leaves broadest above the middle **Smooth Alder, *Alnus rugosa*.**
- 9a. Tree; fruit a cluster of bladder-like sacs each containing a small achene **Ironwood, *Ostrya virginiana*.**
- 9b. Shrubs; fruit a nut within a close-fitting involucre —10.
 - 10a. Involucre of 2 broad bracts, almost separate and not much longer than the fruit **Hazel, *Corylus americana*.**
 - 10b. Involucre of united bracts, prolonged into a bristly beak beyond the fruit **Beaked Hazel, *Corylus rostrata*.**

FAGACEAE, the Beech Family

Trees (or 1 species shrubby), with alternate simple leaves and monoecious flowers, the staminate flowers in catkins, and the pistillate solitary or in small clusters. Fruit a nut (or acorn) enclosed in a cup or bur.

- 1a. Leaves serrate with numerous sharp-pointed teeth —2.
- 1b. Leaves serrate, lobed, or entire, but never serrate with sharp-pointed teeth; fruit an acorn; pith 5-angled in the young twigs (Oak) —3.
 - 2a. Bark gray, smooth; buds 3-4 times longer than wide; nut
 - 2b. Bark rough; buds relatively thicker; nut rounded
- 3a. Leaves entire, except for a bristle at the tip
- 3b. Leaves toothed or lobed, the points bristle-tipped —4.
- 3c. Leaves toothed or lobed, the points without bristles —10.
 - 4a. Leaves entire below the middle, with a few shallow lobes beyond
 - 4b. Leaves deeply lobed throughout —5.
- 5a. Cup of the acorn saucer-shape, covering less than one-third of the acorn —6.
- 5b. Cup of the acorn hemispherical or top-shape, covering one-third or more of the acorn —8.
 - 6a. Length of the lateral leaf-lobes less than one-third the width of the leaf; acorn cup 2-2.5 cm. wide
 - 6b. Length of the lateral leaf-lobes more than one-third the width of the leaf —7.
- 7a. Acorn depressed-globose, about 1 cm. in diameter

Beech, *Fagus grandifolia*.

Chestnut, *Castanea dentata*.

Shingle Oak, *Quercus imbricaria*.

Black Jack Oak, *Quercus marilandica*.

Red Oak, *Quercus rubra*.

Pin Oak, *Quercus palustris*.

7b. Acorn ovoid, 1.5-2 cm. thick	Schneck's Oak, Quercus schneckii.
8a. Leaves pubescent beneath	Black Oak, Quercus velutina.
8b. Leaves glabrous beneath —9.	
9a. Buds glabrous; inner bark of the trunk yellow	Hill's Oak, Quercus ellipsoidalis.
9b. Buds pubescent beyond the middle; inner bark of trunk red	Scarlet Oak, Quercus coccinea.
10a. Leaves deeply pinnately lobed —11.	
10b. Leaves crenate, dentate, or sinuate, not lobed —12.	
11a. Leaf divided nearly to the middle by a pair of deep lateral lobes near the middle of the leaf; acorn more than half covered by the cup	Bur Oak, Quercus macrocarpa.
11b. Leaf without a median pair of deeper lobes; acorn about one-fourth covered by the cup	White Oak, Quercus alba.
12a. Leaves broadest at or near the middle, with numerous (8-13) sharp coarse teeth on each side	Yellow Oak, Quercus muhlenbergii.
12b. Leaves broadest above the middle, with a few shallow, rounded or subacute teeth (7 or less on each side) —13.	
13a. Large tree; leaves densely white-tomentose beneath; acorn on a stalk 3-10 cm. long	Swamp White Oak, Quercus bicolor.
13b. Shrub; leaves thinly white-tomentose beneath; acorn sessile or nearly so	Scrub Oak, Quercus prinoides.

URTICACEAE, the Nettle Family

Herbs or trees, with small inconspicuous apetalous flowers.

1a. Trees or tall shrubs — 2.

1b. Herbs — 7.

2a. Leaves oblong-ovate to lanceolate, serrate — 3.

2b. Leaves broadly ovate to rotund, some of them lobed (Mulberry) — 6.

3a. Leaves thick, coarsely and doubly serrate, broadest near the middle (Elm) — 4.

3b. Leaves thin, simply serrate, broadest distinctly below the middle

Hackberry, *Celtis occidentalis*.

4a. Some of the branches with flat corky wings; leaves smooth above

Cork Elm, *Ulmus racemosa*.

4b. Branches without corky wings; leaves more or less rough above — 5.

5a. Petioles and axillary buds glabrous

White Elm, *Ulmus americana*.

5b. Petioles and axillary buds pubescent with rusty hairs

Slippery Elm, *Ulmus fulva*.

6a. Leaves rough above

Red Mulberry, *Morus rubra*.

6b. Leaves smooth above

White Mulberry, *Morus alba*.

7a. Leaves alternate — 8.

7b. Leaves opposite — 9.

8a. Leaves 2-5 cm. long, stems pubescent	Pellitory, Parietaria pennsylvanica.
8b. Leaves 8-20 cm. long; stem armed with stinging hairs	Wood Nettle, Laportea canadensis.
9a. Twining plant; leaves serrate or cleft	Hop, Humulus lupulus.
9b. Erect plant; leaves palmately compound	Hemp, Cannabis sativa.
9c. Erect plants; leaves not lobed or compound — 10.	
10a. Stems armed with stinging hairs — 11.	
10b. Stems glabrous or rough, but not with stinging hairs — 12.	
11a. Leaves ovate, with a heart-shape base	Stinging Nettle, Urtica dioica.
11b. Leaves lanceolate or ovate-lanceolate, not heart-shape at base	Slender Nettle, Urtica gracilis.
12a. Stems glabrous, pellucid	Clearweed, Pilea pumila.
12b. Stems rough, opaque	False Nettle, Boehmeria cylindrica.

SANTALACEAE, the Sandalwood Family

Low herbs with alternate entire leaves and terminal clusters of small greenish-white bell-shape flowers without petals in spring and early summer.

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| 1a. Inflorescence of several-flowered clusters
terminating the stem and in the upper
axils | Toad-flax,
Comandra
umbellata. |
| 1b. Inflorescence of axillary clusters of 1-5
flowers | Toad-flax,
Comandra livida. |

LORANTHACEAE, the Mistletoe Family

Parasitic plants, attached to the branches of trees.

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| One species in Michigan, a dwarf brown
plant 5-20 mm. long, with minute
scale-like leaves, growing on the
branches of Black Spruce | Dwarf Mistletoe,
Arceuthobium
pusillum. |
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ARISTOLOCHIACEAE, the Birthwort Family

Flowers greenish-brown or reddish-brown, at or near the ground, with inferior 6-celled ovary.

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| 1a. Leaves alternate, on the stem; flowers on
a basal scaly branch (1-4 dm. high;
summer) | Virginia
Snakeroot,
Aristolochia
serpentaria. |
| 1b. Leaves a single basal pair, bearing 1
short-stalked flower between them
(spring)
(Wild Ginger) — 2. | |
| 2a. Lobes of the perianth ending in
a tubular portion 5-8 mm.
long | Wild Ginger,
Asarum
canadense. |
| 2b. Lobes of the perianth ending in
a tubular portion over 1 cm. | Wild Ginger,
Asarum canadense |

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|--------------------------------|-------------------------|
| long | var. acuminatum. |
| 2c. Lobes of the perianth | Wild Ginger, |
| triangular, not tubular at the | Asarum canadense |
| end | var. reflexum. |

POLYGONACEAE, the Buckwheat Family

Herbs with alternate entire leaves, stipules surrounding the stem above the base of each leaf, and small green, white or pink flowers without petals.

- 1a. Erect or ascending or prostrate or floating plants — [2.](#)
- 1b. Scrambling or climbing plants, clinging by sharp recurved prickles on the 4-angled stems (flowers greenish or pink, summer) (Tear-thumb) — [32.](#)
- 1c. Twining vines (flowers white or greenish, summer) — [33.](#)
 - 2a. Sepals 6, the 3 inner ones enlarging in fruit and surrounding the achenes; flowers in panicles — [3.](#)
 - 2b. Sepals 4 or 5 (occasional flowers may be found with 6 sepals, but the flowers are not in panicles) (summer) — [13.](#)
- 3a. Leaves arrow-shape or halberd-shape, with 2 basal lobes (Sorrel) — [4.](#)
- 3b. Leaves without basal lobes (Dock) — [5.](#)
 - 4a. Leaves halberd-shape, the basal lobes directed sidewise **Red Sorrel,**
Rumex acetosella.
 - 4b. Leaves arrow-shape, the basal lobes directed backward **Green Sorrel,**
Rumex acetosa.

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| 5a. The projecting wings of the fruiting calyx (known as valves) with sharp slender teeth | Bitter Dock,
Rumex obtusifolius. |
| 5b. Valves entire or finely dentate, but without sharp slender teeth <u>—6.</u> | |
| 6a. Pedicels straight, thickened toward the end, all regularly deflexed, 3-4 times longer than the fruiting calyx | Swamp Dock,
Rumex verticillatus. |
| 6b. Pedicels slender, flexuous, spreading <u>—7.</u> | |
| 7a. Leaves flat or nearly so <u>—8.</u> | |
| 7b. Leaves with strongly crisped or wavy-curved margins; plants usually of cultivated grounds or waste places (5-10 dm., summer) <u>—12.</u> | |
| 8a. With grain-like tubercles on all 3 valves of the fruit <u>—9.</u> | |
| 8b. With grain-like tubercles on only one valve, or entirely lacking <u>—10.</u> | |
| 9a. Valves broadly cordate, finely toothed | Water Dock,
Rumex brittanica. |
| 9b. Valves triangular-ovate, entire or nearly so | Dock, Rumex mexicanus. |
| 10a. Valves oblong | Bloody Dock,
Rumex sanguineus. |
| 10b. Valves broadly heart-shape <u>—11.</u> | |
| 11a. Grain-like tubercle less than half as long as the valve | Patience Dock,
Rumex patientia. |
| 11b. Grain-like tubercle more than half as long as the valve | Tall Dock, Rumex altissimus. |

- 12a. The grain-like tubercle on the valves of the fruit broadly ellipsoid, with rounded apex **Sour Dock, Rumex crispus.**
- 12b. Tubercle ovoid with tapering apex **Sour Dock, Rumex elongatus.**
- 13a. Flowers inconspicuous, in small axillary clusters; leaves jointed at the base (Knotweed) — [14.](#)
- 13b. Flowers more or less conspicuous, in obvious spikes or racemes which terminate the stems or branches, or arise from the axils of the upper leaves — [18.](#)
 - 14a. Leaves sharply folded lengthwise (1-4 dm. tall) **Knotweed, Polygonum tenue.**
 - 14b. Leaves flat or nearly so — [15.](#)
- 15a. The small sepals pink or white at the margin (stems prostrate or ascending) — [16.](#)
- 15b. Sepals greenish or yellowish throughout (stems erect or ascending) — [17.](#)
 - 16a. Leaves thin; common weed of dooryards and gardens **Knotweed, Polygonum aviculare.**
 - 16b. Leaves thick and fleshy; a plant of sandy shores **Knotweed, Polygonum aviculare var. littorale.**
- 17a. Leaves narrowly lanceolate or linear-oblong; rare species **Knotweed, Polygonum ramosissimum.**
- 17b. Leaves broadly oblong, oval, or elliptical; common weed of yards and gardens **Knotweed, Polygonum erectum.**

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| 18a. Leaves broadly triangular (3-7 dm. high; flowers white) | Buckwheat,
Fagopyrum
esculentum. |
| 18b. Leaves from linear to ovate or oblong — 19. | |
| 19a. Sepals 4; flowers in very long and slender spike-like racemes (4-10 dm. high) | Knotweed,
Polygonum
virginianum. |
| 19b. Sepals 5; flowers in spikes or racemes — 20. | |
| 20a. Flowers on slender pedicels, forming a loose raceme; leaves linear, jointed at the base (1-3 dm. high; flowers pink or white; chiefly near the Great Lakes) | Jointweed,
Polygonella
articulata. |
| 20b. Flowers sessile or nearly so, forming a spike or spike-like raceme — 21. | |
| 21a. Stipular sheaths at the base of the leaves ciliate at their upper margin — 22. | |
| 21b. Stipular sheaths not ciliate at the upper margin — 28. | |
| 22a. Sheaths with spreading borders — 23. | |
| 22b. Sheaths without a spreading border, appressed to the stem (Smartweed) — 24. | |
| 23a. Leaves ovate, acuminate; stem erect (1-2 m. high; flowers pink) | Prince's Feather,
Polygonum
orientate. |
| 23b. Leaves oblong, obtuse or subacute; spreading or ascending plant of wet soil | Water Smartweed,
Polygonum
amphibium var.
hartwrightii. |

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| 24a. Peduncles with glandular hairs (5-15 dm. high) | Smartweed,
Polygonum careyi. |
| 24b. Peduncles not glandular (1-8 dm. high) (Smartweed)
<u>— 25.</u> | |
| 25a. Sepals beset with minute black dots
<u>— 26.</u> | |
| 25b. Sepals white, pink, or red, not black-dotted <u>— 27.</u> | |
| 26a. Racemes drooping or nodding at the tip; achene dull-colored | Smartweed,
Polygonum hydropiper. |
| 26b. Racemes erect; achene smooth and shining | Smartweed,
Polygonum acre. |
| 27a. Sheaths smooth; leaves usually with a dark spot near the base | Smartweed,
Polygonum persicaria. |
| 27b. Sheaths hairy; leaves not dark-spotted | Smartweed,
Polygonum hydropiperoides. |
| 28a. Leaves obtuse or somewhat acute at the apex <u>— 29.</u> | |
| 28b. Leaves acuminate at the apex (5-15 dm. high; flowers white to pink) (Smartweed)
<u>— 30.</u> | |
| 29a. Stem unbranched, erect, bearing a single terminal raceme (5-30 cm. high; flowers pink) | Bistort,
Polygonum viviparum. |
| 29b. Stem branched, submerged in water or creeping on muddy shores (flowers pink) | Water Smartweed,
Polygonum amphibium. |
| 30a. Raceme single or two; leaves broadly ovate-lanceolate, about 3 times as long as wide | Smartweed,
Polygonum muhlenbergii. |

- 30b. Racemes numerous; leaves lanceolate, 4-6 times as long as wide — [31.](#)
 - 31a. Racemes drooping or nodding at the tip **Smartweed, Polygonum lapathifolium.**
 - 31b. Racemes erect **Smartweed, Polygonum pennsylvanicum.**
 - 32a. Leaves arrow-shape, the basal lobes pointing backward **Tear-thumb, Polygonum sagittatum.**
 - 32b. Leaves halberd-shape, the basal lobes pointing sidewise **Tear-thumb, Polygonum arifolium.**
 - 33a. The three outer sepals becoming conspicuously winged in fruit (False Buckwheat) — [34.](#)
 - 33b. The sepals all unchanged in fruit, except in size (Black Bindweed) — [35.](#)
 - 34a. Wings of the fruit with wavy-curved margins **False Buckwheat, Polygonum scandens.**
 - 34b. Wings of the fruit flat **False Buckwheat, Polygonum dumetorum.**
 - 35a. Leaf-sheaths with a ring of bristles at the base **Black Bindweed, Polygonum cilinode.**
 - 35b. Leaf-sheaths without a ring of bristles **Black Bindweed, Polygonum convolvulus.**

CHENOPODIACEAE, the Goosefoot Family

Herbs, with inconspicuous greenish or reddish flowers without petals, in summer.

1a. Leaves linear or nearly so, entire —2.

1b. Leaves of a broader shape, usually toothed or lobed —5.

2a. Leaves rather stiff, narrowly linear or thread-like, with spine-like tips

**Russian Thistle,
Salsola kali var.
tenuifolia.**

2b. Leaves soft, not spine-like —3.

3a. Widely branched, rather diffuse, 1-5 dm. tall; plant of the shore of the Great Lakes

**Bug-seed,
Corispermum
hyssopifolium.**

3b. Erect plants with ascending branches —4.

4a. Leaves glabrous (3-6 dm. tall)

**Goosefoot,
Chenopodium
leptophyllum.**

4b. Leaves minutely ciliate on the margin (bushy branched, 5-10 dm. tall)

**Kochia, Kochia
scoparia.**

5a. Principal leaves with a broad truncate, rounded, or hastate base —6.

5b. Principal leaves narrowed to the base —12.

6a. Leaves broadly ovate, with 1-4 large sharp projecting teeth on each side

**Goosefoot,
Chenopodium
hybridum.**

6b. Leaves hastate or triangular-ovate, entire or with many teeth —7.

7a. Leaves entire or merely undulate —8.

7b. Leaves sharply or sinuately toothed —9.

8a. Stem erect, simple or sparingly branched

**Good King Henry,
Chenopodium**

8b. Stem diffuse or ascending, freely branched	bonus-henricus. Orache, Atriplex patula.
9a. Flowers in small heads, in the axils or in terminal spikes; leaves sinuately toothed or nearly entire	Strawberry Blite, Chenopodium capitatum.
9b. Flowers in terminal panicles; leaves sharply toothed (Goosefoot) — 10.	Goosefoot, Chenopodium murale.
10a. Panicles short, not as long as the subtending leaves	
10b. Panicles long, exceeding the subtending leaves — 11.	Goosefoot, Chenopodium urbicum.
11a. Calyx green	Goosefoot, Chenopodium rubrum.
11b. Calyx red	
12a. Foliage glandular and strongly aromatic — 13.	
12b. Foliage not glandular nor aromatic; sometimes ill- scented — 15.	
13a. Flowers in large loose open spreading panicles; leaves deeply pinnatifid	Jerusalem Oak, Chenopodium botrys.
13b. Flowers clustered in slender axillary or terminal spikes — 14.	
14a. Spikes dense, leafy	Mexican Tea, Chenopodium ambrosioides.
14b. Spikes open, nearly leafless	Wormseed, Chenopodium ambrosioides var. anthelminticum.

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| 15a. Stem erect, 5-20 dm. tall; leaves frequently white-mealy | Lamb's Quarters,
Chenopodium
album. |
| 15b. Stem prostrate or ascending, succulent; leaves glaucous-white beneath | Goosefoot,
Chenopodium
glaucum. |
| 15c. Stem widely and diffusely branched; leaves green, soon deciduous | Cycloloma,
Cycloloma
atriplicifolium. |

AMARANTHACEAE, the Amaranth Family

Herbs, with alternate leaves, and inconspicuous greenish or reddish flowers without petals, which are axillary or in dense clusters, blooming in summer.

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| 1a. Flower-clusters axillary <u>—2.</u> | |
| 1b. Flower-clusters in terminal spikes or panicles, sometimes also axillary <u>—3.</u> | |
| 2a. Plant prostrate or decumbent; seed about 1.5 mm. broad | Pigweed,
Amaranthus
blitoides. |
| 2b. Plant erect or ascending, widely branched; seeds about 1 mm. broad (3-10 dm. high) | Tumble Weed,
Amaranthus
graecizans. |
| 3a. Principal leaves with a pair of spines at their base | Thorny Amaranth,
Amaranthus
spinosus. |
| 3b. Spines none at the base of the leaves <u>—4.</u> | |
| 4a. Weedy plants of cultivated or waste ground; flowers monoecious or | |

- polygamous; pistillate
flowers with a calyx
(Pigweed) —5.
- 4b. Plants of swamps or stream-
banks; flowers dioecious;
pistillate flowers without
calyx (Water Hemp) —7.
- 5a. Spikes short, 1-8 cm. long, crowded in
dense ovoid panicles; the terminal
spike not conspicuously elongated
beyond the appressed or ascending
lower ones **Pigweed,
Amaranthus
retroflexus.**
- 5b. Spikes slender, 1-12 cm. long; the
terminal spike greatly exceeding the
short inconspicuous divergent lower
ones —6.
- 6a. Bracts subulate, sharply awned **Pigweed,
Amaranthus
hybridus.**
- 6b. Bracts merely acuminate **Pigweed,
Amaranthus
paniculatus.**
- 7a. Flowers in leafy spikes, or the lower in
separate clusters **Water Hemp,
Acnida
tuberculata.**
- 7b. Flowers in separate distinct clusters **Water Hemp,
Acnida
tuberculata var.
subnuda.**

PHYTOLACCACEAE, the Pokeweed Family

Herbs with alternate entire leaves, small flowers without petals, and a many-celled ovary.

One species in Michigan, 1-2 m. high, with numerous racemes of whitish flowers, in late summer, followed by dark-purple berries

**Pokeweed,
Phytolacca
decandra.**

NYCTAGINACEAE, the Four-o'Clock Family

Herbs, with opposite entire leaves and flowers in small clusters surrounded by a broad open calyx-like involucre; the true calyx colored like a corolla; petals none (4-8 dm. high; flowers purple, in summer).

1a. Leaves lanceolate or narrower, sessile

**Umbrella-wort,
Oxybaphus
hirsutus.**

1b. Leaves ovate, petioled

**Umbrella-wort,
Oxybaphus
nyctagineus.**

ILLECEBRACEAE, the Knotwort Family

Herbs, with opposite entire leaves, and minute flowers without petals. (Prostrate spreading or freely branched plants, 3 dm. high or less; flowers in summer.)

1a. Stipules none; leaves slightly connate at base, subulate

**Knawel,
Scleranthus
annuus.**

1b. Stipules present, but small; leaves elliptical or oval (Forked Chickweed)
—2.

2a. Stems pubescent; internodes seldom more than 1 cm. long

**Forked
Chickweed,**

2b. Stems smooth; internodes
about 2 cm. long

**Anychia
polygonoides.
Forked
Chickweed,
Anychia
canadensis.**

AIZOACEAE, the Carpet-weed Family

Prostrate herbs, with whorled leaves and small whitish axillary flowers without petals, in summer.

One species in Michigan

**Carpet-weed,
Mollugo
verticillata.**

CARYOPHYLLACEAE, the Pink Family

Herbs, with opposite or whorled entire leaves, and stems frequently swollen at the nodes. Sepals 4 or 5; petals separate, as many as the sepals, or rarely none; stamens twice as many as the petals in plants with conspicuous flowers, sometimes fewer in those with small flowers; ovary 1-celled, with the ovules on a central axis, and with 2-5 styles.

- 1a. Calyx spreading, of separate sepals;
flowers 15 mm. wide or less; petals
sometimes none — 2.
- 1b. Calyx tubular, of united sepals; flowers
in many species more than 15 mm.
wide; petals always present — 22.
 - 2a. Stipules present — 3.
 - 2b. Stipules none — 4.

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| 3a. Leaves opposite; flowers pink (about 1 dm. high; summer) | Sand Spurrey,
Spargularia rubra. |
| 3b. Leaves whorled; flowers white (1-5 dm. high; leaves linear; summer) | Spurrey, Spargula
arvensis. |
| 4a. Leaves subulate or thread-like
<u>— 5.</u> | |
| 4b. Leaves linear to ovate <u>— 7.</u> | |
| 5a. Leaves opposite (1 dm. high or less; flowers white, summer) | Pearlwort, Sagina
procumbens. |
| 5b. Leaves fascicled in the axils <u>— 6.</u> | |
| 6a. Styles 4 or 5 (1 dm. high; terminal white flowers 5 mm. wide, in summer) | Pearlwort, Sagina
nodosa. |
| 6b. Styles 3 (1-4 dm. high; flowers white, nearly 1 cm. wide, summer) | Stitchwort,
Arenaria stricta. |
| 7a. Petals entire (3 dm. high or less; flowers white, in summer) (Stitchwort) <u>— 8.</u> | |
| 7b. Petals notched or 2-cleft at the end, or none <u>— 11.</u> | |
| 8a. Principal leaves 1 cm. long or less <u>— 9.</u> | |
| 8b. Principal leaves 1.5 cm. long or more <u>— 10.</u> | |
| 9a. Petals half as long as the sepals | Stitchwort,
Arenaria
leptoclados. |
| 9b. Petals almost as long as the sepals | Stitchwort,
Arenaria
serpyllifolia. |
| 10a. Leaves oblong-oval, obtuse. | Stitchwort,
Arenaria
lateriflora. |
| 10b. Leaves lanceolate, acute. | Stitchwort,
Arenaria
macrophylla. |

- 11a. Capsule splitting by valves at maturity;
styles usually 3 (Chickweed) — [12](#).
- 11b. Capsule opening by terminal teeth at
maturity; styles usually 5 (tufted or
matted plants, 1-5 dm. high; flowers
white, in spring and summer)
(Mouse-ear Chickweed) — [18](#).
 - 12a. Petals distinctly shorter than
the sepals, or none — [13](#).
 - 12b. Petals as long as the sepals, or
longer — [15](#).
- 13a. Leaves ovate (1-3 dm. high; flowers
white, all summer) **Chickweed,
Stellaria media.**
- 13b. Leaves lanceolate to oblong (in water or
wet places, 1-4 dm. high; flowers
white, in summer) — [14](#).
 - 14a. Flowers in a leafy terminal
branching cluster **Chickweed,
Stellaria borealis.**
 - 14b. Flowers in a lateral cluster
with minute bracts **Chickweed,
Stellaria uliginosa.**
- 15a. Flowers in clusters with leaf-like bracts,
or axillary and solitary (in water or
wet places; 1-3 dm. high; flowers
white, summer) **Chickweed,
Stellaria
crassifolia.**
- 15b. Flowers in clusters with scale-like
bracts — [16](#).
 - 16a. Leaves distinctly linear;
cymes lateral; a common
species in marshes (2-5 dm.
high; flowers white, early
summer) **Chickweed,
Stellaria longifolia.**
 - 16b. Leaves distinctly broadest
near the base; flower-
cluster terminal — [17](#).
- 17a. Pedicels erect; clusters usually few-
flowered; in extreme northern part of **Chickweed,
Stellaria longipes.**

- the state only (1-3 dm. high; flowers white, summer)
- 17b. Pedicels spreading; clusters open, many-flowered (2-6 dm. high; flowers white, in summer) **Chickweed, Stellaria graminea.**
- 18a. Petals distinctly longer than the sepals [— 19.](#)
- 18b. Petals as long as the sepals, or shorter than them [— 21.](#)
- 19a. Flowers much less than 1 cm. wide **Mouse-ear Chickweed, Cerastium nutans.**
- 19b. Flowers more than 1 cm. wide [— 20.](#)
- 20a. Stem-leaves linear or narrowly lanceolate **Mouse-ear Chickweed, Cerastium arvense.**
- 20b. Stem-leaves oblong **Mouse-ear Chickweed, Cerastium arvense var. oblongifolium.**
- 21a. Bracts green; pedicels short and inflorescence crowded **Mouse-ear Chickweed, Cerastium viscosum.**
- 21b. Bracts with transparent white margins; pedicels longer than the calyx and inflorescence open **Mouse-ear Chickweed, Cerastium vulgatum.**
- 22a. Styles 5 [— 23.](#)
- 22b. Styles 3 (3-10 dm. high; flowers in summer) [— 25.](#)
- 22c. Styles 2 [— 29.](#)
- 23a. Calyx-teeth much longer than the calyx-tube (erect, 4-10 dm. high; flowers large, red, late summer) **Corn Cockle, Agrostemma githago.**

- 23b. Calyx-teeth shorter than the calyx-tube
(4-10 dm. high; flowers in summer)
— 24.
- 24a. Flowers crimson **Mullein Pink,
Lychnis coronaria.**
- 24b. Flowers white or pink **White Campion,
Lychnis alba.**
- 25a. Flowers night-blooming, always wilted
during the day **Catchfly, Silene
noctiflora.**
- 25b. Flowers open during the day — 26.
- 26a. Flowers 6 mm. wide or less,
white or pink **Catchfly, Silene
antirrhina.**
- 26b. Flowers 1-2 cm. wide, white
to pink or purple — 27.
- 26c. Flowers 2 cm. wide or more,
crimson **Fire Pink, Silene
virginica.**
- 27a. Principal leaves in whorls of 4 **Starry Campion,
Silene stellata.**
- 27b. Leaves opposite — 28.
- 28a. Calyx globular, much inflated or bladder-like **Bladder Campion,
Silene latifolia.**
- 28b. Calyx club-shape, not inflated **Sweet William
Catchfly, Silene
armeria.**
- 29a. Leaves linear or narrowly lanceolate, 5
mm. wide or less (flowers pink or
white, in summer) — 30.
- 29b. Leaves lanceolate or ovate (flowers
pink, white, or red, in summer)
— 32.
- 30a. Flowers in terminal clusters;
leaves hairy (2-4 dm. high) **Deptford Pink,
Dianthus armeria.**
- 30b. Flowers solitary at the ends of
long pedicels — 31.
- 31a. Flowers 3-4 mm. wide (1-2 dm. high) **Gypsophyll,
Gypsophila**

- | | |
|--|--|
| 31b. Flowers 1 cm. wide or more (1-5 dm. high) | muralis.
Meadow Pink,
Dianthus
deltoides. |
| 32a. Flowers less than 1 cm. broad
<u>— 33.</u> | |
| 32b. Flowers more than 1 cm. broad <u>— 34.</u> | |
| 33a. Flowers white, in large panicles (4-7 dm. high) | Baby's Breath,
Gypsophila
paniculata. |
| 33b. Flowers pale red, in loose clusters (4-10 dm. high) | Cowherb,
Saponaria
vaccaria. |
| 34a. Leaves with 3-5 prominent veins (4-7 dm. high) | Soapwort,
Saponaria
officinalis. |
| 34b. Leaves with one mid-vein (3-6 dm. high) | Sweet William,
Dianthus
barbatus. |

PORTULACACEAE, the Purslane Family

Herbs with opposite or alternate leaves and regular flowers with 2 sepals, 5 petals, and a 1-celled ovary with 2 or 3 styles.

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|--|--|
| 1a. Leaves a single pair on each stem (1-2 dm. high; flowers pink, in racemes in early spring) (Spring Beauty) <u>— 2.</u> | |
| 1b. Leaves numerous (prostrate or spreading; flowers in summer) <u>— 3.</u> | |
| 2a. Leaves lance-ovate to oblong, not more than six times as long as wide | Spring Beauty,
Claytonia
caroliniana. |

- | | |
|---|--|
| 2b. Leaves linear or linear-lanceolate, more than six times as long as wide | Spring Beauty,
Claytonia virginica. |
| 3a. Flowers yellow, about 5 mm. wide | Purslane,
Portulaca oleracea. |
| 3b. Flowers 2-5 cm. wide | Portulaca,
Portulaca grandiflora. |

CERATOPHYLLACEAE, the Hornwort Family

Submerged aquatics, with whorled, finely dissected leaves and inconspicuous flowers with neither calyx nor corolla.

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|-------------------------|--|
| One species in Michigan | Hornwort,
Ceratophyllum demersum. |
|-------------------------|--|

NYMPHAEACEAE, the Water Lily Family

Aquatic plants, with usually large and floating leaves which are round or elliptical and palmately veined.

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|---|------------------------------|
| 1a. Floating and emersed leaves centrally peltate <u>2.</u> | |
| 1b. Leaves rounded but not peltate, with a deep sinus <u>3.</u> | |
| 2a. Leaves round, 3 dm. in diameter or more; flowers very large, pale yellow | Lotus, Nelumbo lutea. |

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|--|--|
| 2b. Leaves oval, 5-15 cm. long;
flowers small, purple | Water Shield,
Brasenia
schreberi. |
| 3a. Flowers yellow (Pond Lily) <u>—4.</u> | |
| 3b. Flowers white or tinged with pink (Water
Lily) <u>—5.</u> | |
| 4a. Leaves more than 1 dm. long | Pond Lily,
Nymphaea
advena. |
| 4b. Leaves less than 1 dm. long | Pond Lily,
Nymphaea
microphylla. |
| 5a. Flowers very fragrant; leaves purplish
beneath | Water Lily,
Castalia odorata. |
| 5b. Flowers not fragrant; leaves green
beneath | Water Lily,
Castalia tuberosa. |

RANUNCULACEAE, the Crowfoot Family

Herbs with alternate (rarely opposite) leaves, acrid watery juice, separate sepals and petals, numerous stamens, and several or many (rarely only 1) simple pistils. Petals present or absent, in the latter case the sepals are usually petal-like in appearance.

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| 1a. Climbing plants with opposite leaves
(flowers in late summer) (Virgin's
Bower) <u>—2.</u> | |
| 1b. Aquatic plants with dissected submerged
leaves (flowers in late spring and
summer) (Water Crowfoot) <u>—3.</u> | |
| 1c. Terrestrial or mud plants, not agreeing
with 1a or 1b <u>—6.</u> | |
| 2a. Flowers white, 2-3 cm. wide | Virgin's Bower,
Clematis |

- | | |
|---|---|
| 2b. Flowers pink-purple, 5-8 cm. wide | virginiana.
Virgin's Bower,
Clematis
verticillaris. |
| 3a. Flowers white <u>—4.</u> | |
| 3b. Flowers yellow <u>—5.</u> | |
| 4a. Leaves rigid, not collapsing when removed from the water | Water Crowfoot,
Ranunculus
circinatus. |
| 4b. Leaves soft, collapsing when removed from the water | Water Crowfoot,
Ranunculus
aquatilis var.
capillaceus. |
| 5a. Submerged leaves divided into hair-like segments | Water Crowfoot,
Ranunculus
delphinifolius. |
| 5b. Submerged leaves palmately divided into linear lobes | Water Crowfoot,
Ranunculus
purshii. |
| 6a. Flowers blue, irregular, with one spur (4-8 dm. high, summer) | Larkspur,
Delphinium ajacis. |
| 6b. The five petals each prolonged into a spur; flowers showy (4-8 dm. tall) (Columbine) <u>—7.</u> | |
| 6c. Flowers regular, without spurs <u>—8.</u> | |
| 7a. Spurs nearly straight; flowers scarlet and yellow (spring) | Wild Columbine,
Aquilegia
canadensis. |
| 7b. Spurs strongly incurved; flowers blue or white (spring, early summer) | Columbine,
Aquilegia vulgaris. |
| 8a. Flowers yellow <u>—9.</u> | |
| 8b. Flowers of various colors, but never yellow <u>—28.</u> | |

- 9a. Petals none; sepals petal-like; leaves crenate or dentate (2-4 dm. high, flowers in early spring) **Cowslip, *Caltha palustris*.**
- 9b. Petals small; sepals petal-like; leaves deeply palmately lobed (4-6 dm. tall; flowers in late spring) **Globe-flower, *Trollius laxus*.**
- 9c. Petals yellow; sepals green or yellowish — 10.
- 10a. Leaves linear to narrowly oblong, entire or with minute teeth (flowers 10-15 mm. wide, in summer) — 11.
- 10b. Leaves broader, some or all of them lobed or divided, or cordate-ovate and not lobed — 12.
- 11a. Stems ascending (4-8 dm. high), rooting at the lower joints; fruits pointed with a long slender beak **Spearwort, *Ranunculus laxicaulis*.**
- 11b. Stems prostrate and trailing, rooting at the joints; fruits tipped with a minute short beak **Spearwort, *Ranunculus flammula* var. *reptans*.**
- 12a. Basal leaves, or most of them, merely serrate or crenate, and not obviously lobed — 13.
- 12b. All the leaves lobed or divided — 16.
- 13a. Stem-leaves resembling the basal ones, and not lobed (1-2 dm. high; summer) **Sea-side Crowfoot, *Ranunculus cymbalaria*.**
- 13b. Stem-leaves deeply divided into oblong or linear segments — 14.

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| 14a. Flowers 1.5 cm. wide or more
(1-3 dm. high; spring) | Buttercup,
Ranunculus
rhomboideus. |
| 14b. Flowers 1 cm. wide or less (2-
5 dm. high; spring) —15. | |
| 15a. Foliage glabrous or minutely pubescent;
basal leaves cordate | Small-flowered
Crowfoot,
Ranunculus
abortivus. |
| 15b. Foliage villous; basal leaves barely
cordate or not at all | Small-flowered
Crowfoot,
Ranunculus
micranthus. |
| 16a. Stem erect or essentially so
—17. | |
| 16b. Stem prostrate, creeping, or
ascending. Early in spring
stems may be found which
appear almost erect.
Common spring-flowering
buttercups are all classified
here —22. | |
| 17a. Flowers 2 cm. broad or larger
(Buttercup) —18. | |
| 17b. Flowers 1 cm. broad or smaller —19. | |
| 18a. Terminal lobe of leaf stalked
(2-5 dm. high, early
summer) | Buttercup,
Ranunculus
bulbosus. |
| 18b. Terminal lobe of leaf sessile
(5-10 dm. high; all
summer) | Buttercup,
Ranunculus acris. |
| 19a. Plant glabrous, succulent; stem hollow
(1-5 dm. high; spring and summer) | Cursed Crowfoot,
Ranunculus
sceleratus. |
| 19b. Plant pubescent (Buttercup) —20. | |

- 20a. Fruits tipped with a prominent recurved beak; plant of shady woods (2-6 dm. high; flowers in late spring) **Buttercup, *Ranunculus recurvatus*.**
- 20b. Fruits tipped with a straight or slightly curved beak; plants of marshes or wet soil (3-6 dm. high; flowers in summer) [— 21.](#)
- 21a. Fruits in a short-cylindric head on a conical receptacle **Buttercup, *Ranunculus pennsylvanicus*.**
- 21b. Fruits in a globose or short-ovoid head, on an obovoid receptacle **Buttercup, *Ranunculus macounii*.**
- 22a. Leaves dissected into numerous linear or narrowly wedge-shape divisions; plants growing in water or very wet places (late spring and summer) **Water Crowfoot, *Ranunculus delphinifolius*.**
- 22b. Leaves palmately lobed, the terminal division not definitely stalked [— 23.](#)
- 22c. Leaves compound, some or all of the divisions on definite stalks (Buttercup) [— 24.](#)
- 23a. Stem-leaves numerous (1-2 dm. tall; late spring and summer) **Crowfoot, *Ranunculus purshii*.**
- 23b. Stem-leaves one or none, the principal leaves all basal (about 1 dm. high; flowers in summer) **Buttercup, *Ranunculus lapponicus*.**
- 24a. Style short, obviously curved [— 25.](#)

- 24b. Style long and slender,
straight or nearly so
(common spring-flowering
buttercups, 2-5 dm. high)
— 26.
- 25a. Stems creeping; flowers 2 cm. wide or
more (spring) **Buttercup,
Ranunculus
repens.**
- 25b. Stems ascending; flowers 1.5 cm. wide
or less (3-6 dm. high; summer) **Buttercup,
Ranunculus
macounii.**
- 26a. The two lateral divisions of
the leaf sessile or nearly so **Buttercup,
Ranunculus
fascicularis.**
- 26b. The two lateral divisions of
the leaf on long stalks
— 27.
- 27a. Roots fibrous; plants of wet soil **Buttercup,
Ranunculus
septentrionalis.**
- 27b. Roots thickened; plants of dry woods or
thickets **Buttercup,
Ranunculus
hispidus.**
- 28a. Leaves dissected into
numerous narrowly linear
acute divisions (4-7 dm.
high; flowers large, bluish,
in summer) **Love-in-a-mist,
Nigella
damascena.**
- 28b. Leaves lobed or divided, but
the divisions not separated
by definite stalks — 29.
- 28c. Leaves truly compound, all
their divisions separated by
distinct stalks — 39.
- 29a. Leaves all basal, their lobes (usually 3)
entire (1-2 dm. high; flowers pink-

- purple, in earliest spring) (Hepatica)
— 30.
- 29b. Lobes of the leaf serrate or incised
— 31.
- 30a. Lobes of the leaf obtuse or rounded
**Hepatica,
Hepatica triloba.**
- 30b. Lobes of the leaf acute
**Hepatica,
Hepatica acutiloba.**
- 31a. Petals none; sepals 3, usually falling away as soon as the flower opens (2-4 dm. high; flowers greenish-white, in spring)
**Golden Seal,
Hydrastis canadensis.**
- 31b. Petals none; sepals petal-like, 4 or more (Anemone) — 32.
- 31c. Petals present, but much smaller than the 5 petal-like sepals — 38.
- 32a. Stem-leaves sessile or nearly so — 33.
- 32b. Stem-leaves on definite petioles — 35.
- 33a. Ovary tipped with a long slender hairy style; flowers bluish-purple (1-4 dm. high; early spring)
**Pasque Flower,
Anemone patens
var. wolfgangiana.**
- 33b. Ovary with a short style, densely woolly; flowers red, greenish, or white (1-4 dm. high; late spring and summer) — 34.
- 33c. Ovary tipped with a short style, glabrous or nearly so; flowers white (4-7 dm. high; late spring and early summer)
**Anemone,
Anemone canadensis.**
- 34a. Stem 1-flowered, sepals white
**Anemone,
Anemone parviflora.**

- | | |
|---|--|
| 34b. Stem 3-flowered; sepals usually red | Anemone,
Anemone multifida. |
| 35a. Achenes densely woolly; flowers appearing in summer — 36. | |
| 35b. Achenes merely pubescent; stems 1-flowered; woodland species blooming in early spring (1-2 dm. high; flowers white) | Wood Anemone,
Anemone quinquefolia. |
| 36a. Segments of the leaf broadly wedge-shape or ovate; flowers white (4-8 dm. high) — 37. | |
| 36b. Segments of the leaf linear-oblong; flower red, greenish, or white; plants growing on the shores of the Great Lakes | Anemone,
Anemone multifida. |
| 37a. Segments of the basal leaves wedge-lanceolate; head of fruit cylindric | Anemone,
Anemone cylindrica. |
| 37b. Segments of the basal leaves ovate-lanceolate; head of fruit ovoid or oblong | Anemone,
Anemone virginiana. |
| 38a. Stem-leaves present — 9b. | |
| 38b. Leaves all basal (1-2 dm. high; leaves 3-divided; flowers white, in early summer) | Gold-thread,
Coptis trifolia. |
| 39a. Flowers numerous, in branching panicles (Meadow Rue) — 40. | |
| 39b. Flowers numerous, in racemes — 43. | |
| 39c. Flowers solitary or few, in loose clusters; woodland plants blooming in early spring (1-3 dm. high; flowers white to purple) — 45. | |

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|---|---|
| 40a. Blooming in spring; stem-leaves with obvious petioles (3-7 dm. high; flowers white or greenish) | Meadow Rue,
Thalictrum
dioicum. |
| 40b. Blooming in late spring or summer, stem-leaves sessile or nearly so (8-15 dm. high; flowers white)
<u>— 41.</u> | |
| 41a. Filaments club-shape, approximately as wide as the anther | Meadow Rue,
Thalictrum
polygamum. |
| 41b. Filaments slender or thread-like <u>— 42.</u> | |
| 42a. Leaves minutely pubescent beneath, but not glandular | Meadow Rue,
Thalictrum
dasycarpum. |
| 42b. Leaves minutely glandular beneath | Meadow Rue,
Thalictrum
revolutum. |
| 43a. Racemes slender, 10-90 cm. long (8-15 dm. high; flowers white, in summer) | Bugbane,
Cimicifuga
racemosa. |
| 43b. Racemes short and stout, 3-8 cm. long (4-8 dm. high; flowers white, in late spring) (Baneberry) <u>— 44.</u> | |
| 44a. Berries white; pedicels strongly thickened at maturity | Baneberry, Actaea
alba. |
| 44b. Berries red; pedicels slender | Baneberry, Actaea
rubra. |
| 45a. Flowers white; stem-leaves alternate | Isopyrum,
Isopyrum
biternatum. |
| 45b. Flowers white to pink or purplish; stem-leaves whorled | Rue Anemone,
Anemonella
thalictroides. |

MAGNOLIACEAE, the Magnolia Family

Trees or shrubs, with alternate leaves, and large, frequently showy flowers.

One species in Michigan, with broad 4-lobed leaves and greenish-yellow flowers in late spring

**Tulip Tree,
Liriodendron
tulipifera.**

ANONACEAE, the Custard Apple Family

Trees or shrubs, with alternate simple entire leaves, 3 sepals, and 6 petals.

One species in Michigan; tall shrub or small tree, with obovate leaves and large dull-purple flowers in spring

**Papaw, Asimina
triloba.**

MENISPERMACEAE, the Moonseed Family

Woody climbers, with alternate leaves, 6-8 petals, and numerous stamens.

One species in Michigan, with 5-7-angled leaves which are peltate near the edge, and small white flowers in early summer

**Moonseed,
Menispermum
canadense.**

BERBERIDACEAE, the Barberry Family

Shrubs or herbs; petals 6 or more; stamens 6-18, frequently opening by two terminal lids; pistil 1.

1a. Leaves simple —2.

1b. Leaves compound —3.

2a. Stem shrubby (flowers yellow, in racemes, in spring) **Barberry, Berberis vulgaris.**

2b. Stem herbaceous, with a single pair of palmately lobed leaves (4-6 dm. tall; flower white, solitary, terminal, in spring) **May Apple, Podophyllum peltatum.**

3a. Leaves all basal; leaflets 2 (2-4 dm. high; the flower-stalks bearing solitary white flowers in spring) **Twin-leaf, Jeffersonia diphylla.**

3b. The stem-leaf ternately compound, with numerous leaflets (4-8 dm. high; with yellowish-green clustered flowers in spring) **Blue Cohosh, Caulophyllum thalictroides.**

LAURACEAE, the Laurel Family

Trees or shrubs, with aromatic taste or odor, and alternate simple leaves; flowers small, imperfect, the anthers opening by lids.

1a. Freely branched shrub; leaves obovate-oblong, entire (flowers yellow, in early spring, before the leaves) **Spice Bush, Benzoin aestivale.**

1b. Tree or tall shrub; some or all of the leaves 2-3-lobed (flowers greenish-yellow, appearing with the leaves) **Sassafras, Sassafras variifolium.**

PAPAVERACEAE, the Poppy Family

Herbs with milky or colored juice, regular flowers, 2 sepals, 4, 6, or 8 petals, numerous stamens, and a 1-celled ovary.

- | | |
|--|---|
| 1a. Leaves palmately lobed; flower 2.5-5 cm. wide, with 8 petals or more (leaf basal; flower white, in early spring) | Bloodroot,
<i>Sanguinaria canadensis</i>. |
| 1b. Leaves pinnately toothed or lobed; flower 7-10 cm. wide, with 4-6 petals (4-8 cm. high; leaves clasping; summer) | Poppy, <i>Papaver somniferum</i>. |
| 1c. Leaves divided pinnately to the mid-rib into several toothed or lobed segments; flower 2.5 cm. broad or less (3-5 cm. high; flowers yellow)
<u>— 2.</u> | |
| 2a. Flowers in clusters of 2-4, about 3 cm. wide (spring) | Celandine Poppy,
<i>Stylophorum diphyllum</i>. |
| 2b. Flowers in umbels of 3-8, about 1.5 cm. wide | Celandine,
<i>Chelidonium majus</i>. |

FUMARIACEAE, the Fumitory Family

Herbs with watery juice, compound or dissected leaves, and irregular flowers; sepals 2, small; petals 4, in two pairs, and one or both of the outer pair spurred at the base; stamens 6.

- | | |
|---|---|
| 1a. Both outer petals spurred or sack-like at the base <u>— 2.</u> | |
| 1b. One outer petal spurred or sack-like at the base <u>— 4.</u> | |
| 2a. A climbing vine with flowers in panicles (white or pinkish flowers in summer) | Climbing
Fumitory,
<i>Adlumia fungosa</i>. |

- 2b. Low herbs (2-4 cm.) with basal leaves and white or pinkish flowers in racemes (early spring) —3.
- 3a. Spurs of the corolla triangular, divergent **Dutchman's Breeches, *Dicentra cucullaria*. Squirrel Corn, *Dicentra canadensis*. Fumitory, *Fumaria officinalis*.**
- 3b. Spurs of the corolla short and rounded
- 4a. Flowers about 5 mm. long, pink-purple tipped with red (3-8 dm. high; summer) **Fumitory, *Fumaria officinalis*.**
- 4b. Flowers 10 mm. long or more, yellow, at least at the tip (2-6 dm. high) (*Corydalis*) —5.
- 5a. Flowers yellow throughout (spring) ***Corydalis*, *Corydalis aurea*.**
- 5b. Flowers pink, tipped with yellow (summer). ***Corydalis*, *Corydalis sempervirens*.**

CRUCIFERAE, the Mustard Family

Herbs, with alternate, frequently lobed or dissected leaves, and regular flowers, usually in racemes; sepals and petals each 4, stamens 6, 4 long and 2 short (or rarely 2 only), ovary 1.

- 1a. Petals yellow or yellowish —2.
- 1b. Petals white, pink, or purple, never yellow —29.
- 2a. Leaves simple, entire or dentate, never lobed —3.

- 2b. Leaves deeply lobed or compound (the bracteal leaves, at or near the flower-clusters, may be simple and unlobed) —10.
- 3a. Leaves clasping the stem —4.
- 3b. Leaves not clasping at base —6.
 - 4a. Clasping base and apex of leaf obtuse or rounded; pod very long and slender (3-8 dm. high; summer) **Hare's Ear, *Conringia orientalis*.**
 - 4b. Clasping base and apex of leaf acute; pod obovoid (3-7 dm. high; early summer) (False Flax) —5.
- 5a. Stem and leaves glabrous **False Flax, *Camelina sativa*.**
- 5b. Leaves and usually the stem pubescent **False Flax, *Camelina microcarpa*.**
- 6a. Flowers about 2 mm. wide (1-3 dm. high; flowers in summer) —7.
- 6b. Flowers 5 mm. wide or more (2-6 dm. high; flowers in summer) —8.
- 7a. Leaves about twice as long as broad, widest near or below the middle **Whitlow Grass, *Draba nemorosa*.**
- 7b. Leaves 3-5 times as long as broad, widest above the middle **Yellow Alyssum, *Alyssum alyssoides*.**
- 8a. Leaves lanceolate, gradually tapering to the base; flowers about 15 mm. wide **Sand Rocket, *Diplotaxis muralis*.**
- 8b. Leaves ovate, acute at base; flowers about 15 mm. wide

— 22a.

8c. Leaves entire or minutely
toothed; flowers 5-10 mm.
wide — 9.

9a. Pods 25 mm. long or less, on slender
pedicels about 8 mm. long

9b. Pods 20 mm. long or more, on stout
pedicels about 4 mm. long

10a. Leaves bipinnate or dissected
into very numerous
divisions (3-8 dm. high;
flowers in summer) — 11.

10b. Leaves simply pinnate — 13.

11a. Flowers about 5 mm. broad; pods about
20 mm. long by 1 mm. broad

11b. Flowers about 3 mm. broad; pods about
8 mm. long by 2 mm. wide (Tansy
Mustard) — 12.

12a. Stems gray with a close fine
pubescence

12b. Stems green

13a. Pod short, not more than 3 times as long
as wide (coarse plants, preferring wet
or sandy ground; flowers in summer)
(Yellow Cress) — 14.

13b. Pod elongated, more than 4 times as
long as wide — 17.

**Worm-seed
Mustard,
Erysimum
cheiranthoides.**

**Worm-seed
Mustard,
Erysimum
parviflorum.**

**Herb Sophia,
Sisymbrium
sophia.**

**Tansy Mustard,
Sisymbrium
canescens.**

**Tansy Mustard,
Sisymbrium
canescens var.
brachycarpon.**

- | | |
|---|---|
| 14a. Stems creeping, with erect or ascending branches; flowers about 8 mm. wide | Yellow Cress,
Radicula
sylvestris. |
| 14b. Stems erect or ascending (3-10 dm. high); flowers about 4 mm. wide — 15. | |
| 15a. Pods about twice as long as the pedicels | Yellow Cress,
Radicula obtusa. |
| 15b. Pods about as long as the pedicels, or shorter than them — 16. | |
| 16a. Plant glabrous or minutely pubescent | Yellow Cress,
Radicula palustris. |
| 16b. Plant hirsute | Yellow Cress,
Radicula palustris
var. hispida. |
| 17a. Petals 7 mm. long or more; pod terminating in a conspicuous beak (coarse, weedy plants, 3-12 dm. high, blooming in summer) — 18. | |
| 17b. Petals of smaller size; pod not terminating in a conspicuous beak — 24. | |
| 18a. Upper stem-leaves clasping at the base | Rutabaga,
Brassica
campestris. |
| 18b. Upper stem-leaves not clasping — 19. | |
| 19a. Pod tipped with a slender cylindrical beak whose base is much narrower than the pod — 20. | |
| 19b. Pod gradually narrowed at its tip into a stout, flattened or angled beak — 22. | |
| 20a. Leaves oblanceolate, rather regularly pinnatifid, the terminal segment about the | Sand Rocket,
Diplotaxis muralis. |

same size as the lateral
ones

20b. Leaves broad, irregularly
pinnatifid especially below
the middle, with a large
terminal segment [—21.](#)

21a. Beak of pod 3-4 mm. long

**Black Mustard,
Brassica nigra.**

21b. Beak of pod 5 mm. long or more

**Indian Mustard,
Brassica juncea.**

22a. Leaves dentate or lobed

**Charlock, Brassica
arvensis.**

22b. Leaves deeply pinnatifid
[—23.](#)

23a. Pod dehiscent when ripe by two valves,
tipped with a flat or angled beak

**White Mustard,
Brassica alba.**

23b. Pod indehiscent, with spongy cross-
partitions between the seeds, tipped
with a conical beak

**Wild Radish,
Raphanus
raphanistrum.**

24a. Terminal segment of the
principal leaves much
larger than the lateral
segments; flowers in spring
and summer [—25.](#)

24b. Terminal segment of the
principal leaves equaling or
smaller than the lateral
ones (5-10 dm. high; pods
very long and slender;
flowers in summer)

**Sisymbrium,
Sisymbrium
altissimum.**

25a. Flowers about 3 mm. wide; pods erect
and closely appressed to the stem (3-
9 dm. high; weed blooming in
summer) (Hedge Mustard) [—26.](#)

25b. Flowers about 7 mm. wide; pods
spreading or ascending (3-6 dm.

high; flowers in spring) (Winter Cress) — 27.

26a. Pods pubescent

**Hedge Mustard,
Sisymbrium
officinale.**

26b. Pods glabrous

**Hedge Mustard,
Sisymbrium
officinale var.
leiocarpum.**

27a. Lateral leaf-segments 5-8 pairs on the principal leaves

**Winter Cress,
Barbarea verna.**

27b. Lateral leaf-segments 1-4 pairs on the principal leaves — 28.

28a. Flowers bright yellow, in racemes; pods spreading or ascending

**Winter Cress,
Barbarea vulgaris.**

28b. Flowers pale yellow, in corymb-like clusters; pods erect and somewhat appressed

**Winter Cress,
Barbarea stricta.**

29a. Principal stem-leaves compound or deeply lobed (the uppermost or bracteal leaves may be simple) — 30.

29b. Principal stem-leaves entire, dentate, serrate, or sometimes shallowly lobed or none (the basal leaves, at the surface of the ground, may be deeply lobed or compound) — 42.

30a. Leaves ternately divided or compound (2-5 dm. high; flowers in spring) (Toothwort) — 31.

30b. Leaves pinnately divided or compound — 33.

31a. Leaf-segments lanceolate or narrowly oblong

**Toothwort,
Dentaria laciniata.**

- 31b. Leaf-segments ovate or ovate-oblong
— [32](#).
- 32a. Stem-leaves 2, opposite or nearly opposite **Toothwort,
Dentaria diphylla.**
- 32b. Stem-leaves 2-5, alternate **Toothwort,
Dentaria maxima.**
- 33a. Plants growing in water, or in mud near water (spring and summer) — [34](#).
- 33b. Plants of dry or moist soil — [35](#).
- 34a. Aerial leaves distinctly compound, with 3-11 leaflets **Water Cress,
Radicula nasturtium-aquaticum.**
- 34b. Aerial leaves merely serrate to pinnatifid; the submerged leaves, if present, dissected **Lake Cress,
Radicula aquatica.**
- 35a. Flowers 5 mm. broad, or less — [36](#).
- 35b. Flowers 6 mm. broad, or more — [39](#).
- 36a. Stem-leaves irregularly pinnatifid or lobed; pod about as broad as long (2-4 dm. high; flowers in summer) **Garden Cress,
Lepidium sativum.**
- 36b. Stem-leaves distinctly pinnatifid, with 3-6 pairs of lateral segments (spring) (Bitter Cress) — [37](#).
- 37a. Leaves chiefly basal, pubescent on the upper side (1-3 dm. tall) **Bitter Cress,
Cardamine hirsuta.**
- 37b. Stem-leaves conspicuous, glabrous on the upper side — [38](#).
- 38a. Plant of dry soil; flowers about 3 mm. wide (1-4 dm. high) **Bitter Cress,
Cardamine parviflora.**

- | | |
|--|---|
| 38b. Plant of moist or wet soil;
flowers about 5 mm. wide
(2-8 dm. high) | Bitter Cress,
Cardamine
pennsylvanica. |
| 39a. Leaves irregularly pinnatifid or lobed,
not segmented into definitely paired
divisions (coarse plants 4-8 dm. high;
flowers in summer) — 40. | |
| 39b. Leaves deeply segmented into 3-10
pairs of divisions — 41. | |
| 40a. Flowers pink or white from
the first | Radish, Raphanus
sativus. |
| 40b. Flowers yellow at first,
turning white with age | Wild Radish,
Raphanus
raphanistrum. |
| 41a. Flowers pink or white, appearing in
spring (2-5 dm. high) | Cuckoo Flower,
Cardamine
pratensis. |
| 41b. Flowers yellowish or cream-color; a
weed blooming in summer (5-10 dm.
high) | Sisymbrium,
Sisymbrium
altissimum. |
| 42a. A fleshy, much-branched plant
of the shores of the Great
Lakes, with a pod
transversely divided into
two joints (2-3 dm. high;
summer) | Sea Rocket, Cakile
edentula. |
| 42b. Pod not transversely divided
into two joints — 43. | |
| 43a. Pod short, its length not more than 3
times its diameter — 44. | |
| 43b. Pod long and slender, its length more
than 3 times its diameter — 55. | |
| 44a. Pods not conspicuously
flattened, thick and plump,
about circular in cross- | |

section (flowers in
summer) — [45.](#)

44b. Pods distinctly flat — [46.](#)

45a. A plant escaped from cultivation in dry
or moist soil, with very large basal
leaves (5-10 dm. high)

**Horse Radish,
*Radicula
armoracia.***

45b. A plant of water or very wet soil, the
largest leaves seldom more than 15
cm. long (1-5 dm. tall)

**Lake Cress,
*Radicula aquatica.***

46a. Stem-leaves clasping the stem
by an auricled base — [47.](#)

46b. Stem-leaves sessile or
petioled, not clasping, or
none — [49.](#)

47a. Stem and leaves glabrous or pubescent;
pod very flat and circular, about 10
mm. wide (1-5 dm. tall; early
summer)

**Penny Cress,
*Thlaspi arvense.***

47b. Stem and leaves glabrous or pubescent;
pod not more than 5 mm. wide (1-6
dm. high; spring and early summer)
— [48.](#)

48a. Pods broadly ovate

**Field Cress,
*Lepidium
campestre.***

48b. Pods triangular, or slightly
indented at the apex

**Shepherd's Purse,
*Capsella bursa-
pastoris.***

49a. Pods about circular, or a very little
longer than broad — [50.](#)

49b. Pods ovoid or oblong, broadest near the
middle, and distinctly longer than
wide (Whitlow Grass) — [53.](#)

50a. Leaves entire (1-3 dm. high;
flowers in summer)

**Yellow Alyssum,
*Alyssum
alyssoides.***

- 50b. Leaves serrate (2-6 dm. high;
flowers in summer) [— 51.](#)
- 51a. Stamens 6 **Garden Cress,
*Lepidium sativum.***
- 51b. Stamens 2 (Pepper Grass) [— 52.](#) **Pepper Grass,
*Lepidium virginicum.***
- 52a. Petals present **Pepper Grass,
*Lepidium virginicum.***
- 52b. Petals none **Pepper Grass,
*Lepidium apetalum.***
- 53a. Petals deeply 2-cleft (about 1 dm. high;
early spring) **Whitlow Grass,
*Draba verna.***
- 53b. Petals entire or barely notched at the tip
[— 54.](#)
- 54a. Leaves all or chiefly at or near
the base (about 1 dm. high;
spring) **Whitlow Grass,
*Draba caroliniana.***
- 54b. Stems leafy up to the flowers
(1-5 dm. high; summer) **Whitlow Grass,
*Draba arabisans.***
- 55a. Stem-leaves cordate or sagittate at the
base and sessile, forming a more or
less clasping leaf (3-10 dm. high)
(Rock Cress) [— 56.](#)
- 55b. Stem-leaves sessile or somewhat
petioled, but not clasping [— 63.](#)
- 56a. Seeds in 2 rows in each cavity
of the pod (early summer)
[— 57.](#)
- 56b. Seeds in 1 row in each cavity
of the pod [— 59.](#)
- 57a. Calyx pubescent; the pods reflexed **Rock Cress,
*Arabis holboellii.***
- 57b. Calyx glabrous; the pods spreading or
ascending [— 58.](#)

- | | |
|--|--|
| 58a. Basal leaves densely pubescent | Rock Cress,
Arabis
brachycarpa. |
| 58b. Basal leaves smooth or nearly so | Rock Cress,
Arabis
drummondii. |
| 59a. Petals conspicuous, about twice as long as the calyx, or longer; straight, erect, mostly unbranched plants — 60 . | |
| 59b. Petals inconspicuous, equaling or but little longer than the calyx — 61 . | |
| 60a. Pods 3-4 cm. long, ascending (summer) | Rock Cress,
Arabis patens. |
| 60b. Pods 8-10 cm. long, recurved (late spring) | Rock Cress,
Arabis laevigata. |
| 61a. Pods widely spreading; stem usually sparingly branched near the base (spring) | Rock Cress,
Arabis dentata. |
| 61b. Pods erect or appressed; stem usually unbranched (summer) — 62 . | |
| 62a. Stem-leaves and stem smooth and glaucous | Rock Cress,
Arabis glabra. |
| 62b. Stem-leaves and stem almost always pubescent, and never glaucous | Rock Cress,
Arabis hirsuta. |
| 63a. Principal stem-leaves 7-10 cm. long, or more — 64 . | |
| 63b. Principal stem-leaves 2-5 cm. long — 65 . | |
| 64a. Leaves lanceolate or oblong; flowers 10 mm. wide or less (3-7 dm. high; summer) | Rock Cress,
Arabis canadensis. |
| 64b. Leaves ovate or ovate-lanceolate; flowers 15-20 | Dame's Rocket,
Hesperis
matronalis. |

- mm. wide (5-8 dm. high;
late spring and summer)
- 65a. Basal leaves ovate to orbicular or
cordate, not more than twice as long
as broad (1-3 dm. high; spring)
(Bitter Cress) [—66.](#)
- 65b. Basal leaves oblong, lanceolate, or
oblanceolate, at least 3 times as long
as broad (1-3 dm. high) [—67.](#)
- 66a. Flowers purple or rose color **Bitter Cress,
Cardamine
douglassii.**
- 66b. Flowers white **Bitter Cress,
Cardamine
bulbosa.**
- 67a. Basal leaves pinnatifid (spring and
summer) **Rock Cress,
Arabis lyrata.**
- 67b. Basal leaves entire or toothed [—68.](#)
- 68a. Leaves narrowly
oblanceolate; rare plant **Rock Cress, Braya
humilis.**
occurring from Mackinac
northward (summer)
- 68b. Leaves obovate or oblong; an **Mouse-ear Cress,
Sisymbrium
thalianum.**
introduced weed (spring)

CAPPARIDACEAE, the Caper Family

Herbs, with alternate compound leaves, 4 petals, and 6 or more stamens,
which are about equal in length; fruit a 1-celled pod.

- One species in Michigan, stamens about 11;
leaflets 3 (2-4 dm. high; flowers
yellowish, in summer) **Clammy-weed,
Polanisia
graveolens.**

RESEDACEAE, the Mignonette Family

Herbs, with alternate leaves and terminal racemes of small yellowish flowers; sepals 6, petals 6, stamens numerous.

One species in Michigan, with divided leaves and irregularly cleft petals, blooming in summer

**Yellow
Mignonette,
Reseda lutea.**

SARRACENIACEAE, the Pitcher Plant Family

Insectivorous plants, with hollow, pitcher-shaped leaves, and large purple flowers at the ends of naked stems.

One species in Michigan, growing in bogs and blooming in late spring

**Pitcher Plant,
*Sarracenia
purpurea*.**

DROSERACEAE, the Sundew Family

Insectivorous herbs, with a rosette of basal leaves bearing gland-tipped bristles on their upper surface, and with slender racemes of small white flowers in summer; inhabitants of bogs and swamps (2 dm. high, or less).

1a. Leaf-blade about as long as wide

**Sundew, *Drosera
rotundifolia*.**

1b. Leaf-blade about 2-3 times as long as wide

**Sundew, *Drosera
longifolia*.**

1c. Leaf-blade about 5-8 times as long as wide

**Sundew, *Drosera
anglica*.**

1d. Leaf-blade narrowly linear, about 10
times as long as wide

**Sundew, *Drosera*
linearis.**

PODOSTEMACEAE, the River Weed Family

Small submerged aquatics, growing attached to stones in running water,
with dissected leaves and minute flowers.

One species in Michigan

**River Weed,
Podostemum
ceratophyllum.**

CRASSULACEAE, the Orpine Family

Herbs, with usually alternate leaves; the sepals, petals, and pistils each 4 or
5, or in one species the petals none, and the stamens as many or twice as
many as the sepals.

1a. Leaves entire (Stonecrop) —2.

1b. Leaves toothed —3.

2a. Leaves 3-5 mm. long, very
thick and fleshy (tufted
plants about 1 dm. high,
with yellow flowers in
summer)

**Stonecrop, *Sedum*
acre.**

2b. Leaves 10-30 mm. long, flat
(tufted plants 1-2 dm. high;
flowers white, in spring)

**Stonecrop, *Sedum*
ternatum.**

3a. Petals present, purple (2-5 dm. high;
summer)

**Live-for-ever,
Sedum
purpureum.**

3b. Petals none (3-6 dm. high; summer)

**Ditch Stonecrop,
Penthorum
sedoides.**

SAXIFRAGACEAE, the Saxifrage Family

Herbs or shrubs, with alternate or opposite leaves; petals and sepals each 5, or the petals none; stamens 5 or 10; styles or stigmas 2-4.

1a. Shrubs with lobed leaves (3-15 dm. high;
flowers in late spring) —2.

1b. Herbs —12.

2a. Stems thorny —3.

2b. Stems not thorny (Currant)
—8.

3a. Flowers and fruits in racemes

**Swamp Currant,
Ribes lacustre.**

3b. Flowers and fruit in short clusters
(Gooseberry) —4.

4a. Ovary and fruit prickly and
bristly

**Gooseberry, Ribes
cynosbati.**

4b. Ovary and fruit smooth, or
sometimes a little glandular
—5.

5a. Stamens equaling the ovate calyx-lobes
in length, or a very little longer —6.

5b. Stamens distinctly longer than the linear
calyx-lobes —7.

6a. Leaves glabrous beneath, or
nearly so

**Gooseberry, Ribes
oxyacanthoides.**

6b. Leaves softly pubescent
beneath

**Gooseberry, Ribes
oxyacanthoides
var. calcicola.**

- | | |
|---|--|
| 7a. Flowers, including the stamens, about 15 mm. long | Gooseberry, <i>Ribes gracile</i>. |
| 7b. Flowers, including the stamens, about 8 mm. long | Gooseberry, <i>Ribes rotundifolium</i>. |
| 8a. Calyx prolonged above the ovary into a tube which is longer than the sepals —9. | |
| 8b. Calyx-tube shorter than the ovary or none —10. | |
| 9a. Flowers greenish-yellow, inconspicuous; calyx-tube narrowly bell-shape | Black Currant, <i>Ribes floridum</i>. |
| 9b. Flowers bright yellow, conspicuous; calyx-tube narrow, with spreading lobes | Golden Currant, <i>Ribes aureum</i>. |
| 10a. Ovary and berry bristly with glandular hairs | Skunk Currant, <i>Ribes prostratum</i>. |
| 10b. Ovary and fruit smooth, or with sessile glands —11. | |
| 11a. Leaves dotted beneath with resinous glands | Black Currant, <i>Ribes hudsonianum</i>. |
| 11b. Leaves glabrous or pubescent beneath, without resinous glands | Red Currant, <i>Ribes triste</i>. |
| 12a. Flowers minute, yellowish, without petals, in the axils of the leaves (1-2 dm. high; flowers in spring) | Golden Saxifrage, <i>Chrysosplenium americanum</i>. |
| 12b. Flowers large, solitary, terminating erect stalks (leaves mostly basal; flowers white, in late summer; flower-stalks 1-5 dm. high) (Grass-of-Parnassus) —13. | |
| 12c. Flowers in terminal racemes, panicles, or clusters —15. | |

- 13a. Flowers less than 2 cm. wide; leaves narrowed to the base
- 13b. Flowers 2-3.5 cm. wide; leaves rounded or cordate at the base [— 14.](#)

**Grass-of-Parnassus,
Parnassia
parviflora.**

- 14a. A 3-cleft scale at the base of each petal

**Grass-of-Parnassus,
Parnassia
caroliniana.**

- 14b. A many-cleft (9-15) scale at the base of each petal

**Grass-of-Parnassus,
Parnassia
palustris.**

- 15a. Leaves linear to oblanceolate, 3 times as long as broad, or more, and pinnately veined (Saxifrage) [— 16.](#)
- 15b. Leaves broadly ovate to nearly circular, frequently cordate at the base, and always palmately veined or lobed [— 20.](#)

- 16a. Leaves basal; the flower-stalk bearing no leaves except small ones at the base of its branches [— 17.](#)

- 16b. Flower-stalk leafy below (1-3 dm. high; flowers yellow, summer) [— 18.](#)

- 17a. Petals white; sepals erect; leaves conspicuously toothed, 3-10 cm. long (1-3 dm. high; spring)
- 17b. Petals greenish; sepals reflexed; leaves minutely toothed or entire, 10-30 cm. long (5-10 dm. high; spring)
- 18a. Leaves with 3 sharp teeth at the apex

**Saxifrage,
Saxifraga
virginiensis.**

**Saxifrage,
Saxifraga
pennsylvanica.**

**Saxifrage,
Saxifraga**

- 18b. Leaves with numerous teeth
or entire — [19](#).
- 19a. Leaves linear, chiefly on the stem
- 19b. Leaves spatulate, chiefly in a basal
rosette
- 20a. Stamens 5 (leaves mostly
basal; flowers greenish or
purplish in late spring, on
stalks 5-10 dm. high)
(Alum Root) — [21](#).
- 20b. Stamens 10 (flowers white, in
spring) — [23](#).
- 21a. Flowers regular
- 21b. Flowers irregular, the calyx oblique,
longer on the upper side than on the
lower — [22](#).
- 22a. Stamens projecting beyond
the calyx
- 22b. Stamens not projecting
beyond the calyx
- 23a. Stem-leaves alternate or none — [24](#).
- 23b. Stem with a pair of opposite leaves (2-4
dm. high)
- 24a. Petals deeply fringed (1-2 dm.
high)
- 24b. Petals entire (1-3 dm. high)
- tricuspidata.**
- Saxifrage,
Saxifraga aizoides.**
- Saxifrage,
Saxifraga aizoon.**
- Alum Root,
Heuchera
americana.**
- Alum Root,
Heuchera
hirsuticaulis.**
- Alum Root,
Heuchera hispida.**
- Bishop's Cap,
Mitella diphylla.**
- Bishop's Cap,
Mitella nuda.**
- False Mitrewort,
Tiarella cordifolia.**

HAMAMELIDACEAE, the Witch Hazel Family

Shrubs, with alternate simple leaves; sepals, petals, and stamens each 4; ovary 2-lobed.

One species in Michigan; tall shrub with
obovate leaves and yellow flowers
appearing late in autumn

**Witch Hazel,
Hamamelis
virginiana.**

PLATANACEAE, the Plane Tree Family

Trees, with broad, palmately veined and lobed leaves, and minute flowers in dense spherical heads.

One species in Michigan

**Sycamore,
Platanus
occidentalis.**

ROSACEAE, the Rose Family

Trees, herbs, or shrubs, with alternate, frequently compound leaves; petals and sepals usually 5, stamens numerous, pistils 1 to many; receptacle expanded into a saucer-shape or cup-shape organ, bearing the sepals, petals, and stamens at its margin, the pistils at its center, and resembling a calyx-tube or flattened calyx.

1a. Shrubs or trees [—2.](#)

1b. Herbaceous plants [—48.](#)

2a. Leaves compound [—3.](#)

2b. Leaves simple [—24.](#)

3a. Flowers in large panicles or corymbs,
each flower 5-10 mm. across; leaflets
7 or more [—4.](#)

3b. Flowers solitary or in small clusters, each
flower usually 20-80 mm. wide;

leaflets frequently only 3 or 5 [—7.](#)

4a. Flowers in a pyramidal or oblong panicle, the ovaries superior (1-2 m. high; flowers white, in summer) — 70b.

4b. Flowers in rounded or hemispheric clusters, the ovary inferior [—5.](#)

5a. Leaves pubescent on the lower surface

Rowan Tree, *Pyrus aucuparia*.

5b. Leaves glabrous beneath when mature (small trees; flowers white, in early summer or late spring) (Mountain Ash) [—6.](#)

6a. Leaves acuminate at the apex

Mountain Ash, *Pyrus americana*.

6b. Leaves obtuse or acute at the apex

Mountain Ash, *Pyrus sitchensis*.

7a. Flowers yellow (5-10 dm. high; summer)

Cinquefoil, *Potentilla fruticosa*.

7b. Flowers pink or red, rarely white, 4-10 cm. across (shrubs, 5-15 dm. high, or climbing; stems usually thorny; flowers in early summer) (Rose) [—8.](#)

7c. Flowers white, 1-3 cm. across (4-20 dm. high; flowers in late spring) [—16.](#)

8a. Leaflets on most of the leaves 3; styles cohering in a column which protrudes from among the stamens

Climbing Rose, *Rosa setigera*.

8b. Leaflets 5-11; styles not cohering in a protruding column [—9.](#)

- 9a. Sepals persistent on the fruit after flowering — [10](#).
- 9b. Sepals soon deciduous from the young fruit after flowering — [14](#).
 - 10a. A pair of spines below each leaf larger than the other spines — [11](#).
 - 10b. Spines all alike in size or nearly so, or absent completely — [12](#).
- 11a. Sepals entire **Wild Rose, *Rosa woodsii*.**
- 11b. Sepals pinnatifid **Dog Rose, *Rosa canina*.**
 - 12a. Stems with few thorns or none at all **Wild Rose, *Rosa blanda*.**
 - 12b. Stems prickly — [13](#).
- 13a. Fruit somewhat pear-shape, narrowed toward the base. **Wild Rose, *Rosa acicularis*.**
- 13b. Fruit globose, rounded at the base **Wild Rose, *Rosa acicularis* var. *bourgeauiana*.**
 - 14a. The pair of spines at the base of each leaf straight or nearly so **Wild Rose, *Rosa humilis*.**
 - 14b. The pair of spines at the base of each leaf distinctly recurved or hooked — [15](#).
- 15a. Leaves densely glandular-pubescent beneath **Sweetbrier, *Rosa rubiginosa*.**
- 15b. Leaves glabrous or minutely pubescent **Swamp Rose, *Rosa carolina*.**
 - 16a. Stems trailing or creeping — [17](#).
 - 16b. Stems erect, ascending, or arched — [19](#).

- 17a. Stems distinctly shrubby and thorny
(Dewberry) — [18](#).
- 17b. Stems almost herbaceous, without thorns
 - 18a. Leaves thin, dull above; fruit black, large and juicy
Dwarf Raspberry, *Rubus triflorus*.
 - 18b. Leaves firm or thick, shining above; fruit reddish, small, consisting of a few sour drupelets
Dewberry, *Rubus villosus*.
- 19a. Ripe fruit dropping away from the white receptacle or core; terminal leaflet of each leaf with a long stalk, while the lateral leaflets are sessile or nearly so
(Raspberry) — [20](#).
- 19b. Ripe fruit and receptacle or core dropping together; all the leaflets on stalks which are approximately equal in length (Blackberry) — [22](#).
 - 20a. Stem very glaucous with a whitish or bluish waxy deposit; fruit black
Black Raspberry, *Rubus occidentalis*.
 - 20b. Stem not glaucous; fruit red
— [21](#).
- 21a. Calyx velvety-pubescent
Red Raspberry, *Rubus idaeus*.
- 21b. Calyx bristly-hispid
Red Raspberry, *Rubus idaeus* var. *aculeatissimus*.
- 22a. Pedicels with gland-tipped hairs, but no prickles
Blackberry, *Rubus allegheniensis*.
- 22b. Pedicels with prickles
Blackberry, *Rubus nigricans*.

- 22c. Pedicels with neither prickles
nor gland-tipped hairs
— 23.
- 23a. Leaves downy beneath **Blackberry, Rubus frondosus.**
- 23b. Leaves smooth beneath **Blackberry, Rubus canadensis.**
- 24a. Ovaries 1 or more, superior
(attached to the surface of
the receptacle, but not
concealed within it or
united to it) — 25.
- 24b. Ovary 1, inferior
(permanently enclosed
within the receptacle, with
only the styles protruding)
— 38.
- 25a. Ovaries more than 1 — 26.
- 25b. Ovary 1 — 30.
- 26a. Leaves serrate — 27.
- 26b. Leaves lobed — 28.
- 27a. Leaves glabrous or very nearly so (8-20
dm. high; flowers white or pinkish,
summer) **Meadow-sweet, Spiraea salicifolia.**
- 27b. Leaves closely pubescent beneath (5-15
dm. high; flowers pink, summer) **Hardhack, Spiraea tomentosa.**
- 28a. Flowers showy, purple or
white, 3-4 cm. broad — 29.
- 28b. Flowers white, about 1 cm.
wide (1-3 m. high; flowers
in early summer) **Ninebark, Physocarpus opulifolius.**
- 29a. Flowers purple **Flowering Raspberry, Rubus odoratus.**
- 29b. Flowers white **Salmonberry, Rubus parviflorus.**

- 30a. Flowers in racemes (trees or tall shrubs; flowers white, in late spring) — [31](#).
- 30b. Flowers in small umbels or corymbs (flowers white, in spring) — [32](#).
- 31a. Leaves oblong, the points of their teeth incurved **Black Cherry, *Prunus serotina*.**
- 31b. Leaves obovate, the points of their teeth spreading **Choke Cherry, *Prunus virginiana*.**
- 32a. Flowers about 1 cm. wide — [33](#).
- 32b. Flowers about 1.5-2.5 cm. wide — [36](#).
- 33a. Low shrubs, with the spatulate or oblong leaves widest above the middle — [34](#).
- 33b. Erect tall shrubs or small trees, with the leaves widest below the middle — [35](#).
- 34a. An erect shrub (5-10 dm. high) **Appalachian Cherry, *Prunus cuneata*.**
- 34b. A prostrate or ascending shrub (3-15 dm. high) **Sand Cherry, *Prunus pumila*.**
- 35a. Leaves very broadly ovate, almost as wide as long (small tree) **Perfumed Cherry, *Prunus mahaleb*.**
- 35b. Leaves oblong-lanceolate, about 3 times as long as broad (shrub or small tree, 2-10 m. high) **Pin Cherry, *Prunus pennsylvanica*.**
- 36a. Sepals glandular-serrate (tall shrub or small tree) **Wild Plum, *Prunus nigra*.**
- 36b. Sepals entire — [37](#).
- 37a. Leaves with sharp teeth, frequently bristle-tipped; a native species (tall **Wild Plum, *Prunus americana*.**

- shrub or small tree, frequently
growing in thickets)
- 37b. Leaves with obtuse teeth; a species
escaped from cultivation (widely
branched tree)
- 38a. Trees, in cultivation or
escaped from cultivation
near roads or dwellings,
with showy flowers 2.5-5
cm. across, edible fruits,
and no thorns (spring)
[— 39.](#)
- 38b. Native species, trees or
shrubs, growing in woods,
fields, or thickets;
frequently with thorns
(spring) [— 40.](#)
- 39a. Leaves finely serrulate or entire
- 39b. Leaves coarsely serrate or somewhat
lobed
- 40a. Shrubs or small trees, without
thorns [— 42.](#)
- 40b. Bushy trees or shrubs, with
thorns or stiff thorn-like
branches, and with flowers
generally 1.5-2.5 cm.
across [— 41.](#)
- 41a. Flowers pink, very fragrant
- 41b. Flowers white (Hawthorn, the genus
Crataegus). Several species of this
genus occur in the state, for the
identification of which the Manual
must be used.
- Cherry, *Prunus*
cerasus.**
- Pear, *Pyrus*
communis.**
- Apple, *Pyrus*
malus.**
- Wild Crab, *Pyrus*
coronaria.**

- 42a. Mid-vein glandular above
(shrubs 1-3 m. tall; flowers
white or pink)
(Chokeberry) — [43.](#)
 - 42b. Mid-vein not glandular
(shrubs or trees, 1-10 m.
tall; flowers white)
(Juneberry) — [44.](#)
 - 43a. Leaves glabrous beneath
 - 43b. Leaves tomentose beneath
 - 44a. Petals 15-25 mm. long — [45.](#)
 - 44b. Petals 5-12 mm. long — [46.](#)
 - 45a. Mature leaves glabrous
 - 45b. Mature leaves pubescent beneath
 - 46a. Flowers in racemes — [47.](#)
 - 46b. Flowers solitary, or in small
clusters of 2-4
 - 47a. Leaves coarsely dentate, with about 1
tooth for each lateral vein
 - 47b. Leaves finely serrate, with about 2-3
teeth for each lateral vein
 - 48a. Flowers yellow — [49.](#)
 - 48b. Flowers white, pink, purple,
or rose, never yellow
— [63.](#)
- Chokeberry, *Pyrus melanocarpa*.**
- Chokeberry, *Pyrus arbutifolia* var. *atropurpurea*.**
- Juneberry, *Amelanchier canadensis*.**
- Juneberry, *Amelanchier canadensis* var. *botryapium*.**
- Juneberry, *Amelanchier oligocarpa*.**
- Juneberry, *Amelanchier spicata*.**
- Juneberry, *Amelanchier oblongifolia*.**

49a. Plant with basal trifoliate leaves,
resembling strawberry (1-3 dm. high;
late spring)

**Barren
Strawberry,
Waldsteinia
fragarioides.**

49b. Plants with leafy stems —50.

50a. Flowers solitary in the axils of
foliage leaves, on long
peduncles (trailing or
creeping plants; flowers in
late spring and summer)
—51.

50b. Flowers in narrow terminal
spike-like racemes (3-8 dm.
high; summer) (Agrimony)
—52.

50c. Flowers in irregular or
spreading clusters —55.

51a. Leaflets 5

**Five-finger,
Potentilla
canadensis.**

51b. Leaflets 7-25

**Silver Weed,
Potentilla
anserina.**

52a. Principal leaflets more than 3
times (about 3-1/2) as long
as wide

**Agrimony,
Agrimonia
parviflora.**

52b. Principal leaflets less than 3
times (about 2-1/2) as long
as wide —53.

53a. Leaves nearly glabrous beneath, or with
scattered spreading hairs

**Agrimony,
Agrimonia
gryposepala.**

53b. Leaves softly pubescent beneath —54.

54a. Leaves distinctly glandular
beneath

**Agrimony,
Agrimonia striata.**

- 54b. Leaves not glandular beneath **Agrimony,
Agrimonia mollis.**
- 55a. Principal leaves palmately compound
with 5-7 leaflets (Cinquefoil) — 56.
- 55b. Principal stem-leaves with 3 leaflets, or
pinnately compound with several
leaflets — 58.
- 56a. Leaves silvery-white beneath,
laciniately toothed (1-4 dm.
high; late spring and
summer) **Silvery Cinquefoil,
Potentilla
argentea.**
- 56b. Leaves not silvery-white
beneath (3-10 dm. high;
summer) — 57.
- 57a. Terminal leaflet more than 3 times as
long as wide **Cinquefoil,
Potentilla recta.**
- 57b. Terminal leaflet less than 3 times as
long as wide **Cinquefoil,
Potentilla
intermedia**
- 58a. Flowers about 4 mm. wide (2-
6 dm. high; spring) **Spring Avens,
Geum verum.**
- 58b. Flowers 6 mm. wide, or wider
— 59.
- 59a. Principal leaves with lobed leaflets, of
which the terminal is the largest; leaf-
axis bearing also some small leaflets
between those of usual size (4-12 dm.
high; late spring and summer)
(Avens) — 60.
- 59b. Principal leaves with toothed or
pinnately cleft leaflets, the lateral
ones about equaling the terminal one
in size, and without any small
scattered leaflets (3-8 dm. tall;
summer) (Cinquefoil) — 61.

- | | |
|--|--|
| 60a. Terminal leaflet cordate at base | Avens, Geum macrophyllum. |
| 60b. Terminal leaflet wedge-shape or acute at base | Avens, Geum strictum. |
| 61a. Leaflets 3 | Cinquefoil, Potentilla monspeliensis. |
| 61b. Leaflets 5-15 — 62. | |
| 62a. Leaflets crenate | Cinquefoil, Potentilla paradoxa. |
| 62b. Leaflets deeply incised | Cinquefoil, Potentilla pennsylvanica. |
| 63a. Leaves all basal, the flowers on leafless stalks — 64. | |
| 63b. Stem-leaves present — 66. | |
| 64a. Leaves simple (1-2 dm. high; summer) | Dalibarda, Dalibarda repens. |
| 64b. Leaves trifoliate (1-2 dm. high; spring) (Strawberry) — 65. | |
| 65a. Leaflets thick and firm, the petioles and pedicels pubescent with spreading or ascending hairs; fruit subglobose, the achenes embedded in pits on its surface | Strawberry, Fragaria virginiana. |
| 65b. Leaflets thin, the petioles and pedicels nearly glabrous or with appressed hairs; fruit conic, the achenes on its surface | Wood Strawberry, Fragaria americana. |
| 66a. Leaves pinnate with numerous leaflets — 67. | |
| 66b. Stem-leaves with 3-5 leaflets — 72. | |

- 67a. Leaflets laciniate or deeply lobed
(flowers pink or purple, early
summer) — [68](#).
- 67b. Leaflets merely toothed — [69](#).
- 68a. Stem-leaves few, small and
opposite (2-4 dm. tall)
- 68b. Stem-leaves large and
alternate (5-20 dm. tall)
- 69a. Individual flowers small, not exceeding
6 mm. across, in large clusters or
spikes — [70](#).
- 69b. Individual flowers more than 10 mm.
wide, in few-flowered clusters
(Cinquefoil) — [71](#).
- 70a. Flowers in dense spikes (5-15
dm. high; late summer)
- 70b. Flowers in panicles
- 71a. Flowers red or purple (3-6 dm. high;
summer)
- 71b. Flowers white (5-10 dm. high; early
summer)
- 72a. Pistils 5 (5-10 dm. high;
flowers white or pink, early
summer) — [73](#).
- 72b. Pistils 10, in a ring (flowers
pink or purple) — 68b.
- 72c. Pistils numerous, in a head or
close group — [74](#).
- 73a. Stipules linear or subulate, 5-8 mm.
long
- 73b. Stipules leaf-like, 10-25 mm. long,
serrate
- Purple Avens,
Geum triflorum.
Queen of the
Prairie,
Filipendula rubra.**
- Burnet,
Sanguisorba
canadensis.
Sorbaria, Sorbaria
sorbifolia.
Marsh Cinquefoil,
Potentilla
palustris.
Cinquefoil,
Potentilla arguta.**
- Bowman's Root,
Gillenia trifoliata.
American Ipecac,
Gillenia stipulata.**

- 74a. Flowers red or purple — [75.](#)
- 74b. Flowers white — [76.](#)
- 75a. Leaflets sharply and irregularly toothed or lobed; petals erect, narrowed at the base (3-9 dm. high; early summer) **Purple Avens, Geum rivale.**
- 75b. Leaflets finely and regularly toothed, oblong; petals spreading — 71a.
- 76a. Leaflets entire below, 3-toothed at the apex (1-3 dm. high; summer) **Cinquefoil, Potentilla tridentata.**
- 76b. Leaflets toothed all around the margin — [77.](#)
- 77a. Leaves all trifoliate (2-5 dm. high; late spring) — 17b.
- 77b. Some of the upper leaves merely lobed or dentate (5-8 dm. high) (Avens) — [78.](#)
- 78a. Stem bristly-hairy (early summer) **Avens, Geum virginianum.**
- 78b. Stem softly and finely pubescent (summer) **Avens, Geum canadense.**

LEGUMINOSAE, the Pulse Family

Trees, shrubs, or herbs, with alternate compound (except 3 species with simple) leaves and stipules; flowers usually irregular (except in a few species), with a large upper petal and 4 smaller ones, the 2 lower enclosing the stamens and pistil; stamens almost always 10, and generally united by their filaments; pistil 1, simple, ripening into a pod.

- 1a. Shrubs or trees — [2.](#)
- 1b. Herbs, twining, but without tendrils — [9.](#)
- 1c. Herbs; the leaves, or some of them, tipped with tendrils — [12.](#)

- 1d. Herbs, not climbing or twining; tendrils none [—21.](#)
- 2a. Leaves simple [—3.](#)
- 2b. Leaves compound [—4.](#)
- 3a. Leaves broadly cordate (tall shrub or small tree; flowers pink, early spring) **Redbud, *Cercis canadensis*.**
- 3b. Leaves lanceolate or elliptical (3-6 dm. high; flowers yellow, summer) **Dyer's Greenweed, *Genista tinctoria*.**
- 4a. Twigs or branches thorny [—5.](#)
- 4b. Thorns none [—7.](#)
- 5a. Thorns branched, scattered on the stem (tall tree; flowers greenish, early summer) **Honey Locust, *Gleditsia triacanthos*.**
- 5b. Thorns unbranched, a pair of them at the base of each leaf (late spring) [—6.](#)
- 6a. Branches glabrous or nearly so (tree; flowers white) **Black Locust, *Robinia pseudo-acacia*.**
- 6b. Branches glandular-pubescent (tall shrub; flowers pinkish) **Clammy Locust, *Robinia viscosa*.**
- 6c. Branches bristly (shrub. 1-3 m. high; flowers pink) **Bristly Locust, *Robinia hispida*.**
- 7a. Trees; leaves 2-3-pinnate (flowers greenish-white, spring) **Kentucky Coffee-tree, *Gymnocladus dioica*.**
- 7b. Low shrubs; leaves once-pinnate (3-6 dm. high; summer) [—8.](#)
- 8a. Flowers bright-blue, each with a single petal **Lead Plant, *Amorpha canescens*.**
- 8b. Flowers yellowish and pink-purple, petals 5 — 26a.
- 9a. Leaflets 5-7; flowers in racemes (purplish, late summer) **Wild Bean, *Apios tuberosa*.**
- 9b. Leaflets 3 (flowers greenish, purple, or white, late summer) [—10.](#)

- | | |
|---|--|
| 10a. Flowers in small capitate clusters; lower 2 petals strongly incurved | Wild Bean,
Strophostyles
helvola. |
| 10b. Flowers in racemes (Hog Peanut) — 11. | |
| 11a. Stem pubescent or glabrate; leaflets seldom more than 5 cm. long | Hog Peanut,
Amphicarpa
monoica. |
| 11b. Stem villous with retrorse hairs; leaflets usually longer than 5 cm. | Hog Peanut,
Amphicarpa
pitcheri. |
| 12a. Style with a tuft of hairs at the apex; lateral petals of the corolla adherent to the lower ones as far as the middle; stipules less than 10 mm. long, and usually less than one-fourth the length of the lower leaflets (spring and summer) (Vetch) — 13. | |
| 12b. Style hairy along the inner side; lateral petals of the corolla free from the lower ones or adherent only at the very base; stipules more than 8 mm. long and usually one-third or more the length of the lower leaflets (late spring and summer) — 17. | |
| 13a. Flowers axillary, sessile or nearly so (flowers purple) — 14. | |
| 13b. Flowers in peduncled racemes — 15. | |
| 14a. Upper leaves oblong-obovate, truncate or notched, and | Spring Vetch,
Vicia sativa. |

- mucronate at the apex
 - 14b. Upper leaves lance-linear, sharply acute
- 15a. Flowers 15-20 mm. long, 4-8 in a cluster (flowers purple)
- 15b. Flowers 8-12 mm. long [— 16.](#)
 - 16a. Racemes one-sided; flowers blue
 - 16b. Racemes loosely flowered, not one-sided; flowers whitish, the lower petals tipped with blue
- 17a. Flowers yellowish-white
- 17b. Flowers purple [— 18.](#)
 - 18a. Stipules nearly regularly halberd-shape, almost as large as the leaflets
 - 18b. Stipules half-sagittate, apparently attached laterally near the middle [— 19.](#)
- 19a. Leaflets 4-8 pairs, ovate; racemes with 10 or more flowers
- 19b. Leaflets 2-4 pairs, linear to oblong or elliptical; racemes with 2-9 flowers (Marsh Pea) [— 20.](#)
 - 20a. Stems with a membranous wing on the margins
 - 20b. Stems angled, but not winged
- 21a. Leaves simple (2-3 dm. high; flowers yellow, summer)

**Common Vetch,
Vicia angustifolia.**

Vetch, *Vicia americana*.

Vetch, *Vicia cracca*.

Vetch, *Vicia caroliniana*.

**Vetchling,
Lathyrus ochroleucus.**

**Beach Pea,
Lathyrus maritimus.**

**Wild Pea,
Lathyrus venosus.**

**Marsh Pea,
Lathyrus palustris.**

**Marsh Pea,
Lathyrus palustris
var. *myrtifolius*.**

**Rattlebox,
*Crotalaria***

- 21b. Leaves palmately compound; leaflets 7-11 (3-6 dm. high; flowers blue, late spring)
- 21c. Leaves pinnately compound; leaflets 5 to many — [22.](#)
- 21d. Leaves compound; leaflets 3 — [28.](#)
- 22a. Leaflets 5; flowers rose-purple in a spike-like head (5-8 dm. high, late summer)
- 22b. Leaflets more than 5 — [23.](#)
- 23a. Flowers bright-blue, in a dense spike (3-6 dm. high; summer) — 8a.
- 23b. Flowers bright-yellow; stamens not united (summer) — [24.](#)
- 23c. Flowers white, cream-color, or yellowish, or marked with purple — [26.](#)
- 24a. Leaflets linear-oblong, 2 cm. long or less; stamens 5 or 10 (3-6 dm. high) (Partridge Pea) — [25.](#)
- 24b. Leaflets lanceolate-oblong, 2-5 cm. long; 7 stamens with normal anthers and 3 with imperfect anthers (8-15 dm. high)
- 25a. Anthers 10; flowers 2-4 cm. wide
- 25b. Anthers 5; flowers 5-10 mm. wide
- 26a. Silky-hairy with whitish hairs; flowers marked with purple (3-5 dm. high; summer)
- sagittalis.**
- Lupine, *Lupinus perennis*.**
- Prairie Clover, *Petalostemum purpureum*.**
- Wild Senna, *Cassia marilandica*.**
- Partridge Pea, *Cassia chamaecrista*.**
- Partridge Pea, *Cassia nictitans*.**
- Goat's Rue, *Tephrosia virginiana*.**

- 26b. Glabrous or nearly so
(summer) (Milk Vetch)
— 27.
- 27a. Flowers greenish cream-color (4-10 dm.
high)
- 27b. Flowers white (3-5 dm. high)
- 28a. Flowers in heads, umbels, or
short dense spikes — 29.
- 28b. Flowers in loose racemes or
panicles — 42.
- 29a. Flowers bright-yellow; decumbent or
ascending plants (spring and
summer) — 30.
- 29b. Flowers white, cream, purple, or red;
never yellow — 34.
- 30a. Whole flower only about 2
mm. long; pod coiled
— 31.
- 30b. Flowers larger, each one 3-6
mm. long; pod straight (1-4
dm. high) (Hop Clover)
— 32.
- 31a. Flowers numerous in each head
- 31b. Flowers in clusters of 2 — 20
- 32a. Stipules linear
- 32b. Stipules ovate — 33.
- 33a. Heads densely flowered; flowers 20 or
more; upper petal striate when dry
- Milk Vetch,
Astragalus
canadensis.
Milk Vetch,
Astragalus
neglectus.**
- Black Medick,
Medicago
lupulina.
Bur Clover,
Medicago hispida.
Hop Clover,
Trifolium
agrarium.**
- Hop Clover,
Trifolium**

- procumbens.
- 33b. Heads loosely flowered; flowers usually 10 or fewer; upper petal scarcely striate or not at all **Hop Clover, *Trifolium dubium*.**
- 34a. Leaves palmately compound, the 3 leaflets all from the same point (late spring and summer) (Clover) [—35.](#)
- 34b. Leaves pinnately compound, the terminal leaflet on a distinct stalk [—39.](#)
- 35a. Individual flowers sessile, or on very short pedicels [—36.](#)
- 35b. Individual flowers distinctly pedicelled [—37.](#)
- 36a. Heads oblong, on distinct peduncles; calyx longer than the corolla (flowers nearly white; 1-4 dm. tall) **Stone Clover, *Trifolium arvense*.**
- 36b. Heads nearly globose, almost sessile, closely subtended by the leaves; corolla longer than the calyx (2-8 dm. high; flowers red-purple) **Red Clover, *Trifolium pratense*.**
- 37a. Stems prostrate or creeping; heads long-peduncled, arising from the creeping branches (flower-stalks 1-2 dm. high; flowers white) **White Clover, *Trifolium repens*.**
- 37b. Some or all of the stems erect; heads arising from the leafy stems (flowers white or pink) [—38.](#)
- 38a. Plants with long basal runners; flowers 10-13 mm. long (2-3 dm. high) **Buffalo Clover, *Trifolium stoloniferum*.**

- | | |
|--|--|
| 38b. Basal runners none; flowers
6-8 mm. long (3-8 dm.
high) | Alsike Clover,
Trifolium
hybridum. |
| 39a. Prostrate; leaflets broadly ovate; flowers
3-10 in a cluster — 10a. | |
| 39b. Erect; leaflets broadest near the middle;
flowers numerous (5-12 dm. high;
flowers yellowish-white, late
summer) (Bush Clover) — 40. | |
| 40a. Leaflets less than twice as
long as broad | Bush Clover,
Lespedeza hirta. |
| 40b. Leaflets more than twice as
long as broad — 41. | |
| 41a. Leaflets linear, 5 mm. wide or less;
heads with obvious peduncles | Bush Clover,
Lespedeza
angustifolia. |
| 41b. Leaflets narrowly elliptical, the
principal ones more than 5 mm.
wide; heads sessile or nearly so | Bush Clover,
Lespedeza
capitata. |
| 42a. Leaflets finely toothed — 43. | |
| 42b. Leaflets entire — 44. | |
| 43a. Flowers violet or blue (3-6 dm. high;
summer) | Alfalfa, Medicago
sativa. |
| 43b. Flowers yellow (1-2 m. high; summer) | Yellow Sweet
Clover, Melilotus
officinalis. |
| 43c. Flowers white (1-3 m. high; summer) | White Sweet
Clover, Melilotus
alba. |
| 44a. Flowers yellow (5-10 dm.
high; summer) | Wild Indigo,
Baptisia tinctoria. |
| 44b. Flowers white, the leaflets all
from the same point (5-10
dm. high; summer) | Wild Indigo,
Baptisia
leucantha. |
| 44c. Flowers blue, purple, or pink
(rarely white, and then the | |

terminal leaflet stalked)

— 45.

45a. Racemes arising from the base of the plant, leafless (4-8 dm. high; summer)

**Tick Trefoil,
Desmodium
nudiflorum.**

45b. Racemes terminal or a few of them axillary; leaflets generally more than 3 cm. long; pod (usually to be seen at the base of the raceme) transversely segmented into 2 or more joints (summer) (Tick Trefoil; the genus *Desmodium*. Pods are usually necessary for satisfactory identification) — 46.

45c. Racemes short, loose, chiefly axillary; leaflets generally less than 3 cm. long; the short ovate or ovoid pod not transversely jointed (5-10 dm. high; flowers in summer) (Bush Clover; the genus *Lespedeza*. Reference to the Manual is recommended) — 58.

46a. Leaves clustered near the summit of the stem (4-12 dm. high)

**Tick Trefoil,
Desmodium
grandiflorum.**

46b. Leaves scattered on the stem — 47.

47a. Plants prostrate; racemes paniced; leaflets nearly circular

**Tick Trefoil,
Desmodium
rotundifolium.**

47b. Plants decumbent or ascending; racemes short, simple, few-flowered; stipules ovate (stems 4-8 dm. long)

**Tick Trefoil,
Desmodium
pauciflorum.**

47c. Plants erect or ascending; racemes paniced — 48.

48a. Leaflets of an oblong type, broadest at or near the

middle, and about 4 times
as long as broad (5-10 dm.
tall) — [49.](#)

48b. Leaflets of an ovate or
lanceolate type, broadest
below the middle, and not
more than 3 times as long
as wide — [50.](#)

49a. Stem pubescent; leaves sessile or nearly
so

49b. Stem glabrous or nearly so; leaves
obviously petioled

50a. Stipules lanceolate to ovate, 1
cm. long or more (6-15 dm.
high) — [51.](#)

50b. Stipules narrowly lanceolate
or subulate, less than 1 cm.
long — [54.](#)

51a. Stems glabrous or minutely pubescent;
leaves acuminate — [52.](#)

51b. Stems hispid or densely pubescent;
leaves ovate or ovate-lanceolate,
obtuse or barely acute — [53.](#)

52a. Leaves glabrous on both sides

52b. Leaves rough above, hairy
beneath

53a. Leaflets broadly ovate

**Tick Trefoil,
Desmodium
sessilifolium.**

**Tick Trefoil,
Desmodium
paniculatum.**

**Tick Trefoil,
Desmodium
bracteosum.**

**Tick Trefoil,
Desmodium
bracteosum var.
longifolium.**

**Tick Trefoil,
Desmodium
canescens.**

53b. Leaflets ovate-lanceolate

54a. Flowers 10-12 mm. long (1-2 m. high)

54b. Flowers 5-8 mm. long (5-8 dm. tall) [— 55.](#)

54c. Flowers 3-4 mm. long (4-8 dm. tall) [— 56.](#)

55a. Leaflets broadly ovate

55b. Leaflets oblong-ovate

56a. Leaflets 3-5 cm. long, oblong-ovate, scabrous above

56b. Leaflets 1-2.5 cm. long, broadly ovate or oval, not scabrous above [— 57.](#)

57a. Stem and leaves glabrous or very nearly so

57b. Stem and leaves conspicuously pubescent

58a. Flower-clusters sessile, or on peduncles shorter than the subtending leaves [— 59.](#)

58b. Flower-clusters, or many of them, on peduncles longer than the leaves [— 61.](#)

59a. Leaves linear-oblong

**Tick Trefoil,
Desmodium
illinoense.**

**Tick Trefoil,
Desmodium
canadense.**

**Tick Trefoil,
Desmodium
viridiflorum.**

**Tick Trefoil,
Desmodium
dillenii.**

**Tick Trefoil,
Desmodium
rigidum.**

**Tick Trefoil,
Desmodium
marilandicum.**

**Tick Trefoil,
Desmodium
obtusum.**

**Bush Clover,
Lespedeza**

- virginica.**
- 59b. Leaves ovate or oval — [60.](#)
- 60a. Leaves and stem velvety or downy **Bush Clover, Lespedeza stuvei.**
- 60b. Leaves and stem glabrous, or with close appressed pubescence **Bush Clover, Lespedeza frutescens.**
- 61a. Leaves ovate or broadly elliptical; corolla conspicuously exceeding the calyx — [62.](#)
- 61b. Leaves linear-oblong; calyx about as long as the corolla **Bush Clover, Lespedeza manniana.**
- 62a. Stem erect or ascending — [63.](#)
- 62b. Stem trailing; peduncles much exceeding the leaves **Bush Clover, Lespedeza procumbens.**
- 63a. Villous-pubescent; inflorescence dense; some peduncles shorter than the leaves **Bush Clover, Lespedeza nuttallii.**
- 63b. Slightly pubescent or glabrous; inflorescence loose, on peduncles much longer than the leaves **Bush Clover, Lespedeza violacea.**

LINACEAE, the Flax Family

Herbs with simple leaves, and regular flowers, having 5 sepals, 5 yellow or blue petals, 5 stamens, and 5 styles.

- 1a. Flowers blue (3-6 dm. high; summer) **Flax, Linum usitatissimum.**
- 1b. Flowers yellow (3-8 dm. high; summer)
(Wild Flax) — [2.](#)

- | | |
|---|---|
| 2a. Middle stem-leaves below the branches opposite | Wild Flax, <i>Linum striatum</i>. |
| 2b. Middle stem-leaves below the branches alternate <u>—3.</u> | |
| 3a. Leaves narrowly lanceolate to linear, 1-4 mm. wide <u>—4.</u> | |
| 3b. Leaves oblanceolate to oblong, 4-6 mm. wide | Wild Flax, <i>Linum virginianum</i>. |
| 4a. Leaves entire | Wild Flax, <i>Linum medium</i>. |
| 4b. Upper leaves glandular-ciliate | Wild Flax, <i>Linum sulcatum</i>. |

OXALIDACEAE, the Wood Sorrel Family

Herbs, with alternate or basal compound leaves with 3 reverse heart-shaped leaflets; sepals, petals, and styles each 5; stamens 10. (Wood Sorrel)

- | | |
|---|--|
| 1a. Leaves all basal; flowers white to pink-purple (1-2 dm. high; late spring) <u>—2.</u> | |
| 1b. Stem-leaves present; flowers yellow (1-5 dm. high; spring and summer) <u>—3.</u> | |
| 2a. Flower-stalks bearing a single flower | Wood Sorrel, <i>Oxalis acetosella</i>. |
| 2b. Flower-stalks bearing an umbel of several flowers | Wood Sorrel, <i>Oxalis violacea</i>. |
| 3a. Stem prostrate and creeping Wood Sorrel, <i>Oxalis repens</i> . | |
| 3b. Stem erect or ascending <u>—4.</u> | |
| 4a. Pedicels with spreading pubescence | Wood Sorrel, <i>Oxalis corniculata</i>. |
| 4b. Pedicels with appressed pubescence | Wood Sorrel, <i>Oxalis stricta</i>. |

GERANIACEAE, the Geranium Family

Herbs, with deeply lobed or divided leaves; flowers regular, with 5 sepals, 5 petals, 5 or 10 stamens, and a 5-celled ovary.

- 1a. Anthers 5 (spreading or ascending plants,
2-4 dm. high; flowers pink or purple,
spring and summer) — [2](#).
- 1b. Anthers 10 — [3](#).

2a. Leaves pinnately dissected

**Stork's-bill,
Erodium
cicutarium.**

2b. Leaves palmately divided into
cuneate lobes

**Crane's-bill,
Geranium
pusillum.**

- 3a. Leaves ternately divided, the lobes
pinnatifid (2-4 dm. high; flowers
purple, late spring and summer)

**Herb Robert,
Geranium
robertianum.**

- 3b. Leaves palmately 3-11-lobed — [4](#).

4a. Petals 12 mm. long or more (3-
6 dm. high; flowers pale
purple, spring)

**Wild Geranium,
Geranium
maculatum.**

4b. Petals less than 10 mm. long
(Crane's-bill) — [5](#).

- 5a. Seed-bearing portion of the pistil smooth,
glabrous or nearly so (low spreading
plant; flowers purple, late spring and
summer)

**Crane's-bill,
Geranium
columbinum.**

- 5b. Seed-bearing portion of the pistil
transversely wrinkled (widely
branching. 1-3 dm. tall; flowers
purple, summer)

**Crane's-bill,
Geranium molle.**

5c. Seed-bearing portion of the pistil
pubescent (widely branching. 1-5 dm.
tall) —6.

6a. Leaves divided almost to the
base (flowers pinkish or
white, spring and summer)
—7.

6b. Leaves divided one-half to
two-thirds the way to the
base (flowers purple,
summer)

**Crane's-bill,
Geranium
rotundifolium.**

7a. Petals white or pale pink; flowers in
compact clusters

**Crane's-bill,
Geranium
carolinianum.**

7b. Petals pink-purple; flowers in loose
clusters

**Crane's-bill,
Geranium
bicknellii.**

RUTACEAE, the Rue Family

Shrubs or low trees, with compound leaves frequently dotted with
translucent glands; flowers small, greenish-white, with 3-5 sepals, petals,
and stamens.

1a. Leaflets 5-9; stems thorny

**Prickly Ash,
Zanthoxylum
americanum.**

1b. Leaflets 3; stems not thorny

**Hop Tree, Ptelea
trifoliata.**

SIMARUBACEAE, the Quassia Family

Trees, with pinnately compound leaves and small greenish-yellow flowers in large panicles in early summer, ripening into winged fruits.

One species in Michigan, escaped from cultivation chiefly in towns

**Tree of Heaven,
Ailanthus
glandulosa.**

POLYGALACEAE, the Milkwort Family

Small herbs, with alternate or whorled simple leaves, and small irregular flowers; sepals 5, petals 3, stamens 6 or 8, more or less united with each other and with the petals.

1a. All of the leaves alternate [—2.](#)

1b. Some or all of the leaves in whorls (1-4 dm. high; flowers greenish, purple, or white; summer) (Milkwort) [—6.](#)

2a. Flowers few, loosely clustered, 15-20 mm. long (1-3 dm. high; flowers purple; early summer)

**Flowering
Wintergreen,
Polygala
paucifolia.**

2b. Flowers numerous, in a spike or raceme [—3.](#)

3a. Stem-leaves minute, linear-subulate; stem slender, erect, 3-7 dm. high (flowers pink; summer)

**Milkwort,
Polygala
incarnata.**

3b. Stem-leaves narrowly oblong or broader; stem generally 1-4 dm. high [—4.](#)

4a. Flowers in a short thick obtuse very dense spike (flowers greenish or purple; summer)

**Milkwort,
Polygala
sanguinea.**

4b. Flowers in a slender tapering spike [—5.](#)

- | | |
|--|--|
| 4c. Flowers in a raceme; plants
with subterranean flowers
also (flowers purple; early
summer) | Milkwort,
Polygala
polygama. |
| 5a. Leaves linear or nearly so (flowers
purple; summer) — 7b. | |
| 5b. Leaves lanceolate or oblong-lanceolate,
2-6 cm. long (flowers white; late
spring) | Seneca Snakeroot,
Polygala senega. |
| 5c. Leaves ovate to ovate-lanceolate (flowers
white; late spring) | Seneca Snakeroot,
Polygala senega
var. latifolia. |
| 6a. Spike oval, thick, obtuse | Milkwort,
Polygala cruciata. |
| 6b. Spike acute <u>—7.</u> | |
| 7a. Spike densely flowered, 1-2 cm. long | Milkwort,
Polygala
verticillata. |
| 7b. Spike loosely flowered, 2-5 cm. long | Milkwort,
Polygala
verticillata var.
ambigua. |

EUPHORBIACEAE, the Spurge Family

Herbs, with alternate, opposite, or whorled leaves and usually milky juice. Flowers small or minute and inconspicuous, without petals and frequently without calyx. In our commoner species, several staminate flowers, each consisting of a single stamen only, and one pistillate flower, consisting of a single pedicelled 3-lobed ovary only, are included within a 4-5-lobed involucre, which is sometimes colored and resembles a calyx or corolla.

- 1a. Stem-leaves alternate; inflorescence
axillary; flowers with calyx and

several stamens (3-8 dm. tall; flowers greenish or purplish; summer)
(Three-seeded Mercury) — [2.](#)

1b. Stem-leaves opposite, usually inequilateral at base; flowers as described for the family; apparent flowers in axillary clusters (summer and autumn) (Spurge) — [3.](#)

1c. Stem-leaves alternate; inflorescence a terminal umbel-like cluster, with its branches subtended by opposite or whorled leaves; flowers as described for the family (Spurge) — [9.](#)

2a. Leaves ovate or ovate-lanceolate; flower-clusters shorter than the subtending bract

**Three-seeded
Mercury,
Acalypha
virginica.**

2b. Leaves lanceolate to oblong; flower-clusters equaling or exceeding the subtending bract

**Three-seeded
Mercury,
Acalypha
gracilens.**

3a. Stem and foliage glabrous — [4.](#)

3b. Stem and foliage more or less pubescent (stems prostrate or ascending, 1-4 dm. long) — [7.](#)

4a. Erect or ascending, usually without basal branches (2-4 dm. tall)

**Spurge,
Euphorbia preslii.**

4b. Prostrate or spreading, branched from the base (stems 1-4 dm. long) — [5.](#)

5a. Leaves entire; plants of the shores of the Great Lakes

**Spurge,
Euphorbia
polygonifolia.**

5b. Leaves serrulate — [6.](#)

- | | |
|--|---|
| 6a. Leaves broadly oblong or obovate; seeds obscurely wrinkled | Spurge,
Euphorbia
serpyllifolia. |
| 6b. Leaves narrowly oblong; seeds with prominent transverse ridges | Spurge,
Euphorbia
glyptosperma. |
| 7a. Seeds black | Spurge,
Euphorbia
hirsuta. |
| 7b. Seeds red — 8. | |
| 8a. Leaves oblong | Spurge,
Euphorbia
maculata. |
| 8b. Leaves elliptical to obovate; involucre split down one side | Spurge,
Euphorbia
humistrata. |
| 9a. Flowers subtended by conspicuous petal-like white appendages (part of the involucre) (4-10 dm. tall; summer) | Spurge,
Euphorbia
corollata. |
| 9b. Flowers not subtended by petal-like appendages — 10. | |
| 10a. Stem-leaves below the inflorescence serrulate (2-5 dm. high; summer) — 11. | |
| 10b. Stem-leaves below the inflorescence entire — 13. | |
| 11a. Upper leaves acute | Spurge,
Euphorbia
platyphylla. |
| 11b. Upper leaves obtuse, rounded, or notched at the apex — 12. | |
| 12a. Leaves of the involucre broadly triangular-ovate, widest near the base | Spurge,
Euphorbia
obtusata. |
| 12b. Leaves of the involucre broadly obovate to nearly | Spurge,
Euphorbia |

- | | | |
|------|---|---|
| | circular, widest near or above the middle | helioscopia. |
| 13a. | Stem-leaves narrowly linear, less than 3 mm. wide (2-4 dm. high; late spring and summer) | Cypress Spurge, Euphorbia cyparissias. |
| 13b. | Stem-leaves narrowly oblong-spatulate, more than 5 mm. wide, and more than 3 times as long as wide (2-6 dm. high; summer) — 14. | |
| 13c. | Stem-leaves obovate to nearly circular, not more than twice as long as wide (1-4 dm. high) — 15. | |
| | 14a. Leaves at base of umbel narrow, resembling those on the stem | Spurge, Euphorbia esula. |
| | 14b. Leaves at base of umbel broad, resembling those of the inflorescence | Spurge, Euphorbia lucida. |
| 15a. | Upper stem-leaves distinctly narrowed at the base; introduced species of waste places (summer) | Spurge, Euphorbia peplus. |
| 15b. | Upper stem-leaves rounded at the sessile base; native species of woodlands (spring and early summer) | Spurge, Euphorbia commutata. |

CALLITRICHACEAE, the Water Starwort Family

Small herbs growing in water or in mud, with opposite entire leaves and small inconspicuous axillary flowers, with neither calyx nor corolla. (Flowers in summer).

- | | | |
|-----|---|--|
| 1a. | Completely submerged; leaves all linear | Water Starwort, Callitriche autumnalis. |
|-----|---|--|

1b. Submerged leaves linear, emersed and
floating leaves obovate

**Water Starwort,
Callitriche
palustris.**

EMPETRACEAE, the Crowberry Family

Low evergreen shrubs, with the linear leaves completely rolled into a tube,
and inconspicuous flowers without petals, in the axils of the leaves.

One species in Michigan, 1-3 dm. high;
leaves less than 1 cm. long; flowers
in summer

**Crowberry,
Empetrum
nigrum.**

LIMNANTHACEAE, the False Mermaid Family

Low herbs with alternate compound leaves and minute axillary flowers;
sepals 3, petals 3, stamens 6.

One species in Michigan, with stems 1-3 dm.
long, and flowers in late spring

**False Mermaid,
Floerkea
proserpinacoides.**

ANACARDIACEAE, the Cashew Family

Shrubs or small trees, with milky or resinous juice, alternate compound
leaves sometimes poisonous to the touch, and small clustered greenish or
yellowish flowers.

1a. Leaflets 7 to many (1-5 m. high)
(Sumach) 2.

1b. Leaflets 3-5.

- | | |
|---|---|
| 2a. Axis of the leaves wing-margined between the leaflets | Sumach, <i>Rhus copallina</i>. |
| 2b. Axis of the leaves not margined — <u>3.</u> | |
| 3a. Leaflets entire | Poison Sumach, <i>Rhus vernix</i>. |
| 3b. Leaflets serrate — <u>4.</u> | |
| 4a. Bark of the older stems glabrous | Sumach, <i>Rhus glabra</i>. |
| 4b. Bark of the older stems densely velvety-hairy | Sumach, <i>Rhus typhina</i>. |
| 5a. Terminal leaflet narrowed to a sessile base (5-20 dm. high) | Sumach, <i>Rhus canadensis</i>. |
| 5b. Terminal leaflet on a definite stalk, round or acute at base (3-8 dm. high, or climbing by hold-fast roots) | Poison Ivy, <i>Rhus toxicodendron</i>. |

AQUIFOLIACEAE, the Holly Family

Shrubs, with alternate simple leaves and small white or greenish axillary flowers in late spring and early summer; sepals, petals, and stamens each 4-6; fruit a berry.

- | | |
|---|--|
| 1a. Leaves entire or nearly so, 1-3 cm. long (1-2 m. tall) | Mountain Holly, <i>Nemopanthus mucronata</i>. |
| 1b. Leaves sharply serrate, 5-8 cm. long (2-5 m. high) (Black Alder) — 2. | |
| 2a. Leaves downy on the veins beneath; fruit red | Black Alder, <i>Ilex verticillata</i>. |
| 2b. Leaves nearly or quite glabrous; fruit orange | Black Alder, <i>Ilex verticillata</i> var. <i>tenuifolia</i>. |

CELASTRACEAE, the Staff Tree Family

Shrubs with simple leaves and inconspicuous flowers; sepals and petals each 4 or 5, the stamens of the same number and attached to a disk which fills the center of the flower; fruit showy, orange and red.

- | | |
|---|---|
| 1a. Leaves alternate (climbing vine; flowers in racemes; late spring) | Bitter-sweet, Celastrus scandens. |
| 1b. Leaves opposite (flowers in axillary clusters) — 2. | |
| 2a. Prostrate, with short erect branches; leaves broadest above the middle (spring) | Creeping Wahoo, Evonymus obovatus. |
| 2b. Tall shrub, with leaves broadest below or near the middle (early summer) | Wahoo, Evonymus atropurpureus. |

STAPHYLEACEAE, the Bladder Nut Family

Shrubs with opposite trifoliate leaves and small axillary clusters of white flowers in spring; sepals, petals, and stamens each 5; ovary 3-celled, ripening into a large inflated 3-celled pod.

- | | |
|---------------------------------------|---|
| One species in Michigan (2-5 m. high) | Bladder Nut, Staphylea trifolia. |
|---------------------------------------|---|

ACERACEAE, the Maple Family

Trees or shrubs, with opposite, lobed or compound leaves and inconspicuous flowers; sepals about 5; petals the same number, or none; stamens 4-12; ovary 2-lobed, ripening into a pair of winged fruits.

1a. Leaves compound (tree; flowers appearing before the leaves)

Box Elder, *Acer negundo*.

1b. Leaves simple (Maple) 2.

2a. Shrubs or small trees; leaves 3-5-lobed; the lobes with regularly serrate margins (flowers greenish-yellow, appearing later than the leaves) 3.

2b. Trees; leaves 3-7-lobed; margins of the lobes entire or incised, but never regularly serrate 4.

3a. Leaves finely and sharply serrate; twigs smooth; bark conspicuously striped with white lines

Striped Maple, *Acer pennsylvanicum*.

3b. Leaves coarsely and bluntly serrate; young twigs pubescent; bark not striped

Mountain Maple, *Acer spicatum*.

4a. Angles between the leaf-lobes rounded (flowers greenish-yellow, appearing with the leaves) 5.

4b. Angles between the leaf-lobes acute or obtuse, but not rounded (flowers purple, red, or yellowish, appearing before the leaves) 6.

5a. Leaves glabrous beneath, or minutely pubescent on the veins

Sugar Maple, *Acer saccharum*.

5b. Leaves downy beneath

Black Maple, *Acer saccharum* var.

nigrum.

6a. Middle leaf-lobe usually more than half the length of the leaf, narrowed at its base; broken twigs with a strong odor

Silver Maple, *Acer saccharinum*.

6b. Middle leaf-lobe usually less than half the length of the leaf, its sides parallel or broadened at the base; broken twigs without strong odor

Red Maple, *Acer rubrum*.

SAPINDACEAE, the Soapberry Family

Trees, with opposite palmately compound leaves, and showy white or yellowish flowers in panicles in spring; sepals 5; petals 4 or 5; stamens about 7; fruit a smooth brown nut.

1a. Leaflets 7; buds viscid; corolla of 5 petals

Horse Chestnut, *Aesculus hippocastanum*.

1b. Leaflets 5; buds smooth; corolla of 4 petals

Buckeye, *Aesculus glabra*.

BALSAMINACEAE, the Touch-me-not Family

Smooth herbs, with alternate simple leaves and showy flowers; one petal-like sepal prolonged into a spur; fruit explosive when ripe (5-10 dm. high; summer).

1a. Flowers pale-yellow, with a few red-brown spots

Touch-me-not, *Impatiens pallida*.

1b. Flowers orange, thickly spotted with red-
brown

**Touch-me-not,
Impatiens biflora.**

RHAMNACEAE, the Buckthorn Family

Shrubs, with simple leaves and small flowers in axillary or terminal clusters in early summer; sepals, petals, and stamens each 4 to 5, or petals none.

1a. Leaves with a single mid-vein; flowers in axillary clusters, greenish
(Buckthorn) [2.](#)

1b. Leaves with 3-5 principal veins; flowers in dense terminal clusters, white
(Red-root) [3.](#)

2a. Lateral veins 3-4 pairs (stout shrub, frequently thorny, escaped from cultivation) **Buckthorn, Rhamnus cathartica.**

2b. Lateral veins 6-9 pairs (1 m. high or less; in swamps and bogs) **Buckthorn, Rhamnus alnifolia.**

3a. Leaves ovate, rounded or cordate at the base, 2-5 cm. wide or more (4-8 dm. high) **Red-root, Ceanothus americanus.**

3b. Leaves elliptical-lanceolate, 2 cm. wide or less (3-8 dm. high) **Red-root, Ceanothus ovatus.**

VITACEAE, the Grape Family

Shrubs, climbing by tendrils or hold-fast roots, with palmately lobed or palmately compound leaves and small greenish flowers in panicles or flattened clusters; petals and sepals each 4 or 5; fruit a berry.

1a. Leaves compound (summer) (Virginia Creeper) [2.](#)

- 1b. Leaves simple (late spring) (Grape) —4.
 - 2a. Branches of the tendrils chiefly ending in adhesive disks —3.
 - 2b. Branches of the tendrils twining, or rarely with a few disks
 - Virginia Creeper, Psedera vitacea.**
- 3a. Stem and foliage glabrous
 - Virginia Creeper, Psedera quinquefolia.**
- 3b. Stem and foliage pubescent, at least when young
 - Virginia Creeper, Psedera quinquefolia var. hirsuta.**
- 4a. Leaves conspicuously pubescent beneath —5.
- 4b. Leaves glabrous beneath when mature, or pubescent on the veins only —6.
- 5a. A tendril or flower-cluster opposite each leaf
 - Fox Grape, Vitis labrusca.**
- 5b. No tendril opposite each third leaf
 - Summer Grape, Vitis aestivalis.**
- 6a. Pith continuous through the joints of the stem
 - Fox Grape, Vitis rotundifolia.**
- 6b. Pith interrupted by the solid joints —7.
- 7a. Leaf-lobes with rounded angles between them
 - Summer Grape, Vitis bicolor.**
- 7b. Leaf-lobes with sharp angles between them —8.
 - 8a. Leaves coarsely toothed, unlobed or slightly 3-lobed
 - Frost Grape, Vitis cordifolia.**
 - 8b. Leaves sharply toothed, prominently lobed
 - Frost Grape, Vitis vulpina.**

TILIACEAE, the Linden Family

Trees, with alternate, simple, palmately veined leaves, and clusters of fragrant white flowers in late spring arising from the middle of a leaf-like bract; sepals and petals each 5; stamens numerous, but united into 5 sets.

One species in Michigan

Basswood, *Tilia americana*.

MALVACEAE, the Mallow Family

Herbs with alternate leaves; sepals and petals each 5; stamens numerous, united by their filaments to form a tube surrounding the styles; ovary many-celled.

1a. Flowers yellow (summer and autumn)

—2.

1b. Flowers pale-yellow, with a dark center
(2-4 dm. high; late summer)

**Flower-of-an-hour,
Hibiscus trionum.**

1c. Flowers white to red or blue, never
yellow —3.

2a. Leaves broadly heart-shape
(10-15 dm. tall)

**Velvet Leaf,
Abutilon theophrasti.**

2b. Leaves ovate-lanceolate (2-5
dm. tall)

Sida, *Sida spinosa*.

3a. Calyx subtended by 6 to many bractlets
which are sometimes united at base
(summer) —4.

3b. Calyx subtended by 3 bractlets, or by
none —6.

- | | |
|--|--|
| 4a. Flowers 2-4 cm. wide (5-10 dm. high; flowers pink) | Marsh Mallow,
<i>Althaea officinalis</i>. |
| 4b. Flowers 7-15 cm. wide (8-15 dm. high; flowers pink to nearly white) (Rose Mallow) —5. | |
| 5a. Leaves densely pubescent below | Rose Mallow,
<i>Hibiscus moscheutos</i>. |
| 5b. Leaves glabrous | Rose Mallow,
<i>Hibiscus militaris</i>. |
| 6a. Petals prominently notched at the end or reverse heart-shape (Mallow) —7. | |
| 6b. Petals obtuse or truncate (summer) —11. | |
| 7a. Flowers 1-1.5 cm. wide —8. | |
| 7b. Flowers 2-5 cm. wide (3-8 dm. high; flowers in summer) —9. | |
| 8a. Stems procumbent, prostrate, or spreading (spring, summer, and autumn) | Mallow, <i>Malva rotundifolia</i>. |
| 8b. Stems erect (10-15 dm. high; summer) | Mallow, <i>Malva verticillata</i>. |
| 9a. Leaves with prominent but shallow lobes; flowers axillary | Mallow, <i>Malva sylvestris</i>. |
| 9b. Leaves deeply lobed or cleft; flowers in the upper axils, producing a raceme-like cluster —10. | |
| 10a. Lobes of the leaf dentate or incised | Mallow, <i>Malva alcea</i>. |
| 10b. Lobes of the leaf pinnately cleft into linear or narrowly oblong divisions | Mallow, <i>Malva moschata</i>. |
| 11a. Flowers white (1-2 m. high) | Virginia Mallow,
<i>Sida</i> |

11b. Flowers purple or pink (3-5 dm. high,
spreading)

hermaphrodita.
Poppy Mallow,
Callirhoe
triangulata.

HYPERICACEAE, the St. John's-wort Family

Herbs or shrubs, with opposite entire leaves dotted with translucent glands; flowers usually yellow (or pink); sepals and petals each 5; stamens 5 to many; ovary with 3-5 styles. (St. John's-wort.)

1a. Shrubs (4-8 dm. high; flowers yellow,
summer) [—2.](#)

1b. Herbs (flowers in summer) [—3.](#)

2a. Styles 5

St. John's-wort,
Hypericum
kalmianum.

2b. Styles 3

St. John's-wort,
Hypericum
prolificum.

3a. Flowers pinkish, 15 mm. broad (3-5 dm.
high, in swamps)

Marsh St. John's-
wort, Hypericum
virginicum.

3b. Flowers yellow [—4.](#)

4a. Flowers about 4 cm. wide;
principal leaves 5-10 cm.
long (7-15 dm. tall)

St. John's-wort,
Hypericum
ascyron.

4b. Flowers 8-25 mm. wide;
stamens 15 or more [—5.](#)

4c. Flowers 1-10 mm. wide;
stamens 12 or fewer (1-6
dm. high) [—7.](#)

5a. Petals dotted with black (4-8 dm. high)
[—6.](#)

5b. Petals without black dots (2-5 dm. high)	St. John's-wort, Hypericum ellipticum.
6a. Flowers 20-25 mm. wide; leaves of an oblong type, broadest near the middle	St. John's-wort, Hypericum perforatum.
6b. Flowers 10-15 mm. wide; leaves of an ovate type, broadest below the middle	St. John's-wort, Hypericum punctatum. Pineweed, Hypericum gentianoides.
7a. Leaves minute, subulate, 1-3 mm. long	St. John's-wort, Hypericum canadense.
7b. Leaves linear, with 1-3 principal veins, broadest near or above the middle	St. John's-wort, Hypericum majus.
7c. Leaves lanceolate, 4-6 times as long as broad, with 5-7 principal veins	
7d. Leaves oblong, elliptic, or ovate, 1.5-3 times as long as broad <u>— 8.</u>	
8a. Uppermost bracts linear	St. John's-wort, Hypericum mutilum.
8b. Uppermost bracts resembling the leaves in shape, but smaller	St. John's-wort, Hypericum boreale.

ELATINACEAE, the Waterwort Family

Small marsh herbs, with opposite leaves without translucent dots, and inconspicuous axillary flowers. (Stems 2-5 cm. long; flowers in summer.)

One species in Michigan

**Waterwort,
Elatine americana.**

CISTACEAE, the Rock-rose Family

Small herbs or shrubs, with opposite or alternate entire leaves; flowers regular, with 5 sepals, 3 or 5 petals, and 3 to many stamens.

1a. Flowers yellow (early summer) —2.

1b. Flowers greenish or purplish, minute, in panicles (late summer) (Pinweed) —4.

2a. Leaves crowded, closely appressed to the branches; flowers 7 mm. wide (2-4 dm. high)

**False Heather,
Hudsonia
tomentosa.**

2b. Leaves spreading; flowers 15-30 mm. wide (3-6 dm. high) (Frostweed) —3.

3a. Petal-bearing flowers solitary

**Frostweed,
Helianthemum
canadense.**

3b. Petal-bearing flowers few, racemose

**Frostweed,
Helianthemum
majus.**

4a. Stem-leaves linear, 4 or more times as long as wide —5.

4b. Stem-leaves oblong or elliptical, about 3 times as long as wide (2-6 dm. tall) —8.

5a. Plant pale with dense appressed pubescence (2-4 dm. high)

**Pinweed, Lechea
stricta.**

5b. Plant green, pubescence sparse or none —6.

- | | |
|---|---|
| 6a. Leaves thread-like, seldom exceeding 1 mm. in width (1-3 dm. tall) | Pinweed, <i>Lechea tenuifolia</i>. |
| 6b. Leaves 1-5 mm. wide (2-6 dm. high) —7. | |
| 7a. Leaves on the basal shoots narrowly lanceolate | Pinweed, <i>Lechea intermedia</i>. |
| 7b. Leaves on the basal shoots oblong-elliptic, about twice as long as wide | Pinweed, <i>Lechea racemulosa</i>. |
| 8a. Pubescence of spreading hairs | Pinweed, <i>Lechea villosa</i>. |
| 8b. Pubescence of appressed hairs | Pinweed, <i>Lechea minor</i>. |

VIOLACEAE, the Violet Family

Herbs with simple, alternate or basal leaves, and conspicuous irregular flowers with a spur (except in the first species); sepals, petals, and stamens each 5; ovary 1-celled.

- | | |
|--|--|
| 1a. Flowers regular or nearly so, greenish-white, axillary; erect plant with leafy stem (3-5 dm. high; spring) | Green Violet, <i>Hybanthus concolor</i>. |
| 1b. Flowers irregular, blue, yellow, or white, conspicuous (Violet) —2. | |
| 2a. Plant stemless, the flowers all on leafless stalks and the leaves all basal (spring or early summer) —3. | |
| 2b. Stems leafy (spring and summer) —17. | |
| 3a. Petals yellow | Round-leaved Violet, <i>Viola rotundifolia</i>. |

- 3b. Petals blue, violet, or white —4.
 - 4a. Principal leaves at time of flowering deeply lobed —5.
 - 4b. Leaves oblong, ovate, or triangular, not narrowed to the petiole, and frequently sharply toothed or incised near the base —7.
 - 4c. Leaves narrowly lanceolate, tapering to the base **Violet, Viola lanceolata.**
 - 4d. Leaves heart-shape or kidney-shape, not lobed —8.
- 5a. Lateral petals bearded —6.
- 5b. Lateral petals not bearded **Bird-foot Violet, Viola pedata.**
 - 6a. Leaves divided to the base into linear segments **Bird-foot Violet, Viola pedatifida.**
 - 6b. Leaves irregularly divided into broader segments **Hand-leaf Violet, Viola palmata.**
- 7a. Leaves ovate-oblong, pubescent **Violet, Viola fimbriatula.**
- 7b. Leaves triangular-lanceolate, usually somewhat dilated at base, nearly or quite glabrous **Violet, Viola sagittata.**
 - 8a. Flowers violet or blue (rarely white-flowered plants are found with the typical blue-flowered ones) (Blue Violets) —9.
 - 8b. Flowers white, the 3 lower petals marked with purple (White Violets) —14.
- 9a. Lateral petals bearded —10.
- 9b. Lateral petals beardless **Great-spurred Violet, Viola**

selkirkii.

- 10a. Foliage glabrous — 11.
- 10b. Petioles and lower surface of leaves pubescent — 13.
- 11a. Beard of the lateral petals with a knob at the tip of each hair **Blue Violet, Viola cucullata.**
- 11b. Beard of the lateral petals not knobbed — 12.
- 12a. Spurred petal hairy **Wood Violet, Viola affinis.**
- 12b. Spurred petal glabrous **Blue Violet, Viola papilionacea.**
- 13a. Spurred petal villous **Blue Violet, Viola septentrionalis.**
- 13b. Spurred petal glabrous, or with a few scattered hairs **Common Blue Violet, Viola sororia.**
- 14a. Leaf-blade obviously pubescent — 15.
- 14b. Leaf-blade glabrous or very nearly so — 16.
- 15a. Lateral petals bearded **Sweet White Violet, Viola incognita.**
- 15b. Lateral petals not bearded **White Violet, Viola renifolia.**
- 16a. Leaf-blades strictly glabrous **Sweet White Violet; Viola pallens.**
- 16b. Leaf-blades with some minute white hairs on the upper surface near the base **Sweet White Violet, Viola blanda.**
- 17a. Stipules large and leaf-like, deeply pinnatifid and nearly or quite as long as the petioles — 18.

- 17b. Stipules small, inconspicuous, entire or toothed, and much shorter than the petiole — [19](#).
- 18a. Leaves serrate; flowers 1.5-2.5 cm. wide (1-3 dm. high; flowers of various colors) **Pansy, *Viola tricolor*.**
- 18b. Upper leaves entire or nearly so; flowers about 1 cm. wide (1-2 dm. high; flowers bluish-white) **Wild Pansy, *Viola rafinesquii*.**
- 19a. Petals yellow (1-4 dm. high) (Yellow Violet) — [20](#).
- 19b. Petals violet, blue, or white — [21](#).
- 20a. Foliage villous-pubescent **Yellow Violet, *Viola pubescens*.**
- 20b. Foliage nearly or quite glabrous **Yellow Violet, *Viola scabriuscula*.**
- 21a. Stipules entire (2-4 dm. high) **Canada Violet, *Viola canadensis*.**
- 21b. Stipules toothed — [22](#).
- 22a. Lateral petals not bearded (1-2 dm. high) **Long-spurred Violet, *Viola rostrata*.**
- 22b. Lateral petals bearded — [23](#).
- 23a. Flowers white or nearly white (1-3 dm. high) **Pale Violet, *Viola striata*.**
- 23b. Flowers blue (about 1 dm. high) — [24](#).
- 24a. Leaves smooth **Dog Violet, *Viola conspersa*.**
- 24b. Leaves pubescent **Sand Violet, *Viola arenaria*.**

CACTACEAE, the Cactus Family

Fleshy, jointed leafless plants, armed with numerous thorns; flowers large (5-10 cm. wide), yellow, with about 10 petals and numerous stamens.

One species in Michigan, on the shores of
Lake Michigan; flowers in summer

**Prickly Pear,
Opuntia
rafinesquii.**

THYMELAEACEAE, the Mezereum Family

Shrubs, with simple alternate entire leaves, and small yellowish flowers in clusters, opening before the leaves; petals none, the sepals somewhat petal-like.

One species in Michigan, 5-15 dm. tall, with
very tough bark

**Leatherwood,
Dirca palustris.**

ELAEAGNACEAE, the Oleaster Family

Shrubs, with opposite, silvery-pubescent, simple, entire leaves, and small clusters of inconspicuous yellow flowers in spring.

One species in Michigan, 1-2 m. high

**Buffalo Berry,
Shepherdia
canadensis.**

LYTHRACEAE, the Loosestrife Family

Herbs or shrubs, with opposite or alternate entire leaves; receptacle cup-shape or tubular, bearing the 5-7 petals and sepals at its margin, and the 6-12 stamens on its inner surface; ovary superior.

- | | |
|--|--|
| 1a. Stem shrubby (aquatic, stems 1-3 m. long; flowers pink, summer) | Water Loosestrife,
Decodon
verticillatus. |
| 1b. Stem herbaceous (4-10 dm. high; flowers purple, summer) <u>—2.</u> | |
| 2a. Flowers solitary in the axils; leaves mostly alternate | Loosestrife,
Lythrum alatum. |
| 2b. Flowers in terminal panicles; leaves opposite or whorled | Loosestrife,
Lythrum salicaria. |

MELASTOMACEAE, the Melastoma Family

Herbs, with opposite leaves with 3-5 principal veins; receptacle urn-shape, bearing 4 sepals and 4 petals at its edge; stamens 8; ovary 4-celled, superior.

- | | |
|---|---|
| One species in Michigan, 3-4 dm. high, with purple flowers in late summer | Meadow Beauty,
Rhexia virginica. |
|---|---|

ONAGRACEAE, the Evening Primrose Family

Herbs with opposite or alternate simple leaves and regular flowers; sepals and petals each 4 (or 2 in one genus), stamens 8 (or 2 in one genus), attached to the summit or inside of a tubular receptacle; ovary 2-4-celled, inferior.

- | | |
|--|--|
| 1a. Aquatic plant of shallow water or muddy ground, with prostrate stem (flowers minute, axillary; petals small or none, summer) | Water Purslane,
Ludvigia palustris. |
| 1b. Land plants with erect or ascending stems <u>—2.</u> | |

- | | |
|--|--|
| 2a. Petals minute, greenish (3-8 dm. high; late summer) | False Loosestrife,
Ludvigia polycarpa. |
| 2b. Petals yellow <u>—3.</u> | |
| 2c. Petals white, pink, purple, or red <u>—9.</u> | |
| 3a. Sepals borne at the summit of the ovary (5-10 dm. high; summer) | False Loosestrife,
Ludvigia alternifolia. |
| 3b. Sepals borne at the summit of the slender tubular receptacle, which is prolonged beyond the ovary <u>—4.</u> | |
| 4a. Stamens all equal in length (Evening Primrose) <u>—5.</u> | |
| 4b. The alternate stamens longer (3-8 dm. high; summer) (Sundrops) <u>—8.</u> | |
| 5a. Leaves deeply dentate or pinnatifid (2-5 dm. high; early summer) | |
| 5b. Leaves entire or undulate or finely toothed (5-12 dm. high; summer) <u>—6.</u> | Evening Primrose,
Oenothera laciniata. |
| 6a. Hairs on the stem with broad reddish bases | Evening Primrose,
Oenothera muricata. |
| 6b. Hairs on the stem none, or without swollen bases <u>—7.</u> | |
| 7a. Stem and foliage glabrous, or with sparse spreading hairs | Evening Primrose,
Oenothera biennis. |
| 7b. Stem and foliage densely but closely appressed-pubescent | Evening Primrose,
Oenothera rhombipetala. |
| 8a. Petals 5-10 mm. long | Sundrops,
Oenothera pumila. |

- | | |
|---|--|
| 8b. Petals 14 mm. long or more | Sundrops,
Oenothera
fruticosa. |
| 9a. Petals 2, reverse heart-shape, stamens 2
(flowers small, white, summer)
(Enchanter's Nightshade) <u>—10.</u> | |
| 9b. Petals 4; stamens 4 or 8 <u>—12.</u> | |
| 10a. Leaves rounded at the base,
denticulate (3-8 dm. high;
fruit prickly) | Enchanter's
Nightshade,
Circaea lutetiana. |
| 10b. Leaves cordate at the base
<u>—11.</u> | |
| 11a. Fruit 2-celled, bristly (2-4 dm. high) | Enchanter's
Nightshade,
Circaea
intermedia. |
| 11b. Fruit 1-celled, with soft hairs (delicate
plant 2 dm. high, or less) | Enchanter's
Nightshade,
Circaea alpina. |
| 12a. Petals entire (summer) <u>—13.</u> | |
| 12b. Petals notched at the end
(flowers white or pinkish,
less than 1 cm. broad, in
summer) (Willow Herb)
<u>—15.</u> | |
| 13a. Flowers 2-3 cm. wide, purple (7-20 dm.
high) | Fireweed,
Epilobium
angustifolium. |
| 13b. Flowers about 1 cm. wide <u>—14.</u> | |
| 14a. Flowers red (2-5 dm. high) | Gaura, Gaura
coccinea. |
| 14b. Flowers white, turning pink
when old | Gaura, Gaura
biennis. |
| 15a. Leaves entire, the margins usually
somewhat revolute <u>—16.</u> | |

15b. Leaves toothed, flat (4-9 dm. high)

— 18.

16a. Plant densely pubescent with
spreading hairs (3-8 dm.
high)

**Willow Herb,
Epilobium molle.**

16b. Plant pubescent with
appressed or incurved hairs

— 17.

17a. Leaves linear, the margin revolute (3-5
dm. high)

**Willow Herb,
Epilobium
densum.**

17b. Leaves narrowly lanceolate, not
revolute (2-4 dm. high)

**Willow Herb,
Epilobium
palustre.**

18a. Seeds tipped with a tuft of
reddish-brown hairs

**Willow Herb,
Epilobium
coloratum.**

18b. Seeds tipped with a tuft of
white hairs

**Willow Herb,
Epilobium
adenocaulon.**

HALORAGIDACEAE, the Water Milfoil Family

Aquatic or marsh herbs, with alternate, opposite, or whorled leaves, and small, inconspicuous terminal or axillary flowers, frequently without petals (summer).

1a. Leaves none, or else very small and
inconspicuous

**Water Milfoil,
Myriophyllum
tenellum.**

1b. Leaves alternate (1-4 dm. high)

**Mermaid Weed,
Proserpinaca
palustris.**

1c. Leaves opposite or whorled — 2.

- | | |
|--|---|
| 2a. Leaves entire (2-4 dm. high) | Mare's-tail,
Hippuris vulgaris. |
| 2b. Leaves toothed or dissected
(Water Milfoil) <u>— 3.</u> | |
| 3a. Flowers in the axils of foliage leaves
<u>— 4.</u> | |
| 3b. Flowers in terminal spikes, subtended by
bracts <u>— 5.</u> | |
| 4a. Flowers above water,
subtended by toothed or
entire leaves | Water Milfoil,
Myriophyllum
heterophyllum. |
| 4b. Flowers submerged, subtended
by dissected leaves | Water Milfoil,
Myriophyllum
farwellii. |
| 5a. Flowers solitary or in pairs at each joint
of the spike | Water Milfoil,
Myriophyllum
alternifolium. |
| 5b. Flowers several at each joint of the spike
<u>— 6.</u> | |
| 6a. Bracts deeply pinnatifid | Water Milfoil,
Myriophyllum
verticillatum var.
pectinatum. |
| 6b. Bracts entire or toothed | Water Milfoil,
Myriophyllum
spicatum. |

ARALIACEAE, the Sarsaparilla Family

Herbs or thorny shrubs, with alternate or whorled leaves, and small flowers in umbels; sepals 5, minute; petals and stamens each 5; ovary inferior, with 2-5 styles, ripening into a berry.

- | | |
|--|--|
| 1a. Leaves simple, palmately lobed (thorny shrub; flowers greenish-white, in panicles, in June) | Devil's Club,
Fatsia horrida. |
| 1b. Leaves once compounded, whorled (umbel one, terminal) <u>—2.</u> | |
| 1c. Leaves twice or thrice compounded (umbels several) <u>—3.</u> | |
| 2a. Leaflets sessile; flowers white, in spring (1-2 dm. high) | Dwarf Ginseng,
Panax trifolium. |
| 2b. Leaflets stalked; flowers greenish, in summer (2-5 dm. high) | Ginseng, Panax
quinquefolium. |
| 3a. Stem and petioles spiny or bristly (flowers white, summer) <u>—4.</u> | |
| 3b. Stem and petioles smooth or a little pubescent (flowers greenish-white) <u>—5.</u> | |
| 4a. Shrubby, with stout thorns (1-3 m. high) | Hercules' Club,
Aralia spinosa. |
| 4b. Herbaceous, with slender bristles (4-10 dm. high) | Bristly
Sarsaparilla,
Aralia hispida. |
| 5a. Stem-leaves present; leaflets cordate at the base (8-15 dm. high; summer) | Spikenard, Aralia
racemosa. |
| 5b. Leaf and flower-stalk arising from the ground; leaflets acute at the base (2-4 dm. high; spring) | Wild Sarsaparilla,
Aralia nudicaulis. |

UMBELLIFERAE, the Parsley Family

Herbs, with alternate, usually compound leaves, the petioles dilated at the base; flowers small, in umbels or heads; sepals 5, minute or even wanting; petals and stamens each 5; ovary inferior, with 2 styles, ripening into a dry fruit.

1a. Leaves simple (flowers in summer) —2.

1b. Leaves compound, or at least deeply cleft
—4.

2a. Leaves linear, sword-shape (4-10 dm. tall; flowers greenish-white)

**Rattlesnake
Master, *Eryngium
yuccifolium.***

2b. Leaves kidney-shape or almost circular (stems creeping, about 1 dm. high; flowers white) (Water Pennywort)
—3.

3a. Leaves peltate, attached by the center

**Water Pennywort,
*Hydrocotyle
umbellata.***

3b. Leaves not peltate, attached by the margin

**Water Pennywort,
*Hydrocotyle
americana.***

4a. Flowers yellow or purple —5.

4b. Flowers white or greenish
—13.

5a. Leaf-segments entire (4-8 dm. high) —6.

5b. Leaf-segments toothed or incised —7.

6a. Leaf-segments filiform (summer)

**Fennel,
*Foeniculum
vulgare.***

6b. Leaf-segments ovate to lanceolate

**Golden Alexander,
*Taenidia
integerrima.***

7a. Leaves pinnately compound; some of the leaflets incised or pinnatifid —8.

7b. Leaves ternately compound; the segments crenate or serrate —9.

7c. Leaves deeply palmately cleft or divided; flowers in head-like umbels — 18a.

8a. Leaf-segments obtuse, rounded, or cordate at the

**Wild Parsnip,
*Pastinaca sativa.***

- base (6-15 dm. high;
summer)
- Prairie Parsley,
Polytaenia
nuttallii.**
- 8b. Leaf-segments narrowed to the
base (4-8 dm. high; spring)
- 9a. Terminal leaflets conspicuously stalked,
their total length, including stalk, at
least 50% greater than the length of
the lateral leaflets (Meadow Parsnip)
[— 10.](#)
- 9b. Terminal leaflets not conspicuously
stalked, their total length, including
stalk, about equaling the lateral
leaflets (4-8 dm. high; late spring)
(Golden Alexander) [— 12.](#)
- Meadow Parsnip,
Thaspium aureum
var.
atropurpureum.**
- 10a. Flowers purple (4-8 dm. high;
early summer)
- 10b. Flowers yellow [— 11.](#)
- 11a. Stem-leaves once-ternate; leaflets finely
serrate (4-8 dm. high; early summer)
- Meadow Parsnip,
Thaspium
aureum.**
- 11b. Many stem-leaves 2-3-ternate; leaflets
coarsely serrate or incised (6-12 dm.
high; early summer)
- Meadow Parsnip,
Thaspium
barbinode.**
- 12a. Basal and lower stem-leaves
2-3-ternate
- Golden Alexander,
Zizia aurea.**
- 12b. Basal leaves simple; stem-
leaves once-ternate
- Golden Alexander,
Zizia cordata.**
- 13a. Leaves once-pinnate (or the submerged
leaves decompose, if present)
(summer) [— 14.](#)
- 13b. Leaves ternately, palmately, or 2-3-
pinnately compound [— 16.](#)

- 14a. Leaflets mostly ovate or ovate-lanceolate, some of them coarsely incised (3-9 dm. high) **Water Parsnip, *Berula erecta*.**
- 14b. Leaflets linear to oblong, serrate to nearly entire, not incised (6-15 dm. high) — 15.
- 15a. Leaflets entire, or with a few low remote teeth **Cowbane, *Oxypolis rigidior*.
Water Parsnip, *Sium cicutaefolium*.**
- 15b. Leaflets finely but sharply serrate **Harbinger of Spring, *Erigenia bulbosa*.**
- 16a. Leaves principally basal, decompose; flowers in early spring (1-2 dm. high)
- 16b. Leaves principally on the stem — 17.
- 17a. Leaves palmately or ternately once-compound — 18.
- 17b. Leaves 2-3 times compound or decompose — 24.
- 18a. Flowers short-pedicelled, crowded in head-like umbels, greenish; ovary bristly (4-9 dm. high; early summer) (Black Snakeroot) — 19.
- 18b. Flowers in open umbels, white — 22.
- 19a. Styles short, not projecting beyond the bristles of the mature fruit — 20.
- 19b. Styles long, projecting beyond the bristles of the fruit, and recurved — 21.

- 20a. Staminate flowers on pedicels
3-4 mm. long, equaling or
barely exceeding the fruit **Black Snakeroot,**
***Sanicula trifoliata*.**
- 20b. Staminate flowers short-
pedicelled, concealed
among the fruits **Black Snakeroot,**
Sanicula
***canadensis*.**
- 21a. Fruit short-stalked, 4 mm. long or less **Black Snakeroot,**
***Sanicula gregaria*.**
- 21b. Fruit sessile, 6-7 mm. long **Black Snakeroot,**
Sanicula
***marilandica*.**
- 22a. Umbel unsymmetrical, its
branches irregular in
length; plant slender (3-8
dm. tall; early summer) **Honewort,**
Cryptotaenia
***canadensis*.**
- 22b. Umbel symmetrical with
regular branches; plants tall
and stout [—23.](#)
- 23a. Stem and leaves very pubescent (10-25
dm. high; summer) **Cow Parsnip,**
Heracleum
***lanatum*.**
- 23b. Stem and leaves glabrous or nearly so
(5-15 dm. high; early summer) **Masterwort,**
Imperatoria
***ostruthium*.**
- 24a. Ovary and fruit bristly (4-10
dm. high) [—25.](#)
- 24b. Ovary and fruit smooth or
winged, never bristly
[—27.](#)
- 25a. Umbels loose, open, few-flowered;
woodland plants blooming in spring
(Sweet Cicely) [—26.](#)
- 25b. Umbels densely flowered; weedy plants
blooming from summer to fall **Wild Carrot,**
***Daucus carota*.**
- 26a. Stem villous-pubescent **Sweet Cicely,**
Osmorhiza

- claytoni.
Sweet Cicely,
Osmorhiza
longistylis.
- 26b. Stem glabrous except at the joints
- 27a. Leaflets merely serrate (flowers in summer) — [28.](#)
- 27b. Leaflets coarsely incised, so that the leaf appears dissected — [30.](#)
- 28a. Umbel densely pubescent (8-15 dm. high) **Angelica, Angelica villosa.**
- 28b. Umbel smooth — [29.](#)
- 29a. Leaf-segments broadly ovate (8-15 dm. high) **Angelica, Angelica atropurpurea.**
- 29b. Leaf-segments lanceolate (8-15 dm. high) **Water Hemlock, Cicuta maculata.**
- 29c. Leaf-segments linear (4-10 dm. high) **Water Hemlock, Cicuta bulbifera.**
- 30a. Principal branches of the umbel 2-5; fruit linear-oblong; woodland plants blooming in spring (2-4 dm. high) **Chervil, Chaerophyllum procumbens.**
- 30b. Principal branches of the umbel 7 or more; fruit ovate to broadly elliptical (summer) — [31.](#)
- 31a. Native plants, growing in swamps (5-15 dm. high) **Hemlock Parsley, Conioselinum chinense.**
- 31b. Introduced plants, in waste places and along roads — [32.](#)
- 32a. Stems conspicuously spotted with purple (5-15 dm. high) **Poison Hemlock, Conium maculatum.**
- 32b. Stems not spotted with purple (2-5 dm. high) **Caraway, Carum carvi.**

CORNACEAE, the Dogwood Family

Trees, shrubs, or herbs, with alternate leaves and small flowers in rather crowded rounded or flattened clusters; sepals 4, minute; petals and stamens each 4; ovary inferior, ripening into a berry. In one genus the flowers are minute and greenish, with 5 sepals and petals minute or none.

1a. Leaves alternate [—2.](#)

1b. Leaves opposite [—3.](#)

2a. Flowers white, conspicuous, in flattened clusters (shrubs 2-4 m. high; flowers in late spring) **Dogwood, *Cornus alternifolia*.**

2b. Flowers greenish, inconspicuous, in small axillary clusters (tree; flowers in spring) **Sour Gum, *Nyssa sylvatica*.**

3a. Flower clusters small and dense, surrounded by a showy involucre of 4 bracts, resembling a corolla of 4 petals [—4.](#)

3b. Flowers in open flattened clusters, without petal-like involucre (shrubs 1-4 m. high; late spring) [—5.](#)

4a. Herbaceous, 3 dm. high or less (flowers in late spring) **Dwarf Dogwood, *Cornus canadensis*.**

4b. Tall shrub or tree (flowers in late spring) **Flowering Dogwood, *Cornus florida*.**

5a. Leaves distinctly pubescent beneath with woolly or spreading hairs [—6.](#)

- 5b. Leaves smooth beneath, or pubescent
with short appressed hairs —9.
 - 6a. Leaves rough above; fruit
white **Dogwood, *Cornus asperifolia.***
 - 6b. Leaves smooth or finely soft-
hairy above —7.
- 7a. Leaves at least twice as long as wide;
branches brownish or purplish —8.
- 7b. Leaves less than twice as long as wide;
branches greenish; fruit blue **Dogwood, *Cornus circinata.***
 - 8a. Branches purplish; fruit blue **Dogwood, *Cornus amomum.***
 - 8b. Branches brownish; fruit white **Dogwood, *Cornus baileyi.***
- 9a. Branches bright red or reddish-purple **Dogwood, *Cornus stolonifera.***
- 9b. Branches grayish **Dogwood, *Cornus paniculata.***

ERICACEAE, the Heath Family

Herbs or shrubs, frequently with evergreen leaves; sepals 4-5; corolla regular, with 4-5 petals; stamens as many or twice as many; ovary 3-10-celled, with 1 style.

- 1a. Plants without green color; leafless or
with scale leaves only —2.
- 1b. Plants with green leaves —4.
 - 2a. Flowers solitary (1-2 dm. high;
summer) **Indian Pipe, *Monotropa uniflora.***
 - 2b. Flowers in clusters —3.
- 3a. Petals united into a bell-shape corolla (3-
9 dm. high; summer) **Pine Drops, *Pterospora***

	andromedea. Beech Drops, Monotropa hypopitys.
3b. Petals all separate (1-3 dm. high; summer)	
4a. Leaves all basal; herbaceous plants with terminal racemes (1-4 dm. high; summer) (Shin-leaf) <u>—5.</u>	
4b. Stem-leaves present <u>—12.</u>	
5a. Style straight <u>—6.</u>	
5b. Style bent near the apex <u>—8.</u>	
6a. Racemes one-sided, the flowers all turned in one direction (flowers white or greenish- white) <u>—7.</u>	
6b. Raceme regular, the flowers not all pointing in the same direction (flowers white or pink)	Shin-leaf, Pyrola minor.
7a. Flowers numerous in each raceme	Shin-leaf, Pyrola secunda.
7b. Flowers only 3-7 in each raceme	Shin-leaf, Pyrola secunda var. obtusata.
8a. Flowers pink or purple <u>—9.</u>	
8b. Flowers white or greenish <u>—10.</u>	
9a. Leaves cordate at base	Shin-leaf, Pyrola asarifolia.
9b. Leaves rounded at base, not cordate	Shin-leaf, Pyrola asarifolia var. incarnata.
10a. Leaves shining on the upper side; sepals one-third as long as the petals	Shin-leaf, Pyrola americana.

- 10b. Leaves dull on the upper side;
sepals one-fourth as long as
the petals, or a little shorter
[— 11.](#)
- 11a. Leaf-blades mostly shorter than their
petioles, thick and firm **Shin-leaf, *Pyrola*
chlorantha.**
- 11b. Leaf-blades thin, usually longer than
their petioles **Shin-leaf, *Pyrola*
elliptica.**
- 12a. Petals nearly or quite separate
from each other [— 13.](#)
- 12b. Petals united into a
gamopetalous corolla, the
tube of which is as long as
or longer than the lobes
[— 18.](#)
- 13a. Leaves opposite or whorled; stems
herbaceous or nearly so (summer)
[— 14.](#)
- 13b. Leaves alternate; stems shrubby (early
summer) [— 16.](#)
- 14a. Flowers solitary; leaves
broadly ovate to nearly
circular (1 dm. high; flower
white) **One-flowered
Wintergreen,
Moneses uniflora.**
- 14b. Flowers in clusters; leaves
narrow (stems trailing, 1-3
dm. high; flowers white or
pinkish) [— 15.](#)
- 15a. Leaves broadest above the middle,
green **Prince's Pine,
Chimaphila
umbellata.
Spotted**
- 15b. Leaves broadest below the middle,
spotted with white **Wintergreen,
Chimaphila
maculata.**

- | | |
|---|---|
| 16a. Leaves 2-5 cm. long, densely woolly beneath (5-10 dm. high; flowers white) | Labrador Tea,
Ledum groenlandicum. |
| 16b. Leaves 1-2 cm. long, pale beneath but not wholly (creeping; flowers pink) (Cranberry) — 17. | |
| 17a. Leaves acute | Cranberry,
Vaccinium oxycoccos. |
| 17b. Leaves obtuse | Cranberry,
Vaccinium macrocarpon. |
| 18a. Leaves opposite or whorled; corolla saucer-shape (shrubs 3-8 dm. high; flowers purple, summer) — 19. | |
| 18b. Leaves alternate; corolla bell-shape or salver-form — 20. | |
| 19a. Branches and twigs cylindrical, not angled | Sheep Laurel,
Kalmia angustifolia. |
| 19b. Branches and twigs with 2 sharp angles | Swamp Laurel,
Kalmia polifolia. |
| 20a. Plants prostrate, or with a few ascending branches only (flowers white or pink) — 21. | |
| 20b. Plants erect or ascending — 23. | |
| 21a. Flowers 10-20 mm. long, very fragrant (early spring) | Trailing Arbutus,
Epigaea repens. |
| 21b. Flowers 4-5 mm. long (late spring) — 22. | |

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|--|--|
| 22a. Leaves spatulate, broadest beyond the middle | Bearberry,
Arctostaphylos
uva-ursi. |
| 22b. Leaves oval, broadest at the middle | Snowberry,
Chiogenes
hispidula. |
| 23a. Leaves linear, white beneath, their margins strongly revolute (shrub 3-8 dm. high; flowers white, late spring) | Bog Rosemary,
Andromeda
glaucophylla. |
| 23b. Leaves oblong, scurfy beneath with rusty scales (bog shrub 4-10 dm. high; flowers white, in spring) | Leatherleaf,
Chamaedaphne
calyculata. |
| 23c. Leaves smooth, pubescent, or resinous beneath, but not scurfy nor white
<u>— 24.</u> | |
| 24a. Low shrubs 10-15 cm. high, erect from a creeping rootstock; leaves with the taste of wintergreen (flowers white or pink, summer) | Wintergreen,
Gaultheria
procumbens. |
| 24b. Bushy shrubs 3-8 dm. high; leaves dotted beneath with yellowish resinous dots; ovary 10-celled (flowers greenish-pink, spring) | Huckleberry,
Gaylussacia
baccata. |
| 24c. Shrubs 1 dm. to 3 m. high; leaves not resinous-dotted beneath; ovary 5-celled (flowers white or greenish-pink, spring or early summer) <u>— 25.</u> | |
| 25a. Corolla bell-shape, the stamens projecting beyond it (5-15 dm. high) | Deerberry,
Vaccinium
stamineum. |

- 25b. Corolla cylindrical or urn-shape, the
stamens not projecting — [26.](#)
- 26a. Filaments hairy (Blueberry)
— [27.](#)
- 26b. Filaments glabrous (Bilberry)
— [32.](#)
- 27a. Low bushy shrubs, usually less than 5
dm. and never more than 1 m. high
— [28.](#)
- 27b. Tall erect shrubs, 1-4 m. high — [31.](#)
- 28a. Foliage pubescent
- 28b. Foliage glabrous — [29.](#)
- 29a. Leaves pale-green and glaucous, entire
or nearly so
- 29b. Leaves bright-green, distinctly serrulate
— [30.](#)
- 30a. Fruit blue
- 30b. Fruit black
- 31a. Leaves downy beneath; fruit black
- 31b. Leaves smooth or minutely pubescent
beneath; fruit blue
- 32a. Full-grown leaves less than
2.5 cm. long; low much-
branched shrubs, mostly
less than 5 dm. high — [33.](#)

**Blueberry,
Vaccinium
canadense.**

**Blueberry,
Vaccinium
vacillans.**

**Blueberry,
Vaccinium
pennsylvanicum.**

**Blueberry,
Vaccinium
pennsylvanicum
var. nigrum.**

**Blueberry,
Vaccinium
atrococcum.**

**Blueberry,
Vaccinium
corymbosum.**

- 32b. Full-grown leaves more than 2.5 cm. long; shrubs usually a meter high or more — [34.](#)
- 33a. Leaves entire; petals usually 4. **Bilberry, Vaccinium uliginosum.**
- 33b. Leaves finely serrulate; petals 5 **Bilberry, Vaccinium caespitosum.**
- 34a. Leaves serrulate, green beneath, acute; corolla globular **Bilberry, Vaccinium membranaceum.**
- 34b. Leaves entire, pale beneath, obtuse; corolla ovoid **Bilberry, Vaccinium ovalifolium.**

PRIMULACEAE, the Primrose Family

Herbs, with alternate or opposite simple leaves and regular flowers; petals more or less united; stamens attached one in front of each petal; ovary 1-celled with 1 style.

- 1a. Leaves all basal; flowers on leafless stalks — [2.](#)
- 1b. Stem-leaves present — [5.](#)
- 2a. Flowers nodding, the petals reflexed (3-6 dm. high; flowers showy, white or pink, in spring) **Shooting Star, Dodecatheon meadia.**
- 2b. Flowers erect or spreading; petals not reflexed — [3.](#)
- 3a. Corolla not longer than the calyx; flowers small and inconspicuous (1 dm. high; **Androsace, Androsace**

- flowers white or pink, spring) **occidentalis.**
- 3b. Corolla conspicuous, much longer than the calyx (flowers pink or purple, summer) (Primrose) —4.
- 4a. Leaves white-mealy beneath (1-4 dm. high) **Primrose, Primula farinosa.**
- 4b. Leaves green beneath (2 dm. high or less) **Primrose, Primula mistassinica.**
- 5a. All the stem-leaves in one whorl just below the flower-cluster —6.
- 5b. Stem-leaves several or many, scattered over the stem —7.
- 6a. Stem-leaves about 1 cm. long — 3a.
- 6b. Stem-leaves 5-10 cm. long **Star Flower, Trientalis americana.**
- 7a. Flowers red, blue, or white (summer) —8.
- 7b. Flowers yellow (summer) —9.
- 8a. Leaves opposite; flowers axillary (stems spreading, 1-4 dm. long; flowers blue or red) **Pimpernel, Anagallis arvensis.**
- 8b. Leaves alternate; flowers racemose (1-4 dm. high; flowers minute, white) **Water Pimpernel, Samolus floribundus.**
- 9a. Stem creeping **Moneywort, Lysimachia nummularia.**
- 9b. Stem erect (Loosestrife) —10.
- 10a. Flowers in dense spike-like racemes (3-8 dm. high) **Loosestrife, Lysimachia thysiflora.**
- 10b. Flowers axillary or racemose (3-9 dm. high) —11.

11a. Corolla dotted or streaked with purple or brown; leaves punctate with dark spots — [12](#).

11b. Corolla plain yellow; leaves not dark-dotted — [14](#).

12a. Flowers in racemes — [13](#).

12b. Flowers all axillary

13a. Flowers all in racemes; leaves opposite or some of them alternate

13b. The lowest flowers axillary; leaves opposite or whorled

14a. Leaves ovate, on slender ciliate petioles

14b. Leaves lanceolate, sessile or short-petioled, pinnately veined

14c. Leaves linear, with one mid-vein

**Loosestrife,
Lysimachia
quadrifolia.**

**Loosestrife,
Lysimachia
terrestris.**

**Loosestrife,
Lysimachia
producta.**

**Loosestrife,
Steironema
ciliatum.**

**Loosestrife,
Steironema
lanceolatum.**

**Loosestrife,
Steironema
quadriflorum.**

OLEACEAE, the Olive Family

Trees or shrubs, with opposite leaves and regular flowers; sepals 4, or calyx none; petals 4, united, or none; stamens usually 2; ovary 2-celled, superior.

1a. Leaves simple (shrub 2-5 m. high; flowers blue or white, in showy clusters in spring)

**Lilac, Syringa
vulgaris.**

- 1b. Leaves compound (trees; flowers greenish, inconspicuous, in spring) (Ash) —2.
 - 2a. Lateral leaflets sessile **Black Ash, Fraxinus nigra.**
 - 2b. Lateral leaflets stalked —3.
- 3a. Twigs sharply 4-angled **Blue Ash, Fraxinus quadrangulata.**
- 3b. Twigs not distinctly angled —4.
 - 4a. Leaves pubescent beneath **Red Ash, Fraxinus pennsylvanica.**
 - 4b. Leaves glabrous beneath —5.
- 5a. Leaves pale-green beneath, obscurely serrulate **White Ash, Fraxinus americana.**
- 5b. Leaves bright-green beneath, sharply serrulate **Green Ash, Fraxinus pennsylvanica var. lanceolata.**

GENTIANACEAE, the Gentian Family

Herbs, with opposite or basal, entire, usually simple leaves and regular flowers; sepals, petals, and stamens equal in number, 4-12; ovary superior, 1-celled.

- 1a. Leaves reduced to small scales (1-4 dm. high; flowers small, greenish-yellow, in summer) **Bartonia, Bartonia virginica.**
- 1b. Leaves rounded, floating (flowers white, summer) **Floating Heart, Nymphoides lacunosum.**
- 1c. Leaves compound (2-4 dm. high; flowers white or bluish, early summer) **Buckbean, Menyanthes**

- 1d. Leaves simple, whorled (1-2 m. high; flowers yellowish-white, summer) **trifoliata.**
American Columbo, Frasera caroliniensis.
- 1e. Leaves simple, opposite —2.
- 2a. Corolla rotate, with spreading lobes, 2-4 cm. broad, pink (5-8 dm. high; summer) **Rose Pink, Sabbatia angularis.**
- 2b. Corolla bell-shape, each petal with a spur at the base, purplish or white, and not over 1 cm. long (1-4 dm. high; summer) **Spurred Gentian, Halenia deflexa.**
- 2c. Corolla bell-shape, tubular, funnel-form, or salver-form, not spurred —3.
- 3a. Corolla-lobes fringed (flowers bright-blue) (Fringed Gentian) —4.
- 3b. Corolla-lobes entire —5.
- 4a. Leaves lanceolate (2-8 dm. high; autumn) **Fringed Gentian, Gentiana crinita.**
- 4b. Leaves linear (1-4 dm. high; late summer) **Fringed Gentian, Gentiana procera.**
- 5a. Corolla 2 cm. long or a little less —6.
- 5b. Corolla 2.5-5 cm. long (late summer and autumn) (Gentian) —7.
- 6a. Upper leaves linear or narrowly lanceolate (2-4 dm. high; flowers pink-purple, late summer) **Centaury, Centaurium umbellatum.**
- 6b. Upper leaves ovate, with several principal veins (1-5 dm. high; flowers blue, late summer and autumn) **Gentian, Gentiana quinquefolia.**
- 7a. Calyx-lobes rough or ciliate at the margin (flowers blue, or rarely white) —8.

7b. Calyx-lobes smooth (2-8 dm. high)

— 10.

8a. Corolla-lobes spreading; leaves
narrowly lanceolate,
indistinctly veined (2-5 dm.
high)

**Gentian, Gentiana
puberula.**

8b. Corolla-lobes erect or
incurved; leaves ovate to
ovate-lanceolate, with 3-7
principal veins (3-6 dm.
high) — 9.

9a. Calyx-lobes equaling or exceeding the
calyx-tube

**Gentian, Gentiana
saponaria.**

9b. Calyx-lobes shorter than the calyx-tube

**Gentian, Gentiana
andrewsii.**

10a. Leaves ovate or ovate-
lanceolate, somewhat
cordate at base (flowers
greenish-white or
yellowish-white)

**Gentian, Gentiana
flavida.**

10b. Leaves narrowly lanceolate or
nearly linear, not cordate
(flowers blue)

**Gentian, Gentiana
linearis.**

APOCYNACEAE, the Dogbane Family

Herbs, with opposite simple entire leaves and regular flowers; sepals, petals, and stamens each 5; petals united; stamens attached to the corolla; ovaries 2, with a single style or stigma.

1a. Plant creeping or trailing; flowers blue,
axillary, 2-3 cm. broad (spring)

**Periwinkle, Vinca
minor.**

1b. Plant erect or essentially so; flowers 1
cm. broad or less (4-12 dm. high)

—2.

2a. Corolla pinkish, about 8 mm.
long by 6-8 mm. broad
(early summer)

**Dogbane,
Apocynum
androsaemifolium.**

2b. Corolla white or greenish,
about 6 mm. long by 4 mm.
broad (summer) —3.

3a. Leaves petioled, acute at the base

**Indian Hemp,
Apocynum
cannabinum.**

3b. Leaves sessile, rounded or truncate at the
base

**Indian Hemp,
Apocynum
cannabinum var.
hypericifolium.**

ASCLEPIADACEAE, the Milkweed Family

Herbs, with simple entire leaves and regular flowers; juice usually milky; except in the first species, which is a twining vine. The flowers have an unusual structure: calyx of 5 sepals; petals 5, united with each other, and spreading or reflexed so that they conceal the calyx; stamens 5, united with each other and with the stigma to form a complex organ in the center of the flower; ovaries 2; on the back of each stamen is a colored projecting hood, which is frequently the most conspicuous part of the flower, and may be mistaken for the corolla.

1a. Twining vine, with dark purple flowers
(summer)

**Black Swallow-
wort, Cynanchum
nigrum.**

1b. Stems not twining —2.

2a. Leaves whorled (3-6 dm. high;
summer) (Milkweed) —3.

2b. Leaves opposite or alternate
—4.

- | | |
|---|--|
| 3a. Leaves in whorls of 4, lanceolate
(flowers pink) | Milkweed,
Asclepias
quadrifolia. |
| 3b. Leaves in whorls of 4-7, linear (flowers
greenish-white) | Milkweed,
Asclepias
verticillata. |
| 4a. Umbel sessile (4-8 dm. high;
flowers green, summer) | Green Milkweed,
Acerates
viridiflora. |
| 4b. Umbel peduncled <u>—5.</u> | |
| 5a. Leaves linear or narrowly linear-
lanceolate (4-8 dm. high; flowers
greenish-white, summer) | Green Milkweed,
Acerates
floridana. |
| 5b. Leaves lanceolate or broader (flowers in
summer) <u>—6.</u> | |
| 6a. Leaves pubescent beneath <u>—7.</u> | |
| 6b. Leaves glabrous or nearly so
<u>—10.</u> | |
| 7a. Flowers brilliant orange (3-6 dm. high) | Butterfly Weed,
Asclepias
tuberosa. |
| 7b. Flowers red or purple <u>—8.</u> | |
| 8a. Reflexed lobes of corolla
merely purple-tinged (1-2
m. high) | Milkweed,
Asclepias syriaca. |
| 8b. Reflexed lobes of corolla
bright-red or purple <u>—9.</u> | |
| 9a. The erect hoods of each flower about 5
mm. long (7-12 dm. high) | Milkweed,
Asclepias
purpurascens. |
| 9b. The erect hoods of each flower about 3
mm. long (6-10 dm. high) | Swamp Milkweed,
Asclepias
incarnata var.
pulchra. |
| 10a. Leaves broadly rounded and
almost sessile at base | |

- (flowers purplish) — 11.
- 10b. Leaves narrowed at the base,
distinctly petioled (8-15
dm. high) — 12.
- 11a. Umbel solitary, terminal and erect on a
long peduncle (4-8 dm. high) **Milkweed,
Asclepias
amplexicaulis.**
- 11b. Umbels terminal or lateral, bent toward
one side (7-12 dm. high) **Milkweed,
Asclepias
sullivantii.**
- 12a. Corolla (not hoods) red (1-2
m. high) **Swamp Milkweed,
Asclepias
incarnata.**
- 12b. Corolla (not hoods) greenish
(8-15 dm. high) **Milkweed,
Asclepias exaltata.**

CONVOLVULACEAE, the Morning Glory Family

Twining or trailing herbs (except one species), with regular flowers; sepals 5; corolla 5-angled or 5-lobed; stamens 5, attached to the corolla; ovary superior, 2-3-celled.

- 1a. Plants with green foliage and
conspicuous flowers (summer) — 2.
- 1b. Leafless brown or yellow plants, with
very small flowers (Dodder) — 7.
- 2a. Style divided at the top into
linear or oblong stigmas
(flowers white or pink)
(Bindweed) — 3.
- 2b. Style not divided at the top;
stigmas sessile, capitate
(Morning Glory) — 6.

- | | |
|---|---|
| 3a. Stem erect; leaves rounded or somewhat cordate at base, not hastate or sagittate (1-3 dm. high) | Bindweed,
Convolvulus
spithamaeus. |
| 3b. Stem trailing or twining; leaves sagittate or hastate <u>—4.</u> | |
| 4a. Calyx almost concealed by two large heart-shape bracts <u>—5.</u> | |
| 4b. Bracts at base of calyx none | Bindweed,
Convolvulus
arvensis. |
| 5a. Leaves triangular-hastate, with sharp basal lobes | Bindweed,
Convolvulus
sepium. |
| 5b. Leaves oblong-ovate, the basal lobes obtuse | Bindweed,
Convolvulus
sepium var.
pubescens. |
| 6a. Stem smooth or nearly so; ovary 2-celled (flowers white) | Wild Potato Vine,
Ipomoea
pandurata. |
| 6b. Stem with reflexed hairs; ovary 3-celled (flowers of various colors) | Morning Glory,
Ipomoea
purpurea. |
| 7a. Introduced weed, growing as a parasite on clover | Dodder, Cuscuta
epithymum. |
| 7b. Native species, on various shrubs and herbs <u>—8.</u> | |
| 8a. Flowers sessile <u>—9.</u> | |
| 8b. Flowers distinctly pedicelled <u>—12.</u> | |
| 9a. Sepals united below into a gamosepalous calyx <u>—10.</u> | |
| 9b. Sepals separate from each other <u>—11.</u> | |
| 10a. Calyx-lobes obtuse | Dodder, Cuscuta
arvensis. |

10b. Calyx-lobes acute	Dodder, Cuscuta obtusiflora.
11a. Flowers in dense rope-like twists on various species of herbs	Dodder, Cuscuta paradoxa.
11b. Flowers in dense clusters on various species of shrubs	Dodder, Cuscuta compacta.
12a. Tips of the petals inflexed	Dodder, Cuscuta coryli.
12b. Tips of the petals erect or spreading —13.	
13a. Capsule depressed at the summit	Dodder, Cuscuta cephalanthi.
13b. Capsule pointed at the summit	Dodder, Cuscuta gronovii.

POLEMONIACEAE, the Polemonium Family

Herbs with alternate or opposite leaves and conspicuous regular flowers; sepals 5, united; petals 5, united and bearing the 5 stamens in the corolla-tube; ovary superior, 3-celled.

1a. Leaves pinnately compound and alternate (2-4 dm. high; flowers blue, in spring)	Greek Valerian, Polemonium reptans.
1b. Leaves fascicled, narrowly linear (about 1 dm. high; flowers pink-purple, in spring)	Moss Pink, Phlox subulata.
1c. Leaves simple, strictly opposite —2.	
2a. Corolla-lobes deeply 2-cleft to the middle (1-2 dm. high; flowers pink, in spring)	Cleft Phlox, Phlox bifida.
2b. Corolla-lobes entire and rounded, or somewhat notched at the apex —3.	

- | | |
|---|---|
| 3a. Flowers in summer (8-15 dm. high; flowers purple) | Garden Phlox,
Phlox paniculata. |
| 3b. Flowers in spring (3-6 dm. high) <u>—4.</u> | |
| 4a. Corolla blue-purple; stems ascending | Sweet William,
Phlox divaricata. |
| 4b. Corolla pink or red-purple; stems erect | Sweet William,
Phlox pilosa. |

HYDROPHYLLACEAE, the Water-leaf Family

Herbs with alternate lobed or divided leaves and regular flowers; sepals 5; petals 5, united; stamens 5, attached to the corolla-tube and projecting beyond it; ovary 1-celled.

- | | |
|---|---|
| 1a. Leaves palmately veined and lobed (4-8 dm. high; flowers purple, early summer) | Water-leaf,
Hydrophyllum canadense. |
| 1b. Leaves pinnately veined and lobed (2-6 dm. high; flowers blue or purple, varying to white) <u>—2.</u> | |
| 2a. Corolla-lobes much shorter than the corolla-tube (summer) | Phacelia, Phacelia franklinii. |
| 2b. Corolla-lobes much longer than the corolla-tube (late spring and summer) (Water-leaf) <u>—3.</u> | |
| 3a. Calyx with a small reflexed appendage between each pair of sepals | Water-leaf,
Hydrophyllum appendiculatum. |
| 3b. Calyx without appendages | Water-leaf,
Hydrophyllum virginicum. |

BORAGINACEAE, the Borage Family

Herbs with alternate entire leaves; sepals 5; petals 5, united, corolla generally regular; stamens 5, attached to the corolla-tube; ovary deeply 4-lobed with a single style.

1a. Corolla reddish-purple, about 8 mm.
wide (4-10 dm. high; spring)

**Hound's Tongue,
Cynoglossum
officinale.**

1b. Corolla blue with a yellow center, 4-8
mm. wide (1-5 dm. high; spring and
early summer) (Forget-me-not)
—13.

1c. Corolla deep orange, salver-form (2-6
dm. high; spring) (Puccoon) —15.

1d. Corolla white or blue, or lightly tinged
with yellow or red —2.

2a. Corolla rotate, with a very
short tube, bright-blue,
about 20 mm. broad (3-8
dm. high; summer)

**Borage, Borago
officinalis.**

2b. Corolla tubular, funnel-form,
or salver-form —3.

3a. Corolla 10 mm. long or more; its tube
distinctly longer than the calyx —4.

3b. Corolla less than 10 mm. long; its tube
equaling or shorter than the calyx
—8.

4a. Flowers yellowish-white, or
somewhat tinged with pink
or greenish —5.

4b. Flowers blue or purple —6.

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| 5a. Corolla-lobes erect; leaves sessile (3-8 dm. high; early summer) | False Gromwell;
Onosmodium
occidentale. |
| 5b. Corolla-lobes spreading; leaves decurrent (6-10 dm. high; summer) | Comfrey,
Symphytum
officinale. |
| 6a. Stem and leaves glabrous (3-6 dm. high; spring) | Bluebell,
Mertensia
virginica. |
| 6b. Stem and leaves pubescent (4-8 dm. high) <u>7.</u> | |
| 7a. Corolla regular; leaves ovate or ovate-lanceolate (summer) | Bluebell,
Mertensia
paniculata. |
| 7b. Corolla irregular; leaves linear-oblong (summer) | Blueweed, Echium
vulgare. |
| 8a. Ovary and fruit covered with hooked prickles <u>9.</u> | |
| 8b. Ovary and fruit not prickly <u>12.</u> | |
| 9a. Principal leaves 2.5 cm. wide or more <u>10.</u> | |
| 9b. Principal leaves 2 cm. wide or less (3-8 dm. high; flowers blue or white, summer) (Stickseed) <u>11.</u> | |
| 10a. Leaves chiefly basal, the racemes on long leafless peduncles (4-8 dm. high; flowers pale blue, early summer) | Wild Comfrey,
Cynoglossum
boreale. |
| 10b. Stems leafy (8-12 dm. high; flowers white, summer) | Beggar Lice,
Lappula
virginiana. |
| 11a. A bract at the base of each flower | Stickseed, Lappula
echinata. |

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|--|---|
| 11b. Racemes without bracts at the base of each flower | Stickseed, Lappula deflexa var. americana. |
| 12a. Racemes bractless, or bracted only at the base (1-4 dm. high) — 14. | |
| 12b. Raceme with a bract at the base of each flower (flowers white or yellowish) — 17. | |
| 13a. Corolla 4 mm. wide | Forget-me-not, Myosotis laxa. |
| 13b. Corolla 6-8 mm. wide | Forget-me-not, Myosotis scorpioides. |
| 14a. Calyx-lobes all of equal length (summer) | Scorpion Grass, Myosotis arvensis. |
| 14b. Calyx-lobes distinctly unequal in length (spring) | Scorpion Grass, Myosotis virginica. |
| 15a. Corolla-lobes denticulate | Puccoon, Lithospermum angustifolium. |
| 15b. Corolla-lobes entire — 16. | |
| 16a. Flowers sessile; stem softly pubescent | Puccoon, Lithospermum canescens. |
| 16b. Flowers on pedicels 2-5 mm. long; stem hispid or bristly | Puccoon, Lithospermum gmelini. |
| 17a. Corolla white; fruit brown and wrinkled (weed 2-4 dm. high; spring and summer) | Corn Gromwell, Lithospermum arvense. |
| 17b. Corolla yellowish-white; fruit white and smooth (5-10 dm. high) — 18. | |
| 18a. Corolla distinctly surpassing the calyx in length (spring | Corn Gromwell, Lithospermum |

- and summer)
- 18b. Corolla equaling or shorter
than the calyx (spring)

officinale.
Wild Gromwell,
Lithospermum
latifolium.

LABIATAE, the Mint Family

Herbs with opposite leaves, square stems, and usually aromatic odor; flowers irregular, with united petals, or almost regular; stamens 2 or 4, attached to the tube of the corolla; ovary deeply 4-lobed, with a single style.

- 1a. Stamens 2 —2.
- 1b. Stamens 4 —15.
- 2a. Corolla regular or nearly so;
flowers white, in dense
axillary clusters; plants
usually of wet grounds (2-
8 dm. high; summer and
autumn) —3.
- 2b. Corolla distinctly irregular and
more or less 2-lipped —7.
- 3a. Calyx-teeth short, triangular, acute or
obtuse (Bugle Weed) —4.
- 3b. Calyx-teeth narrow, acuminate or
cuspidate (Water Hoarhound) —5.
- 4a. Stems and stolons bearing
tubers
- 4b. Stems and stolons not bearing
tubers
- 5a. Leaves serrate; calyx-teeth sharp-
pointed —6.
- 5b. Leaves coarsely incised; calyx-teeth
awn-tipped

Bugle Weed,
Lycopus uniflorus.
Bugle Weed,
Lycopus virginicus.

Water Hoarhound,
Lycopus
americanus.

- 6a. Corolla twice as long as the calyx; leaves narrowed at the base **Water Hoarhound, *Lycopus rubellus*.**
- 6b. Corolla barely longer than the calyx; leaves sessile or nearly so **Water Hoarhound, *Lycopus lucidus* var. *americanus*.**
- 7a. Corolla blue, 3-4 mm. long; flowers in loose axillary clusters (1-4 dm. high; summer) (Pennyroyal) —8.
- 7b. Corolla 8-40 mm. long —9.
 - 8a. Leaves serrate **Pennyroyal, *Hedeoma pulegioides*.**
 - 8b. Leaves linear, entire **Pennyroyal, *Hedeoma hispida*.**
 - 9a. Lower lobe of the corolla fringed, very much longer than the upper (5-15 dm. high; corolla pale-yellow; late summer) **Horse Balm, *Collinsonia canadensis*.**
 - 9b. Lower lobe of the corolla nearly or quite as long as the upper and not fringed —10.
 - 10a. Calyx narrowly tubular; its teeth about equal in size (5-10 dm. high; flowers in dense terminal heads, in summer) —11.
 - 10b. Calyx campanulate, 2 of its teeth different in size from the other 3 (4-8 dm. high; flowers pink-purple, in terminal clusters in summer) —14.
 - 11a. Corolla scarlet **Oswego Tea, *Monarda didyma*.**

- 11b. Corolla bright crimson or rose-red **Wild Bergamot,
Monarda fistulosa
var. rubra.**
- 11c. Corolla white, pink, pale-purple, or
yellowish — [12.](#)
- 12a. Flower-clusters all terminal
— [13.](#)
- 12b. Flower-clusters both terminal
and axillary **Horse Mint,
Monarda punctata.**
- 13a. Leaves and stem with soft spreading
pubescence **Wild Bergamot,
Monarda fistulosa.**
- 13b. Leaves and stem grayish with fine
appressed pubescence **Wild Bergamot,
Monarda mollis.**
- 14a. Upper calyx-teeth about 3
times as long as the lower **Blephilia, Blephilia
hirsuta.**
- 14b. Upper calyx-teeth but little
longer than the lower **Blephilia, Blephilia
ciliata.**
- 15a. Calyx with a distinct protuberance on
the back of the upper side (Skullcap)
— [16.](#)
- 15b. Calyx without a distinct protuberance
— [20.](#)
- 16a. Corolla 5-8 mm. long;
flowers in axillary racemes **Mad-dog Skullcap,
Scutellaria
lateriflora.**
(3-8 dm. high; flowers
blue, in summer)
- 16b. Corolla 6-10 mm. long;
flowers axillary, solitary **Skullcap,
Scutellaria parvula.**
(1-3 dm. high; flowers
violet, early summer)
- 16c. Corolla 12-30 mm. long;
flowers axillary or in
terminal racemes (4-8 dm.
high; flowers blue,
summer) — [17.](#)

- | | |
|---|--|
| 17a. Stem-leaves cordate | Skullcap,
Scutellaria
versicolor. |
| 17b. Stem-leaves not distinctly cordate
<u>— 18.</u> | |
| 18a. Stem-leaves sessile or nearly
so; plant of swamps and
river-banks | Skullcap,
Scutellaria
galericulata. |
| 18b. Stem-leaves with petioles 1
cm. or more long; plants
of dry or moist woods
<u>— 19.</u> | |
| 19a. Stem glandular-pubescent toward the
summit; corolla 16 mm. long or less | Skullcap,
Scutellaria pilosa. |
| 19b. Stem not glandular; corolla 20 mm.
long | Skullcap,
Scutellaria incana. |
| 20a. Calyx-teeth 5, all equal or
nearly so at the time of
flowering <u>— 28.</u> | |
| 20b. Calyx-teeth 5, one of them
different in size and shape
from the other four (2-6
dm. high; flowers light
blue, summer) | Dragon Head,
Dracocephalum
parviflorum. |
| 20c. Calyx-teeth 5, two of them
different in size and shape
from the other three <u>— 21.</u> | |
| 20d. Calyx-teeth 10, subulate
(woolly plant 4-10 dm.
high, with whitish flowers
in axillary clusters in
summer) | Hoarhound,
Marrubium
vulgare. |
| 21a. Corolla deeply split on the upper side
and the stamens protruding; upper
lip of the calyx much shorter than
the lower (5-10 dm. high; flowers | |

- pink-purple, in terminal spikes,
summer) (Wood Sage) [—22.](#)
- 21b. Corolla not deeply split on the upper
side [—23.](#)
- 22a. Calyx canescent
- 22b. Calyx villous
- 23a. Flowers in dense terminal head-like
spikes, none axillary (1-5 dm. high;
flowers pink-purple or blue, in
summer)
- 23b. Some or all of the flowers in axillary
clusters [—24.](#)
- 24a. Leaves linear, entire (1-4 dm.
high; flowers purple,
summer) [—25.](#)
- 24b. Leaves oblong to ovate
(summer) [—26.](#)
- 25a. Pedicels shorter than the calyx
- 25b. Pedicels much longer than the calyx
- 26a. Leaves 1 cm. long or less,
entire (stems growing in
mats, 1-3 dm. long;
flowers purple, in
summer)
- 26b. Leaves dentate (flowers
purple, summer) [—27.](#)
- 27a. Flowers subtended by bracts as long as
the calyx (2-5 dm. high)
- 27b. Flowers with minute bracts or none (1-
3 dm. high)
- Wood Sage,
Teucrium
canadense.**
- Wood Sage,
Teucrium
occidentale.**
- Self-heal, Prunella
vulgaris.**
- Summer Savory,
Satureja hortensis.**
- Calamint, Satureja
glabra.**
- Wild Thyme,
Thymus serpyllum.**
- Basil, Satureja
vulgaris.**
- Basil-thyme,
Satureja acinos.**

- 28a. Corolla 2-lipped or nearly regular, the upper lip flattened, not conspicuously arched over the stamens — [29.](#)
- 28b. Corolla conspicuously 2-lipped, the stamens ascending under the concave upper lip — [42.](#)
- 29a. Flowers in dense terminal spikes; corolla 2-lipped (8-15 dm. high; summer) — [30.](#)
- 29b. Flowers peduncled, 1-4 in the axils of linear leaves — 25b.
- 29c. Flowers in many-flowered whorls, which are axillary or terminal, or aggregated into terminal spikes or racemes — [31.](#)
- 30a. Corolla yellowish
- 30b. Corolla purplish
- 31a. Corolla distinctly irregular, the lower lip longer than the upper — [32.](#)
- 31b. Corolla almost regular, the lobes nearly uniform in size — [35.](#)
- 32a. Stem-leaves sessile or very nearly so (flowers blue) — [33.](#)
- 32b. Stem-leaves long-petioled — [34.](#)
- 33a. Leaves linear-oblong, acute at both ends (3-8 dm. high; summer)

**Giant Hyssop,
Agastache
nepetoides.**

**Giant Hyssop,
Agastache
scrophulariaefolius.**

**Hyssop, Hyssopus
officinalis.**

- 33b. Leaves oblong to ovate, rounded at the ends (2-4 dm. high; late spring) **Bugle, Ajuga reptans.**
- 34a. Leaves ovate to oblong, acute; flowers pink, white, or pale purple (6-15 dm. high; summer) **Catnip, Nepeta cataria.**
- 34b. Leaves nearly circular or kidney-shape; flowers blue (creeping; flowers in spring and summer) **Ground Ivy, Nepeta hederacea.**
- 35a. Flowers in terminal spikes, or the lower axillary (3-8 dm. high; flowers pink-purple or white, summer) (Mint) [— 36.](#)
- 35b. Flowers all in axillary whorls (flowers pink-purple or white, summer) (Mint) [— 38.](#)
- 35c. Flowers in terminal capitate corymbed clusters (4-8 dm. high; flowers white or dotted with purple, summer) (Mountain Mint) [— 41.](#)
- 36a. Leaves sessile or with very short petiole **Spearmint, Mentha spicata.**
- 36b. Leaves with manifest petioles [— 37.](#)
- 37a. Principal leaves less than half as broad as long **Peppermint, Mentha piperita.**
- 37b. Principal leaves more than half as broad as long **Bergamot Mint, Mentha citrata.**
- 38a. Stem glabrous; leaves ovate to obovate (4-8 dm. high) **Downy Mint, Mentha gentilis.**
- 38b. Stem pubescent, at least on the angles (1-6 dm. high) [— 39.](#)
- 39a. Principal leaves distinctly petioled and somewhat rounded at base **Wild Mint, Mentha arvensis.**

39b. Leaves tapering to the base — [40.](#)

40a. Leaves and stem pubescent

Wild Mint, *Mentha arvensis* var. *canadensis*.

40b. Leaves glabrous; stem pubescent on the angles only

Wild Mint, *Mentha arvensis* var. *glabrata*.

41a. Leaves linear; calyx-teeth awl-shape

Mountain Mint, *Pycnanthemum flexuosum*.

41b. Leaves narrowly lanceolate; calyx-teeth triangular-ovate

Mountain Mint, *Pycnanthemum virginianum*.

42a. Stems decumbent to diffuse; leaves cordate to nearly circular (stems 2-5 dm. long or high; flowers in spring and summer) (Dead Nettle) — [43.](#)

42b. Stem erect; leaves palmately cleft; calyx-teeth spiny (6-15 dm. tall; flowers pink, in summer)

Motherwort, *Leonurus cardiaca*.

42c. Stems erect or ascending; leaves ovate-lanceolate to linear (summer) — [45.](#)

43a. Upper leaves closely sessile (flowers red-purple)

Dead Nettle, *Lamium amplexicaule*.

43b. Leaves all petioled — [44.](#)

44a. Flowers red or purple

Dead Nettle, *Lamium maculatum*.

44b. Flowers white

Dead Nettle, *Lamium album*.

- 45a. Flowers 2-2.5 cm. long, in loose terminal spikes (5-10 dm. tall; flowers rose-color)
- 45b. Flowers 1-2 cm. long, in axillary and terminal spiked whorls [— 46.](#)
- 46a. Calyx-teeth spiny pointed (flowers pink or pale-purple) (Hemp Nettle) [— 47.](#)
- 46b. Calyx-teeth acute to awl-shape, but not spiny (3-10 dm. high; flowers pale-purple) (Hedge Nettle) [— 48.](#)
- 47a. Leaves ovate (3-8 dm. high)
- 47b. Leaves linear to lanceolate (1-4 dm. high)
- 48a. Leaves glabrous [— 49.](#)
- 48b. Leaves distinctly pubescent [— 50.](#)
- 49a. Leaves ovate-lanceolate, serrate
- 49b. Leaves linear-oblong, entire or nearly so
- 50a. Stem pubescent on the angles alone; leaves petioled
- 50b. Stem pubescent on both sides and angles; leaves nearly sessile [— 51.](#)
- 51a. Leaves oblong or oblong-lanceolate, more than 1 cm. wide
- 51b. Leaves linear-lanceolate, 1 cm. wide or less
- False Dragon Head, Physostegia virginiana.**
- Hemp Nettle, Galeopsis tetrahit.**
- Hemp Nettle, Galeopsis ladanum.**
- Hedge Nettle, Stachys tenuifolia.**
- Hedge Nettle, Stachys hyssopifolia.**
- Hedge Nettle, Stachys tenuifolia var. aspera.**
- Hedge Nettle, Stachys palustris.**
- Hedge Nettle, Stachys arenicola.**

VERBENACEAE, the Verbena Family

Herbs, with simple opposite leaves and slightly irregular flowers in spikes or heads; petals 5, united and bearing the 4 stamens in the corolla-tube; ovary 1, 2-celled or 4-celled, with 1 style.

1a. Plants prostrate or spreading —2.

1b. Plants erect (flowers in summer)

(Vervain) —3.

2a. Leaves serrate; flowers in short dense spikes (flowers pale-blue, summer)

Fog Fruit, Lippia lanceolata.

2b. Leaves pinnatifid; flowers in loose bracted spikes (flowers light-purple, summer)

Vervain, Verbena bracteosa.

3a. Spikes dense, continuous (flowers purple or blue, varying to white) —4.

3b. Spikes slender, interrupted, the flowers scattered (corolla white or pale-blue) —6.

4a. Leaves lanceolate, manifestly petioled (1-2 m. high)

Vervain, Verbena hastata.

4b. Leaves sessile, not lanceolate (5-8 dm. high) —5.

5a. Leaves narrowly oblanceolate, tapering at the entire base

Vervain, Verbena angustifolia.

5b. Leaves oblong to obovate, not tapering at the base

Vervain, Verbena stricta.

6a. Leaves incised, tapering to a sessile base (5-10 dm. high)

Vervain, Verbena, officinalis.

- 6b. Leaves serrate, petioled (1-2 m. high) **Vervain, Verbena
urticaefolia.**

SOLANACEAE, the Nightshade Family

Herbs or shrubs, with alternate leaves and regular or slightly irregular flowers; sepals 5, united; corolla of 5 united petals, bearing the 5 stamens attached; ovary 1, 2-5 (usually 2)-celled, with a slender style.

- 1a. Corolla rotate; anthers close together
(flowers in summer) —2.
- 1b. Corolla not rotate; anthers separate —5.
 - 2a. Stem and leaves prickly (3-8
dm. high) —3.
 - 2b. Stem and leaves not prickly
—4.
- 3a. Flowers white or bluish
- 3b. Flowers yellow
 - 4a. Climbing vine; leaves
frequently lobed (flowers
blue)
 - 4b. Not climbing; leaves toothed
(flowers white)
- 5a. Climbing or trailing shrub, with purplish,
white, or greenish flowers about 1
cm. wide (frequently thorny; flowers
in summer)
- 5b. Herbaceous plants, not climbing —6.
 - 6a. Flowers white, red, or blue, 2.5
cm. or more wide

**Horse Nettle,
Solanum
carolinense.
Buffalo Bur,
Solanum
rostratum.
Bittersweet,
Solanum
dulcamara.
Nightshade,
Solanum nigrum.
Matrimony Vine,
Lycium
halimifolium.**

- (summer) —7.
- 6b. Flowers yellow, yellowish-white, or greenish-yellow (summer) —12.
- 7a. Corolla-tube 10 cm. long or more (5-12 dm. high) —8.
- 7b. Corolla-tube 5 cm. long or less —10.
- 8a. Stem finely pubescent; leaves entire or nearly so **Thorn-apple, Datura metel.**
- 8b. Stem glabrous; leaves coarsely toothed (Jimson Weed) —9.
- 9a. Stem green; corolla white **Jimson Weed, Datura stramonium.**
- 9b. Stem purple; corolla light-blue or purple **Jimson Weed, Datura tatula.**
- 10a. Corolla pale-blue (5-10 dm. high) **Apple of Peru, Nicandra physalodes.**
- 10b. Corolla, red or violet (2-4 dm. high) **Petunia, Petunia violacea.**
- 10c. Corolla white —11.
- 11a. Corolla all white (2-4 dm. high) **Petunia, Petunia axillaris.**
- 11b. Corolla with yellow center **White Ground Cherry, Physalis grandiflora.**
- 12a. Corolla 30 mm. wide or more, somewhat irregular; stamens declined to one side (3-6 dm. high) **Henbane, Hyoscyamus niger.**
- 12b. Corolla smaller, strictly regular —13.
- 13a. Flowers in terminal panicles; corolla tubular, with slightly spreading lobes **Wild Tobacco, Nicotiana rustica.**

(5-10 dm. high)

13b. Flowers solitary in the axils; corolla short, widely spreading (3-8 dm. high) (Ground Cherry) — [14.](#)

14a. Annuals with branching slender roots — [15.](#)

14b. Perennials with thickened roots and rootstocks — [16.](#)

15a. Plants pubescent

**Ground Cherry,
Physalis
pubescens.**

15b. Plants smooth, or with a few scattered hairs

**Ground Cherry,
Physalis ixocarpa.**

16a. Stem viscid-pubescent

**Ground Cherry,
Physalis
heterophylla.**

16b. Stem glabrous or slightly pubescent, not viscid — [17.](#)

17a. Leaves and stem distinctly pubescent

**Ground Cherry,
Physalis
virginiana.**

17b. Leaves and stem almost glabrous

**Ground Cherry,
Physalis
subglabrata.**

SCROPHULARIACEAE, the Figwort Family

Herbs with opposite or alternate leaves and usually irregular flowers; corolla of united petals, bearing the 2 or 4 (or rarely 5) stamens attached; petals actually 5, but sometimes apparently only 2 or 4; a sterile fifth stamen sometimes present; ovary superior, 2-celled.

- 1a. Anther-bearing stamens 5 (6-15 dm.
high; flowers in summer) (Mullein)

—2.

- 1b. Anther-bearing stamens 4; a sterile fifth
stamen may or may not be present

—3.

- 1c. Anther-bearing stamens 2 —37.

- 2a. Leaves densely white-woolly;
flowers yellow, in dense
spikes

**Mullein,
Verbascum
thapsus.**

- 2b. Leaves smooth or nearly so;
flowers yellow or white, in
loose racemes

**Moth Mullein,
Verbascum
blattaria.**

- 3a. Flowers (not the bracts) greenish-yellow,
yellow, or orange —4.

- 3b. Flowers blue, purple, brown, red, pink,
or white, never yellow —16.

- 4a. Flowers in dense terminal
leafy-bracted spikes —5.

- 4b. Flowers in loose racemes or
axillary —9.

- 5a. Corolla 7 mm. long or less, or none —6.

- 5b. Corolla 12 mm. long or more —7.

- 6a. Leaves alternate (3-6 dm. high;
early summer)

**Synthyris,
Synthyris bullii.**

- | | |
|---|--|
| 6b. Leaves opposite (1-2 dm. high; summer) | Eyebright,
Euphrasia arctica. |
| 7a. Stem-leaves entire (2-6 dm. high; summer) | Painted Cup,
Castilleja pallida
var.
septentrionalis. |
| 7b. Stem-leaves palmately lobed, bracteal leaves scarlet (3-6 dm. high; early summer) | Painted Cup,
Castilleja coccinea. |
| 7c. Stem-leaves pinnately lobed or incised (Lousewort) — 8. | |
| 8a. Flowers in spring (2-4 dm. high) | Lousewort,
Pedicularis canadensis. |
| 8b. Flowers in late summer (3-8 dm. high) | Lousewort,
Pedicularis lanceolata. |
| 9a. Upper lip of the corolla very different in size and shape from the lower lip — 10. | |
| 9b. Upper lip of the corolla resembling the lower lip in shape, and not very different in size (5-12 dm. high; summer) (False Foxglove) — 13. | |
| 10a. Leaves alternate (2-5 dm. high; summer) | Butter-and-eggs,
Linaria vulgaris. |
| 10b. Leaves opposite — 11. | |
| 11a. Stem erect; leaves narrowed at the base — 30b. | |
| 11b. Stem creeping or spreading (summer) — 12. | |
| 12a. Leaves pinnately veined, ovate | Musk Flower,
Mimulus moschatus. |
| 12b. Leaves palmately veined, circular or nearly so | Yellow Monkey Flower, Mimulus |

- glabratus var.
jamesii.**
- 13a. Stem glabrous — [14.](#)
- 13b. Stem pubescent — [15.](#)
- 14a. Principal stem-leaves
pinnatifid
- 14b. Principal stem-leaves entire
- 15a. Corolla hairy on the outside
- 15b. Corolla smooth on the outside
- 16a. Leaves all basal; flowers on
leafless stalks (1 dm. high
or less; flowers pink or
white, summer)
- 16b. Leaves opposite (those
subtending the flowers may
be alternate) — [20.](#)
- 16c. Leaves alternate or irregularly
scattered — [17.](#)
- 17a. Leaves entire — [18.](#)
- 17b. Leaves pinnately lobed or incised — 8a.
- 17c. Leaves palmately veined and lobed;
stem trailing (flowers blue, summer)
- 18a. Corolla-tube less than 1 cm.
long, spurred — [19.](#)
- 18b. Corolla more than 2 cm. long,
not spurred (3-8 dm. high;
flowers red-purple,
summer)
- False Foxglove,
Gerardia
virginica.**
- False Foxglove,
Gerardia
laevigata.**
- False Foxglove,
Gerardia
pedicularia.**
- False Foxglove,
Gerardia flava.**
- Mudwort,
Limosella aquatica
var. tenuifolia.**
- Kenilworth Ivy,
Linaria
cymbalaria.**
- Snapdragon,
Antirrhinum
majus.**

- | | |
|--|---|
| 19a. Stem and foliage pubescent (1-3 dm. high; flowers blue, summer) | Small Snapdragon, <i>Linaria minor</i>. |
| 19b. Stem and foliage glabrous (2-6 dm. high; flowers blue, summer) | Toad-flax, <i>Linaria canadensis</i>. |
| 20a. Leaves with 1 or 2 lobes near the base (3-5 dm. high; flowers purple, summer) | <i>Gerardia</i>, <i>Gerardia auriculata</i>. |
| 20b. Leaves linear (2-6 dm. high; flowers rose-purple, summer and autumn) (<i>Gerardia</i>) — 21. | |
| 20c. Leaves lanceolate or broader, not lobed — 26. | |
| 21a. Pedicels equaling or but little longer than the calyx, and conspicuously shorter than the subtending leaf — 22. | |
| 21b. Pedicels much longer than the calyx, and generally equaling or exceeding the subtending leaf — 24. | |
| 22a. Plants of moist ground, bogs, and shores — 23. | |
| 22b. Plants of dry uplands | <i>Gerardia</i>, <i>Gerardia aspera</i>. |
| 23a. Corolla about 25 mm. long | <i>Gerardia purpurea</i>. |
| 23b. Corolla less than 20 mm. long | <i>Gerardia paupercula</i>. |
| 24a. Stem rough on the angles — 25. | |
| 24b. Stem glabrous | <i>Gerardia tenuifolia</i>. |

25a. Leaves 2-5 mm. wide

**Gerardia,
Gerardia
tenuifolia var.
macrophylla.**

25b. Leaves thread-like, 1 mm. wide or less

**Gerardia,
Gerardia
skinneriana.**

26a. Corolla 16 mm. long, or
shorter [—27.](#)

26b. Corolla 20 mm. long, or
longer [—32.](#)

27a. Corolla dull-purple, brown, or greenish;
one sterile stamen present (1-2.5 m.
high; flowers in summer) (Figwort)
[—28.](#)

27b. Corolla blue or white (1-4 dm. high)
[—29.](#)

28a. Sterile stamen purple

**Figwort,
Scrophularia
marilandica.**

28b. Sterile stamen yellow

**Figwort,
Scrophularia
leporella.**

29a. Flowers nearly or quite sessile
(summer) [—30.](#)

29b. Flowers on pedicels 10 mm. long or
more (spring) [—31.](#)

30a. Foliage-leaves prominently
toothed — 6b.

30b. Foliage-leaves entire, or with
1 or 2 small teeth at the
base

**Cow Wheat,
Melampyrum
lineare.**

31a. Corolla more than 10 mm. long, blue
and white

**Blue-eyed Mary,
Collinsia verna.**

31b. Corolla 5-8 mm. long, blue and white

**Collinsia, Collinsia
parviflora.**

- 32a. Flowers solitary in the axils of the upper foliage-leaves (4-8 dm. high; flowers blue, in summer) (Monkey Flower) — 33.
- 32b. Flowers in dense terminal or subterminal spikes (3-9 dm. high; summer) — 34.
- 32c. Flowers in loose terminal panicles (flowers white or pale-violet) (Beard-tongue) — 35.
- 33a. Leaves clasping at the base **Monkey Flower, Mimulus ringens.**
- 33b. Leaves petioled, not clasping **Monkey Flower, Mimulus alatus.**
- 34a. Stem and foliage glabrous (flowers white) **Turtlehead, Chelone glabra.**
- 34b. Stem and foliage pubescent (flowers purple) **Blue Hearts, Buchnera americana.**
- 35a. Stem finely pubescent (3-6 dm. high; flowers pale-violet, late spring) **Beard-tongue, Pentstemon hirsutus.**
- 35b. Stem glabrous below the inflorescence (6-12 dm. high) — 36.
- 36a. Corolla-tube gradually enlarged from base to tip (flowers pale-violet, in early summer) **Beard-tongue, Pentstemon laevigatus.**
- 36b. Corolla-tube abruptly enlarged just beyond the calyx (flowers white, early summer) **Beard-tongue, Pentstemon laevigatus var. digitalis.**
- 37a. Corolla distinctly irregular, 2-lipped (1-4 dm. high; flowers yellowish or

- white, summer) — 38.
- 37b. Corolla regular or nearly so and 2-lobed, or none — 6a.
- 37c. Corolla regular or nearly so, 4-lobed — 40.
- 38a. Leaves narrowed at the base, with mid-vein **Hedge Hyssop, Gratiola virginiana.**
- 38b. Leaves rounded or somewhat clasping at the base, with 3-5 principal veins (False Pimpernel) — 39.
- 39a. Peduncles longer than the subtending leaves **False Pimpernel, Ilysanthes anagallidea.**
- 39b. Peduncles shorter than the subtending leaves **False Pimpernel, Ilysanthes dubia.**
- 40a. Leaves whorled (8-20 dm. high; flowers white or pale-blue, in spikes, summer) **Culver's Root, Veronica virginica.**
- 40b. Leaves alternate or opposite (Speedwell) — 41.
- 41a. Flowers in racemes, which arise from the axils of the opposite leaves (flowers pale-blue to nearly white, late spring and summer) — 42.
- 41b. Flowers solitary in the axils of leaf-like bracts, or in terminal bracted racemes (1-4 dm. high; spring and summer) — 46.
- 42a. Stem and foliage glabrous; swamp plants 2-7 dm. high — 43.
- 42b. Stem and foliage pubescent; plants of dry ground, 1-3 dm. high — 45.

- | | |
|---|--|
| 43a. Leaves linear or narrowly lanceolate | Marsh Speedwell,
Veronica
scutellata. |
| 43b. Leaves ovate or ovate-lanceolate <u>— 44.</u> | |
| 44a. Stem-leaves sessile and
somewhat clasping | Water Speedwell,
Veronica anagallis-
aquatica. |
| 44b. Stem-leaves on short petioles | Brooklime,
Veronica
americana.
Speedwell,
Veronica
officinalis.
Speedwell,
Veronica
chamaedrys. |
| 45a. Leaves narrowed at base into a petiole | |
| 45b. Leaves rounded or heart-shape at the
base | |
| 46a. Bracteal leaves entire; stem
glabrous or minutely
pubescent <u>— 47.</u> | |
| 46b. All leaves serrate; foliage
pubescent (flowers blue)
<u>— 48.</u> | |
| 47a. Flowers white, about 2 mm. wide | Speedwell,
Veronica
peregrina. |
| 47b. Flowers pale-blue with darker stripes,
3-4 mm. wide | Speedwell,
Veronica
serpyllifolia. |
| 48a. Flowers nearly sessile, about
2 mm. wide | Speedwell,
Veronica arvensis. |
| 48b. Flowers on slender pedicels,
5-8 mm. wide | Speedwell,
Veronica
tournefortii. |

LENTIBULARIACEAE, the Bladderwort Family

Small herbs, growing on rocks, in mud, or in water; calyx and corolla both 2-lipped; stamens 2, attached to the corolla; ovary 1-celled.

1a. Corolla purple —2.

1b. Corolla yellow (flowers in summer)

(Bladderwort) —4.

2a. Leaves oval to elliptical, entire
(about 1 dm. high, on
rocks; flowers in summer)

**Butterwort,
Pinguicula
vulgaris.**

2b. Leaves dissected or none,
submerged (flowers in
summer) (Bladderwort)
—3.

3a. Flower-stalk with a single bract near the
middle

**Bladderwort,
Utricularia
resupinata.**

3b. Flower-stalk without bracts, except at the
base of the pedicels

**Bladderwort,
Utricularia
purpurea.**

4a. Stem and numerous dissected
leaves floating in water

**Bladderwort,
Utricularia
vulgaris var.
americana.**

4b. Stem and minute leaves
creeping on the bottom of
ponds or in mud, while the
flowers are borne on erect
stalks, easily detached from
the delicate stems —5.

5a. Upper lip of corolla conspicuous, as long
or nearly as long as the lower lip;
lower lip with a prominent raised
palate —6.

- | | |
|---|---|
| 5b. Upper lip of corolla half as long as the lower lip, or less <u>7.</u> | |
| 6a. Spur of corolla very short and blunt | Bladderwort,
Utricularia gibba. |
| 6b. Spur of corolla very long and slender | Bladderwort,
Utricularia cornuta. |
| 7a. Spur of corolla very short and blunt, or almost none | Bladderwort,
Utricularia minor. |
| 7b. Spur of corolla long and slender | Bladderwort,
Utricularia intermedia. |

OROBANCHACEAE, the Broom-rape Family

Parasitic plants without green color and with scales in place of leaves; corolla 2-lipped, of united petals; stamens 4, attached to the corolla.

- | | |
|---|--|
| 1a. Flowers in a widely branching panicle, numerous; growing under beech trees (1-5 dm. high; flowers white and purple, late summer) | Beech Drops,
Epifagus virginiana. |
| 1b. Flowers sessile in a dense bracted spike (1-2 dm. high; flowers pale-yellow, early summer) | Squaw-root,
Conopholis americana. |
| 1c. Flowers 1-15, each on a long erect naked peduncle (1-2 dm. high; flowers yellowish or pale-violet, spring and summer) (Cancer-root) <u>2.</u> | |
| 2a. Stem erect and scaly, 5-10 cm. high | Cancer-root,
Orobanche fasciculata. |
| 2b. Stem very short, almost below the surface of the ground, | Cancer-root,
Orobanche |

with long erect peduncles

uniflora.

ACANTHACEAE, the Acanthus Family

Herbs with opposite simple leaves; corolla of united petals, 2-lipped or almost regular; stamens 2 or 4, attached to the corolla; ovary 2-celled.

- | | |
|--|---|
| 1a. Corolla about 10 mm. long; flowers in dense heads (4-10 dm. high; flowers blue or white, summer) | Water Willow,
Dianthera
americana. |
| 1b. Corolla about 30 mm. long; flowers axillary (3-8 dm. high; flowers blue, in summer) (Ruellia) <u>—2.</u> | |
| 2a. Foliage glabrous or slightly pubescent | Ruellia, Ruellia
strepens. |
| 2b. Foliage densely hirsute | Ruellia, Ruellia
ciliosa. |

PHRYMACEAE, the Lopseed Family

Herb with opposite leaves and irregular flowers in long slender spikes; petals united, corolla 2-lipped; stamens 4, attached to the corolla; ovary 1-celled.

- | | |
|--|--|
| One species only, 5-10 dm. high; flowers purple, in summer | Lopseed, Phryma
leptostachya. |
|--|--|

PLANTAGINACEAE, the Plantain Family

Herbs with basal leaves and small white flowers in spikes; sepals 4; petals 4, united; stamens 4; ovary 2-celled.

1a. Leaves linear (1-4 dm. high; summer)

—2.

1b. Leaves broader, lanceolate to broadly ovate or cordate (summer) —3.

2a. Spikes mixed with bracts
several times longer than
the flowers

**Buckhorn,
Plantago aristata.**

2b. Bracts about as long as the
flowers

**Plantain, Plantago
purshii.**

3a. Leaves cordate, pinnately veined; plant
of wet ground and marshes (4-8 dm.
tall)

**Plantain, Plantago
cordata.**

3b. Leaves with 3 to many longitudinal ribs
or veins —4.

4a. Leaves densely pubescent with
grayish hairs —5.

4b. Leaves smooth or slightly
pubescent —6.

5a. Flower-stalks 3-6 dm. high

**Plantain, Plantago
media.**

5b. Flower-stalks less than 3 dm. high

**Plantain, Plantago
virginica.**

6a. Flower-stalks 3-6 dm. high;
spikes not over 10 cm. long

**English Plantain,
Plantago
lanceolata.**

6b. Scapes 1-4 dm. high; spikes
long and slender, usually
equaling or longer than the
peduncle; dooryard
plantains —7.

7a. Leaves green at the base

**Plantain, Plantago
major.**

7b. Leaves reddish at the base

Plantain, *Plantago rugelii*.

RUBIACEAE, the Madder family

Herbs or shrubs, with opposite or whorled leaves and regular flowers; sepals 4, or minute or almost wanting; petals 4, united; stamens 4; ovary inferior.

1a. Shrub (1-3 m. tall; flowers white, in spherical heads, summer)

Button Bush, *Cephalanthus occidentalis*.

1b. Herbaceous —2.

2a. Leaves opposite —3.

2b. Leaves whorled; flowers white, green, or purple (Bedstraw) —6.

2c. Leaves whorled; flowers yellow

Bedstraw, *Galium verum*.

3a. Leaves about as long as wide (trailing; flowers paired, white, in spring)

Partridge Berry, *Mitchella repens*.

3b. Leaves at least twice as long as wide —4.

4a. Corolla salver-form, about 1 cm. wide; peduncles 1-flowered (about 1 dm. high; flowers blue or white, in spring)

Bluets, *Houstonia coerulea*.

4b. Corolla funnel-form, about 5 mm. wide; flowers in clusters (1-2 dm. high; flowers white or pale-purple, summer) (*Houstonia*) —5.

5a. Basal leaves strongly ciliate	Houstonia, Houstonia ciliolata.
5b. Basal leaves smooth	Houstonia, Houstonia longifolia.
6a. Leaves in whorls of 4-7.	
6b. Leaves in whorls of 6-8 <u>—16.</u>	
7a. Ovary and fruit hispid with hooked bristles (3-7 dm. high; summer) <u>—8.</u>	
7b. Ovary and fruit not bristly (early summer) <u>—11.</u>	
8a. Leaves with 1 principal vein (flowers dull purple)	Bedstraw, Galium pilosum.
8b. Leaves with 3 principal veins <u>—9.</u>	
9a. Flowers bright-white	Bedstraw, Galium boreale.
9b. Flowers greenish, yellowish, or purplish <u>—10.</u>	
10a. Leaves acuminate	Bedstraw, Galium lanceolatum.
10b. Leaves acute or obtuse	Bedstraw, Galium circaezans.
11a. Corolla-lobes 3 (2-6 dm. high; flowers white or greenish) <u>—12.</u>	
11b. Corolla-lobes 4 <u>—13.</u>	
12a. Flowers in clusters of 2 or 3	Bedstraw, Galium claytoni.
12b. Flowers solitary in the axils, on long hair-like pedicels	Bedstraw, Galium trifidum.
13a. Corolla brownish or purple (3-6 dm. high)	Bedstraw, Galium latifolium.
13b. Corolla white (1-4 dm. high) <u>—14.</u>	

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|---|--|
| 14a. Flowers rather numerous in small cymes | Bedstraw, <i>Galium palustre</i>. |
| 14b. Flowers in clusters of 2 or 3, or solitary — 15 . | |
| 15a. Principal leaves spreading or ascending | Bedstraw, <i>Galium tinctorium</i>. |
| 15b. Principal leaves recurved or reflexed | Bedstraw, <i>Galium labradoricum</i>. |
| 16a. Ovary and fruit bristly or hispid — 17 . | |
| 16b. Ovary and fruit not bristly (summer) — 18 . | |
| 17a. Leaves narrowly lanceolate to linear, mostly 6-8 in a whorl (stem 5-15 dm. long; flowers white; spring and summer) | Bedstraw, <i>Galium aparine</i>. |
| 17b. Leaves narrowly oval or elliptical, mostly in whorls of 6; flowers in clusters of 3 (1-5 dm. high; flowers greenish, summer) | Bedstraw, <i>Galium triflorum</i>. |
| 18a. Leaves cuspidate or mucronate at the apex (flowers white) — 19 . | |
| 18b. Leaves obtuse at the apex (flowers white or greenish; 2-6 dm. high) — 20 . | |
| 19a. Flowers very numerous in terminal panicles (stem 3-8 dm. long) | Bedstraw, <i>Galium mollugo</i>. |
| 19b. Flowers in axillary clusters (1-3 dm. high) | Bedstraw, <i>Galium tricornis</i>. |
| 19c. Flowers few, in small loose terminal cymes — 20 . | |
| 20a. Stem smooth or nearly so (2-4 dm. high) | Bedstraw, <i>Galium concinnum</i>. |
| 20b. Stem hispid with reflexed bristles (5-15 dm. long) | Bedstraw, <i>Galium asprellum</i>. |

CAPRIFOLIACEAE, the Honeysuckle Family

Shrubs or herbs, with opposite leaves; corolla regular or irregular, petals 4 or 5, united; stamens 4 or 5; ovary inferior, 1-5-celled.

- 1a. Leaves compound (shrubs 1-4 m. high; flowers white, in large clusters in early summer) (Elder) [—2.](#)
- 1b. Leaves simple [—3.](#)
 - 2a. Pith of the twigs white; inflorescence flattened or convex **Elder, *Sambucus canadensis*.**
 - 2b. Pith of the twigs brown; inflorescence pyramidal **Elder, *Sambucus racemosa*.**
- 3a. Plant trailing; flowers nodding, in pairs (1 dm. high; flowers pink, summer) **Twin Flower, *Linnaea borealis* var. *americana*.**
- 3b. Erect herbs (6-12 dm. high; flowers dull-red, axillary, early summer) (Feverwort) [—4.](#)
- 3c. Shrubs, small trees, or woody vines [—5.](#)
 - 4a. Leaf-bases broadly connate and 2-5 cm. wide **Feverwort, *Triosteum perfoliatum*.**
 - 4b. Leaf-bases narrowly connate, not over 1 cm. wide **Feverwort, *Triosteum aurantiacum*.**
- 5a. Climbing vines (spring and early summer) (Honeysuckle) [—6.](#)
- 5b. Erect or spreading shrubs or small trees [—11.](#)
 - 6a. Flowers in 2-flowered axillary clusters (flowers white or **Honeysuckle, *Lonicera japonica*.**

- pink)
- 6b. Flowers in terminal clusters
—[7.](#)
- 7a. Leaves distinctly pubescent beneath
(flowers yellow) —[8.](#)
- 7b. Leaves glabrous beneath, or very
minutely puberulent —[9.](#)
- 8a. Leaves pubescent above
- 8b. Leaves glabrous above
- 9a. Corolla purple on the outside, glabrous
within
- 9b. Corolla yellow on the outside (or slightly
tinged with purple), pubescent within
—[10.](#)
- 10a. Corolla-tube 6-8 mm. long
- 10b. Corolla-tube 11-14 mm. long
- 11a. Corolla tubular at base; style long and
slender —[12.](#)
- 11b. Corolla rotate or somewhat bell-shape,
style very short (flowers white, late
spring or early summer) —[22.](#)
- 12a. Flowers yellow or yellowish
(spring and early summer)
—[13.](#)
- 12b. Flowers white, pink, or red
—[18.](#)
- 13a. Leaves serrate (5-10 dm. tall)
- Honeysuckle,
Lonicera hirsuta.**
- Honeysuckle,
Lonicera
glaucescens.**
- Honeysuckle,
Lonicera
caprifolium.**
- Honeysuckle,
Lonicera dioica.**
- Honeysuckle,
Lonicera
sullivantii.**
- Bush Honeysuckle,
Diervilla lonicera.**

- 13b. Leaves entire; flowers in pairs
(Honeysuckle) — [14.](#)
- 14a. Each pair of flowers
subtended by 2 broad leaf-
like bracts (1-3 m. high) **Honeysuckle,
Lonicera
involucrata.**
- 14b. Bracts at the base of each pair
of flowers linear or
narrowly lanceolate — [15.](#)
- 15a. Native species of woods and bogs
— [16.](#)
- 15b. Introduced species, growing mostly
near dwellings; leaves very pubescent
beneath **Honeysuckle,
Lonicera
xylosteum.**
- 16a. Peduncles 15 mm. long or
more (1-4 m. high) — [17.](#)
- 16b. Peduncles about 5 mm. long
(1 m. high, or less) **Honeysuckle,
Lonicera coerulea
var. villosa.
Honeysuckle,
Lonicera
canadensis.
Honeysuckle,
Lonicera
oblongifolia.
Honeysuckle,
Lonicera tatarica.**
- 17a. Leaves ciliate
- 17b. Leaves not ciliate
- 18a. Corolla irregular, over 1 cm.
long (1-4 m. high; spring)
- 18b. Corolla regular, less than 1
cm. long (5-15 dm. high;
flowers white or pink, in
axillary clusters, early
summer) — [19.](#)
- 19a. Flowers in axillary spikes **Wolfberry,
Symphoricarpos
occidentalis.**
- 19b. Flowers almost sessile in the axils
— [20.](#)

20a. Flowers numerous in each axil	Indian Currant, Symphoricarpos orbiculatus.
20b. Flowers 1 or 2 in each axil (Snowberry) —21.	
21a. Leaves green beneath	Snowberry, Symphoricarpos racemosus.
21b. Leaves whitened beneath	Snowberry, Symphoricarpos racemosus var. pauciflorus.
22a. Leaves palmately lobed —23.	
22b. Leaves not lobed —25.	
23a. Outermost flowers of the cluster enlarged and imperfect (1-4 m. high)	Cranberry Tree, Viburnum opulus var. americanum.
23b. All flowers of the cluster alike —24.	
24a. Flower-clusters 4-10 cm. broad (1-2 m. high)	Arrow Wood, Viburnum acerifolium.
24b. Flower-clusters 2-3 cm. broad	Squashberry, Viburnum pauciflorum.
25a. Outer flowers of the cluster enlarged and imperfect (1-3 m. high)	Hobble-bush, Viburnum alnifolium.
25b. All flowers of the cluster alike —26.	
26a. Leaves finely serrate; the veins not prominent —27.	
26b. Leaves coarsely serrate, all or most of the teeth terminating in a prominent vein —29.	

- | | |
|--|---|
| 27a. Peduncle of the flower-cluster, below its branches, at least 2 cm. long (1-3 m. high) | Withe-rod,
Viburnum
cassinoides. |
| 27b. Peduncle of the cluster 1 cm. long, or even shorter (3-8 m. high) — 28. | |
| 28a. Leaves distinctly acuminate | Sheep-berry,
Viburnum lentago. |
| 28b. Leaves obtuse or barely acute | Black Haw,
Viburnum
prunifolium. |
| 29a. Leaves densely pubescent beneath (6-15 dm. high) | Arrow-wood,
Viburnum
pubescens. |
| 29b. Leaves glabrous beneath, or with tufts of hairs in the forks of the veins (1-4 m. high) | Arrow-wood,
Viburnum
dentatum. |

VALERIANACEAE, the Valerian Family

Herbs with opposite leaves and small nearly or quite regular flowers; petals 5, united; stamens 3; sepals minute or wanting; ovary inferior.

- | | |
|---|---|
| 1a. Stem-leaves pinnately cleft (3-10 dm. high; flowers white or pinkish, summer) — 2. | |
| 1b. Stem-leaves entire or dentate (2-6 dm. high; flowers white, summer) (Corn Salad) — 3. | |
| 2a. Leaf-segments parallel-veined
Valerian, Valeriana edulis. | |
| 2b. Leaf-segments net-veined | Swamp Valerian,
Valeriana
uliginosa. |

3a. Upper stem-leaves entire

**Corn Salad,
Valerianella
chenopodifolia.**

3b. Upper stem-leaves dentate

**Corn Salad,
Valerianella
radiata.**

DIPSACACEAE, the Teasel Family

Herbs with opposite leaves, and small pale blue flowers aggregated in dense heads; calyx minute; petals 4, united; stamens 4, attached to the corolla; ovary inferior.

One species in Michigan, 1-2 m. high, with
prickly leaves and stem, blooming in
summer

**Teasel, Dipsacus
sylvestris.**

CUCURBITACEAE, the Gourd Family

Herbs, climbing by tendrils, with alternate palmately lobed leaves and imperfect flowers; staminate flowers in showy clusters, with 5-6 petals and 3 stamens; pistillate flowers small.

1a. Leaves 5-angled or shallowly 5-lobed
(flowers white, summer)

**Bur Cucumber,
Sicyos angulatus.**

1b. Leaves 5-lobed to about the middle
(commonly cultivated and frequently
wild; flowers white, summer)

**Wild Cucumber,
Echinocystis
lobata.**

CAMPANULACEAE, the Bellflower Family

Herbs with alternate simple leaves and milky juice; sepals 5; petals 5, united; stamens 5, attached at the very base of the corolla; ovary inferior.

- | | |
|---|--|
| 1a. Stem-leaves circular or nearly so, cordate-clasping at base (2-6 dm. high; flowers blue, axillary, in summer) | Venus' Looking-Glass, <i>Specularia perfoliata</i>. |
| 1b. Stem-leaves linear or nearly so, not over 1 cm. wide (summer) <u>2.</u> | |
| 1c. Stem-leaves ovate to lanceolate, 2 cm. wide or more (flowers blue, in a terminal spike or raceme, summer) (Bellflower) <u>4.</u> | |
| 2a. Stem and leaves glabrous (or rarely pubescent) (1-6 dm. high; flowers blue) | Harebell, <i>Campanula rotundifolia</i>. |
| 2b. Stem and leaves rough with reflexed bristles (marsh plants, with weak slender stems 3-10 dm. long; flowers white or pale-blue) (Marsh Bellflower) <u>3.</u> | |
| 3a. Corolla 5-8 mm. long | Marsh Bellflower, <i>Campanula aparinoides</i>. |
| 3b. Corolla 10-12 mm. long | Marsh Bellflower, <i>Campanula uliginosa</i>. |
| 4a. Corolla rotate; flowers in spikes (6-15 dm. high) | Bellflower, <i>Campanula americana</i>. |
| 4b. Corolla bell-shape; flowers in one-sided racemes (4-10 dm. high) | Bellflower, <i>Campanula rapunculoides</i>. |

LOBELIACEAE, the Lobelia Family

Herbs with alternate simple leaves and milky juice; flowers irregular; petals 5, united; corolla split down the upper side; stamens 5, united by their anthers into a ring or tube surrounding the style; ovary 2-celled, inferior. Flowers in summer and autumn.

- | | |
|--|---|
| 1a. Leaves all basal, tubular; flowers on leafless stalks (aquatic, 1-4 dm. high; flowers blue) | Water Lobelia,
Lobelia dortmanna. |
| 1b. Leaves normal, on the stem <u>—2.</u> | |
| 2a. Flowers more than 2 cm. long (5-10 dm. high) <u>—3.</u> | |
| 2b. Flowers about 1 cm. long, or shorter (flowers light blue) <u>—4.</u> | |
| 3a. Flowers scarlet | Cardinal Flower,
Lobelia cardinalis. |
| 3b. Flowers blue | Great Lobelia,
Lobelia siphilitica. |
| 4a. Flowers in loose racemes, pedicelled <u>—5.</u> | |
| 4b. Flowers in slender terminal spike-like racemes, nearly sessile (4-10 dm. high, usually unbranched) | Lobelia, Lobelia
spicata. |
| 5a. Foliage pubescent (3-8 dm. high) | Indian Tobacco,
Lobelia inflata. |
| 5b. Foliage glabrous (1-4 dm. high) | Lobelia, Lobelia
kalmii. |

COMPOSITAE, the Composite Family

Herbs, with various types of foliage, but with flowers of characteristic structure, resembling a sunflower, a thistle, or a dandelion. Each apparent flower is a head of numerous small flowers, attached side by side to the expanded end of the stem, and subtended and partly enclosed by a series of bracts, called the involucre, which resembles a calyx.

The calyx of the individual flower is minute or actually wanting, and is usually modified to aid in seed dispersal. It appears at the base of the corolla, at the summit of the inferior ovary, and is known as pappus. The structure of the pappus is best observed in the ripe fruit.

The corolla of the individual flowers consists of 5 (or rarely 4) united petals. In some flowers the petals are united to form a tubular or bell-shape corolla. In others they are united to form a flat or strap-shape corolla. The stamens are attached to the corolla, and are united by their anthers into a tube which surrounds the style, and above which the 2-lobed stigma protrudes.

The apparent flower of a Composite, composed of several or many individual flowers, is termed a head. It may be composed entirely of tubular flowers, as the thistle or bone-set; or entirely of strap-shape flowers, as the dandelion; or of both sorts together, as the aster or sunflower. In the latter case, the tubular flowers invariably occupy the center of the head, called the disk, and the larger strap-shape flowers are at the margin, where their projecting corollas, called rays, may be very conspicuous. Such heads are called radiate.

In a few composites (see 1a below) the flowers have minute corollas without colored parts.

In identifying a composite, determine first whether the heads are composed of tubular flowers, of strap-shape flowers, or of both sorts together; and, secondly, observe the nature of the pappus, using preferably the ripe heads, or at least the oldest flower-heads available. No further difficulties will be encountered.

- 1a. Flowers without petal-like or brightly colored parts; staminate and pistillate flowers in separate heads (or rarely in

the same heads); coarse weeds with inconspicuous flowers (summer and autumn) — 2.

1b. Flowers with some petal-like parts, usually brightly colored or white — 10.

2a. Leaves toothed or lobed — 3.

2b. Leaves deeply pinnatifid or dissected (4-15 dm. high; flowers in erect spikes) (Ragweed) — 7.

3a. Fruit or pistillate flowers thickly covered with sharp hooked spines (3-10 dm. high) (Cocklebur) — 4.

3b. Fruit not spiny — 8.

4a. With spines on the stem at the base of the leaves

4b. Without any spines on the stem — 5.

5a. Body of the bur smooth or slightly hairy

5b. Body of the bur and the spines densely pubescent — 6.

6a. Body of the bur more than twice as long as thick; a common weed

6b. Body of the bur less than twice as long as thick

7a. Leaves twice-pinnatifid

7b. Leaves once-pinnatifid

**Cocklebur,
Xanthium
spinosum.**

**Cocklebur,
Xanthium
canadense.**

**Cocklebur,
Xanthium
commune.**

**Cocklebur,
Xanthium
echinatum.**

**Ragweed,
Ambrosia
artemisiifolia.**

**Ragweed,
Ambrosia**

- 8a. Leaves deeply 3-lobed (1-5 m. high)
 8b. Leaves serrate or obscurely lobed [— 9.](#)
- 9a. Stem simple or sparingly branched; pistillate heads in the axils of the upper leaves (1-3 m. high)
 9b. Stem much branched; heads all alike, in panicles
- 10a. Flowers all strap-shape; juicy milky. (The central flowers must be examined carefully, since they are frequently much smaller than the marginal ones)
[— 11.](#)
- 10b. Flowers all tubular, with regular. 4-5-lobed corollas
[— 45.](#)
- 10c. Flowers both tubular and strap-shape; heads radiate (in a few species the rays are small and may be overlooked by mistake)
[— 108.](#)
- 11a. Flowers blue (summer and autumn)
[— 12.](#)
- 11b. Flowers orange, yellow, white, or purplish [— 15.](#)
- 12a. Heads 2.5 cm. wide, or larger
[— 13.](#)
- 12b. Heads 1.5 cm. wide, or smaller [— 14.](#)
- 13a. Leaves linear; bracts longer than the flowers, heads 5-10 cm. wide (6-15

psilostachya.
Giant Ragweed,
Ambrosia trifida.

Giant Ragweed,
Ambrosia trifida
var. integrifolia.
Marsh Elder, Iva
xanthifolia.

Salsify,
Tragopogon

- dm. high)
- 13b. Leaves broader, mostly serrate; bracts shorter than the flowers; heads 2.5-4 cm. wide
- 14a. Heads in a narrow crowded cluster (5-15 dm. high)
— 30b.
- 14b. Heads in a spreading open panicle (Wild Lettuce)
[— 22.](#)
- 15a. Heads solitary at the summit of leafless stalks [— 16.](#)
- 15b. Heads several, on leafy, naked, or scaly stalks [— 19.](#)
- 16a. Basal leaves strictly entire; heads about 2.5 cm. wide (summer and autumn)
— 35a.
- 16b. Basal leaves toothed, lobed, or pinnatifid (spring and summer) [— 17.](#)
- 17a. Heads 8-14 mm. wide (1-4 dm. high)
- 17b. Heads 25-50 mm. wide (1-6 dm. high) (Dandelion) [— 18.](#)
- 18a. Outer involucral bracts reflexed
- 18b. Outer involucral bracts erect or spreading
- 19a. Pappus none; heads about 1 cm. wide (4-10 dm. high; summer)
- 19b. Pappus of an inner row of bristles and an outer row of short scales; heads
- porrifolius.**
Chicory,
Cichorium
intybus.
- Dwarf Dandelion,**
Krigia virginica.
- Dandelion,**
Taraxacum
officinale.
Dandelion,
Taraxacum
erythrospermum.
Nipplewort,
Lapsana
communis.
Cynthia, Krigia
amplexicaulis.

- about 3 cm. wide (3-8 dm. high; early summer)
- 19c. Pappus of feathery bristles (summer) — 20.
- 19d. Pappus of simple bristles — 25.
- 20a. Flower-stalk scaly, without foliage leaves (2-6 dm. high)
- 20b. Stem leafy (3-10 dm. high) — 21.
- 21a. Leaves entire, linear-lanceolate
- 21b. Leaves serrate, oblong-lanceolate
- 22a. Pappus tawny in color (1-3 m. high) — 23.
- 22b. Pappus white — 24.
- 23a. Leaves pinnatifid
- 23b. Leaves undivided, dentate
- 24a. Upper leaves entire; heads about 1.5 cm. wide (5-10 dm. high)
- 24b. Upper leaves dentate or lobed; heads about 1 cm. wide (1-3 m. high)
- 25a. Achene tipped with a slender beak, bearing the pappus at its summit (summer) (Wild Lettuce) — 26.
- 25b. Achene without a beak — 29.
- 26a. Leaves hirsute or hispid on the mid-veins beneath — 27.
- Fall Dandelion,
Leontodon
autumnalis.**
- Meadow Salsify,
Tragopogon
pratensis.
Picris, Picris
hieracioides.**
- Wild Lettuce,
Lactuca spicata.
Wild Lettuce,
Lactuca spicata
var. integrifolia.**
- Wild Lettuce,
Lactuca pulchella.**
- Wild Lettuce,
Lactuca floridana.**

- 26b. Leaves glabrous — [28.](#)
- 27a. Leaves pubescent on both sides (1-2 m. high) **Wild Lettuce, Lactuca hirsuta.**
- 27b. Leaves glabrous, except on the mid-vein (5-15 dm. high) **Wild Lettuce, Lactuca scariola var. integrata.**
- 28a. Leaves entire or sparsely toothed (1-2 m. high) **Wild Lettuce, Lactuca sagittifolia.**
- 28b. Leaves chiefly sinuate-pinnatifid (1-3 m. high) **Wild Lettuce, Lactuca canadensis.**
- 29a. Flowers white, cream-color, or purplish (summer and autumn) (Rattlesnake Root) — [30.](#)
- 29b. Flowers bright-yellow or orange — [33.](#)
- 30a. Heads nodding (6-20 dm. high) — [31.](#)
- 30b. Heads pointing in various directions, in spike-like panicles; involucre pubescent (5-15 dm. high) **Rattlesnake Root, Prenanthes racemosus.**
- 31a. Heads with 5-7 flowers in each **Rattlesnake Root, Prenanthes altissima.**
- 31b. Heads with 8-12 flowers — [32.](#)
- 31c. Heads with 20 or more flowers **Rattlesnake Root, Prenanthes crepidinea.**
- 32a. Pappus dark reddish-brown **Rattlesnake Root, Prenanthes alba.**
- 32b. Pappus pale-brown or nearly white **Rattlesnake Root, Prenanthes trifoliata.**
- 33a. Pappus tawny or brown in color (summer and autumn) (Hawkweed)

- 34.
- 33b. Pappus white — 42.
- 34a. Heads 2.5 cm. in diameter, or larger — 35.
- 34b. Heads 1-2 cm. in diameter (4-10 dm. high) — 37.
- 35a. Leaves all basal (1-4 dm. high)
- 35b. Stem-leaves present (4-15 dm. high) — 36.
- 36a. Leaves rounded at the sessile base
- 36b. Leaves narrowed toward the base
- 37a. A rosette of basal leaves conspicuous at flowering time — 38.
- 37b. No rosette of basal leaves at time of flowering — 41.
- 38a. Leaves glabrous on the upper side — 39.
- 38b. Leaves hairy on the upper side — 40.
- 39a. Stem glabrous, leafless or with one or two leaves
- 39b. Stem with several leaves, hairy below
- 40a. Leaves with short scattered hairs above
- 40b. Leaves and stem densely covered with very long
- Hawkweed,
Hieracium
pilosella.**
- Hawkweed,
Hieracium
canadense.**
- Hawkweed,
Hieracium
umbellatum.**
- Hawkweed,
Hieracium
venosum.**
- Hawkweed,
Hieracium
marianum.**
- Hawkweed,
Hieracium
gronovii.**
- Hawkweed,
Hieracium**

- hairs
- 41a. Leaves glabrous **longipilum.
Hawkweed,
Hieracium
paniculatum.**
- 41b. Leaves very hairy **Hawkweed,
Hieracium
scabrum.**
- 42a. Bracts of the involucre
smooth (5-20 dm. tall;
summer and autumn) (Sow
Thistle) [— 43.](#)
- 42b. Bracts of the involucre hairy
[— 44.](#)
- 43a. The clasping leaf-bases acute **Sow Thistle,
Sonchus oleraceus.**
- 43b. The clasping leaf-bases rounded **Sow Thistle,
Sonchus asper.**
- 44a. Heads 2.5-5 cm. broad;
involucre 2 cm. long (4-10
dm. high) **Sow Thistle,
Sonchus arvensis.**
- 44b. Heads 1-2 cm. broad;
involucre 6-10 mm. long
(3-6 dm. high; summer) **Hawksbeard,
Crepis tectorum.**
- 45a. Leaves or involucre or both spiny
(thistles, burdock, etc.) [— 46.](#)
- 45b. Neither leaves nor involucre spiny
[— 60.](#)
- 46a. Leaves 1-4 dm. broad, not
spiny (flowers purple or
white; summer) (Burdock)
[— 47.](#)
- 46b. Leaves narrower, not spiny
[— 48.](#)
- 47a. Diameter of involucre at flowering time **Burdock, Arctium
3-5 cm. (1-3 m. high) lappa.**

- 47b. Diameter of involucre at flowering time 1.5-3 cm. (5-15 dm. high) **Burdock, *Arctium minus*.**
- 48a. Each head 1-flowered; heads aggregated in a globular head-like cluster (1-2 m. high; flowers blue or white, summer) **Globe Thistle, *Echinops sphaerocephalus*.**
- 48b. Each head many-flowered — 49.
- 49a. Principal involucre bracts with stout spreading spines 2-4 cm. long (5-15 dm. high; flowers purple, summer) **Milk Thistle, *Silybum marianum*.**
- 49b. Principal involucre bracts with slender spines or none — 50.
- 50a. Pappus feathery; receptacle bristly (summer and autumn) (Thistle) — 51.
- 50b. Pappus not feathery (flowers purple, summer) — 59.
- 51a. Heads subtended by a circle of large leafy bracts (5-15 dm. high; flowers pale-yellow) **Thistle, *Cirsium spinoissimum*.**
- 51b. Heads not subtended by several leafy bracts — 52.
- 52a. Leaves conspicuously white-woolly on both sides (4-10 dm. high) — 53.
- 52b. Leaves conspicuously white-woolly or brown-woolly below, not above (flowers purple or pink) — 54.
- 52c. Leaves green on both sides (flowers pink or purple, rarely white) — 56.
- 53a. Leaves deeply pinnately parted with linear divisions; flowers almost white **Thistle, *Cirsium pitcheri*.**

- 53b. Leaves irregularly pinnatifid; flowers purple-pink
 54a. Stem-leaves entire or shallowly lobed (1-3 m. high)
 54b. Stem-leaves obviously pinnatifid (1-2 m. high)
 — [55.](#)
- 55a. Leaves decurrent on the stem
 Thistle, *Cirsium undulatum*.
- 55b. Leaves not decurrent
 56a. Outer and middle involucre bracts appressed, pointless or with weak short prickles
 — [57.](#)
 56b. Outer and middle bracts erect, not appressed, acuminate into a long slender more or less prickly tip (4-10 dm. high) — [58.](#)
- 57a. Heads numerous 2-2.5 cm. broad, in close clusters (5-12 dm. high)
 Canada Thistle, *Cirsium arvense*.
- 57b. Heads few or solitary, 3-5 cm. broad (1-2 m. high)
 58a. Principal bracts with a conspicuous viscid stripe down the middle; heads 6-19 cm. broad, solitary or few
 Thistle, *Cirsium hillii*.
- 58b. Principal bracts not viscid
 Thistle, *Cirsium pumilum*.
- 59a. Receptacle not bristly; heads 3-5 cm. wide (1-3 m. high; flowers pale-purple, summer)
 Cotton Thistle, *Onopordum acanthium*.

59b. Receptacle bristly; heads 2-2.5 cm.
wide (5-12 dm. high; flowers purple
to white, late summer)

**Thistle, *Carduus*
crispus.**

60a. Leaves basal; stem-leaves
none or reduced to scales
(2-8 dm. high; flowers
whitish, in spring) — [61](#).

60b. Stem-leaves present; basal
leaves present or absent
— [62](#).

61a. Leaves toothed or lobed; flower-stalk
not scaly

**Adenocaulon,
Adenocaulon
bicolor.**

61b. Leaves deeply cleft; flower-stalk scaly
— 197a.

62a. Leaves compound or
dissected (flowers in
summer and autumn)
— [63](#).

62b. Leaves merely lobed, never
truly compound or
dissected — [72](#).

62c. Leaves entire or serrate — [78](#).

63a. Some of the involucre bracts leaf-like,
longer than the heads (3-20 dm. high;
flowers yellow or greenish, summer
and autumn) (Beggar Ticks) — [126](#).

63b. Bracts short and not leaf-like — [64](#).

64a. Heads 7-20 mm. wide, in a
flat-topped or convex
cluster (3-10 dm. high;
flowers yellow, summer)
(Tansy) — [65](#).

64b. Heads smaller, in spikes,
racemes, or panicles
(flowers yellow or

greenish, late summer and autumn) (Wormwood)

— 66.

- | | |
|---|--|
| 65a. Heads 7-10 mm. wide, numerous in a dense cluster | Tansy, <i>Tanacetum vulgare</i>. |
| 65b. Heads 10-20 mm. wide, few, 2-10 in a loose open cluster | Tansy, <i>Tanacetum huronense</i>. |
| 66a. Heads 2-3 mm. broad (4-15 dm. high) <u>— 67.</u> | |
| 66b. Heads 4-6 mm. broad (3-10 dm. high) <u>— 69.</u> | |
| 67a. Leaf-lobes narrowly linear, strictly entire | Wormwood, <i>Artemisia caudata</i>. |
| 67b. Leaf-lobes serrate <u>— 68.</u> | |
| 68a. Heads in a loose spreading panicle | Wormwood, <i>Artemisia annua</i>. |
| 68b. Heads in axillary clusters, producing a leafy spike-like panicle | Wormwood, <i>Artemisia biennis</i>. |
| 69a. Leaf-lobes narrowly linear <u>— 70.</u> | |
| 69b. Leaf-lobes oblong to obovate, not linear <u>— 71.</u> | |
| 70a. Shrubby; involucre pubescent | Southernwood, <i>Artemisia abrotanum</i>. |
| 70b. Herbaceous; involucre glabrous or rarely pubescent | Wormwood, <i>Artemisia canadensis</i>. |
| 71a. Leaves finely gray-pubescent on both sides | Wormwood, <i>Artemisia absinthium</i>. |
| 71b. Leaves smooth or nearly so above, densely white-woolly beneath | Mugwort, <i>Artemisia vulgaris</i>. |
| 72a. Heads 2-4 cm. broad, purple, blue, or rarely white (3-6 | |

- dm. high; summer) — [92.](#)
- 72b. Heads 1 cm. wide or less
— [73.](#)
- 73a. Leaves densely white-woolly beneath
(flowers yellowish, late summer)
— [74.](#)
- 73b. Leaves smooth or hairy, never white-
woolly — [75.](#)
- 74a. Heads 6-8 mm. wide (4-8 dm.
high)
- 74b. Heads 3-4 mm. wide (5-10
dm. high)
- 75a. Principal bracts of the involucre 5, with
frequently a few much smaller ones
— [76.](#)
- 75b. Principal bracts of the involucre
numerous — [77.](#)
- 76a. Heads few in small terminal
clusters; foliage somewhat
viscid-pubescent (6-15 dm.
high; flowers yellow,
summer) — 114a.
- 76b. Heads very numerous in flat-
topped clusters; foliage
never viscid-pubescent (1-2
m. high; flowers white, late
summer) — 105c.
- 77a. Leaves broadly halberd-shape, 3-lobed
(1-2 m. high; flowers white, late
summer) — 105b.
- 77b. Leaves lobed only at the base (5-10 dm.
high; flowers yellow, summer)
— 93a.

**Wormwood,
Artemisia
stelleriana.**

**Wormwood,
Artemisia
ludoviciana.**

77c. Leaves pinnatifid (2-8 dm. high; flowers yellow) — [172.](#)

78a. Bracts of the involucre dry and chaffy, at least at the tip; plants always pubescent and usually white-woolly — [79.](#)

78b. Bracts of the involucre green or colored, but never dry and chaffy — [90.](#)

79a. Pappus none; heads 3-4 mm. wide, in ample paniced spikes (flowers yellowish, late summer) — 74b.

79b. Pappus a minute ring or crown; leaves crenate (5-10 dm. high; flowers yellow, summer) — 93a.

79c. Pappus of hairs; heads in flat-topped clusters or slender spikes — [80.](#)

80a. Heads sessile or subsessile in small flat-topped clusters; flowering in spring or early summer; principal leaves basal (1-4 dm. high; flowers white or purplish) (Everlasting) — [81.](#)

80b. Heads in terminal spikes (2-6 dm. high; flowers purplish, summer)

80c. Heads in small or large flat-topped clusters, flowering in summer or autumn; principal leaves on the stem — [88.](#)

81a. Stolons from the basal rosette of leaves leafy throughout and ascending at the tip — [82.](#)

**Cudweed,
Gnaphalium
purpureum.**

81b. Stolons prostrate throughout, leafy only
at the tip [—87.](#)

82a. Basal leaves 2-5 cm. long, 1-
nerved [—83.](#)

82b. Basal leaves 5-12 cm. long, 3-
nerved [—84.](#)

83a. Basal leaves spatulate or oblanceolate,
smooth above

83b. Basal leaves obovate, pubescent above

84a. Basal leaves smooth above

84b. Basal leaves dull green and
pubescent above [—85.](#)

85a. Heads 6-8 mm. high

85b. Heads 8-11 mm. high [—86.](#)

86a. Leaf-blade ovate or obovate

86b. Leaf-blade spatulate, with
rounded tip

87a. Styles crimson

87b. Styles pale yellow

88a. Erect; involucre bracts pearly
white (4-9 dm. high)

**Everlasting,
Antennaria
canadensis.**

**Everlasting,
Antennaria
neodioica.**

**Everlasting,
Antennaria
parlinii.**

**Everlasting,
Antennaria
plantaginifolia.**

**Everlasting,
Antennaria fallax.**

**Everlasting,
Antennaria
occidentalis.**

**Everlasting,
Antennaria
neglecta.**

**Everlasting,
Antennaria
petaloidea.**

**Pearly
Everlasting,
Anaphalis
margaritacea.**

- 88b. Erect; involucre bracts dull white or pale brown, somewhat pubescent (4-8 dm. high) (Cudweed) [— 89.](#)
- 88c. Diffusely branched; heads in dense clusters; bracts brown (1-2 dm. high) **Cudweed, Gnaphalium uliginosum.**
- 89a. Leaves decurrent on the stem **Cudweed, Gnaphalium decurrens.**
- 89b. Leaves not decurrent on the stem **Cudweed, Gnaphalium polycephalum.**
- 90a. Twining vine (flowers white, summer) **Hemp Weed, Mikania scandens.**
- 90b. Not twining or climbing [— 91.](#)
- 91a. Involucre bracts deeply fringed at the tip (flowers purple, blue, or rarely white, summer; 3-6 dm. high) [— 92.](#)
- 91b. Involucre bracts entire or nearly so [— 93.](#)
- 92a. Upper leaves linear or narrowly lanceolate **Corn Flower, Centaurea cyanus.**
- 92b. Upper leaves oblong or oblong-lanceolate **Knapweed, Centaurea nigra.**
- 93a. Pappus none or a short ring or crown (5-10 dm. high; flowers yellow, summer) **Costmary, Chrysanthemum balsamita var. tanacetoides.**
- 93b. Pappus of 2-4 stiff awns (2-15 dm. high; flowers yellow, late summer) (Bur Marigold) [— 130.](#)
- 93c. Pappus of hairs or bristles [— 94.](#)

- 94a. Leaves linear or narrowly lanceolate, entire; heads never in a large flat-topped cluster — [95.](#)
- 94b. Leaves not linear — [99.](#)
- 95a. Heads showy, purple, in a long spike or raceme (late summer) (Blazing Star) — [96.](#)
- 95b. Heads not showy, in a loose panicle or raceme — 240b.
 - 96a. Involucral bracts rounded at the tip, appressed (5-15 dm. high) — [97.](#)
 - 96b. Involucral bracts pointed (3-6 dm. high) — [98.](#)
- 97a. Heads 8-12-flowered **Blazing Star,
*Liatris spicata.***
- 97b. Heads with 25 flowers or more **Blazing Star,
*Liatris scariosa.***
 - 98a. Involucral bracts long-acuminate, spreading **Blazing Star,
*Liatris squarrosa.***
 - 98b. Involucral bracts mucronate, appressed **Blazing Star,
*Liatris cylindracea.***
- 99a. Flowers yellow (2-8 dm. high) — 172b.
- 99b. Flowers bright-red or purple, in flat-topped clusters (8-20 dm. high; late summer) (Ironweed) — [100.](#)
- 99c. Flowers blue (3-8 dm. high; late summer) **Mist Flower,
*Eupatorium coelestinum.***
- 99d. Flowers flesh-color, pink, cream-color, or white (flowers in summer) — [103.](#)
 - 100a. Leaves glabrous beneath or minutely pubescent; heads 15-30-flowered — [101.](#)

- 100b. Leaves tomentose beneath;
heads 30-50-flowered
[— 102.](#)
- 101a. Inflorescence densely crowded;
usually 1 m. or less high
Ironweed,
Vernonia
fasciculata.
- 101b. Inflorescence loose and open, 15-30
cm. wide; 1-2 m. high
Ironweed,
Vernonia
altissima.
- 102a. Pappus tawny in color
Ironweed,
Vernonia
missurica.
- 102b. Pappus purple
Ironweed,
Vernonia
illinoensis.
- 103a. Leaves alternate [— 104.](#)
- 103b. Leaves opposite [— 106.](#)
- 103c. Leaves whorled (1-3 m. high; flowers
pink or purple, late summer) (Joe-Pye
Weed) [— 107.](#)
- 104a. Heads 5-flowered (5-20 dm.
high; flowers white or
pinkish, late summer)
(Indian Plantain) [— 105.](#)
- 104b. Heads 10 25-flowered (5-10
dm. high; flowers white,
late summer)
False Boneset,
Kuhnia
eupatorioides.
- 104c. Heads with more than 50
flowers (3-20 dm. high;
flowers white, summer)
Fireweed,
Erechtites
hieracifolia.
- 105a. Leaves entire, with many veins from
base to apex
Indian Plantain,
Cacalia tuberosa.
- 105b. Leaves sharply serrate
Indian Plantain,
Cacalia
suaveolens.

- 105c. Leaves broadly triangular or kidney-shape, sinuate or entire
Indian Plantain,
Cacalia
atriplicifolia.
- 106a. Leaves united at the base (5-15 dm. high)
Boneset,
Eupatorium
perfoliatum.
- 106b. Leaves sessile but not united at the base (5-15 dm. high)
Upland Boneset,
Eupatorium
sessilifolium.
- 106c. Leaves petioled (4-12 dm. high)
White Snakeroot,
Eupatorium
urticaefolium.
- 107a. Inflorescence ovoid or pyramidal
Joe-Pye Weed,
Eupatorium
purpureum.
- 107b. Inflorescence depressed or flattened
Joe-Pye Weed,
Eupatorium
purpureum var.
maculatum.
- 108a. Rays yellow or brown
[— 109.](#)
- 108b. Rays white to blue or red, never yellow or brown
[— 197.](#)
- 109a. Principal leaves basal, the stem merely with bract-like scales [— 110.](#)
- 109b. Principal leaves on the stem, opposite or whorled [— 111.](#)
- 109c. Principal leaves on the stem, alternate, or with smaller ones clustered in their axils [— 132.](#)
- 110a. Flower-stalk 1-5 dm. high, 1-flowered (spring)
Coltsfoot,
Tussilago farfara.
- 110b. Flower-stalk 1-3 m. high, several-flowered (summer)
Prairie Dock,
Silphium
terebinthinaceum.

- 111a. Ray-flowers pistillate (the 2-lobed style protrudes from their base)
[— 112.](#)
- 111b. Ray-flowers with neither stamens nor pistil [— 118.](#)
 - 112a. Principal leaves lobed (summer) (Leafcup)
[— 113.](#)
 - 112b. Principal leaves toothed or entire, not lobed [— 115.](#)
- 113a. Rays 10 or more (1-2 m. high) **Leafcup, Polymnia uvedalia.**
- 113b. Rays 5 (5-15 dm. high) [— 114.](#)
 - 114a. Rays shorter than the involucre or none **Leafcup, Polymnia canadensis.**
 - 114b. Rays about 1 cm. long **Leafcup, Polymnia canadensis var. radiata.**
- 115a. Stem 6 dm. high or less; pappus of slender hairs (spring) **Arnica, Arnica cordifolia.**
- 115b. Stem usually 8-20 dm. high; pappus of short scales or none (summer)
[— 116.](#)
 - 116a. Leaves united at base into a cup surrounding the stem **Cup Plant, Silphium perfoliatum.**
 - 116b. Leaves closely sessile with a rounded base **Rosin Weed, Silphium integrifolium.**
 - 116c. Leaves tapering to a short petiole; principal leaves whorled **Rosin Weed, Silphium trifoliatum.**
 - 116d. Leaves abruptly rounded at the sessile base, all opposite (Ox-eye) [— 117.](#)

- | | |
|--|---|
| 117a. Leaves smooth | Ox-eye, <i>Heliopsis helianthoides</i>. |
| 117b. Leaves rough | Ox-eye, <i>Heliopsis scabra</i>. |
| 118a. Principal stem-leaves lobed or divided — 119. | |
| 118b. Principal stem-leaves entire or serrate — 127. | |
| 119a. Submerged aquatic; leaf-segments filiform | Water Marigold, <i>Bidens beckii</i>. |
| 119b. Terrestrial plants; leaves merely 3-lobed (3-8 dm. high; late spring and summer) (Tickseed) — 120. | |
| 119c. Terrestrial plants; leaves compound or dissected (summer and autumn) — 121. | |
| 120a. Leaf-lobes linear-oblong, all about equal | Tickseed, <i>Coreopsis palmata</i>. |
| 120b. Lateral leaf-lobes very much smaller than the terminal | Tickseed, <i>Coreopsis lanceolata</i>. |
| 121a. Leaf-segments entire (Tickseed) — 122. | |
| 121b. Leaf-segments serrate (5-15 dm. high) (Tickseed Sunflower) — 124. | |
| 122a. Leaf-segments numerous, linear or nearly so (4-10 dm. high) — 123. | |
| 122b. Leaf-segments 3-5, lanceolate (1-3 m. high) | Tickseed, <i>Coreopsis tripteris</i>. |
| 123a. Rays yellow throughout | Tickseed, <i>Coreopsis verticillata</i>. |

- 123b. Rays brown, at least at the base **Tickseed,
Coreopsis
tinctoria.**
- 124a. Achenes wedge-shape, the
inner ones less than 2 mm.
wide — [125.](#)
- 124b. Achenes obovate, the inner
ones more than 2 mm. wide **Tickseed
Sunflower, Bidens
aristosa.**
- 125a. Leaf-lobes lanceolate **Tickseed
Sunflower, Bidens
trichosperma.**
- 125b. Leaf-lobes linear **Tickseed
Sunflower, Bidens
trichosperma var.
tenuiloba.**
- 126a. Outer leaf-like bracts 10-16;
achenes brown **Beggar Ticks,
Bidens vulgata.**
- 126b. Outer leaf-like bracts 5-8;
achenes black **Beggar Ticks,
Bidens frondosa.**
- 126c. Outer leaf-like bracts about 4 **Beggar Ticks,
Bidens discoidea.**
- 127a. Bracts of the involucre all essentially
alike in form and texture (flowers in
summer and autumn) (Sunflower)
— [179.](#)
- 127b. Bracts of the involucre in two distinct
sets, differing in form or consistency
or both — [128.](#)
- 128a. Leaves entire (3-8 dm. high;
late spring and summer)
— 120b.
- 128b. Leaves serrate (late summer
and autumn) (Bur
Marigold) — [129.](#)

- | | |
|--|---|
| 129a. Rays large and conspicuous, 2-3 cm. long (3-10 dm. high) | Bur Marigold,
<i>Bidens laevis</i>. |
| 129b. Rays 1 cm. long or less — 130 . | |
| 130a. Outer bracts leaf-like, serrate, 3-8 cm. long (4-15 dm. high) | Bur Marigold,
<i>Bidens comosa</i>. |
| 130b. Outer bracts 1-2.5 cm. long (2-15 dm. high) — 131 . | |
| 131a. Heads nodding after flowering | Bur Marigold,
<i>Bidens cernua</i>. |
| 131b. Heads permanently erect | Bur Marigold,
<i>Bidens connata</i>. |
| 132a. Heads small, seldom more than 1 cm. wide, including the rays, blooming in late summer and autumn; flowers numerous, crowded in spikes, racemes, corymbs, or panicles (Goldenrod) — 133 . | |
| 132b. Heads medium size or large, more than 1 cm. and usually exceeding 2 cm. in width, including the rays — 165 . | |
| 133a. Heads chiefly in clusters or short racemes in the axils of ordinary foliage leaves, or occasionally the upper compacted into a leafy cluster terminating the stem — 134 . | |
| 133b. Heads crowded at or near the ends of the branches at about the same distance from the base of the panicle, forming a rounded or flat-topped inflorescence — 140 . | |

133c. Heads more or less uniformly distributed along the length of the branches, forming a cylindrical or pyramidal inflorescence, never flat-topped — [146.](#)

134a. Stem and both sides of the leaves more or less pubescent or rough (4-10 dm. high) — [135.](#)

134b. Stem and both sides of the leaves essentially smooth or with very short hairs (3-10 dm. high) — [136.](#)

135a. Rays white

**Goldenrod,
Solidago bicolor.**

135b. Rays yellow

**Goldenrod,
Solidago hispida.**

136a. Basal leaves abruptly narrowed to winged petioles — [137.](#)

136b. Basal leaves not abruptly narrowed to winged petioles — [138.](#)

137a. Involucre 2-5 mm. long

**Goldenrod,
Solidago latifolia.**

137b. Involucre 8-12 mm. long

**Goldenrod,
Solidago
macrophylla.**

138a. Lower leaves broadly oval, obtuse, thickish, crenate; achenes glabrous

**Goldenrod,
Solidago erecta.**

138b. Lower leaves lanceolate, acuminate, thin, sharply serrate; achenes hairy — [139.](#)

- | | |
|--|---|
| 139a. Stem usually simple; heads few in very small clusters | Goldenrod,
Solidago caesia
var. axillaris. |
| 139b. Stem usually diffusely branched; heads numerous | Goldenrod,
Solidago caesia. |
| 140a. Lower leaves ovate, oblong, or oval, pinnately veined (5-15 dm. high) — 141. | |
| 140b. Lower leaves linear-lanceolate. 3-5-veined (3-12 dm. high) — 142. | |
| 141a. Stem and leaves rough-hairy | Goldenrod,
Solidago rigida. |
| 141b. Stem and leaves smooth | Goldenrod,
Solidago ohioensis. |
| 142a. Heads very few in a small cluster; leaves few and scattered | Goldenrod,
Solidago
houghtonii. |
| 142b. Heads very many, in a large cluster; stem very leafy — 143. | |
| 143a. Leaves hairy | Goldenrod,
Solidago
graminifolia var.
nuttallii. |
| 143b. Leaves smooth — 144. | |
| 144a. Leaves folded, 8-20 mm. wide | Goldenrod,
Solidago riddellii. |
| 144b. Leaves flat, 1-8 mm. wide — 145. | |
| 145a. Leaves 4-8 mm. wide, distinctly 3-5-ribbed | Goldenrod,
Solidago
graminifolia. |
| 145b. Leaves 1-4 mm. wide, usually with 1 mid-vein | Goldenrod,
Solidago
tenuifolia. |

- 146a. Only 2-5 stem-leaves below the inflorescence (1-3 dm. high)
- 146b. Stem-leaves numerous
— [147.](#)
- 147a. Basal leaves much larger than the greatly reduced or bract-like upper ones — [148.](#)
- 147b. Leaves essentially uniform in size from base to summit of stem — [157.](#)
- 148a. Racemes or branches of the panicle either short and arranged along a more or less elongated central axis, or elongated and ascending, scarcely recurved, forming a narrow, more or less elongated panicle — [149.](#)
- 148b. Racemes or branches of the panicle usually elongated, spreading outwards, usually recurved, forming a widened panicle — [153.](#)
- 149a. Leaves mostly entire, the upper ones with smaller leaves fascicled in the axils (5-20 dm. high)
- 149b. Leaves mostly serrate, at least the basal ones — [150.](#)
- 150a. Heads on pedicels 5-15 mm. long; achenes pubescent; stems usually clustered (1-5 dm. high, or prostrate)
— [151.](#)
- 150b. Heads on pedicels not over 5 mm. long; achenes smooth or nearly so; stems usually

**Goldenrod,
*Solidago cutleri.***

**Goldenrod,
*Solidago speciosa.***

- single (6-12 dm. high)
— 152.
- 151a. Basal leaves 7-12 cm. long **Goldenrod,
Solidago
racemosa.**
- 151b. Basal leaves 15-30 cm. long **Goldenrod,
Solidago racemosa
var. gillmani.**
- 152a. Leaves pinnately veined **Goldenrod,
Solidago uliginosa.**
- 152b. Leaves 3-5-ribbed **Goldenrod,
Solidago neglecta.**
- 153a. Both sides of the leaf pubescent or
rough — 154.
- 153b. Leaf not pubescent or rough on both
sides — 155.
- 154a. Stem closely pubescent (2-8
dm. high) **Goldenrod,
Solidago
nemoralis.**
- 154b. Stem glabrous (5-12 dm.
high) **Goldenrod,
Solidago juncea
var. scabrella.**
- 155a. Leaves rough above, smooth below (6-
15 dm. high) **Goldenrod,
Solidago patula.**
- 155b. Leaves smooth on both sides (5-12
dm. high) — 156.
- 156a. Branches of the panicle
spreading or recurved **Goldenrod,
Solidago juncea.**
- 156b. Branches of the panicle
upright **Goldenrod,
Solidago juncea
var. ramosa.**
- 157a. Stem more or less pubescent or hairy
throughout (5-20 dm. high) — 158.
- 157b. Stem smooth, at least below the
inflorescence — 161.

- | | |
|--|---|
| 158a. Involucre 2-2.7 mm. long | Goldenrod,
Solidago canadensis. |
| 158b. Involucre 3-5 mm. long
— 159. | |
| 159a. Leaves pinnately veined, scabrous above | Goldenrod,
Solidago rugosa. |
| 159b. Leaves 3-5-ribbed, pubescent but not scabrous above — 160. | |
| 160a. Stem and lower side of leaves covered with short hairs; common species | Goldenrod,
Solidago altissima. |
| 160b. Stem and lower side of leaves with distinct, loose, soft hairs (shore of Lake Superior) | Goldenrod,
Solidago altissima
var. procera. |
| 161a. Involucre 2-2.7 mm. long (5-20 dm. high) — 158a. | |
| 161b. Involucre 3-6 mm. long — 162. | |
| 162a. Racemes or branches of the panicle either short and arranged along a more or less elongated axis, or elongated and ascending, scarcely recurved, forming a narrow more or less elongated panicle (5-10 dm. high) | Goldenrod,
Solidago speciosa
var. angustata. |
| 162b. Racemes or branches of the panicle usually elongated, spreading outward, usually recurved, forming a widened panicle; leaves distinctly serrate — 163. | |
| 163a. Leaves pinnately veined (5-12 dm. high) | Goldenrod,
Solidago ulmifolia. |

163b. Leaves 3-5-ribbed (5-20 dm. high)

— 164.

164a. Leaves glabrous on both sides

164b. Leaves slightly pubescent beneath

**Goldenrod,
Solidago serotina.**

**Goldenrod,
Solidago serotina
var. gigantea.**

165a. Ray-flowers pistillate (the 2-lobed style protrudes from their base)

— 166.

165b. Ray-flowers with neither stamens nor pistil — 174.

166a. Principal leaves more than 2 dm. long (1-3 m. high; summer) — 167.

166b. Principal leaves less than 1.5 dm. long — 168.

**Compass Plant,
Silphium
laciniatum.**

167a. Leaves deeply lobed

167b. Leaves toothed or serrate

**Elecampane, Inula
helenium.**

168a. Leaves narrowly linear (3-6 dm. high; late summer)

**Sneezeweed,
Helenium
tenuifolium.**

168b. Leaves of a broader shape — 169.

169a. Heads 1-2 cm. wide; flowers in spring and early summer (2-8 dm. high) (Ragwort) — 170.

169b. Heads 2-5 cm. wide; flowers in late summer and autumn — 173.

170a. Basal leaves cordate at base

**Ragwort, Senecio
aureus.**

170b. Basal leaves narrowed to the base — 171.

- | | |
|---|---|
| 171a. Basal leaves obovate | Ragwort, Senecio obovatus. |
| 171b. Basal leaves oblong | Ragwort, Senecio balsamitae. |
| 172a. Introduced annual in waste places (1-4 dm. high; spring and summer) | Groundsel, Senecio vulgaris. |
| 172b. Native biennial in moist ground (3-8 dm. high; summer) | Squaw Weed, Senecio discoideus. |
| 173a. Leaves 2-5 cm. long, sharply spinulose-serrate; involucre viscid (3-6 dm. high; summer) | Gum Plant, Grindelia squarrosa. |
| 173b. Leaves 5-12 cm. long, merely serrate; involucre gray-pubescent (5-15 dm. high; late summer) | Sneeze Weed, Helenium autumnale. |
| 174a. Disk hemispherical or oblong-cylindrical (Summer) —175. | |
| 174b. Disk flat or somewhat convex (Sunflower) (summer and autumn) —179. | |
| 175a. Disk yellow or greenish-yellow (1-3 dm. high) —176. | |
| 175b. Disk gray-brown or purple (5-15 dm. high) —177. | |
| 176a. Principal stem-leaves pinnately divided | Golden Glow, Rudbeckia laciniata. |
| 176b. Principal stem-leaves merely serrate | Yellow Ironweed, Actinomeris alternifolia. |
| 177a. Rays drooping; leaves pinnately divided | Gray-headed Coneflower, Lepachys pinnata. |

177b. Rays spreading when in bloom [— 178.](#)

178a. Lower leaves deeply 3-lobed

**Coneflower,
Rudbeckia triloba.**

178b. Stem-leaves sharply serrate

**Coneflower,
Rudbeckia
speciosa var.
sullivantii.**

178c. Stem-leaves entire or
sparingly serrate

**Black-eyed Susan,
Rudbeckia hirta.**

179a. Disk-flowers brown or purple [— 180.](#)

179b. Disk-flowers yellow [— 182.](#)

180a. Stem-leaves ovate or ovate-
lanceolate; petioles
prominent, not winged
[— 181.](#)

180b. Stem-leaves ovate or ovate-
lanceolate, contracted at the
base into a winged petiole
(6-15 dm. high)

**Sunflower,
Helianthus
atrorubens.**

180c. Stem-leaves oblong-
lanceolate, very thick and
rigid, gradually narrowed
to a sessile or short-
petioled base (5-20 dm.
high)

**Sunflower,
Helianthus
scaberrimus.**

181a. Disk less than 2 cm. wide (3-10 dm.
high)

**Sunflower,
Helianthus
petiolaris.**

181b. Disk more than 2.5 cm. wide (1-3 m.
high)

**Sunflower,
Helianthus
annuus.**

182a. Leaves all or chiefly at the
base (5-10 dm. high)

**Sunflower,
Helianthus
occidentalis.**

182b. Leaves chiefly scattered on
the stem [— 183.](#)

183a. Leaves mainly or all alternate, and not definitely 3-ribbed (1-4 m. high)

[— 184.](#)

183b. Leaves mainly or all opposite, lanceolate to ovate, and 3-ribbed

[— 186.](#)

184a. Stem glabrous

**Sunflower,
Helianthus grosse-
serratus.**

184b. Stem hairy or rough [— 185.](#)

185a. Leaves hairy beneath, rough above, lanceolate

**Sunflower,
Helianthus
giganteus.**

185b. Leaves rough on both sides, elongated

**Sunflower,
Helianthus
maximiliani.**

186a. Leaves sessile (5-15 dm. high) [— 187.](#)

186b. Leaves petioled, or narrowed at the base into a petiole (5-30 dm. high) [— 189.](#)

187a. Leaves wedge-shape at the base

**Sunflower,
Helianthus
doronicoides.**

187b. Leaves rounded at the base [— 188.](#)

188a. Stem glabrous or nearly so

**Sunflower,
Helianthus
divaricatus.**

188b. Stem densely and softly hirsute

**Sunflower,
Helianthus mollis.**

189a. Stems rough, pubescent, or hispid [— 190.](#)

189b. Stems glabrous or nearly so [— 193.](#)

190a. Leaves narrowly lanceolate, more than 5 times as long as wide

**Sunflower,
Helianthus**

	giganteus var. subtuberosus.
190b. Leaves ovate-lanceolate, not more than 4 times as long as wide — 191.	
191a. Leaves rounded at base, above the petiole	Sunflower, Helianthus hirsutus.
191b. Leaves narrowed to the base — 192.	
192a. Bracts of the involucre spreading	Jerusalem Artichoke, Helianthus tuberosus. Sunflower, Helianthus laetiflorus.
192b. Bracts all appressed	Sunflower, Helianthus laetiflorus.
193a. Heads 3 cm. wide or less, including the rays	Sunflower, Helianthus microcephalus.
193b. Heads 4 cm. wide or more, including the rays — 194.	
194a. Leaves narrowed at the base into a winged petiole — 195.	
194b. Petiole slender, not winged	Sunflower, Helianthus decapetalus.
195a. Leaves green on both sides; bracts longer than the disk	Sunflower, Helianthus tracheliifolius.
195b. Leaves paler below than above; bracts not longer than the disk — 196.	
196a. Leaves minutely pubescent beneath	Sunflower, Helianthus strumosus.

- | | |
|---|---|
| 196b. Leaves conspicuously
downy beneath | Sunflower,
Helianthus
strumosus var.
mollis. |
| 197a. Leaves all basal, the flowers on scaly
stalks (2-8 dm. high; flowers whitish,
in spring) | Coltsfoot, Petasites
palmata. |
| 197b. Stem-leaves present, opposite — 198. | |
| 197c. Stem-leaves present, alternate — 200. | |
| 198a. Leaves ovate, dentate, 2-6
cm. long (2-8 dm. high;
summer) (Galinsoga)
— 199. | |
| 198b. Leaves lobed, 10-25 cm.
long — 113b. | |
| 199a. Pubescence sparse, appressed | Galinsoga,
Galinsoga
parviflora.
Galinsoga,
Galinsoga
parviflora var.
hispida. |
| 199b. Pubescence abundant, spreading | |
| 200a. Leaves dissected or deeply
lobed or pinnatifid; pappus
never capillary; rays white
to pink (3-10 dm. high;
summer and autumn)
— 201. | |
| 200b. Leaves entire or serrate
— 206. | |
| 201a. Heads 4-8 mm. wide (Yarrow) — 202. | |
| 201b. Heads 12-50 mm. wide — 203. | |
| 202a. Flower-clusters flat-topped | Yarrow, Achillea
millefolium. |
| 202b. Flower-clusters very convex | Yarrow, Achillea
lanulosa. |

- 203a. Principal leaves pinnatifid — 213a.
- 203b. Principal leaves 1-3 times pinnately parted or dissected — [204](#).
- 204a. Leaf-segments very narrowly linear; leaves 2-3-pinnate — [205](#).
- 204b. Leaf-segments linear or lanceolate; heads 2.5-5 cm. wide **Camomile, Anthemis arvensis.**
- 204c. Leaf-segments ovate to ovate-oblong; heads 1-2 cm. wide **Feverfew, Chrysanthemum parthenium.**
- 205a. Foliage strongly scented **Dog Fennel, Anthemis cotula.**
- 205b. Foliage not ill-scented **Wild Camomile, Matricaria inodora.**
- 206a. Heads 3-6 mm. broad, including the rays (summer and autumn) — [207](#).
- 206b. Heads 7 mm. broad or larger, including the rays — [209](#).
- 207a. Rays purple (1-4 dm. high) **Horse Weed, Erigeron divaricatus.**
- 207b. Rays white — [208](#).
- 208a. Leaves obovate to oblong (3-10 dm. high) — 135a.
- 208b. Leaves linear or narrowly lanceolate (2-25 dm. high) **Horse Weed, Erigeron canadensis.**
- 209a. Pappus none, or minute and not of hairs (summer and autumn) — [210](#).
- 209b. Pappus of hairs — [214](#).

- 210a. Disk-flowers purple or brown (4-12 dm. high; rays pink) (Purple Coneflower) [— 211.](#)
- 210b. Disk-flowers yellow or nearly white [— 212.](#)
- 211a. Leaves ovate or ovate-lanceolate, most of them serrate **Purple Coneflower, *Brauneria purpurea*.**
- 211b. Leaves narrowly lanceolate, gradually narrowed at the base, entire **Purple Coneflower, *Brauneria pallida*.**
- 212a. Rays broadly obovate; heads 1-2 cm. wide (3-6 dm. high) **Sneezewort, *Achillea ptarmica*.**
- 212b. Rays oblong or narrowly elliptical [— 213.](#)
- 213a. Leaves serrate (3-10 dm. high) **Ox-eye Daisy, *Chrysanthemum leucanthemum* var. *pinnatifidum*.**
- 213b. Leaves entire (8-25 dm. high) ***Boltonia*, *Boltonia asteroides*.**
- 214a. Involucral bracts all the same length or nearly so and narrow, or with a few short outer ones; plants blooming in spring and summer, or a few plants persisting in bloom until autumn (Fleabane) [— 215.](#)
- 214b. Involucral bracts unequal, the outer successively shorter (or rarely nearly equal), loosely or closely

overlapping; plants 3-15
dm. high, blooming in late
summer and autumn
(Aster) — [221.](#)

215a. Rays short and inconspicuous, barely
longer than the pappus (1-5 dm. high; summer) **Fleabane,
Erigeron acris var.
asteroides.**

215b. Rays conspicuous, spreading, 3 mm.
long or more — [216.](#)

216a. Rare plants of the Northern
Peninsula, with entire
leaves and stems 1-5 dm.
high, from a thick woody
root (flowers white or
purple, summer) — [217.](#)

216b. Common species, with erect
stems from fibrous roots;
leaves toothed (except in
one species) — [218.](#)

217a. Heads 3-5 cm. wide; rays about 100 **Fleabane,
Erigeron glabellus.**

217b. Heads 1-2 cm. wide; rays 20-30 **Fleabane,
Erigeron
hyssopifolius.**

218a. Stem unbranched, except for
the peduncles; leaves
chiefly basal; heads 1-9 (2-
5 dm. high; flowers pale-
purple, spring)

**Fleabane,
Erigeron
pulchellus.**

218b. Stem branched; principal
leaves on the stem; heads
usually numerous (3-12
dm. high; spring and
summer) — [219.](#)

219a. Stem-leaves linear, entire **Fleabane,
Erigeron ramosus.**

- 219b. Stem-leaves ovate-lanceolate, the principal ones toothed — [220](#).
- 220a. Rays 100 or more, light-purple or pink
- 220b. Rays much fewer, white
- 221a. Basal leaves petioled and heart-shape at the base — [222](#).
- 221b. Basal leaves not petioled; stem-leaves with heart-shape clasping bases — [231](#).
- 221c. Basal and stem-leaves sessile or petioled, but never heart-shape or clasping — [240](#).
- 222a. Rays white or violet — [223](#).
- 222b. Rays blue — [225](#).
- 223a. Plant glandular, especially on the pedicels and branches of the inflorescence
- 223b. Plant not glandular — [224](#).
- 224a. Leaves rough above
- 224b. Leaves smooth above
- 225a. Stem-leaves clasping the stem by a cordate base
- 225b. Stem-leaves not cordate-clasping — [226](#).
- 226a. Leaves entire — [227](#).
- 226b. Leaves serrate — [228](#).
- 227a. Leaves glabrous above
- 227b. Leaves rough above
- Fleabane,
Erigeron
philadelphicus.
Fleabane,
Erigeron annuus.**
- Aster, Aster
macrophyllus.**
- Aster, Aster
schreberi.
Aster, Aster
divaricatus.
Aster, Aster
undulatus.**
- Aster, Aster
shortii.
Aster, Aster
azureus.**

- 228a. Involucre 4-6 mm. long
— [229.](#)
- 228b. Involucre 6-10 mm. long
— [230.](#)
- 229a. Leaves rough; petioles mostly winged **Aster, Aster lowrieanus.**
- 229b. Leaves smooth; petioles slender, not winged **Aster, Aster cordifolius.**
 - 230a. Heads few, seldom more than 10, in a loose spreading cluster **Aster, Aster lindleyanus.**
 - 230b. Heads numerous, in a rather elongate crowded cluster **Aster, Aster sagittifolius.**
- 231a. Stem hirsute or rough-pubescent
— [232.](#)
- 231b. Stem smooth, or essentially so — [236.](#)
 - 232a. Leaves conspicuously serrate **Aster, Aster puniceus.**
 - 232b. Leaves entire or nearly so
— [233.](#)
 - 233a. Leaves narrowed toward the base and barely clasping, linear or oblong-linear — [234.](#)
 - 233b. Leaves ovate-oblong or lanceolate, with a broad conspicuously clasping base — [235.](#)
 - 234a. Involucre pubescent but not glandular **Aster, Aster amethystinus.**
 - 234b. Involucre glandular **Aster, Aster oblongifolius.**
- 235a. Involucre very glandular and viscid; rays very numerous, violet-purple; leaves lanceolate **Aster, Aster novae-angliae.**
- 235b. Involucre slightly glandular or not at all; rays 20-30, generally blue-purple; leaves ovate-oblong **Aster, Aster patens.**

- 236a. Leaves of a linear type
— [237.](#)
- 236b. Leaves broader than linear,
at least 1 cm. wide — [238.](#)
- 237a. Bracts narrow, approximately equal in
length — 253a.
- 237b. Bracts of several lengths, the outer
successively shorter — 256a.
- 238a. Leaves smooth above **Aster, Aster laevis.**
- 238b. Leaves rough above — [239.](#)
- 239a. Leaves contracted below the middle
and then abruptly dilated to the
clasping base **Aster, Aster
prenanthoides.**
- 239b. Leaves gradually narrowed toward the
base **Aster, Aster
puniceus.**
- 240a. Rays conspicuous — [241.](#)
- 240b. Rays minute or wanting **Aster, Aster
angustus.**
- 241a. Stems and leaves gray with a silky
pubescence **Aster, Aster
sericeus.**
- 241b. Stem and leaves green, not silky
— [242.](#)
- 242a. Bracts glandular-viscid; rays
violet **Aster, Aster
oblongifolius.**
- 242b. Bracts bristly-ciliate — [243.](#)
- 242c. Bracts smooth or pubescent,
not glandular or bristly-
ciliate — [244.](#)
- 243a. Leaves crowded, rigid; rays white **Aster, Aster
multiflorus.**
- 243b. Leaves not crowded and rigid; rays
blue — 234a.
- 244a. Bracts narrowed at the tip
into thickened firm green
awl-shape points — [245.](#)

- 244b. Bracts acute or obtuse at the flattened tip — [247.](#)
- 245a. Involucre 4-5 mm. long — [246.](#)
- 245b. Involucre 7-8 mm. high
 - 246a. Stem smooth
 - 246b. Stem hairy; leaves linear
 - 246c. Stem densely white-woolly
- 247a. Leaves at most 4.5 cm. long — [248.](#)
- 247b. Leaves larger, at least the principal ones — [249.](#)
- 248a. Stems in clusters; leaves rigid, linear, with 1 vein; flowers blue
 - 248b. Stem solitary; leaves not rigid; flowers rose-pink
- 249a. Heads solitary at the end of minutely leafy branchlets; leaves linear
- 249b. Heads in flat-topped clusters; leaves lanceolate or broader — [250.](#)
- 249c. Heads in more or less one-sided racemes — [251.](#)
- 249d. Heads in panicles or irregular clusters — [253.](#)
 - 250a. Leaves rigid, linear-lanceolate
 - 250b. Leaves not rigid, lanceolate
- 251a. Leaves lanceolate, sharply serrate — [252.](#)

Aster, Aster polyphyllus.

Aster, Aster ericoides.

Aster, Aster ericoides var. villosus.

Aster, Aster ericoides var. platyphyllus.

Aster, Aster linariifolius.

Aster, Aster nemoralis.

Aster, Aster dumosus.

Aster, Aster ptarmicoides.

Aster, Aster umbellatus.

- | | |
|--|--|
| 251b. Leaves linear or narrowly linear-lanceolate, only the larger ones with a few teeth near the middle | Aster, Aster vimineus. |
| 252a. Stem glabrous or somewhat pubescent | Aster, Aster lateriflorus. |
| 252b. Stem woolly with long hairs | Aster, Aster lateriflorus var. hirsuticaulis. |
| 253a. Bracts narrow, approximately equal in length | Aster, Aster longifolius. |
| 253b. Bracts of several lengths, the outer successively shorter — 254. | |
| 254a. Heads 10-15 mm. wide, including the rays | Aster, Aster tradescanti. |
| 254b. Heads 15-25 mm. wide, including the rays — 255. | |
| 255a. Bracts with conspicuous dilated or subrhombic tips | Aster, Aster salicifolius. |
| 255b. Bracts without conspicuous green tips — 256. | |
| 256a. Rays purple or rose; bog plant with linear leaves | Aster, Aster junceus. |
| 256b. Rays white, or slightly tinged with blue; leaves oblong to narrowly lanceolate | Aster, Aster paniculatus. |

GLOSSARY

Achene. A small, dry, hard, seed-like fruit containing a single seed.

Acuminate. Taper-pointed.

Acute. Ending with an acute angle.

Alternate. Located singly on the stem, with other leaves above or below.

Annual. Living but a single season.

Anther. The (usually) enlarged end of a stamen, bearing the pollen.

Ascending. Rising or curving obliquely upward.

Auricle. An ear-shape appendage at the base of a leaf or other organ.

Auricled, auriculate. Furnished with auricles.

Awl-shape. Tapering to a slender stiff point.

Awn. An awl-shape or bristle-shape appendage.

Axil. The point on a stem just above the base of a leaf or branch.

Axillary. Arising from or produced in the axil.

Basal. Arising from or produced at the base.

Beak. Ending in a prominent slender point.

Bract. A small leaf near the base of a flower or flower-stalk, or in a flower-cluster.

Bracteal. An adjective derived from bract.

Bipinnate. A leaf with a pinnately branched axis, bearing leaflets on the sides of the branches.

Calyx. The outer portion of the flower, usually green in color. In some plants it is colored to resemble (or replace) the corolla, and in others may be minute or wanting.

Capitate. Shaped like a head; or arranged in a dense compact cluster.

Capsule. A dry fruit with usually several seeds, opening at maturity.

Catkin. A cylindrical or ovoid cluster of inconspicuous flowers, for example, the "pussy willow."

Cells of ovary. The cavity or cavities within an ovary, in which the seeds are produced.

Ciliate. Provided with hairs at the margin.

Clasping. With the base of a leaf or other organ wholly or partly surrounding the stem.

Cleft. Deeply divided toward the base or the mid-rib.

Closed sheath. A leaf-sheath in which the margins are united to form a tube.

Composite. A flower-cluster containing several or many small flowers, closely crowded together and provided with calyx-like bracts, so that the whole cluster resembles a single flower.

Compound. Composed of 2 or more similar parts united, as a compound ovary.

Compound leaf. A leaf with two or more separate leaflets on a single petiole.

Connate. Grown together.

Cordate. Heart-shape. A whole leaf-blade may be cordate, or the term may be applied to the base of a leaf only.

Cordate-sagittate. Intermediate in shape between cordate and sagittate.

Corm. An enlarged stem-base, of solid structure and usually underground.

Corolla. The portion of a flower next to the calyx (in ordinary cases). It is generally the most conspicuous part of the flower, but may be completely absent, or inconspicuous, or replaced by the calyx.

Corymb. A flat-topped or convex-topped flower-cluster.

Creeping. With stems prostrate on the ground and rooting at intervals.

Crenate. With round-pointed teeth at the margin.

Crenulate. Finely or minutely crenate.

Cuspidate. Ending with a short sharp stiff point.

Deciduous. Not persistent for a long time; not evergreen.

Decompound. Repeatedly branched with numerous leaflets.

Decurrent. Extending with wing-like expansions down the stem.

Decumbent. A stem prostrate at the base, but with the tip more or less ascending.

Dehiscent. Breaking open at maturity to discharge the contents.

Deltoid. Broadly triangular.

Dioecious. Bearing staminate and pistillate flowers upon separate plants.

Dissected. Finely divided into numerous small or narrow segments.

Divided. With deep segments or lobes.

Elliptical. Having the shape of an ellipse.

Elliptical-lanceolate. Intermediate in shape between elliptical and lanceolate.

Entire. With an unbroken margin, without teeth or lobes.

Epiphyte. A plant growing attached to the bark of another plant, and without connection with the soil.

Erect. Growing in nearly or quite a vertical position.

Evenly pinnate. A compound leaf terminating in a pair of leaflets.

Filament. The (usually) slender basal portion of a stamen, supporting the anther at its tip.

Floweret. A small flower.

Gamopetalous. Composed of united petals.

Gamosepalous. Composed of united sepals.

Glabrous. Smooth; without hairs.

Glandular. Bearing glands.

Glaucous. Covered with a thin bluish or whitish deposit, easily rubbed off.

Glume. A bract at the base of a spikelet of a grass.

Half recurved. Curved half-way backward.

Hastate. Shaped like an arrow-head, but with the basal lobes pointing outwards instead of backward.

Head. A dense cluster of flowers, about as broad as long.

Hirsute. With stiff coarse hairs.

Imperfect. Flowers which contain either pistil or stamens, not both.

Incised. With deep, sharp, irregular, divisions.

Indehiscent. Not breaking open at maturity to discharge the contents.

Inflorescence. A cluster of flowers.

Internode. A section of stem between two joints, or nodes.

Involucre. A collection of bracts at the base of a flower-cluster.

Irregular. Possessing similar parts of different size or form.
An irregular flower is generally distinguished by petals of unequal size or shape.

Laciniate. Cut into narrow pointed lobes or divisions.

Lanceolate. Shaped like a lance-head, several times longer than wide,
and broadest below the middle.

Linear. Long and narrow, but with about uniform width.

Linear-lanceolate. Intermediate in shape between linear and lanceolate;
narrowly lanceolate.

Lip. The largest and most conspicuous petal in an irregular corolla,
usually applied to the lower petal of an orchid.

Lobe. A segment or division of any organ.

Leaflet. One portion of the blade of a compound leaf.

Lemma. One of the bracts in the spikelet of a grass, and described in the treatment of that family.

Membranous. Thin or membrane-like in texture.

Monoecious. Bearing stamens and pistils in separate flowers, but on the same plant.

Mucronate. Tipped with a short small abrupt tip.

Node. A joint of a stem, at which leaves are borne and branches appear.

Ob lanceolate. Reversed lanceolate in shape.

Oblique. With unequal sides.

Oblong. Somewhat rectangular in shape, with parallel sides.

Oblong-lanceolate. Intermediate in shape between oblong and lanceolate.

Oblong-spatulate. Intermediate in shape between oblong and spatulate.

Obovate. Reversed ovate in shape.

Obtuse. Blunt-tipped; terminating in an obtuse angle.

Odd-pinnate. A compound leaf terminating in a single leaflet.

Once-compound. A compound leaf bearing leaflets at the end or along the sides of the main axis.

Once-pinnate. A compound leaf bearing leaflets along the sides of the axis.

Open sheath. A leaf-sheath with separate margins.

Opposite. Situated in pairs on opposite sides of the stem or axis.

Ovary. The basal, usually swollen portion of the pistil, within which the seeds are produced.

Ovate. Egg-shape in outline.

Ovate-lanceolate. Intermediate in shape between ovate and lanceolate; broadly lanceolate or narrowly ovate.

Ovate-oblong. Intermediate in shape between ovate and oblong.

Ovoid. Egg-shape.

Palmate. With several organs or structures attached at or proceeding from the same point; applied chiefly to the arrangement of principal veins in a leaf and of leaflets in a compound leaf.

Panicle. A loose, more or less irregular, branching cluster of pedicelled flowers, usually much longer than thick.

Parallel-veined. With the principal veins of the leaf paralleling each other from the base to the apex, or (rarely) from the mid-rib to the margin.

Pedicel. The stalk of a single flower.

Parasite. A plant which grows attached to another and derives its nourishment from it.

Peduncle. The stalk of a flower-cluster, or of a solitary flower.

Peltate. Attached to the stalk by the lower surface, instead of the margin.

Perennial. Living through several seasons.

Perfect. Bearing stamens and pistils in the same flower.

Perfoliate. Clasping the stem so completely that the stem seems to pass through it.

Perianth. The calyx and corolla of a flower.

Perigynium. A sac-like structure surrounding the achene of a sedge.

Persistent. Remaining attached for a considerable time.

Petal. One member or segment of the corolla.

Petiole. The stalk of a leaf.

Pinnate. With several organs or structures attached at the sides of an axis or stalk; applied chiefly to the arrangement of the principal veins in a leaf and of leaflets in a compound leaf.

Pinnatifid. Deeply pinnately cut or divided.

Pistil. The central portion of a flower, consisting of ovary, style, and stigma; the seed-bearing part of the flower.

Pistillate. Bearing pistils.

Polygamous. Applied to plants in which some flowers are perfect and others either staminate or pistillate.

Pubescent. Hairy.

Raceme. A more or less elongated flower-cluster, bearing pedicelled flowers along a single axis.

Racemose. Arranged in racemes.

Receptacle. The end of a peduncle or pedicel upon which the organs of a flower, or the flowers of a head, are attached.

Recurved. Curved back.

Reflexed. Abruptly bent back or down.

Regular. Uniform in shape or structure. Flowers are generally considered regular when all the petals are of the same size and shape.

Retrorse. Directed backward or downward.

Revolute. Rolled backward or under.

Rootstock. A horizontal subterranean stem, sending up leaves or stems.

Rotate. Wheel-shape; essentially flat and circular.

Sac-like. Inflated; sack-like.

Sagittate. Shaped like an arrow-head.

Salver-form. A corolla having a slender tube abruptly expanded at the summit into a flat or spreading portion.

Scape. A peduncle arising directly from the base of the plant, leafless or bearing bracts only.

Segment. One member or portion of an organ.

Sepal. One member or portion of the calyx.

Serrate. With sharp teeth at the margin.

Serrulate. Finely or minutely serrate.

Sessile. Without a stalk, petiole, or pedicel.

Sheathing. Inclosing.

Simple. In one piece; not compound; usually applied to leaves with a single blade.

Sinuate. Wavy-margined.

Sinus. The angle between two lobes or divisions.

Spadix. A short fleshy spike.

Spathe. A large bract or pair of bracts enclosing a flower-cluster.

Spatulate. Shaped like a spatula, with a narrow base and an enlarged, more or less rounded summit.

Spike. An elongated flower-cluster having sessile flowers upon an unbranched axis.

Spike-like. Resembling a spike.

Spinulose-serrate. Provided with teeth tipped with minute spines.

Spur. A hollow projection from the calyx or corolla, usually slender in shape, and generally directed backward.

Stamen. One of the organs of a flower, consisting of a filament and anther.

Staminate. Bearing stamens.

Stolon. A short stem arising from the base of a plant, prostrate or nearly so, and eventually taking root.

Striate. Marked with fine stripes or ridges.

Style. A portion of the pistil, usually slender, and connecting the ovary and stigma.

Superior. A superior ovary occupies the center of the flower and is not attached to any other floral organs.

Subtending. Situated at the base of an organ.

Subulate. Awl-shape.

Ternately. Divided by threes.

Tomentose. Densely hairy with matted or tangled hairs.

Trifoliate. With three leaflets.

Truncate. Cut straight across at the tip, or nearly so.

Tube. The more or less cylindrical portion of a gamosepalous calyx or a gamopetalous corolla, distinguished from the expanded or lobed terminal portion.

Tubular. Shaped like a tube.

Twice-pinnate. Same as bipinnate.

Two-lipped. A calyx or corolla in which the upper half is decidedly different in size or shape from the lower.

Umbel. A flower-cluster with several or many pedicelled flowers all arising from the same point.

Undulate. With a wavy margin.

Viscid. Sticky.

Villous. With long soft hairs.

Whorl. An arrangement of 3 or more leaves or flowers in a circle around a node.

Whorled. In a whorl.

Wing. A thin flat expansion on the sides or edge of an organ.

INDEX

Abies, [1](#)

Abutilon, [72](#)

Acalypha, [66](#)

Acanthaceae, [107](#)

Acanthus Family, [107](#)

Acer, [70](#)

Aceraceae, [70](#)

Acerates, [90](#)

Achillea, [130](#), [131](#)

Acnida, [30](#)

Acorus, [8](#)

Actaea, [39](#)

Actinomeris, [128](#)

Adder's Mouth, [17](#)

Adenocaulon, [118](#)

Adlumia, [41](#)

Aesculus, [70](#)

Agastache, [97](#)

Agrimonia, [55](#)

Agrimony, [55](#)

Agropyron, [6](#)

Agrostemma, [33](#)

Agrostis, [6](#)

Ailanthus, [65](#)

Aizoaceae, [31](#)

Alder, [22](#)

Alder, Black, [69](#)

Aletris, [13](#)

Alfalfa, [61](#)

Alisma, [3](#)

Alismaceae, [3](#)

Allium, [13](#)

Alnus, [22](#)

Alopecurus, [4](#)

Alsike Clover, [61](#)

Althaea, [72](#)

Alum Root, [50](#), [51](#)

Alyssum, [42](#), [46](#)

Alyssum, Yellow, [42](#), [46](#)

Amaranth Family, [30](#)

Amaranthaceae, [30](#)

Amaranthus, [30](#)

Amaryllidaceae, [14](#)

Amaryllis Family, [14](#)

Ambrosia, [114](#)

Amelanchier, [55](#)

American Columbo, [88](#)

American Ipecac, [57](#)

Ammophila, [4](#)

Amorpha, [58](#)

Amphicarpa, [59](#)

Anacardiaceae, [68](#)

Anagallis, [87](#)

Anaphalis, [120](#)

Andromeda, [86](#)

Andropogon, [5](#)

Androsace, [87](#)

Anemone, [38](#), [39](#)

Anemone, Rue, [39](#)

Anemone, Wood, [38](#)

Anemonella, [39](#)

Angelica, [82](#)

Anonaceae, [40](#)

Antennaria, [120](#)

Anthemis, [130](#)

Antirrhinum, [103](#)

Anychia, [31](#)

Apios, [58](#)

Aplectrum, [16](#)

Apocynaceae, [90](#)

Apocynum, [90](#)

Appalachian Cherry, [54](#)

Apple, [54](#)

Apple of Peru, [101](#)

Aquifoliaceae, [69](#)

Aquilegia, [36](#)

Arabis, [46](#), [47](#)

Araceae, [8](#)

Aralia, [80](#)

Araliaceae, [80](#)

Arbutus, Trailing, [85](#)

Arceuthobium, [25](#)

Arctium, [117](#)

Arctostaphylos, [85](#)

Arenaria, [32](#)

Arethusa, [16](#)

Arisaema, [8](#)

Aristolochia, [25](#)

Aristolochiaceae, [25](#)

Arnica, [123](#)

Arrow Arum, [9](#)

Arrow Grass, [3](#)

Arrow Grass Family, [3](#)

Arrow-head, [3](#)

Arrow Wood, [111](#)

Artemisia, [119](#)

Artichoke, Jerusalem, [129](#)

Arum Family, [8](#)

Asarum, [25](#)

Asclepias, [90](#), [91](#)

Asclepiadaceae, [90](#)

Ash, [88](#)

Ash, Mountain, [52](#)

Ash, Prickly, [65](#)

Asimina, [40](#)

Asparagus, [10](#)

Aspen, [19](#)

Aster, [132-134](#)

Atriplex, [29](#)

Avena, [6](#)

Avens, [56](#), [57](#)

Baby's Breath, [34](#)

Balm of Gilead, [19](#)

Balsam, [1](#)

Balsam Poplar, [19](#)

Balsaminaceae, [71](#)

Baneberry, [39](#)

Baptisia, [61](#)

Barbarea, [44](#)

Barberry, [40](#)

Barberry Family, [40](#)

Barnyard Grass, [5](#)

Barren Strawberry, [55](#)

Bartonia, [88](#)

Basil, [97](#)

Basil-thyme, [97](#)

Basswood, [72](#)

Bayberry, [21](#)

Beach Grass, [4](#)

Beach Pea, [59](#)

Beaked Hazel, [22](#)

Bearberry, [85](#)

Beard Grass, [5](#)

Beard-tongue, [104](#)

Bedstraw, [108](#), [109](#)

Beech, [22](#)

Beech Drops, [84](#), [106](#)

Beech Family, [22](#)

Beggar Lice, [94](#)

Beggar Ticks, [124](#)

Bellflower, [112](#)

Bellflower Family, [112](#)

Bellwort, [12](#)

Benzoin, [41](#)

Berberidaceae, [40](#)

Berberis, [40](#)

Bergamot Mint, [98](#)

Berula, [81](#)

Betula, [21](#), [22](#)

Betulaceae, [21](#)

Bidens, [123](#), [124](#)

Bilberry, [86](#), [87](#)

Bindweed, [91](#), [92](#)

Bindweed, Black, [28](#)

Birch, [21](#), [22](#)

Birch Family, [21](#)

Bird-foot Violet, [75](#)

Birthwort Family, [25](#)

Bishop's Cap, [51](#)

Bistort, [28](#)

Bitter Cress, [45](#), [47](#)

Bitter Dock, [26](#)

Bitter Nut, [21](#)

Bittersweet, [100](#)

Bitter-sweet, [69](#)

Black Alder, [69](#)

Black Ash, [88](#)

Blackberry, [53](#)

Black Bindweed, [28](#)

Black Cherry, [54](#)

Black Currant, [49](#), [50](#)

Black-eyed Susan, [128](#)

Black Haw, [111](#)

Black Jack Oak, [22](#)

Black Locust, [58](#)

Black Maple, [70](#)

Black Medick, [60](#)

Black Mustard, [44](#)

Black Oak, [23](#)

Black Raspberry, [53](#)

Black Snakeroot, [82](#)

Black Spruce, [1](#)

Black Swallow-wort, [90](#)

Black Walnut, [21](#)

Black Willow, [20](#)

Bladder Campion, [34](#)

Bladder Nut, [69](#)

Bladder Nut Family, [69](#)

Bladderwort, [106](#)

Bladderwort Family, [105](#)

Blazing Star, [13](#), [121](#)

Blephilia, [96](#)

Blite, [29](#)

Bloodroot, [41](#)

Bloody Dock, [26](#)

Blue Ash, [88](#)

Bluebell, [93](#), [94](#)

Blueberry, [86](#)

Blue Cohosh, [40](#)

Blue-eyed Grass, [15](#)

Blue-eyed Mary, [104](#)

Blue Flag, [15](#)

Blue Grass, [7](#)

Blue Hearts, [104](#)

Blue-joint, [5](#)

Bluets, [108](#)

Blue Violet, [76](#)

Blueseed, [94](#)

Boehmeria, [24](#)

Bog Rosemary, [86](#)

Boltonia, [131](#)

Boneset, False, [122](#)

Borage, [93](#)

Borage Family, [93](#)

Boraginaceae, [93](#)

Borago, [93](#)

Bowman's Root, [57](#)

Box Elder, [70](#)

Brasenia, [35](#)

Brassica, [43](#), [44](#)

Brauneria, [131](#)

Braya, [47](#)

Bristly Locust, [58](#)

Bristly Sarsaparilla, [80](#)

Brome-grass, [7](#)

Bromus, [7](#)

Broom-rape Family, [106](#)

Buchnera, [104](#)

Buckbean, [88](#)

Buckeye, [70](#)

Buckhorn, [107](#)

Buckthorn, [71](#)

Buckthorn Family, [71](#)

Buckwheat, [27](#)

Buckwheat, False, [28](#)

Buckwheat Family, [25](#)

Buffalo Berry, [77](#)

Buffalo Bur, [100](#)

Buffalo Clover, [61](#)

Bugbane, [39](#)

Bugle, [98](#)

Bugle Weed, [95](#)

Bug-seed, [28](#)

Bulrush, [7](#)

Bur Clover, [60](#)

Bur Cucumber, [112](#)

Burdock, [117](#)

Bur Marigold, [124](#)

Burnet, [57](#)

Bur Oak, [23](#)

Bur-reed, [2](#)

Bur-reed Family, [2](#)

Bush Clover, [61](#), [63](#)

Bush Honeysuckle, [110](#)

Butter-and-eggs, [102](#)

Buttercup, [36](#), [37](#), [38](#)

Butterfly Weed, [91](#)

Butternut, [21](#)

Butterwort, [105](#)

Button Bush, [108](#)

Cacalia, [122](#)

Cactaceae, [77](#)

Cactus Family, [77](#)

Cakile, [45](#)

Calamagrostis, [5](#)

Calamint, [97](#)

Calla, [9](#)

Callirhoe, [73](#)

Callitrichaceae, [68](#)

Callitriche, [68](#)

Calopogon, [16](#)

Caltha, [36](#)

Calypso, [17](#)

Camassia, [14](#)

Camelina, [42](#)

Camomile, [130](#)

Campanula, [112](#)

Campanulaceae, [112](#)

Campion, [33](#), [34](#)

Canada Thistle, [118](#)

Canada Violet, [76](#)

Canadian Blue Grass, [7](#)

Cancer-root, [106](#)

Cannabis, [24](#)

Caper Family, [47](#)

Capparidaceae, [47](#)

Caprifoliaceae, [109](#)

Capsella, [46](#)

Caraway, [83](#)

Cardamine, [45](#), [47](#)

Cardinal Flower, [113](#)

Carduus, [118](#)

Carex, [8](#)

Carpet-weed, [31](#)

Carpet-weed Family, [31](#)

Carpinus, [21](#)

Carrion-flower, [11](#)

Carrot, Wild, [82](#)

Carum, [83](#)

Carya, [21](#)

Caryophyllaceae, [31](#)

Cashew Family, [68](#)

Cassia, [60](#)

Castalia, [35](#)

Castanea, [22](#)

Castilleja, [102](#)

Catchfly, [33](#), [34](#)

Catnip, [98](#)

Cat-tail, [2](#)

Cat-tail Family, [2](#)

Caulophyllum, [40](#)

Ceanothus, [71](#)

Cedar, [1](#)

Celandine, [41](#)

Celandine Poppy, [41](#)

Celastraceae, [69](#)

Celastrus, [69](#)

Celtis, [24](#)

Cenchrus, [5](#)

Centaurea, [121](#)

Centaureum, [89](#)

Centaury, [89](#)

Cephalanthus, [108](#)

Cerastium, [33](#)

Ceratophyllaceae, [34](#)

Ceratophyllum, [34](#)

Cercis, [58](#)

Chaerophyllum, [83](#)

Chamaedaphne, [86](#)

Chamaelirium, [13](#)

Charlock, [44](#)

Cheat, [7](#)

Chelidonium, [41](#)

Chelone, [104](#)

Chenopodiaceae, [28](#)

Chenopodium, [28](#), [29](#)

Cherry, [54](#)

Cherry, Ground, [101](#)

Chervil, [83](#)

Chestnut, [22](#)

Chickweed, [32](#), [33](#)

Chickweed, Mouse-ear, [33](#)

Chicory, [115](#)

Chimaphila, [85](#)

Chiogenes, [85](#)

Chives, Wild, [13](#)

Chokeberry, [55](#)

Choke Cherry, [54](#)

Chrysanthemum, [121](#), [130](#), [131](#)

Chrysosplenium, [50](#)

Cichorium, [115](#)

Cicuta, [82](#)

Cimicifuga, [39](#)

Cinquefoil, [52](#), [56](#), [57](#)

Circaea, [78](#)

Cirsium, [117](#), [118](#)

Cistaceae, [74](#)

Cladium, [8](#)

Clammy Locust, [58](#)

Clammy-weed, [47](#)

Claytonia, [34](#)

Clearweed, [24](#)

Cleft Phlox, [92](#)

Clematis, [35](#)

Climbing Fumitory, [41](#)

Climbing Rose, [52](#)

Clintonia, [13](#)

Clover, [61](#)

Clover, Bush, [61](#), [63](#)

Clover, Hop, [60](#)

Clover, Prairie, [60](#)

Clover, Sweet, [61](#)

Cocklebur, [114](#)

Cockle, Corn, [33](#)

Coffee-tree, [58](#)

Cohosh, Blue, [40](#)

Colic-root, [13](#)

Collinsia, [104](#)

Collinsonia, [95](#)

Coltsfoot, [122](#), [130](#)

Columbine, [36](#)

Comandra, [24](#)

Comfrey, [93](#), [94](#)

Commelina, [9](#)

Commelinaceae, [9](#)

Common Blue Violet, [76](#)

Common Cat-tail, [2](#)

Common Vetch, [59](#)

Compass Plant, [127](#)

Compositae, [113](#)

Composite Family, [113](#)

Coneflower, Gray-headed, [128](#)

Coneflower, Purple, [131](#)

Conioselinum, [83](#)

Conium, [83](#)

Conopholis, [106](#)

Conringia, [42](#)

Convolvulaceae, [91](#)

Convolvulus, [91](#), [92](#)

Coptis, [39](#)

Corallorhiza, [16](#)

Coral Root, [16](#)

Coreopsis, [123](#), [124](#)

Corispermum, [28](#)

Cork Elm, [24](#)

Cornaceae, [83](#)

Corn Cockle, [33](#)

Corn Flower, [121](#)

Corn Gromwell, [94](#)

Corn Salad, [111](#)

Cornus, [83](#), [84](#)

Corydalis, [42](#)

Corylus, [22](#)

Costmary, [121](#)

Cotton Grass, [8](#)

Cotton Thistle, [118](#)

Cottonwood, [19](#)

Cowbane, [81](#)

Cowherb, [34](#)

Cow Parsnip, [82](#)

Cowslip, [36](#)

Cow Wheat, [104](#)

Crab, [54](#)

Crab Grass, [5](#)

Crack Willow, [20](#)

Cranberry, [85](#)

Cranberry Tree, [111](#)

Crane-fly Orchis, [16](#)

Crane's-bill, [64](#), [65](#)

Crassulaceae, [48](#)

Creeping Cedar, [1](#)

Creeping Wahoo, [69](#)

Crepis, [117](#)

Cress, [43-47](#)

Cress, Bitter, [45](#), [47](#)

Cress, Field, [46](#)

Cress, Garden, [45](#), [46](#)

Cress, Lake, [45](#), [46](#)

Cress, Mouse-ear, [47](#)

Cress, Penny, [46](#)

Cress, Rock, [46](#), [47](#)

Cress, Water, [45](#)

Cress, Winter, [44](#)

Cress, Yellow, [43](#)

Crotalaria, [59](#)

Crowberry, [68](#)

Crowberry Family, [68](#)

Crowfoot, [35](#)

Crowfoot, Cursed, [37](#)

Crowfoot Family, [35](#)

Crowfoot, Sea-side, [36](#)

Crowfoot, Small-flowered, [36](#)

Crowfoot, Water, [35](#), [37](#)

Cruciferae, [42](#)

Cryptotaenia, [82](#)

Cuckoo Flower, [45](#)

Cucurbitaceae, [112](#)

Cudweed, [120](#), [121](#)

Culver's Root, [104](#)

Cup Plant, [123](#)

Currant, [49](#), [50](#)

Currant, Indian, [110](#)

Cursed Crowfoot, [37](#)

Cuscuta, [92](#)

Custard Apple Family, [40](#)

Cut-grass, [5](#)

Cycloloma, [29](#)

Cynanchum, [90](#)

Cynoglossum, [93](#), [94](#)

Cynthia, [115](#)

Cyperaceae, [7](#)

Cyperus, [8](#)

Cypress Spurge, [67](#)

Cypripedium, [18](#)

Dactylis, [6](#)

Daisy, Ox-eye, [131](#)

Dalibarda, [56](#)

Dame's Rocket, [47](#)

Dandelion, [115](#)

Dandelion, Dwarf, [115](#)

Datura, [100](#)

Daucus, [82](#)

Day-flower, [9](#)

Day Lily, [11](#)

Dead Nettle, [99](#)

Decodon, [77](#)

Deerberry, [86](#)

Dentaria, [44](#)

Deptford Pink, [34](#)

Desmodium, [62](#), [63](#)

Devil's Club, [80](#)

Dewberry, [53](#)

Dianthera, [107](#)

Dianthus, [34](#)

Dicentra, [41](#)

Diervilla, [110](#)

Digitaria, [5](#)

Dioscorea, [14](#)

Dioscoreaceae, [14](#)

Diploaxis, [42](#), [43](#)

Dipsacaceae, [112](#)

Dipsacus, [112](#)

Dirca, [77](#)

Ditch Stonecrop, [49](#)

Dock, [26](#)

Dock, Prairie, [122](#)

Dodder, [92](#)

Dodecatheon, [87](#)

Dogbane, [90](#)

Dogbane Family, [90](#)

Dog Fennel, [130](#)

Dog Rose, [52](#)

Dog's-tooth Violet, [12](#)

Dog Violet, [76](#)

Dogwood, [83](#), [84](#)

Dogwood Family, [83](#)

Downy Mint, [78](#)

Draba, [42](#), [46](#)

Dracocephalum, [96](#)

Dragon Head, [96](#)

Dragon Head, False, [99](#)

Dragon Root, [8](#)

Drop-seed, [5](#)

Drosera, [48](#)

Droseraceae, [48](#)

Duckweed, [9](#)

Duckweed Family, [9](#)

Dulichium, [7](#)

Dutchman's Breeches, [41](#)

Dwarf Birch, [22](#)

Dwarf Dandelion, [115](#)

Dwarf Dogwood, [83](#)

Dwarf Ginseng, [80](#)

Dwarf Iris, [15](#)

Dwarf Mistletoe, [25](#)

Dwarf Raspberry, [53](#)

Dwarf Water Plantain, [3](#)

Dwarf White Trillium, [12](#)

Dyer's Greenweed, [58](#)

Echinochloa, [5](#)

Echinocystis, [112](#)

Echinodorus, [3](#)

Echinops, [117](#)

Echium, [94](#)

Eel Grass, [3](#)

Elaeagnaceae, [77](#)

Elatinaceae, [74](#)

Elatine, [74](#)

Elder, [109](#)

Elder, Box, [70](#)

Elder, Marsh, [114](#)

Elecampane, [127](#)

Eleocharis, [7](#)

Eleusine, [6](#)

Elm, [24](#)

Elodea, [3](#)

Elymus, [6](#)

Empetraceae, [68](#)

Empetrum, [68](#)

Enchanter's Nightshade, [78](#)

English Plantain, [107](#)

Epifagus, [106](#)

Epigaea, [85](#)

Epilobium, [79](#)

Epipactis, [17](#)

Eragrostis, [7](#)

Erechtites, [122](#)

Ericaceae, [84](#)

Erigenia, [81](#)

Erigeron, [130-132](#)

Eriocaulaceae, [9](#)

Eriocaulon, [9](#)

Eriophorum, [8](#)

Erodium, [64](#)

Eryngium, [80](#)

Erysimum, [43](#)

Erythronium, [12](#)

Eupatorium, [121](#), [122](#)

Euphorbia, [67](#), [68](#)

Euphrasia, [102](#)

Evening Primrose, [78](#)

Evening Primrose Family, [78](#)

Everlasting, [120](#)

Everlasting, Pearly, [120](#)

Evonymus, [69](#)

Eyebright, [102](#)

Fagaceae, [22](#)

Fagopyrum, [27](#)

Fagus, [22](#)

Fall Dandelion, [115](#)

False Asphodel, [14](#)

False Boneset, [122](#)

False Buckwheat, [28](#)

False Dragon Head, [99](#)

False Flax, [42](#)

False Foxglove, [102](#)

False Gromwell, [93](#)

False Heather, [74](#)

False Loosestrife, [78](#)

False Mermaid, [68](#)

False Mermaid Family, [68](#)

False Mitrewort, [51](#)

False Nettle, [24](#)

False Pimpernel, [104](#)

False Solomon's Seal, [14](#)

Fatsia, [80](#)

Fennel, [81](#)

Fennel, Dog, [130](#)

Fescue Grass, [7](#)

Festuca, [7](#)

Feverfew, [130](#)

Feverwort, [109](#)

Field Cress, [46](#)

Field Garlic, [13](#)

Figwort, [103](#)

Figwort Family, [101](#)

Filipendula, [57](#)

Fire Pink, [33](#)

Fireweed, [79](#), [122](#)

Five-finger, [55](#)

Flax, [63](#)

Flax, False, [42](#)

Flax Family, [63](#)

Fleabane, [131](#), [132](#)

Floating Foxtail, [4](#)

Floating Heart, [88](#)

Floerkea, [68](#)

Flowering Dogwood, [83](#)

Flowering Raspberry, [54](#)

Flowering Wintergreen, [65](#)

Flower-of-an-hour, [72](#)

Foeniculum, [81](#)

Fog Fruit, [99](#)

Forget-me-not, [94](#)

Forked Chickweed, [31](#)

Four-o'Clock Family, [31](#)

Foxglove, False, [102](#)

Fox Grape, [72](#)

Foxtail, [4](#)

Foxtail, Floating, [4](#)

Fragaria, [56](#)

Frasera, [88](#)

Fraxinus, [88](#)

Fringed Gentian, [89](#)

Fringed Orchis, [19](#)

Frog's Bit Family, [3](#)

Frost Grape, [72](#)

Frostweed, [74](#)

Fumaria, [42](#)

Fumariaceae, [41](#)

Fumitory, [42](#)

Fumitory, Climbing, [41](#)

Fumitory Family, [41](#)

Galeopsis, [99](#)

Gale, Sweet, [21](#)

Galinsoga, [130](#)

Galium, [108](#), [109](#)

Garden Cress, [45](#), [46](#)

Garden Phlox, [92](#)

Garlic, Field, [13](#)

Gaultheria, [86](#)

Gaura, [79](#)

Gaylussacia, [86](#)

Genista, [58](#)

Gentian, [89](#)

Gentiana, [89](#)

Gentianaceae, [88](#)

Gentian Family, [88](#)

Gentian, Spurred, [89](#)

Geraniaceae, [64](#)

Geranium, [64](#), [65](#)

Geranium Family, [64](#)

Gerardia, [102](#), [103](#)

Geum, [56](#), [57](#)

Giant Hyssop, [97](#)

Giant Ragweed, [114](#)

Gillenia, [57](#)

Ginseng, [80](#)

Gleditsia, [58](#)

Globe-flower, [36](#)

Globe Thistle, [117](#)

Glyceria, [7](#)

Gnaphalium, [120](#), [121](#)

Goat's Rue, [60](#)

Golden Alexander, [81](#)

Golden Currant, [49](#)

Golden Glow, [128](#)

Goldenrod, [125-127](#)

Golden Saxifrage, [50](#)

Golden Seal, [38](#)

Gold-thread, [39](#)

Good King Henry, [29](#)

Gooseberry, [49](#)

Goosefoot, [28](#), [29](#)

Goosefoot Family, [28](#)

Gourd Family, [112](#)

Gramineae, [4](#)

Grape, [72](#)

Grape Family, [71](#)

Grape Hyacinth, [14](#)

Grass Family, [4](#)

Grass of Parnassus, [50](#)

Grass, Star, [14](#)

Gratiola, [104](#)

Gray-headed Coneflower, [128](#)

Great Lobelia, [113](#)

Great Solomon's Seal, [11](#)

Great-spurred Violet, [75](#)

Greek Valerian, [92](#)

Green Ash, [88](#)

Green Brier, [11](#)

Green Foxtail, [4](#)

Green Milkweed, [90](#)

Green Sorrel, [25](#)

Green Violet, [75](#)

Grindelia, [128](#)

Ground Cherry, [101](#)

Ground Hemlock, [2](#)

Ground Ivy, [98](#)

Groundsel, [128](#)

Gum Plant, [128](#)

Gymnocladus, [58](#)

Gypsophila, [34](#)

Gypsophyll, [34](#)

Habenaria, [18](#), [19](#)

Hackberry, [24](#)

Hair Grass, [6](#)

Halenia, [89](#)

Haloragidaceae, [79](#)

Hamamelidaceae, [51](#)

Hamamelis, [51](#)

Hand-leaf Violet, [75](#)

Harbinger of Spring, [81](#)

Hardhack, [53](#)

Harebell, [112](#)

Hare's Ear, [42](#)

Hawksbeard, [117](#)

Hawkweed, [116](#), [117](#)

Hazel, [22](#)

Heather, False, [74](#)

Heath Family, [84](#)

Hedeoma, [95](#)

Hedge Hyssop, [104](#)

Hedge Mustard, [44](#)

Hedge Nettle, [99](#)

Helenium, [127](#), [128](#)

Helianthemum, [74](#)

Helianthus, [128-130](#)

Heliopsis, [123](#)

Hemerocallis, [11](#)

Hemlock, [1](#)

Hemlock, Ground, [2](#)

Hemlock Parsley, [83](#)

Hemlock, Poison, [83](#)

Hemlock, Water, [82](#)

Hemp, [24](#)

Hemp, Indian, [90](#)

Hemp Nettle, [99](#)

Hemp, Water, [30](#)

Hemp Weed, [131](#)

Henbane, [101](#)

Hepatica, [38](#)

Heracleum, [82](#)

Herb Robert, [64](#)

Herb Sophia, [43](#)

Hercules' Club, [80](#)

Hesperis, [47](#)

Heteranthera, [10](#)

Heuchera, [50](#), [51](#)

Hibiscus, [72](#)

Hickory, [21](#)

Hieracium, [116](#), [117](#)

Hill's Oak, [23](#)

Hippuris, [79](#)

Hoarhound, [96](#)

Hoarhound, Water, [95](#)

Hobble-bush, [111](#)

Hog Peanut, [59](#)

Holly Family, [69](#)

Holly, Mountain, [69](#)

Honewort, [82](#)

Honey Locust, [58](#)

Honeysuckle, [110](#)

Honeysuckle, Bush, [110](#)

Honeysuckle Family, [109](#)

Hop, [24](#)

Hop Clover, [60](#)

Hop Tree, [65](#)

Hordeum, [4](#)

Hornbeam, [21](#)

Horned Pondweed, [2](#)

Hornwort, [34](#)

Hornwort Family, [34](#)

Horse Balm, [95](#)

Horse Chestnut, [70](#)

Horse Mint, [96](#)

Horse Nettle, [100](#)

Horse Radish, [46](#)

Horse Weed, [130](#), [131](#)

Hound's Tongue, [93](#)

Houstonia, [108](#)

Huckleberry, [86](#)

Hudsonia, [74](#)

Humulus, [24](#)

Hyacinth, Wild, [14](#)

Hybanthus, [75](#)

Hydrocharitaceae, [3](#)

Hydrocotyle, [80](#)

Hydrophyllaceae, [93](#)

Hydrophyllum, [93](#)

Hyoscyamus, [101](#)

Hypericaceae, [73](#)

Hypericum, [73](#), [74](#)

Hypoxis, [14](#)

Hyssop, [98](#)

Hyssop, Hedge, [104](#)

Hyssopus, [98](#)

Ilex, [69](#)

Illecebraceae, [31](#)

Ilysanthes, [104](#)

Impatiens, [71](#)

Imperatoria, [82](#)

Indian Cucumber-root, [13](#)

Indian Currant, [110](#)

Indian Hemp, [90](#)

Indian Mustard, [44](#)

Indian Pipe, [84](#)

Indian Plantain, [122](#)

Indian Tobacco, [113](#)

Indian Turnip, [8](#)

Indigo, Wild, [61](#)

Inula, [127](#)

Ipomoea, [92](#)

Iridaceae, [15](#)

Iris, [15](#)

Iris Family, [15](#)

Ironweed, [122](#)

Ironweed, Yellow, [128](#)

Ironwood, [22](#)

Isopyrum, [39](#)

Iva, [114](#)

Jack Pine, [1](#)

Jeffersonia, [40](#)

Jerusalem Artichoke, [129](#)

Jerusalem Oak, [29](#)

Jimson Weed, [100](#)

Joe-Pye Weed, [122](#)

Jointweed, [27](#)

Juglandaceae, [21](#)

Juglans, [21](#)

Juncaceae, [10](#)

Juncaginaceae, [3](#)

Juncus, [10](#)

Juneberry, [55](#)

Juniper, [1](#)

Juniperus, [1](#)

Kalmia, [85](#)

Kenilworth Ivy, [103](#)

Kentucky Coffee-tree, [58](#)

King-nut Hickory, [21](#)

Knapweed, [121](#)

Knawel, [31](#)

Knotweed, [26](#), [27](#)

Knotwort Family, [31](#)

Kochia, [28](#)

Koeleria, [6](#)

Krigia, [115](#)

Kuhnia, [122](#)

Labiatae, [95](#)

Labrador Tea, [85](#)

Lactuca, [115](#), [116](#)

Ladies' Tresses, [16](#)

Lady's Slipper, [18](#)

Lake Cress, [45](#), [46](#)

Lamb's Quarters, [29](#)

Lamium, [99](#)

Laportea, [24](#)

Lappula, [94](#)

Lapsana, [115](#)

Large-toothed Aspen, [19](#)

Larix, [1](#)

Larkspur, [36](#)

Lathyrus, [59](#)

Lauraceae, [41](#)

Laurel Family, [41](#)

Lead Plant, [58](#)

Leafcup, [123](#)

Leatherleaf, [86](#)

Leatherwood, [77](#)

Lechea, [74](#), [75](#)

Ledum, [85](#)

Leek, Wild, [13](#)

Leersia, [5](#)

Leguminosae, [58](#)

Lemna, [9](#)

Lemnaceae, [9](#)

Lentibulariaceae, [105](#)

Leontodon, [115](#)

Leonurus, [98](#)

Lepachys, [128](#)

Lepidium, [45](#), [46](#)

Lespedeza, [61](#), [63](#)

Lettuce, [115](#), [116](#)

Lettuce, Wild, [115](#), [116](#)

Liatris, [121](#)

Lilac, [88](#)

Liliaceae, [10](#)

Lilium, [12](#)

Lily, [12](#)

Lily Family, [10](#)

Lily of the Valley, Wild, [12](#)

Lily, Pond, [35](#)

Lily, Water, [35](#)

Limnanthaceae, [68](#)

Limosella, [103](#)

Linaceae, [63](#)

Linaria, [102](#), [103](#)

Linden Family, [72](#)

Linnaea, [109](#)

Linum, [63](#)

Liparis, [17](#)

Lippia, [99](#)

Liriodendron, [40](#)

Listera, [17](#)

Lithospermum, [94](#)

Live-for-ever, [49](#)

Lizard's Tail, [19](#)

Lobelia, [113](#)

Lobeliaceae, [113](#)

Lobelia Family, [113](#)

Locust, [58](#)

Locust, Honey, [58](#)

Lombardy Poplar, [19](#)

Long-spurred Violet, [76](#)

Lonicera, [110](#)

Loosestrife, [77](#), [87](#), [88](#)

Loosestrife, False, [78](#)

Loosestrife Family, [77](#)

Lopseed, [107](#)

Lopseed Family, [107](#)

Loranthaceae, [25](#)

Lotus, [35](#)

Lousewort, [102](#)

Love Grass, [7](#)

Love-in-a-mist, [38](#)

Low Juniper, [1](#)

Ludvigia, [78](#)

Lupine, [59](#)

Lupinus, [59](#)

Luzula, [10](#)

Lychnis, [33](#)

Lysimachia, [87](#), [88](#)

Lycium, [100](#)

Lycopus, [95](#)

Lythraceae, [77](#)

Lythrum, [77](#)

Madder Family, [108](#)

Mad-dog Skullcap, [96](#)

Magnoliaceae, [40](#)

Magnolia Family, [40](#)

Maianthemum, [13](#)

Mallow, [73](#)

Mallow Family, [72](#)

Mallow, Marsh, [72](#)

Mallow, Poppy, [73](#)

Mallow, Rose, [72](#)

Mallow, Virginia, [73](#)

Malva, [73](#)

Malvaceae, [72](#)

Manna Grass, [7](#)

Maple, [70](#)

Maple Family, [70](#)

Mare's-tail, [79](#)

Marigold, Bur, [124](#)

Marigold, Water, [123](#)

Marrubium, [96](#)

Marsh Cinquefoil, [57](#)

Marsh Elder, [114](#)

Marsh Harebell, [112](#)

Marsh Mallow, [72](#)

Marsh Pea, [59](#)

Marsh Speedwell, [105](#)

Masterwort, [82](#)

Matricaria, [130](#)

Matrimony Vine, [100](#)

May Apple, [40](#)

Meadow Beauty, [77](#)

Meadow Parsnip, [81](#)

Meadow Pink, [34](#)

Meadow Rue, [39](#)

Meadow Salsify, [115](#)

Meadow-sweet, [53](#)

Medeola, [13](#)

Medicago, [60](#), [61](#)

Melampyrum, [104](#)

Melastomaceae, [77](#)

Melastoma Family, [77](#)

Melilotus, [61](#)

Menispermaceae, [40](#)

Menispermum, [40](#)

Mentha, [98](#)

Menyanthes, [88](#)

Mermaid Weed, [79](#)

Mertensia, [93](#), [94](#)

Mexican Tea, [29](#)

Mezereum Family, [77](#)

Microstylis, [17](#)

Mignonette Family, [48](#)

Mignonette, Yellow, [48](#)

Mikania, [121](#)

Milfoil, Water, [79](#)

Milk Thistle, [117](#)

Milk Vetch, [60](#)

Milkweed, [90](#), [91](#)

Milkweed Family, [90](#)

Milkweed, Green, [90](#)

Milkwort, [66](#)

Milkwort Family, [65](#)

Millet, [4](#)

Mimulus, [102](#), [104](#)

Mint, [98](#)

Mint Family, [95](#)

Mist Flower, [121](#)

Mistletoe, Dwarf, [25](#)

Mistletoe Family, [25](#)

Mitchella, [108](#)

Mitella, [51](#)

Mitrewort, False, [51](#)

Mocker-nut Hickory, [21](#)

Mollugo, [31](#)

Monarda, [96](#)

Moneses, [85](#)

Moneywort, [87](#)

Monkey Flower, [104](#)

Monkey Flower, Yellow, [102](#)

Monotropa, [84](#)

Moonseed, [40](#)

Moonseed Family, [40](#)

Morning Glory, [92](#)

Morning Glory Family, [91](#)

Morus, [24](#)

Moss Pink, [92](#)

Motherwort, [98](#)

Moth Mullein, [101](#)

Mountain Alder, [22](#)

Mountain Ash, [52](#)

Mountain Holly, [69](#)

Mountain Maple, [70](#)

Mountain Mint, [98](#)

Mouse-ear Chickweed, [33](#)

Mouse-ear Cress, [47](#)

Mud Plantain, [10](#)

Mudwort, [103](#)

Mugwort, [119](#)

Muhlenbergia, [5](#), [6](#)

Mulberry, [24](#)

Mullein, [101](#)

Mullein Pink, [33](#)

Muscari, [14](#)

Musk Flower, [102](#)

Mustard, [43](#), [44](#)

Mustard Family, [42](#)

Myosotis, [94](#)

Myrica, [20](#), [21](#)

Myricaceae, [20](#)

Myriophyllum, [79](#)

Naiad, [2](#)

Naias, [2](#)

Najadaceae, [2](#)

Narrow-leaved Cat-tail, [2](#)

Nelumbo, [35](#)

Nemopanthus, [69](#)

Nepeta, [98](#)

Nettle, [24](#)

Nettle, Dead, [99](#)

Nettle Family, [23](#)

Nettle, Hedge, [99](#)

Nettle, Hemp, [99](#)

Nettle, Horse, [100](#)

Nicandra, [101](#)

Nicotiana, [101](#)

Nigella, [38](#)

Nightshade, [100](#)

Nightshade, Enchanter's, [78](#)

Nightshade Family, [100](#)

Ninebark, [53](#)

Nipplewort, [115](#)

Nodding Pogonia, [18](#)

Norway Pine, [1](#)

Nut Grass, [8](#)

Nyctaginaceae, [31](#)

Nymphaea, [35](#)

Nymphaeaceae, [35](#)

Nymphoides, [88](#)

Nyssa, [83](#)

Oak, [22](#), [23](#)

Oakesia, [12](#)

Oats, [6](#)

Oenothera, [78](#)

Oleaceae, [88](#)

Oleaster Family, [77](#)

Olive Family, [88](#)

Onagraceae, [78](#)

One-flowered Wintergreen, [85](#)

Onion, Wild, [13](#)

Onosmodium, [93](#)

Opuntia, [77](#)

Orache, [29](#)

Orchard Grass, [6](#)

Orchidaceae, [15](#)

Orchis, [17](#)

Orchis, Crane-fly, [16](#)

Orchis Family, [15](#)

Orchis, Showy, [17](#)

Ornithogalum, [14](#)

Orobanchaceae, [106](#)

Orobanche, [106](#)

Orpine Family, [48](#)

Osmorhiza, [82](#)

Ostrya, [22](#)

Oswego Tea, [96](#)

Oxalidaceae, [64](#)

Oxalis, [64](#)

Ox-eye, [123](#)

Ox-eye Daisy, [131](#)

Oxybaphus, [31](#)

Oxypolis, [81](#)

Painted Cup, [102](#)

Painted Trillium, [12](#)

Pale Violet, [76](#)

Panax, [80](#)

Panic-grass, [5](#)

Panicum, [5](#)

Pansy, [76](#)

Papaveraceae, [41](#)

Papaver, [41](#)

Papaw, [40](#)

Paper Birch, [21](#)

Parietaria, [24](#)

Parnassia, [50](#)

Parsley Family, [80](#)

Parsley, Hemlock, [83](#)

Parsnip, Cow, [82](#)

Parsnip, Meadow, [81](#)

Parsnip, Prairie, [81](#)

Parsnip, Water, [81](#)

Parsnip, Wild, [81](#)

Partridge Berry, [108](#)

Partridge Pea, [60](#)

Pasque Flower, [38](#)

Pastinaca, [81](#)

Patience Dock, [26](#)

Pea, [59](#)

Peach-leaved Willow, [20](#)

Pear, [54](#)

Pear, Prickly, [77](#)

Pearlwort, [32](#)

Pearly Everlasting, [120](#)

Pedicularis, [102](#)

Pellitory, [24](#)

Peltandra, [9](#)

Penny Cress, [46](#)

Pennyroyal, [95](#)

Pennywort, Water, [80](#)

Penthorum, [49](#)

Pentstemon, [104](#)

Pepper Family, [19](#)

Pepper Grass, [46](#)

Peppermint, [98](#)

Perfumed Cherry, [54](#)

Periwinkle, [90](#)

Petalostemum, [60](#)

Petasites, [130](#)

Petunia, [101](#)

Phacelia, [93](#)

Phleum, [4](#)

Phlox, [92](#)

Phragmites, [6](#)

Phryma, [107](#)

Phrymaceae, [107](#)

Physalis, [101](#)

Physocarpus, [53](#)

Physostegia, [99](#)

Phytolacca, [30](#)

Phytolaccaceae, [30](#)

Picea, [1](#)

Pickrel-weed, [10](#)

Pickrel-weed Family, [10](#)

Picris, [115](#)

Pignut Hickory, [21](#)

Pigweed, [30](#)

Pilea, [24](#)

Pimpernel, [87](#)

Pimpernel, False, [104](#)

Pimpernel, Water, [87](#)

Pinaceae, [1](#)

Pin Cherry, [54](#)

Pine, [1](#)

Pine Drops, [84](#)

Pine Family, [1](#)

Pineweed, [74](#)

Pinguicula, [105](#)

Pink, Deptford, [34](#)

Pink Family, [31](#)

Pink, Fire, [33](#)

Pink, Meadow, [34](#)

Pink, Moss, [92](#)

Pink, Mullein, [33](#)

Pink, Rose, [89](#)

Pin Oak, [23](#)

Pinus, [1](#)

Pinweed, [74](#), [75](#)

Piperaceae, [19](#)

Pipewort, [9](#)

Pipewort Family, [9](#)

Pitcher Plant, [48](#)

Pitcher Plant Family, [48](#)

Plane Tree Family, [51](#)

Plantaginaceae, [107](#)

Plantago, [107](#)

Plantain, [107](#)

Plantain Family, [107](#)

Plantain, Indian, [122](#)

Plantain, Mud, [10](#)

Platanaceae, [51](#)

Platanus, [51](#)

Plum, [54](#)

Poa, [7](#)

Podophyllum, [40](#)

Podostemaceae, [48](#)

Podostemum, [48](#)

Pogonia, [18](#)

Poison Hemlock, [83](#)

Poison Ivy, [69](#)

Poison Sumach, [68](#)

Pokeweed, [30](#)

Pokeweed Family, [30](#)

Polanisia, [47](#)

Polemoniaceae, [92](#)

Polemonium, [92](#)

Polemonium Family, [92](#)

Polygala, [65](#), [66](#)

Polygalaceae, [65](#)

Polygonatum, [11](#)

Polygonaceae, [25](#)

Polygonum, [26-28](#)

Polymnia, [123](#)

Polytaenia, [81](#)

Pond Lily, [35](#)

Pondweed, [2](#)

Pondweed Family, [2](#)

Pontederia, [10](#)

Pontederiaceae, [10](#)

Poplar, [19](#)

Poppy, [41](#)

Poppy, Celandine, [41](#)

Poppy Family, [41](#)

Poppy Mallow, [73](#)

Populus, [19](#)

Portulaca, [34](#)

Portulacaceae, [34](#)

Potamogeton, [2](#)

Potentilla, [52](#), [55-57](#)

Prairie Clover, [60](#)

Prairie Dock, [122](#)

Prairie Fringed Orchis, [19](#)

Prairie June-grass, [6](#)

Prairie Parsnip, [81](#)

Prenanthes, [116](#)

Prickly Ash, [65](#)

Prickly Pear, [77](#)

Primrose, [87](#)

Primrose Family, [87](#)

Primula, [87](#)

Primulaceae, [87](#)

Prince's Feather, [27](#)

Prince's Pine, [85](#)

Proserpinaca, [79](#)

Prunella, [97](#)

Prunus, [54](#)

Psedera, [71](#)

Ptelea, [65](#)

Pterospora, [84](#)

Puccoon, [94](#)

Pulse Family, [58](#)

Purple Avena, [57](#)

Purple Coneflower, [131](#)

Purple Fringed Orchid, [19](#)

Purslane, [34](#)

Purslane Family, [34](#)

Purslane, Water, [78](#)

Putty Root, [16](#)

Pyenanthemum, [98](#)

Pyrola, [84](#), [85](#)

Pyrus, [52](#), [54](#), [55](#)

Quack Grass, [6](#)

Quaking Aspen, [19](#)

Quassia Family, [65](#)

Queen of the Prairie, [57](#)

Quercus, [22](#), [23](#)

Radicula, [43](#), [45](#), [46](#)

Radish, [45](#)

Radish, Horse, [46](#)

Radish, Wild, [44](#), [45](#)

Ragged Fringed Orchis, [19](#)

Ragweed, [114](#)

Ragwort, [127](#), [128](#)

Ram's Head Lady's Slipper, [18](#)

Ranunculaceae, [35](#)

Ranunculus, [35-38](#)

Raphanus, [44](#), [45](#)

Raspberry, [53](#), [54](#)

Rattlebox, [59](#)

Rattlesnake Master, [80](#)

Rattlesnake Plantain, [17](#)

Rattlesnake Root, [116](#)

Red Ash, [88](#)

Redbud, [58](#)

Red Cedar, [1](#)

Red Clover, [61](#)

Red Currant, [50](#)

Red Maple, [70](#)

Red Mulberry, [24](#)

Red Oak, [23](#)

Red Raspberry, [53](#)

Red-root, [71](#)

Red Sorrel, [25](#)

Red-top, [6](#)

Reed, [6](#)

Reed Grass, [5](#)

Rein Orchis, [18](#)

Reseda, [48](#)

Resedaceae, [48](#)

Rhamnaceae, [71](#)

Rhamnus, [71](#)

Rhexia, [77](#)

Rhus, [68](#), [69](#)

Ribes, [49](#), [50](#)

River Weed, [48](#)

River Weed Family, [48](#)

Robinia, [58](#)

Rock Cress, [46](#), [47](#)

Rocket, Sea, [45](#)

Rock-rose Family, [74](#)

Rosa, [52](#)

Rosaceae, [51](#)

Rose, [52](#)

Rose Family, [51](#)

Rose Mallow, [72](#)

Rose Pink, [89](#)

Rosin Weed, [123](#)

Round-leaved Orchis, [17](#)

Round-leaved Violet, [75](#)

Rowan Tree, [52](#)

Rubiaceae, [108](#)

Rubus, [53](#), [54](#)

Rudbeckia, [128](#)

Rue Anemone, [39](#)

Rue Family, [65](#)

Ruellia, [107](#)

Rumex, [25](#), [26](#)

Rush, [10](#)

Rush Family, [10](#)

Rush, Twig, [8](#)

Russian Thistle, [28](#)

Rutabaga, [43](#)

Rutaceae, [65](#)

Sabbatia, [89](#)

Sagina, [32](#)

Sagittaria, [3](#)

Salicaceae, [19](#)

Salix, [20](#)

Salmonberry, [54](#)

Salsify, [115](#)

Salsola, [28](#)

Sambucus, [109](#)

Samolus, [87](#)

Sandalwood Family, [24](#)

Sand Bur, [5](#)

Sand Cherry, [54](#)

Sand Rocket, [42](#), [43](#)

Sand Spurrey, [32](#)

Sand Violet, [76](#)

Sanguinaria, [41](#)

Sanguisorba, [57](#)

Sanicula, [82](#)

Santalaceae, [24](#)

Sapindaceae, [70](#)

Saponaria, [34](#)

Sarracenia, [48](#)

Sarraceniaceae, [48](#)

Sarsaparilla, [80](#)

Sarsaparilla Family, [80](#)

Sassafras, [41](#)

Satureja, [97](#)

Saururus, [19](#)

Saxifraga, [50](#)

Saxifragaceae, [49](#)

Saxifrage, [50](#)

Saxifrage Family, [49](#)

Saxifrage, Golden, [50](#)

Scarlet Oak, [23](#)

Scheuchzeria, [3](#)

Schneck's Oak, [23](#)

Scirpus, [7](#)

Scleranthus, [31](#)

Scorpion Grass, [94](#)

Scrophularia, [103](#)

Scrophulariaceae, [101](#)

Scrub Oak, [23](#)

Scutellaria, [96](#)

Sea Rocket, [45](#)

Sea-side Crowfoot, [36](#)

Sedge, [8](#)

Sedge Family, [7](#)

Sedum, [48](#), [49](#)

Self-heal, [97](#)

Seneca Snakeroot, [66](#)

Senecio, [127](#), [128](#)

Senna, Wild, [60](#)

Setaria, [4](#)

Shag-bark Hickory, [21](#)

Sheep-berry, [111](#)

Sheep Laurel, [85](#)

Shepherdia, [77](#)

Shepherd's Purse, [46](#)

Shingle Oak, [22](#)

Shin-leaf, [84](#), [85](#)

Shooting Star, [87](#)

Showy Lady's Slipper, [18](#)

Showy Orchis, [17](#)

Sicyos, [112](#)

Sida, [72](#), [73](#)

Silene, [33](#), [34](#)

Silphium, [122](#), [123](#), [127](#)

Silver Maple, [70](#)

Silver Weed, [55](#)

Silvery Cinquefoil, [66](#)

Silybum, [117](#)

Simarubaceae, [65](#)

Sium, [81](#)

Sisymbrium, [43-45](#), [47](#)

Sisyrinchium, [15](#)

Skullcap, [96](#)

Skunk Cabbage, [8](#)

Skunk Currant, [49](#)

Slender Nettle, [24](#)

Slippery Elm, [24](#)

Slough Grass, [5](#)

Small-flowered Crowfoot, [36](#)

Small-fruited Hickory, [21](#)

Small Snapdragon, [103](#)

Small Solomon's Seal, [11](#)

Smartweed, [27](#), [28](#)

Smilacina, [14](#)

Smilax, [11](#)

Smooth Alder, [22](#)

Snake Mouth, [18](#)

Snakeroot, Black, [82](#)

Snakeroot, Seneca, [66](#)

Snakeroot, Virginia, [25](#)

Snapdragon, [103](#)

Sneezeweed, [127](#)

Sneeze Weed, [128](#)

Sneezewort, [131](#)

Snowberry, [85](#), [111](#)

Soapberry Family, [70](#)

Soapwort, [34](#)

Solanaceae, [100](#)

Solanum, [100](#)

Solidago, [125-127](#)

Solomon's Seal, [11](#)

Sorbaria, [57](#)

Sorrel, Red, [25](#)

Sorrel, Green, [25](#)

Sorrel, Wood, [64](#)

Sour Dock, [26](#)

Sour Gum, [83](#)

Southernwood, [119](#)

Sparganiaceae, [2](#)

Spartina, [5](#)

Spear Grass, [7](#)

Spearmint, [98](#)

Spearwort, [36](#)

Speckled Alder, [22](#)

Specularia, [112](#)

Speedwell, [105](#)

Spergula, [32](#)

Spergularia, [32](#)

Spice Bush, [41](#)

Spiderwort, [9](#)

Spiderwort Family, [9](#)

Spikenard, [80](#)

Spike Rush, [7](#)

Spiraea, [53](#)

Spiranthes, [16](#)

Spirodela, [9](#)

Spotted Wintergreen, [85](#)

Spurge, [67](#), [68](#)

Spurge Family, [66](#)

Spurred Gentian, [89](#)

Spurrey, [32](#)

Spring Avens, [56](#)

Spring Beauty, [34](#)

Spring Vetch, [59](#)

Spruce, [1](#)

Squashberry, [111](#)

Squaw-root, [106](#)

Squaw Weed, [128](#)

Squirrel Corn, [41](#)

Squirrel-tail, [4](#)

Stachys, [99](#)

Staff Tree Family, [69](#)

Staphylea, [69](#)

Staphyleaceae, [69](#)

Star Flower, [87](#)

Star Grass, [14](#)

Star of Bethlehem, [14](#)

Starry Campion, [33](#)

Steironema, [88](#)

Stellaria, [32](#), [33](#)

Stemless Lady's Slipper, [18](#)

Stickseed, [94](#)

Stinging Nettle, [24](#)

Stitchwort, [32](#)

St. John's-wort, [73](#), [74](#)

St. John's-wort Family, [73](#)

Stone Clover, [61](#)

Stonecrop, [48](#)

Stonecrop, Ditch, [49](#)

Stork's-bill, [64](#)

Strawberry, [56](#)

Strawberry, Barren, [55](#)

Strawberry Blite, [29](#)

Streptopus, [11](#)

Striped Maple, [70](#)

Strophostyles, [58](#)

Stylophorum, [41](#)

Sugar Maple, [70](#)

Sumach, [68](#), [69](#)

Summer Grape, [72](#)

Summer Savory, [97](#)

Sundew, [48](#)

Sundew Family, [48](#)

Sundrops, [78](#)

Sunflower, [128-130](#)

Sunflower, Tickseed, [124](#)

Swamp Birch, [22](#)

Swamp Currant, [49](#)

Swamp Dewberry, [53](#)

Swamp Laurel, [85](#)

Swamp Milkweed, [91](#)

Swamp Rose, [52](#)

Swamp Valerian, [111](#)

Swamp White Oak, [23](#)

Sweet Birch, [22](#)

Sweetbrier, [52](#)

Sweet Cicely, [82](#)

Sweet Fern, [20](#)

Sweet Flag, [8](#)

Sweet Gale, [21](#)

Sweet Gale Family, [20](#)

Sweet White Violet, [76](#)

Sweet William, [34](#), [92](#)

Sweet William Catchfly, [34](#)

Sycamore, [51](#)

Symphoricarpos, [110](#), [111](#)

Symphytum, [93](#)

Symplocarpus, [8](#)

Synthyris, [102](#)

Syringa, [88](#)

Taenidia, [81](#)

Tall Dock, [26](#)

Tamarack, [1](#)

Tanacetum, [119](#)

Tansy, [119](#)

Tansy Mustard, [43](#)

Taraxacum, [115](#)

Taxaceae, [2](#)

Taxus, [2](#)

Tear-thumb, [28](#)

Teasel, [112](#)

Teasel Family, [112](#)

Tephrosia, [60](#)

Teucrium, [97](#)

Thalictrum, [39](#)

Thaspium, [81](#)

Thistle, [117](#), [118](#)

Thistle, Canada, [118](#)

Thistle, Cotton, [118](#)

Thistle, Globe, [117](#)

Thistle, Milk, [117](#)

Thistle, Russian, [28](#)

Thistle, Sow, [117](#)

Thlaspi, [46](#)

Thorn-apple, [100](#)

Thorny Amaranth, [30](#)

Three-seeded Mercury, [66](#)

Three-square, [7](#)

Thuja, [1](#)

Thymelaeaceae, [77](#)

Thyme, Wild, [97](#)

Thymus, [97](#)

Tiarella, [51](#)

Tickseed, [123](#), [124](#)

Tickseed Sunflower, [124](#)

Tick Trefoil, [62](#), [63](#)

Tilia, [72](#)

Tiliaceae, [72](#)

Timothy, [4](#)

Tipularia, [16](#)

Toad-flax, [24](#), [103](#)

Tobacco, Indian, [113](#)

Tobacco, Wild, [101](#)

Tofieldia, [14](#)

Toothwort, [44](#)

Touch-me-not, [71](#)

Touch-me-not Family, [71](#)

Tradescantia, [9](#)

Tragopogon, [115](#)

Trailing Arbutus, [85](#)

Tree of Heaven, [65](#)

Trientalis, [87](#)

Trifolium, [60](#), [61](#)

Triglochin, [3](#)

Trillium, [12](#)

Triosteum, [109](#)

Trollius, [36](#)

Tsuga, [1](#)

Tulip Tree, [40](#)

Tumble Weed, [30](#)

Turk's-cap Lily, [12](#)

Turtlehead, [104](#)

Tussilago, [122](#)

Tway-blade, [17](#)

Twig Rush, [8](#)

Twin-leaf, [40](#)

Twin Flower, [109](#)

Twisted-stalk, [11](#)

Typha, [2](#)

Typhaceae, [2](#)

Ulmus, [24](#)

Umbelliferae, [80](#)

Umbrella-wort, [31](#)

Upland Boneset, [122](#)

Urtica, [24](#)

Urticaceae, [23](#)

Utricularia, [106](#)

Uvularia, [12](#)

Vaccinium, [85-87](#)

Valerian, [111](#)

Valeriana, [111](#)

Valerianaceae, [111](#)

Valerianella, [111](#)

Valerian Family, [111](#)

Valerian, Greek, [92](#)

Vallisneria, [3](#)

Velvet Leaf, [72](#)

Venus' Looking Glass, [112](#)

Verbascum, [101](#)

Verbena, [99-100](#)

Verbenaceae, [99](#)

Verbena Family, [99](#)

Vernonia, [122](#)

Veronica, [104](#), [105](#)

Vervain, [99](#), [100](#)

Vetch, [59](#)

Vetchling, [59](#)

Vetch, Milk, [60](#)

Viburnum, [111](#)

Vicia, [59](#)

Vinca, [90](#)

Viola, [75](#), [76](#)

Violaceae, [75](#)

Violet, [75](#), [76](#)

Violet Family, [75](#)

Violet, Green, [75](#)

Virginia Creeper, [71](#)

Virginia Mallow, [73](#)

Virginia Snakeroot, [25](#)

Virgin's Bower, [35](#)

Vitaceae, [71](#)

Vitis, [72](#)

Wahoo, [69](#)

Waldsteinia, [55](#)

Walnut, Black, [21](#)

Walnut Family, [21](#)

Water Arum, [9](#)

Water Cress, [45](#)

Water Crowfoot, [37](#)

Water Dock, [26](#)

Water Hemlock, [82](#)

Water Hemp, [30](#)

Water Hoarhound, [95](#)

Water-leaf, [93](#)

Water-leaf Family, [93](#)

Water Lily, [35](#)

Water Lily Family, [35](#)

Water Lobelia, [113](#)

Water Loosestrife, [77](#)

Water Marigold, [123](#)

Water Milfoil, [79](#)

Water Milfoil Family, [79](#)

Water Parsnip, [81](#)

Water Pennywort, [80](#)

Water Pimpernel, [87](#)

Water Plantain, [3](#)

Water Plantain Family, [3](#)

Water Purslane, [78](#)

Water Shield, [35](#)

Water Smartweed, [27](#), [28](#)

Water Speedwell, [105](#)

Water Starwort, [68](#)

Water Starwort Family, [68](#)

Water-weed, [3](#)

Water Willow, [107](#)

Waterwort, [74](#)

Waterwort Family, [74](#)

Weeping Willow, [20](#)

White Ash, [88](#)

White Campion, [33](#)

White Cedar, [1](#)

White Clover, [61](#)

White Elm, [24](#)

White Fringed Orchis, [19](#)

White Ground Cherry, [101](#)

White Lady's Slipper, [18](#)

White Mulberry, [24](#)

White Oak, [23](#)

White Pine, [1](#)

White Poplar, [19](#)

White Snakeroot, [122](#)

White Spruce, [1](#)

White Sweet Clover, [61](#)

Whitlow Grass, [42](#), [46](#)

Whorled Pogonia, [18](#)

Wild Bean, [58](#)

Wild Bergamot, [96](#)

Wild Camomile, [130](#)

Wild Carrot, [82](#)

Wild Chives, [13](#)

Wild Columbine, [36](#)

Wild Comfrey, [94](#)

Wild Crab, [54](#)

Wild Cucumber, [112](#)

Wild Flax, [63](#)

Wild Geranium, [64](#)

Wild Ginger, [25](#)

Wild Gromwell, [94](#)

Wild Indigo, [61](#)

Wild Leek, [13](#)

Wild Lettuce, [115](#), [116](#)

Wild Lily of the Valley, [13](#)

Wild Mint, [98](#)

Wild Onion, [13](#)

Wild Pansy, [76](#)

Wild Parsnip, [81](#)

Wild Pea, [59](#)

Wild Plum, [54](#)

Wild Potato Vine, [92](#)

Wild Radish, [44](#), [45](#)

Wild Rice, [5](#)

Wild Rose, [52](#)

Wild Rye, [6](#)

Wild Sarsaparilla, [80](#)

Wild Senna, [60](#)

Wild Thyme, [97](#)

Wild Tobacco, [101](#)

Wild Yam, [14](#)

Willow, [20](#)

Willow Family, [19](#)

Willow Herb, [79](#)

Willow, Water, [107](#)

Winter Cress, [44](#)

Wintergreen, [86](#)

Wintergreen, Flowering, [65](#)

Wintergreen, One-flowered, [85](#)

Wintergreen, Spotted, [85](#)

Witch Grass, [5](#)

Witch Hazel, [51](#)

Witch Hazel Family, [51](#)

Withe-rod, [111](#)

Wolfberry, [110](#)

Wolffia, [9](#)

Wood Anemone, [38](#)

Wood-grass, [6](#)

Wood Lily, [12](#)

Wood Nettle, [24](#)

Wood Rush, [10](#)

Wood Sage, [97](#)

Wood Sorrel, [64](#)

Wood Sorrel Family, [64](#)

Wood Strawberry, [56](#)

Wood Violet, [76](#)

Wormseed, [29](#)

Worm-seed Mustard, [43](#)

Wormwood, [119](#)

Xanthium, [114](#)

Xyridaceae, [9](#)

Xyris, [9](#)

Yam Family, [14](#)

Yam, Wild, [14](#)

Yard Grass, [6](#)

Yarrow, [130](#)

Yellow Alyssum, [42](#), [46](#)

Yellow Birch, [21](#)

Yellow Cress, [43](#)

Yellow-eyed Grass, [9](#)

Yellow-eyed Grass Family, [9](#)

Yellow Foxtail, [4](#)

Yellow Fringed Orchis, [19](#)

Yellow Ironweed, [128](#)

Yellow Lady's Slipper, [18](#)

Yellow Lily, [12](#)

Yellow Mignonette, [48](#)

Yellow Monkey Flower, [102](#)

Yellow Oak, [23](#)

Yellow Sweet Clover, [61](#)

Yellow Violet, [76](#)

Yellow Willow, [20](#)

Yew Family, [2](#)

Zannichellia, [2](#)

Zanthoxylum, [65](#)

Zizania, [5](#)

Zizia, [81](#)

FOOTNOTES:

[1] In most flowers the corolla is the conspicuous portion of the perianth, and is composed of united or separate petals, as the case may be. In some flowers the corolla is absent, and the calyx is the conspicuous portion.

[2] About 30 species of *Potamogeton* occur in Michigan, among which the most conspicuous is *Potamogeton natans*, with elliptical floating leaves. For the identification of the species the Manual must be used.

Transcriber's Notes:

Page xlv: Changed lobel to lobed

Page 10: Changed ocntains to contains

Page 39: Changed second 38a to 38b

Page 89: Changed second 4a to 4b

Page 94: Changed augustifolium to angustifolium

Page 108: Changed second 1a to 1b

Page 109: Changed --21. to --20.

Page 122: Changed (springi) to (spring)

Page 130: Changed autum to autumn

Page 146: Changed Eveyrlasting, Pearly to Everlasting, Pearly

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